

Lexicographic Orders of Well-Orderings of the Surreal Numbers

Set approximations, the singular birthday transition, set-length words,
class orders, and higher-order structure

Research report merged from four independent manuscripts
prepared for Vladimir Reshetnikov (sources 08, 09, 11 and 12 of batch 80)

Sources dated 2 October 2026 (source 11: 3 October 2026, UTC); merged 2 October 2026

Abstract

The phrase *all well-orders of the surreal numbers* has several inequivalent meanings. Well-orders of a fixed birthday fragment form a set; well-orders of arbitrary set-sized subsets form an ordinary class; well-orders of the whole surreal line are themselves classes, and their totality belongs to a higher logical level. We develop these settings and distinguish set-like from unrestricted class well-orders.

For an infinite set-sized alphabet X , the existing ProveIt theory gives $|\mathcal{W}(X)| = 2^{|X|}$ and the exact ordinal embedding spectrum $\beta < (|X|)^+$. We apply it to surreal birthday fragments. For arbitrary set-length surreal words, we give an explicit sign-sequence embedding into $\mathbf{No}$, with birthday bounds, and identify the one-sided cuts created by limit-length words.

For the whole surreal line, we define lexicographic comparison directly on labeled class well-orders, without assuming ordinal enumerations or a class of class order types. Every such well-order has a canonical initial segment of type $\mathbf{Ord}$; its remaining tail determines a convex fiber. In the set-like fragment, bounded-support permutations of one global enumeration form an ordinary class isomorphic to $\mathbf{No}$, and interpolate every uniformly set-indexed strict cut of the entire fragment. The fragment contains every class linear order, admits binary class-word coding, and cannot have an exhaustive uniform class of set-codes. The unrestricted order has adjacent pairs, all arising from finite residual alphabets. An omitted-range construction exhibits a proper-class cut filled by a non-set-like well-order. In a full inaccessible-universe interpretation, the unrestricted order does not embed into its set-like fragment.

The proofs use $\mathbf{ZF}$, $\mathbf{ZFC}$, or $\mathbf{GBC}$, as specified, with a further uniform nonembedding theorem in $\mathbf{KM} + \mathbf{GC}$. We give universe-relative and proof-assistant formulations, and propose further research on the higher-order structure.

[merge] This report merges four manuscripts written independently on 2 October 2026 on the same question and delivered together in batch 80: source 11 (the base, whose abstract is printed above), and sources 12, 09 and 08. They prove the same core up to four times. Each shared theorem is printed once; the other sources' versions are recorded in numbered notes, and their genuinely different arguments are kept as marked routes. Source 12 adds an exact dichotomy at birthday cutoffs: for an infinite cardinal κ and $\mu = 2^{<\kappa}$, the minimal-length enumeration order $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is equimorphic with the binary cube $\mathbb{B}_{\kappa \cdot \kappa}$ (ordinal product) exactly when κ is a singular strong limit, and with $\mathbb{B}_\mu$ otherwise, with $\kappa = \beth_\omega$ as an unconditional example. It also adds a second direct comparison of all class well-orders, by their largest common labelled prefix, and an exact external identification of both global orders in a full-class model at an inaccessible cardinal. Source 09 adds bounded reflection of set-indexed families, an explicit unfilled $(\mathbf{Ord}, \mathbf{Ord})$ configuration, and the embeddability in $\mathbf{No}$ of every uniformly class-coded family. Source 08 adds a separator of exactly prescribed birthday, the failure of finite support, and relation-code

lexicography of all class well-orders. Source 11's choiceless proof answers source 12's question on the weak-choice strength; the direct comparisons of sources 11 and 12 answer source 08's question on comparing non-set-like class well-orders; and source 12's dichotomy answers in part a question of the repository's report on the reals. The whole set-sized general-alphabet layer re-proves that report and is printed with pointers to it, never as new.

Status. This is a mathematical research report with proofs, not a claim of machine verification or bibliographic priority. Existing general set-alphabet results are credited to the repository. The further constructions are proved here and should be evaluated by their arguments. The repository audit was targeted; no Lean build or remote change was made.

[merge] **Status of the merged report.** AI-assisted and unrefereed; not formalized. Sources 12 and 09 call themselves AI-assisted in their title pages; sources 08 and 11 do not use the word (author lines: 08 and 09 "Research manuscript prepared for Vladimir Reshetnikov", 11 "Research report", 12 "Research report prepared for Vladimir Reshetnikov"). None of the four claims historical priority, peer review or machine verification. Their finite checks verify finite instances only, never a transfinite or class-theoretic statement. No statement of this report is formalized; the formal status of the inputs it uses is in [section 1.5](#).

Contents

1	About this report	5
2	The objects being studied	14
3	Relationship to ProveIt and the literature	15
4	The other sources' introductions	16
5	Surreal order facts and foundational conventions	20
I	Sets of surreals	29
6	Well-order codes on a fixed set alphabet	29
7	Birthday cutoffs and their set-sized orders	37
8	Birthday cutoffs and the singular transition (source 12)	40
II	Set-length words	42
9	Set-length surreal words embed explicitly in the surreal line	42
10	Local characters and set suprema of word orders	43
11	The complete obstruction to filling a set cut	44
12	Saturation at a regular stage	47
III	Global choice and all class well-orders	48
13	Global choice and well-ordering the surreal class	48
14	Lexicographic comparison without class ordinals	52
15	The canonical initial set-like part	55
16	All jumps arise from finite residual permutations	57
17	Non-set-like well-orders and alternative coding (source 08)	59
18	Numerically well-ordered subclasses and other interpretations (source 12)	61
IV	Set-like global well-orders	61
19	The lexicographic structure of global enumerations	62

V	Foundations, models and formalization	79
20	Formalizing the subject without changing its meaning	79
21	A full-class model that connects the two size levels (source 12)	83
22	A concrete universe enlargement: all objects become sets (source 09)	85
23	Universes, class models, and paradox avoidance (source 08)	86
24	A formalization architecture with explicit size boundaries (source 12)	88
25	Formalization strategies and paradox avoidance (source 09)	89
26	A Lean formalization plan tied to ProveIt (source 08)	92
VI	Set-sized strata: topology, cut spectra and block types	94
27	Density and cellularity at every fixed type (source 16)	95
28	Countable incompleteness and global characters (source 16)	99
29	The cut spectrum of the alphabet $\text{No}_{<\kappa}$ (source 17)	100
30	The minimal slice $\mathcal{W}_\mu(\text{No}_{<\kappa})$ (source 17; routes of sources 15, 16 and 18)	103
31	Longer strata: every cardinal-block type (source 15)	116
32	Set-sized baselines and warnings in sources 13–18	122
33	What this Part answers	126
	Questions, conclusions and status	127
34	Research questions	127
35	The sources' conclusions	136
36	Results, assumptions, and verification status	138
A	Source crosswalk and reproducibility	140
B	The finite checks as shipped	143
C	The other sources' title pages and abstracts	144
D	Crosswalk of the sources' numbered statements	146

1 About this report

1.1 Four manuscripts, one report

[merge] All four manuscripts arrived in commit 4e270aa46 and were placed in ccc046989 (batch 80, cluster K3). Their batch-80 manuscript numbers are their source numbers here.

Source	Manuscript (archive)	Pin and date line	Labels and files
11 (base)	<i>Lexicographic Orders of Well-Orderings of the Surreal Numbers: Set Approximations, Class Orders, and Higher-Order Structure</i> ; 2,807 lines, 35-page PDF (surreal_well_orders (1).zip)	ae7c1e6aa (2 October 2026, 18:03:53 PDT = 3 October 2026, 01:03:53 UTC); dated “3 October 2026” (the UTC date)	prefix swo::; files 11-raw-orders-*
12	<i>Well-Orders of the Surreal Numbers and Their Lexicographic Geometry: Class-sized common prefixes, exact embedding spectra, a singular-cardinal transition, and size-safe formalization</i> ; 1,689 lines, 24-page PDF (surreal_well_orders.zip)	63bb8b1d9 (2 October 2026, 18:01:32 PDT); dated 2 October 2026	prefix swo:st::; files 12-singular-*
09	<i>Well-orders of the Surreal Numbers: Lexicographic universality, a surreal dense core, and the boundary between sets and classes</i> ; 1,677 lines, 24-page PDF (Surreal_Well_Orders_Research.zip)	63bb8b1d9; dated 2 October 2026	prefix swo:ec::; files 09-core-*
08	<i>Lexicographic Orders of Surreal Well-Orderings: Set-sized spectra, global enumeration orders, and a surreal-sized dense skeleton</i> ; 2,022 lines, 26-page PDF (Surreal_Well_Orders_Research (1).zip)	ae7c1e6aa; dated 2 October 2026	prefix swo:sk::; files 08-skeleton-*

The two pairs of archives sharing a name (08 and 09, 11 and 12) are not editions of one another. Every pair of the four texts has word 8-gram overlap of 0.6–1.1 % (preamble boilerplate), no member file is shared, the pins differ within each pair, and the result sets differ. They are four independent manuscripts on one spine; none supersedes another, and all four are merged (union).

Source 11 is the base. Its global-choice equivalence is proved over GB without set choice, the weakest base theory of the four ([theorem 13.1](#)); it covers set alphabets, set-length words, and set-like and unrestricted class well-orders; and it alone attributes the general set-alphabet theorems to the repository’s report on the reals, [SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/](#) (cited below as the reals report, labels lwo:).

1.2 How the four texts are printed

[merge] Source 11’s text is printed in full and in its own order. The other sources’ material is inserted where it belongs.

- Their introductions ([section 4](#)) and foundational sections are printed whole, as sections headed “Source *N*”.
- A result that a printed result already proves by the same argument is recorded in a numbered *Note*: the note gives the source’s statement, its number in that source, and what differs

(hypotheses, strength, extra remarks). Prose of the source that makes a point not made elsewhere is kept in the note.

- A genuinely different proof is printed in full as a marked route (“Source 12’s route”, “second proof”). A result proved only by one source is printed in full where it belongs, with its source named in the heading or the first sentence.
- Every label of every source survives, with the source’s prefix (`swo:` for 11, `swo:st:` for 12, `swo:ec:` for 09, `swo:sk:` for 08). Numbered statements without a label in their source carry `swo:sub:nnumber`, for example `swo:st:n10.4` for source 12’s Corollary 10.4. [Section D](#) maps every numbered statement of the four sources to its place here.
- Text written for the merge is marked `[merge]`. Dated notes of 2 October 2026 record where one source answers another ([section 1.7](#)).

Statements and sections are numbered by their place in this report; source numbers appear in headings, notes and [section D](#). Apart from the renamings of [section 1.3](#) and the corrections listed in [section 1.6](#), the sources’ text is verbatim. Statements of the reals report and of other repository reports are cited by label name, as the sources do; those reports are not reproduced.

1.3 One notation for four sources

`[merge]` The sources’ symbols collide. The following convention is fixed before any of them is used; every renaming is listed, and the tempting false reading is printed where confusion is likely.

Here	Meaning, and the sources’ symbols
GB, GBC	GB has a ZF set part, elementary class comprehension and class replacement, and <i>no</i> choice principle, as in the foundations report (<code>found:sub:gbconvention</code>) and the collection’s notation guide; GBC = GB + GC. Sources 09 and 12 analyse the necessity of choice “over GB plus set-level AC” (09, proposition 13.7 ; 12, theorem 13.2); that theory is written GB + AC in the merge text. Source 11 works over GB itself. <i>False reading:</i> the letters GB in sources 09 and 12 do not include set choice here.
$\text{bd}(x)$	Birthday, the length of the sign sequence (08, 09; the foundations and birthday-cutoff reports). Source 11’s $\ell(x)$ and source 12’s $b(s)$ are renamed $\text{bd}(x)$, $\text{bd}(s)$. In sources 08 and 09 the letter b is the baseline enumeration, never a birthday.
$\mathbf{No}_{<\kappa}$	$\{x \in \mathbf{No} : \text{bd}(x) < \kappa\}$, the birthday cutoff (foundations report $\mathbf{No}_{<\alpha}$). Renamed: source 08’s X_λ , X_κ ; source 09’s X_θ (its $\kappa_\theta = \mathbf{No}_{<\theta} $); source 11’s X_κ ; source 12’s S_κ . The reals report writes $S_{<\theta}$. <i>False reading:</i> source 12’s S_κ is $\mathbf{No}_{<\kappa} = S_{<\kappa}$, not $S_{<\kappa+}$.
$\ell_2(L)$	Binary coding length $\min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$, as in the reals report (<code>lwo:def:binary-length</code>). Source 12 prints the same symbol ℓ_2 and calls it the “binary coding rank” (its macro was named <code>\rk</code>); its prose is kept. $\text{rank}(x)$ is the von Neumann rank of a set (source 11’s <code>\rk</code>), used only in the proof of theorem 13.1 . <i>False reading:</i> the “rank” of source 12 is ℓ_2 , not rank.
$\mathcal{W}(X)$, $\mathcal{W}_\alpha(X)$	All exhaustive well-order enumerations of a set X (lengths below $ X ^+$), and those of length exactly α , compared at the first unequal entry in the given order of X (reals report’s $\mathcal{W}$, $\mathcal{W}_\lambda$). The minimal slice is $\mathcal{W}_{ X }(X)$. Source 11’s $\mathcal{P}_\lambda(X)$ is renamed $\mathcal{W}_\lambda(X)$; source 12’s script $\mathscr{W}$ is printed $\mathcal{W}$.

Here	Meaning, and the sources' symbols
$\mathcal{I}(X), \mathcal{I}, \mathcal{T}$	$\mathcal{I}(X)$: injective words of length below $ X ^+$ over a set X (source 11). $\mathcal{I} = \mathcal{I}(\mathbf{No})$: the class of all injective set-length surreal words, proper prefixes first (source 11's $\mathcal{I}$; source 08's $\mathcal{I}(\mathbf{No})$; source 09's $\mathcal{T}$, renamed). $\mathcal{T}$: all set-length words, not necessarily injective (source 11's $\mathcal{T}$). <i>False reading</i> : source 09's $\mathcal{T}$ was the injective class. The reals report's $\mathcal{I}$ (injections $\kappa \rightarrow \mathbb{R}$) is unrelated.
$\mathfrak{W}_{\text{sl}}(\mathbf{No}), \mathfrak{W}_{\text{all}}(\mathbf{No})$	Set-like global well-orders of $\mathbf{No}$, equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$; and all class well-orders of $\mathbf{No}$. Both are predicates on classes ("metanotation"), never classes of classes. Source 12's $\mathcal{E}(\mathbf{No})$, written $\text{SL}(\mathbf{No})$ in its macros, and its $\mathcal{W}_{\text{cl}}(\mathbf{No})$; source 09's $\mathcal{E}$; source 08's "virtual" $\mathcal{E}$ are renamed. $\mathfrak{J}$ is source 11's collection of class injections $\text{Ord} \rightarrow \mathbf{No}$.
$\mathbb{B}_\gamma, \mathbb{B}_{\text{Ord}}$	$\mathbb{B}_\gamma = (\{0, 1\}^\gamma, <_{\text{lex}})$ (reals report, 08, 12). $\mathbb{B}_{\text{Ord}}$: class binary words on Ord compared at their least disagreement (source 08's $\mathbb{B}_{\text{Ord}}$; source 12's $\mathfrak{B}$, renamed; source 11 writes $\{0, 1\}^{\text{Ord}}$).
$\simeq_e, \hookrightarrow$	Equipmorphism (embeddings both ways; not isomorphism) and order embedding. Source 11's $\approx_{\text{emb}}$ is printed $\simeq_e$.
$\mathcal{P}_{\text{bd}}(e_0), \mathcal{P}_s(b)$	The <i>core</i> : enumerations that agree with a fixed baseline enumeration outside a set of positions. Source 11 codes them by normalized permutations $\sigma : \lambda \rightarrow \lambda$ ($\mathcal{P}_{\text{bd}}(e_0)$, definition 19.3); sources 09 and 08 by fixed-point-free bijections $\sigma : S \rightarrow S$ of a set S of ordinals (source 09's $\mathcal{C}_b$, source 08's $\mathcal{P}_b$, both written $\mathcal{P}_s(b)$ here; source 08's code carrier $\mathcal{P}$ is $\mathcal{P}_s$, and its power-set $\mathcal{P}(V_\kappa)$ keeps the letter $\mathcal{P}$). The two codings have the same evaluations: restrict the identity extension of a support code to $\lambda = \sup\{\xi + 1 : \xi \in S\}$, and conversely discard fixed points. With the same baseline, every order statement about one is a statement about the other.
e_0, B, b	The fixed baseline class bijection $\text{Ord} \rightarrow \mathbf{No}$ of sources 11, 12, and 09 and 08 respectively.
$d(F, G), \Delta(E, F), \delta(e, f)$	The least ordinal at which two class functions on Ord differ (sources 11, 09, 08).
$\eta_{\text{Ord}}, \eta_\kappa$	The set-cut property, and its version for cuts of fewer than κ elements, empty sides allowed (11, definition 5.1 ; 08, definition 5.8); source 09 writes "set-cut property". The subscript is a cardinal bound, not an $\aleph$ -index.
Termination	Sign sequences compare with $- < \text{termination} < +$ (numerical order); words compare with proper prefixes first. Exhaustive enumerations of one set are prefix-free, so no convention is needed there.
(S08. n)	Source 08's displayed equations (1)–(13) keep their numbers as (S08.1)–(S08.13). (S08.10), an inequality inside a proof recorded in note 19.13 , is described there, not displayed.

1.3.1 Addendum of 3 October 2026: notation for sources 13–18

[merge] The six batch-81 sources ([section 1.8](#)) reuse the symbols above with other meanings, and several reuse this report's macro names for other objects. The convention below is fixed before any of their material is printed; it applies to Parts VI–IX and changes nothing in Parts I–V. Every renaming is listed; the tempting false reading is printed where confusion is likely.

Here	Meaning, and the batch-81 sources' symbols
κ, μ, r, q	κ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} = \mathbf{No}_{<\kappa} $, $r = \text{cf } \kappa$ and $q = \text{cf } \mu$, as in source 17. Renamed: source 18's ν, τ (printed r, q ; in source 18's general principle, ν stays the point character of an abstract order D); source 14's $\tau = \text{cf } \kappa$ (printed r); source 15's $\delta = \text{cf } \kappa$ in its saturation section (printed r). Source 16 writes κ for the cardinality of a general alphabet and θ for the cutoff; in its text these are printed μ and κ . Sources 13–18's X_κ and source 16's X_θ are printed $\mathbf{No}_{<\kappa}$. <i>False reading:</i> r is a cofinality, never a coding length (next row).
$X, X = \mu$	A general alphabet: an infinite linearly ordered set, of cardinality μ (source 18's convention; source 16's κ).
$\rho_\kappa, \mathbb{B}_\sigma \times_{\text{lex}} m$	Source 15's r_κ ($\kappa \cdot \kappa$ at a singular strong-limit cutoff, μ otherwise; by theorem 8.3 it is $\ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$), renamed to keep $r = \text{cf } \kappa$; and source 15's $D_{\sigma,m} = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}$, the binary cube with a finite chain of m points as last coordinate, renamed because source 15's core is $\mathcal{D}$.
d, cell, w	Source 18's letter ρ for a regular cardinal is printed θ , as in source 17. Density, cellularity (a supremum, not necessarily attained) and weight of an order topology, as in the reals report; sources 15–18 write d, c, w . Cellularity is never written c . Source 15's $d(a, b)$ (a first-difference coordinate) and source 18's $d_* = \mu^{<\mu}$ keep their letters.
ci, characters	ci is coinitality. The left and right characters of a point x are $\text{cf}(L_{<x})$ and $\text{ci}(L_{>x})$ (sources 17, 18; source 15 writes $\text{cf}^-(f)$, $\text{ci}^+(f)$). A point character, the global cofinality of the order, and the character of a gap are three different invariants (source 18's warning).
GapChar(L), Reg	The set of characters ($\text{cf } A, \text{ci } B$) of the gaps (A, B) of L , a gap having nonempty sides with no endpoints (principal cuts and the external ends are excluded). Source 17's $\Sigma(L)$ and source 18's $\text{GapSpec}(L)$ are printed with the reals report's symbol GapChar; the report surreal-fields-across-universes writes GapSpec. Reg is the class of infinite regular cardinals (source 17's Reg_∞ ; the reals report writes $\text{Reg}(<\kappa)$ for those below κ).
η_ν	Source 17: for regular ν , every cut $A < B$ with $ A , B < \nu$ is filled; sources 15 and 18: every cut with $ A + B < \nu$ (resp. $ A \cup B < \nu$) is filled, for any infinite cardinal ν . Empty sides are allowed throughout. For regular ν the three definitions agree with this report's η_κ (section 1.3), and every threshold statement printed concerns the regular cardinal r .
$P, [p], C_p$; straddling	$P = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is the minimal slice (sources 17, 18; source 15 writes L). For an injective prefix p , $[p]$ (source 17) and C_p (sources 16, 18, and source 11 in Part IV) are the same cylinder. A cylinder that meets both sides of a cut <i>straddles</i> it; this is the word of sources 16, 17 and 18 and is used in all batch-81 Parts. Sources 13 and 14 say that a prefix <i>crosses</i> a cut, source 15 that it is <i>persistent</i> ; these words are explained where printed.
$\mathcal{T}, \mathcal{I}, \mathcal{T}^{\text{inc}}$	All set-length words, injective set-length words, numerically increasing set-length words. Source 16's $\mathcal{T}, \mathcal{T}^{\text{inj}}, \mathcal{T}^{\text{inc}}$, whose macro <code>\Words</code> names <i>all</i> words while this report's <code>\Words</code> names the injective ones; source 13's $\mathcal{T}, \mathcal{I}$ agree with this report. Source 18's centered classes $\mathcal{C}, \mathcal{C}_{\text{inj}}, \mathcal{C}_{\text{inc}}$ are printed $\mathcal{T}_c, \mathcal{I}_c, \mathcal{T}_c^{\text{inc}}$ (its $\mathcal{C}$ would clash with $\mathcal{C}_b$). Source 16's termination types $T_{\min}, T_\kappa$ are printed $\mathbf{T}_{\min}, \mathbf{T}_\kappa$. <i>False reading:</i> source 16's $\mathcal{T}$ is not the injective class.

Here	Meaning, and the batch-81 sources' symbols
$\mathfrak{W}_{\text{sl}}(\mathbf{No}), \mathfrak{W}_{\text{all}}(\mathbf{No})$	As in section 1.3 . Renamed: $\mathfrak{E}$ (sources 13, 14, 16, 17) for set-like global orders (source 15 already writes $\mathfrak{W}_{\text{sl}}$); source 14's $\mathfrak{W}$ (macro <code>\Hwell</code>), source 16's $\mathfrak{H}$ and source 17's $\mathfrak{W}$ (macro <code>\HW0</code>) for all class well-orders (sources 13 and 15 already write $\mathfrak{W}_{\text{all}}$). Source 18's macros <code>\Hsl</code> , <code>\Hall</code> print script $\mathcal{H}_{\text{sl}}$, $\mathcal{H}_{\text{all}}$ there and $\mathfrak{W}_{\text{sl}}$, $\mathfrak{W}_{\text{all}}$ here.
$\mathfrak{J}$	Class injections $\text{Ord} \rightarrow \mathbf{No}$, as source 11's $\mathfrak{J}$ (sources 13, 14, 17 write $\mathfrak{J}$ with the macro <code>\J</code> ; source 16 writes $\mathfrak{J}$ with the macro <code>\I</code> ; source 15's $\mathfrak{J}$). <i>False reading</i> : source 16's <code>\I</code> is not this report's $\mathcal{I} = \mathcal{I}$, the class of injective set-length words.
$\mathcal{P}_{\text{bd}}(b), \mathcal{P}_{\text{s}}(b)$	The core, as in section 1.3 , in the source's own coding: normalized permutations $\mathcal{P}_{\text{bd}}(b)$ for sources 15 ($\mathcal{D}$) and 16 ($\mathcal{P}_b$); fixed-point-free support codes $\mathcal{P}_{\text{s}}(b)$ for sources 13 ($\mathcal{D}$), 14 ($\mathcal{P}_b$), 17 ($\mathcal{C}_b$) and 18 ($\mathcal{C}_b$, macro <code>\Pbd</code>). <i>False reading</i> : source 18's <code>\Pbd</code> is the support coding, not this report's $\mathcal{P}_{\text{bd}}(e_0)$; the two codings have the same evaluations.
$\mathcal{P}_{<\kappa}(b)$	Source 14's support-cardinality layer: codes whose support has cardinality less than κ . <i>False reading</i> : it is not $\mathcal{P}_{\text{bd}}$, whose subscript means bounded (set-sized) support; source 14 itself warns that a cardinality filtration and an ordinal-height filtration must not share the word "bounded".
$\simeq_e, <_{\text{lex}}$	Equimorphism is $\simeq_e$ for all six sources (source 16's $\simeq_{\text{emb}}$; source 18's macro <code>\equim</code> prints $\rightleftharpoons$ there and $\simeq_e$ here). Source 14's macro <code>\lex</code> prints the relation $<_{\text{lex}}$; this report's <code>\lex</code> is the subscript lex , so source 14's relation is printed $<_{\text{lex}}$. Source 18's bold $\mathbf{B}_\gamma$ is printed $\mathbb{B}_\gamma$.
Cuts of the core	A normalized class cut (L, U) has a lower side L without greatest element; the cut of an injective branch F is written L_F (source 16's L_F , source 17's L_f ; source 14's D_F , $D_p^\pm$, $D_{p,H}$ and source 13's (L, U) are explained where printed).
$(\text{SNN}.n)$, labels	Displayed equations of source NN keep their numbers as $(\text{SNN}.n)$, as for source 08. Labels: <code>swo:fs</code> : (13, replacing its <code>swf</code> :), <code>swo:tc</code> : (14), <code>swo:cb</code> : (15, keeping its internal <code>cce</code> : and <code>new</code> :), since dropping them would merge two of its labels), <code>swo:ds</code> : (16), <code>swo:gs</code> : (17), <code>swo:sh</code> : (18, keeping its internal <code>wv</code> :); unlabelled numbered statements get <code>swo:sub:nnumber</code> ; labels written for the merge in Part VI begin <code>swo:vi</code> .

1.4 What the report claims, and what it does not

[merge] The report claims the written proofs printed below, each in the foundation its statement names (ZF, ZFC, GB, GB + AC, GBC, KM + GC, or ZFC with a strongly inaccessible cardinal). The results new against the repository are: the singular transition at birthday cutoffs ([theorem 8.3](#)); the choiceless global-choice equivalence ([theorem 13.1](#)); the direct comparison of all class well-orders ([theorem 14.2](#), [theorem 14.7](#)), its canonical initial part and convex fibres, and its adjacency criterion ([theorems 15.1](#) and [16.2](#) and [proposition 15.3](#)); the core of bounded-support perturbations and its interpolation theorem ([theorem 19.10](#) and [corollary 19.15](#)); binary-class equimorphism and the diagonal non-coding theorems ([theorems 19.32](#), [19.37](#) and [19.38](#)); omitted ranges ([theorems 19.42](#) and [19.43](#) and [corollary 19.46](#)); bounded reflection ([theorem 19.30](#)); the failure of finite support ([proposition 19.31](#)); relation-code lexicography ([proposition 17.1](#)); explicit word coding and the word-cut classification ([theorems 9.1](#) and [11.1](#)); and the universe models and non-embedding theorems ([theorems 20.1](#), [20.2](#), [21.1](#) and [22.1](#)). None of these is claimed to be historically first.

It does *not* claim the following; every limitation stated by any source is kept.

- *No novelty for re-proofs.* The set-sized general-alphabet layer (prefix-freeness, cardinality $2^{|X|}$,

universality, no $|X|^+$ -chain, ordinal and cube spectra, coding length $|X|^+$, extrema, adjacency, $n!$ blocks, strict cubes) re-proves the reals report for every infinite linear alphabet and is printed with pointers to it (section 6.1). Universality and homogeneity of **No** are classical (Ehrlich [15]; Hamkins [20]); all four sources say so. The coding of sets behind the choice equivalence is, in source 09’s words, a standard well-founded-coding argument; the connection is discussed by Hamkins [18, 19].

- *Source 11* (its Status paragraph, its delivery README’s “Scope and verification” and its audit): no machine verification, no Lean build, no new Lean file; no bibliographic priority for constructions proved in the report; research questions are proposed directions, not a certified list of published open problems; the repository audit was targeted and repository-wide correctness was not assessed; the finite checks do not verify transfinite proofs, cardinal arithmetic or class comprehension. Its README states: “The mathematical proofs underwent a separate proof review.” That is the delivery’s own statement; no such review is recorded in the repository, and this report does not rely on it.
- *Source 12* (title page, its Section 1.2, and 12-singular-RESEARCH_STATUS.md, “What is not claimed”): AI-assisted, unrefereed; historical priority not claimed; no previously catalogued open problem is claimed settled, nor that its exact formulations are new; no new Lean or Rocq verification; the Python checks are not substitutes for the proofs; the bounded theorem is not a construction of a full class model at a singular cardinal; the full collection of class well-orders is not treated as a set or ordinary class at the same foundational level; no universal semantic decoder for formula codes is assumed; its nine research questions are proposed continuations; its source search was not sufficient to establish originality.
- *Source 09* (title page, its Appendix A, and 09-core-RESEARCH_STATUS.md, “Important boundaries”): AI-assisted, not peer-reviewed or Lean-verified, no historical priority; classical ingredients and repository results are distinguished from its constructions; no class whose members are proper classes is formed; the global-enumeration representation covers set-like class well-orders only, and no complete classification of the relation-code comparison is claimed; the core realizes uniform set-sized cuts, not all proper-class cuts (an explicit counterexample is supplied); for fixed alphabets the absence of κ^+ and its reverse is not claimed to be a sufficient embeddability criterion; automorphisms and higher-level cuts are not classified; its research directions are proposals; literature and repository searches were targeted, not exhaustive.
- *Source 08* (its Status box and delivery README): explicit derivations, not a claim of historical priority; not independently refereed and not checked in Lean; no new Lean formalization; repository declarations are separated from proposed extensions; repository sources were read but not rebuilt; the finite tests do not establish any transfinite or class-theoretic theorem; every use of “class” is relative to the stated foundational model.
- *Proposed Lean modules are proposals.* The module names in the Lean plans of sections 20 and 24 to 26 do not exist in the repository; the existing declarations they cite do (section 1.5).
- *Review status.* The repository’s Hilbert’s-tenth research tree holds a written-proof review of the four archives (Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md, commit c338541e3). It found their principal arguments sound under the stated foundations, with one scoped correction to source 08, which is applied here (section 5.6), and it reran the four finite checks. A second note there authenticates the placement (review_surreal_placement_ccc046989.md, commit 3aa123856). That review is pinned to the delivered archives and does not cover the text written for this merge; it is not an entry of the surreal collection’s review record, where this

report’s review is pending. ([merge] Added 3 October 2026: the same tree has since reviewed the merged text as well. Its synthesis review (review_surreal_synthesis_0be9b9134.md, commit 280583d2f) read every passage written for the merge and found no mathematical defect under the stated hypotheses, but two overstatements, both corrected here in place: the choice recursion of [theorem 13.1](#) had been called a “coherent” sequence of well-orders in [remark 13.4](#) and [remark 34.15](#), and the README had understated the largest delivered file. Its preservation audit (review_surreal_transfer_0be9b9134.md, commit afb706742) accounts for all 166 formal statements of the four sources (124 printed with normalized-exact bodies, 41 as duplicate-result notes, one question inside an answered remark). Neither is an entry of the collection’s review record either.)

1.5 Formal status

[merge] No statement of this report is formalized, and no `swo:` label has a Lean mapping in the collection’s formalization ledger (`Algebra/SurrealNumbers/docs/FORMALIZATION.md`). Placement in the surreal collection, beside the Lean foundations of `Algebra/SurrealNumbers/Surreal/Foundations/`, confers no formal status.

Four inputs that the sources use *are* proved in Lean, in the universe-relative form recorded in that ledger, where a “set” of surreals is a family indexed by a `Small.{u}` type and the sign carrier lives in `Type (u+1)`:

- properness and bounded birthdays (found:prop:proper; [proposition 5.4](#), first three clauses): `SignSequence`, `birthdays_bounded`, `small_bounded`, `not_small` in `SignSequence.lean`;
- filling of small cuts (found:sub:cutdata): `small_cut_fillers` in `SignSequenceCut.lean` and `exists_separator_of_small_sets` in `SignSequenceCutOperation.lean`;
- a separator of exactly a prescribed birthday above all option birthdays, [theorem 5.6](#): `exists_cut_separator_of_birthday_lt` in `SignSequenceCut.lean`, by the same route as source 08’s proof (the complete lexicographic lattice of Boolean words of one length, which is [lemma 5.5](#), there taken from Mathlib);
- the inconsistency of unrestricted cuts (found:prop:allcuts): `noUnrestrictedCuts` in `SizeOfInstructions.lean`.

The other declarations the sources cite exist at the current revision: `ordinalOrderEmbedding`, `small_strict_upper_bounds`, `simpler_wellFounded`, `existsUnique_simplest_separator`, `cut_realizes`, `cut_isPrefix`, `birthday_cut_le`, `birthday_cut_le_iSup`, `cut_congr`, `SmallCutData`, `HasSmallCutFillers`, `not_small_of_small_cut_fillers`, `lt_iff`, and the others in their tables. None of them states a result of this report.

1.6 Where the merge had to choose

[merge]

- *Base and order.* Source 11, for the reasons in [section 1.1](#); its section order is kept, and member material follows the base result it duplicates or extends.
- *Strongest statement printed as the statement.* The choice equivalence is source 11’s (over GB; sources 12 and 09 assume set choice, and source 09 treats set-like orders only). The diagonal theorem is printed twice: source 11’s ([theorem 19.37](#), rows of set codes) and source 12’s ([theorem 19.38](#), every uniform family of class well-orders, with a set-like witness); sources 09’s and 08’s are notes (source 08’s indexes by Ord only). The birthday-cutoff statement is

source 12's dichotomy; source 08's regular case is a note.

- *Different routes kept in full:* source 09's stabilization proof of the ordinal spectrum; source 12's route to the set-sized spectra through binary coding length; the second and third proofs that a set can be avoided (lemmas 5.3 and 5.10); source 12's and 09's proofs of the choice equivalence over $\mathbf{GB} + \mathbf{AC}$; source 12's largest common labelled prefix; source 12's residual decomposition and adjacency criterion; source 12's concatenation proof of the binary-class embedding; source 12's prefix extension through the complement of the prefix; source 09's support coding of the core; source 09's explicit unfilled configuration; source 09's general universality and homogeneity theorems; source 12's birthday-bounded universality.
- *Kept whole:* each member's introduction, foundational section, universe and model sections, Lean plan, research questions, conclusion, ledger and reproducibility record.
- *Corrections to source text.* The renamings of section 1.3; one mistyped reference in source 11 (a proposition cited as "Theorem", in section 12); source 08's predecessor citation (section 4.3 and the note after it); source 12's question on the weak-choice strength, printed as an answered remark (remark 34.15); source 08's question 11 annotated as answered; one sentence of source 08 on well-foundedness replaced by the correction of the Hilbert's-tenth review (section 5.6 and the note before it). Statements that name delivery files (build scripts, `finite_checks.py`, `surreal_well_orders.tex`) are kept and annotated with the shipped names (section B).

1.7 Where one source answers another

[merge] Dated notes, 2 October 2026.

- Source 12's question on the weak-choice strength* (remark 34.15, and its remark 13.3): answered by source 11. Over $\mathbf{GB}$ without set choice, a class well-order of $\mathbf{No}$ exists exactly when global choice holds (theorem 13.1); see remark 13.4. The same theorem shows that the set-choice hypothesis of source 09's proposition 13.7 is unnecessary. Source 09's direction on the exact axiomatic strength of *other* universality and homogeneity assertions (section 34.3) is unaffected.
- Source 08's question 11, comparison of non-set-like class well-orders* (section 34.4): answered by sources 11 and 12. Theorems 14.2 and 14.7 compare all class well-orders of a labelled class directly, with elementary comprehension, without $\mathbf{ETR}$ and without comparing abstract class order types; this also supplies the single comparison that source 08's section 17 says does not follow by aligning domains (see the note after it). Source 09's relation-code direction concerns a different comparison and stays open.
- Source 08's question 3, singular birthdays and code lengths:* answered in part by source 12's theorem 8.3: at a singular cutoff the least binary coding length of the fixed-length enumeration order is $\kappa \cdot \kappa$ when κ is a strong limit and $2^{<\kappa}$ otherwise. Which cuts fail at singular cutoffs stays open.
- Source 08's question 2, invariants at regular cutoffs:* partly answered by source 11's proposition 12.1 (the minimal slice is η_κ , with every global and local cofinality and coinitality κ); density, cellularity and the cut cofinalities beyond these stay open.
- The reals report's question "Other ground orders"* (its Section "Further research questions", `two:sec:research`, which names "set-sized suborders of surreal numbers"): answered in part by theorem 8.3 for $X = \mathbf{No}_{<\kappa}$ and the minimal stratum ($n = 0$): $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\mu$, except at singular strong-limit κ , where it is $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$ and not $\simeq_e \mathbb{B}_\kappa$. It stays open for $n > 0$, for longer strata, and for other alphabets (source 12's research question 34.11, source 08's question 4).
- Hamkins's question whether universality of the surreal line is a weak global choice principle*

[21]: not settled here, as source 12 says after its [remark 34.15](#).

[merge] Dated note, 3 October 2026 (batch 81). Items (c) and (d) are completed and item (e) is extended by Part VI: the cut spectrum at singular cutoffs, the density, cellularity and cut cofinalities at regular cutoffs, and the longer strata $\mu \cdot \gamma + n$ of $\mathbf{No}_{<\kappa}$ ([section 33](#)).

1.8 The batch-81 addition: six further manuscripts

[merge] (3 October 2026.) Six manuscripts of batch 81, all dated 3 October 2026 and all continuing this report, are merged into it as one addition, sources 13–18. Sources 13–17 arrived in commit 9a4ce14e9 and were placed in f0cd7032d (batch 81L); source 18 arrived in bd599ac06 and was placed in c4720e1b2 (batch 81L2). Their local numbers continue this report’s sequence; the batch-81 manuscript number is in the second column.

Src.	B81	Manuscript (archive), pages	Pin	Labels, files
13	01	<i>Lexicographic Orders of Surreal Well-Orderings: Further Structure</i> ; 33 pp. (Surreal_Well_Orders_Further_Study.zip)	0ecd158fd	swo:fs;; 13-products-*
14	02	<i>Lexicographic Well-Orders of the Surreals II: cut classification, support thresholds, and class-model absoluteness</i> ; 33 pp. (lexicographic_surreal_well_orders_II.zip)	0ecd158fd	swo:tc;; 14-trichotomy-*
15	03	<i>Lexicographic Orders of Surreal Well-Orders: interpretations, canonical blocks, represented cuts, and fixed-type spectra</i> ; 37 pp. (surreal_lexicographic_orders.zip)	0ecd158fd	swo:cb;; 15-blocks-*
16	04	<i>Lexicographic Orders of Surreal Well-Orders: Cuts, Canonical Blocks, and the Limits of Set-Sized Coding</i> ; 33 pp. (surreal_well_orders_further.zip)	491ed8411	swo:ds;; 16-strata-*
17	05	<i>Lexicographic Well-Orders of the Surreal Numbers, II: exact cut spectra, branch recognition, optimal support reserves, and automorphism-extension obstructions</i> ; 29 pp. (surreal_well_orders_further_study(1).zip)	0ecd158fd	swo:gs;; 17-gaps-*
18	06	<i>Lexicographic Well-Orders of the Surreal Numbers: Cuts, canonical class blocks, and changes of interpretation</i> ; 53 pp. (surreal_lexicographic_orders_further_study.zip)	83befe707	swo:sh;; 18-shuffle-*

The six are independent texts, not editions of one another. Every pair shares 0.6–1.4% of word 8-grams (preamble, this report’s title and address, bibliography), no member file is shared, and the theorem sets differ. Sources 14 and 17 share only the inner directory name `surreal_well_orders_II` and a “II” in the title; source 18’s archive name extends source 15’s, but its pin, structure and results differ. None supersedes another; all six are merged (union). Sources 13–17 read this report at commit 0be9b9134, before the corrections of 0f52c0dca; source 18 read it at 83befe707, after them, and before sources 13–17 arrived, so it never saw them.

They overlap heavily with each other and with this report. The represented-cut theorem that answers source 11’s [research question 34.1](#) is proved in all six by one argument; all six re-prove the core layer of Part IV (fresh separators, completion, interpolation, the core isomorphic to $\mathbf{No}$, the diagonal theorem, direct comparison); and the set-sized layer re-proves the reals report. What is

new is a second wave of results on questions of this report: the topology, cut spectra and coding lengths of the set-sized strata at birthday cutoffs; the termination classification of word orders; the trichotomy, characters, automorphism orbits, absoluteness and support thresholds for cuts of the set-coded core; the canonical block decomposition of class well-orders with explicit condensation histories; product (non)absorption; a Σ_1^1 comprehension bound; and the countable models.

The addition is printed by theme, not by source, because the sources overlap: Part VI ([part VI](#)) prints the set-sized strata. The rules of [section 1.2](#) apply unchanged: each shared result is printed once, with every source that proves it named; the other versions are numbered notes, genuinely different proofs are marked routes, and every label of every source survives with the prefixes of [section 1.3.1](#). Re-proofs of results of this report or of other repository reports are printed as notes with pointers, never with a novelty claim.

Status. Sources 13, 14, 15 and 17 call themselves AI-assisted and unrefereed; source 18’s validation “consists of conventional proofs and independent AI-assisted proof review”; source 16 says its checks are “not external peer review”. None claims historical priority, peer review or a machine-checked formalization, and none is formalized here. The status files of sources 13, 14, 16 and 17 are shipped in this directory under their prefixed names.

2 The objects being studied

There are two orders in this problem. Each object under consideration is itself a well-order of some collection of surreal numbers. We then compare these objects lexicographically using the *numerical* order on their surreal labels. This second order need not be a well-order. Indeed, the orders below contain abundant descending sequences.

Write $L \hookrightarrow M$ for an order embedding and $L \simeq_e M$ for embeddings in both directions. Equimorphism does not imply isomorphism.

Object	Elements	Logical size and interpretation
$\mathcal{W}(X)$	Ordinal-indexed bijections onto a fixed set $X \subseteq \mathbf{No}$	A set; ordinary ZFC order theory
$\mathcal{I}(\mathbf{No})$	Injective set-length ordinal words in $\mathbf{No}$	A class of sets; well-orders of arbitrary set-sized subsets
$\mathfrak{W}_{\text{sl}}(\mathbf{No})$	Set-like class well-orders of all $\mathbf{No}$, equivalently class bijections $\text{Ord} \rightarrow \mathbf{No}$	A higher-order totality; use class variables or uniform families
$\mathfrak{W}_{\text{all}}(\mathbf{No})$	All class well-orders of all $\mathbf{No}$, including non-set-like ones	A higher-order totality; direct labeled comparison is needed

Theaktur notation is deliberate. It does not declare a class with proper classes as members. Statements about $\mathfrak{W}_{\text{sl}}$ and $\mathfrak{W}_{\text{all}}$ are formulas with class variables, or statements relative to a specified larger universe. An embedding of an ordinary class order into one of these totalities is a uniform family: one class relation encodes the object assigned to each set-valued index.

Principal answers. For a fixed infinite alphabet X , the exact ordinal ceiling is $(|X|)^+$. Passing to all set-length surreal words removes that ceiling: there is an explicit embedding of the entire class into $\mathbf{No}$, and the singleton words embed $\mathbf{No}$ in the other direction. The two orders are nevertheless not isomorphic, because limit-length words have set-sized left cofinality.

Under global choice, the set-like global fragment contains a uniformly indexed copy of every class linear order. This includes every set ordinal, Ord , reverse Ord , $\text{Ord} + \text{Ord}$, and the whole numerical

surreal order. Yet the entire fragment cannot be exhausted by one uniform class of set-codes. Class universality of a subfamily is compatible with a larger totality of class objects.

The unrestricted global order is different again. It has adjacent pairs and convex residual fibers. Every raw well-order begins with a canonical Ord-sequence; if that sequence omits labels, those labels form a later well-ordered tail. This decomposition permits a precise study of *all* well-orders without treating them all as set-like. It also supplies a nonembedding obstruction: the unrestricted order has too many disjoint two-point fibers to embed into the set-like order in the full containing-universe model. A related uniform definability obstruction is provable in $\text{KM} + \text{GC}$.

3 Relationship to ProveIt and the literature

We inspected the repository at the immutable revision

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The closest precursor is the report on lexicographic well-orderings of the reals [1], in the subdirectory `lexicographic-well-orderings-of-reals/` of the repository's ordinal-order reports. Its general alphabet theorems already establish the cardinality, universality, and ordinal cutoff for every infinite linearly ordered set X . Their application to birthday fragments is a transfer of existing results, not a new generic alphabet theorem.

That report also proves, with $\mathfrak{c} = 2^{\aleph_0}$,

$$\mathbf{No}_{<\theta} \hookrightarrow \mathcal{W}(\mathbb{R}) \quad (\theta < \mathfrak{c}^+), \quad \mathbf{No}_{<\mathfrak{c}^+} \not\hookrightarrow \mathcal{W}(\mathbb{R}).$$

The negative assertion follows from the all-plus chain of length $\mathfrak{c}^+$. It warns against taking an incoherent union of stagewise embeddings.

The repository's foundational report [3] supplies the relevant size discipline: individual surreals are set-coded, bounded fragments are sets, arbitrary class cuts need not be fillable, and quantifying over class functions does not create a class of those functions. Its local set-valued recursion arguments are directly useful. A nearby vector-space report uses the same distinction to avoid an unnecessary ETR hypothesis [5].

The sign-sequence presentation is standard [12, 14]. Absolute universality and homogeneity of the surreal order are part of the established theory discussed by Ehrlich [15]. We supply the short recursion arguments needed here.

The Kanovei–Shelah connection. There is a methodological connection, but the objects are different. Kanovei and Shelah's actual index consists of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary order on $\mathcal{P}(\mathbb{N})$ [16, Section 1]. The maps need not be injective and do not enumerate the entire real line. Their construction illustrates how a definable linear order on many codes can replace a definably selected well-order. The present problem shares that idea while introducing a different level of class-size complexity.

Contribution and attribution. The central developments here are explicit surreal-word compression, direct comparison of unrestricted class well-orders, their canonical initial-Ord decomposition, the bounded-support interpolation theorem, and the relation between omitted ranges and non-set-like tails. No claim about real-specific countability, separability, or gap spectra is transferred to surreal alphabets without proof.

4 The other sources' introductions

[merge] Each of the other manuscripts opens with its own statement of the problem, of its relation to ProveIt, and of the Kanovei–Shelah connection. These introductions are printed whole, one subsection each, followed by notes where the merge corrects or qualifies them. Their summaries of results are not repeated claims: every result they announce is printed, once, in the body.

4.1 Source 12: scope, predecessors, and the results at a glance

The phrase “well-orders of surreal numbers” hides several distinct objects. The usual numerical order of **No** is not a well-order. One may instead put an unrelated well-order on the entire surreal carrier, or consider numerical well-orders on subclasses, or work with an ordinary set of bounded-birthday surreals. This article treats all three viewpoints, but the principal object is the lexicographic comparison of *exhaustive well-orders of a fixed carrier*.

The supplied ProveIt repository contains a substantial predecessor, *Lexicographic Orders of the Well-Orderings of the Reals*, merging three reports and distinguishing the full enumeration order from its minimal-length slice [1]. The inspected README explicitly marks that report as unrefereed and not formalized. Its real-number spectra, binary coding, finite-tail geometry, and warnings about isomorphism versus equimorphism motivate the present work. The repository also contains a Lean development of simplest elements of convex sets of sign sequences [10]. These are useful foundations, not a formal verification of the new theorems below.

We pinned the audit to repository commit

63bb8b1d99359e7188abf0a33fb152e121858150.

The audit was targeted, not an exhaustive review of the repository. All essential arguments used here are written out, so no unreviewed research claim from the repository is required as a black box.

4.1.1 Three different orders

For an infinite set-sized linear order X of cardinality μ , define

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\}, \quad \mathcal{W}(X) = \bigcup_{\mu \leq \alpha < \mu^+} \mathcal{W}_\alpha(X).$$

They are ordered by their first unequal entry, using the given order of X , not the well-order being represented.

For the proper class **No**, let $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ denote, as a *predicate on classes*, the class bijections $E : \text{Ord} \rightarrow \mathbf{No}$. These represent exactly the set-like global well-orders. Let $\mathfrak{W}_{\text{all}}(\mathbf{No})$ denote the predicate of being an arbitrary class well-order relation on **No**. Neither displayed notation asserts the existence of a class having these class-sized objects as members.

The comparison on $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ uses a least ordinal of disagreement. The comparison on $\mathfrak{W}_{\text{all}}(\mathbf{No})$ uses a largest common *labelled* initial segment. The latter can be a proper class, and this is essential rather than a cosmetic extension.

4.1.2 What is proved, and what is not being claimed

The principal developments are the elementary common-prefix theorem (theorem 14.7), the global-choice equivalence (theorem 13.2), the non-enumerability theorem (theorem 19.38), and the singular transition (theorem 8.3). The full and minimal bounded spectra are proved in sections 6.2 and 8.

The difference between the dense set-like order and finite-tail geometry in the full order is proved in [sections 16.1](#) and [19.2](#).

These are mathematical claims with supplied proofs. We do not claim that they settle a previously catalogued open problem, that their exact formulations are historically new, or that they have been independently refereed. The proposed research questions at the end are precisely that: questions suggested by the results, not an assertion of a complete literature census.

4.1.3 The Kanovei–Shelah connection

Kanovei and Shelah order maps $a : \mathfrak{c} \rightarrow \mathcal{P}(\mathbb{N})$ whose ranges are ultrafilters, using a fixed lexicographic order on $\mathcal{P}(\mathbb{N})$ [[16](#), §1]. Their maps need not be bijections, and their index is not the literal collection of all well-orders of $\mathbb{R}$, still less all class well-orders of **No**. The useful common idea is to compare representations at their first difference instead of choosing one canonical representative. Their index construction remains set-sized. Moving to all global surreal well-orders creates an additional size level that their construction does not remove.

[merge] *On source 12’s introduction.* Source 12 read the reals report’s README and the Lean file `SignSequenceSimplicity.lean`, not the reals report’s article; it therefore re-proved the set-sized patterns it needed. They are printed here as a marked route with pointers to the reals report ([sections 6.1](#) and [6.2](#)). The reals report’s article has not changed since source 12’s pin. Source 12’s principal developments are printed as [theorem 14.7](#) (a second route to [theorem 14.2](#)), [theorem 13.2](#) (a second route to [theorem 13.1](#), over $\text{GB} + \text{AC}$), [theorem 19.38](#) (printed beside [theorem 19.37](#)) and [theorem 8.3](#) (new).

4.2 Source 09: the question and the principal conclusions

A surreal number is a set-sized object, but the totality **No** of surreal numbers is a proper class. Consequently, taking “all well-orders” changes logical level when one passes from $\mathbb{R}$ to **No**. The main task is not merely to replace the real alphabet by a larger alphabet. It is to specify which objects are being ordered and which collections of those objects are available in the foundational theory.

Throughout, the word *embedding* means an order-preserving and order-reflecting injection. It need not preserve cuts, suprema, the order topology, birthdays, simplicity, or field operations. An order is *equimorphic* with another when each embeds into the other; this is weaker than isomorphism.

4.2.1 Three different carriers

For a set $X \subseteq \mathbf{No}$, let $\mathcal{W}(X)$ consist of all bijections $e : \alpha \rightarrow X$, with α an ordinal. These are exactly the exhaustive enumerations associated with well-orders on the fixed labeled set X . For different e, f , comparison takes place at the first unequal value. Because both are exhaustive, neither can be a proper initial segment of the other.

Let $\mathcal{I}$ instead contain every set-length injective word $p : \alpha \rightarrow \mathbf{No}$, allowing its range to vary. Here proper-prefix pairs do occur, so an end-of-word convention is necessary. We use the convention that a proper prefix comes first. This is ordinary dictionary order, *not* the convention used to compare surreal sign sequences.

Finally, let $\mathfrak{W}_{\text{sl}}$ be informal notation for the global class bijections $E : \text{Ord} \rightarrow \mathbf{No}$. Each represents a set-like global well-order. In **GBC**, $\mathfrak{W}_{\text{sl}}$ is notation for a condition on class variables, not a class whose members are these class functions. A precise, ordinary class substitute is the set-coded core $\mathcal{P}_s(b)$ of eventual perturbations of a fixed $b : \text{Ord} \rightarrow \mathbf{No}$.

Main global conclusion. In GBC, for every fixed global enumeration b ,

$$(\mathcal{P}_s(b), <_{\text{lex}}) \cong (\mathbf{No}, <).$$

The core is order-dense among all global enumerations and even supplies interpolants for every uniformly set-indexed separated cut involving arbitrary global enumerations. Nevertheless, there is no uniform class-indexed enumeration of *all* global enumerations, and certain proper-class-sized separated families have no interpolant at all.

4.2.2 How this extends the repository’s real-line investigation

The inspected ProveIt snapshot contains a substantial merged report on lexicographic well-orderings of the reals [1]. That report studies ordinal spectra, binary coding, finite tails, local characters, and gaps. Its set-sized perspective is useful here, but its continuum bounds cannot simply be applied to the entire surreal class. We reprove the set-sized arguments needed below for an arbitrary linearly ordered set, without importing unverified transfinite claims from that report.

The genuinely global development is organized around four concrete constructions: bounded-support extension of an injective prefix; set-indexed lexicographic interpolation; an eventually-equal surreal core; and diagonalization at independent pairs of positions. These constructions are the main contribution of this manuscript. Whether particular formulations have appeared previously would require a more exhaustive literature search; they are not presented as established historical firsts.

Kanovei and Shelah’s nonstandard-model construction provides motivation for lexicographically ordering redundant codes. Their actual index is not the collection of all real well-orders: it consists of maps from the continuum cardinal to $\mathcal{P}(\mathbf{N})$ whose ranges are ultrafilters, ordered at their first disagreement [16, §1]. This distinction remains essential in any proposed surreal analogue.

[merge] *On source 09’s introduction.* Source 09 read the reals report’s README and the Lean files `SignSequence.lean` and `SignSequenceCutOperation.lean`. Its set-sized arguments, which it “reproves ... without importing unverified transfinite claims”, are printed as notes and one marked route (section 6.3). The four constructions it calls its main contribution were found independently by the other sources on the same day: bounded-support prefix completion also by sources 11 and 08 (lemma 19.4 and note 19.8); set-indexed interpolation by sources 11, 12 and 08 (theorem 19.10 and notes 19.11 and 19.13); the eventually-equal core by sources 11 and 08 (corollary 19.15 and note 19.17); and the pair-switch diagonalization by sources 11, 12 and 08 (theorems 19.37 and 19.38 and note 19.41). No priority among the sources is implied by the order of printing. The box’s last sentence is theorem 19.42 and theorem 19.43.

4.3 Source 08: the question, disambiguated

The phrase “well-orders of surreal numbers” has two independent ambiguities. First, is the carrier a set of surreal numbers or the whole proper class? Second, is a well-order required to be the restriction of the numerical order, or is it a new ordering of the labelled carrier? A third choice enters when the well-orders themselves are compared: which data are read lexicographically?

This paper uses the following principal convention. An arbitrary well-order on a labelled carrier is represented by its exhaustive increasing enumeration *in that well-order*. Two enumerations are then compared at their first unequal entry, using the *numerical* order of their surreal values. The enumerations themselves need not be numerically increasing. This distinction is indispensable.

For example, the two lists $(0, 1, -1)$ and $(0, -1, 1)$ represent two well-orders of the same three-element set. The second list is smaller in our lexicographic order, since $-1 < 1$. Both underlying relations have ordinal type 3. Passing to ordinal isomorphism types would identify them and destroy the object being studied.

4.3.1 Five different carriers

Object	Elements or parameters	Foundational size
$\mathcal{W}(X)$	Exhaustive enumerations of one fixed set X	A set in ZFC
$\mathcal{I}(\mathbf{No})$	Injective ordinal-length, set-sized words in $\mathbf{No}$	A class of sets
$\mathfrak{W}_{\text{sl}}$	Class bijections $\text{Ord} \rightarrow \mathbf{No}$	A virtual collection of classes
$\mathcal{P}_s(b)$	Set codes for set-supported changes of a fixed b	A class of sets
Arbitrary class well-orders	Class relations, possibly not set-like	A virtual collection; another comparison rule is needed

A *virtual collection* is notation for a condition on class variables, not a new object whose members are proper classes. Thus “ $e \in \mathfrak{W}_{\text{sl}}$ ” abbreviates “ e is a class bijection from Ord onto $\mathbf{No}$.” All statements involving several such parameters can be translated into the two-sorted language of class set theory. Whenever a family of classes is used, it is represented by one uniform class relation. This convention is made precise in [section 19.1](#).

4.3.2 Main conclusions

The central distinction is between a very large ambient order and its class-sized dense skeleton. Fix a global enumeration $b : \text{Ord} \rightarrow \mathbf{No}$. Write $\mathcal{P}_s(b)$ for canonical set codes for permutations σ of Ord that equal the identity outside a set, ordered by the numerical lexicographic comparison of $b \circ \sigma$. We prove

$$(\mathcal{P}_s(b), <_b) \cong (\mathbf{No}, <). \quad (\text{S08.1})$$

Moreover, if uniformly given set-sized families L, R of arbitrary global enumerations satisfy $L <_{\text{lex}} R$, then some $b \circ \sigma$ strictly separates them. The separator is in the same fixed skeleton regardless of the supports, or lack of supports, of the input enumerations.

This yields a useful answer to the embedding question. Every *set* linear order, every ordinal, and indeed every *class linear order whose elements are sets* embeds into $\mathcal{P}_s(b)$. In particular, Ord , $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, their reversals, $\mathbf{No}$, and each set-indexed lexicographic power of $\mathbf{No}$ embed. This does not say that the full virtual order $\mathfrak{W}_{\text{sl}}$ is itself a class order or is isomorphic to $\mathbf{No}$.

At a fixed set-sized carrier X , the answer is different. If $|X| = \kappa$ is infinite, then

$$|\mathcal{W}(X)| = 2^\kappa, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+. \quad (\text{S08.2})$$

Thus making the carrier larger changes the allowed ordinal spectrum; there is no legitimate substitution of a “cardinal $|\mathbf{No}|$ ” into [\(S08.2\)](#).

4.3.3 Relationship to existing work

The set-sized part extends the exhaustive-enumeration viewpoint of the prior working manuscript on the continuum [2]. Its results are reproved here in a form that does not assume separability of the alphabet. The standard surreal background is the sign-sequence and cut-filling viewpoint associated

with Conway, Gonshor, and Alling [11, 12, 13]; Ehrlich’s work places the surreal continuum in a broader universality framework [15].

The immediate formalization base is the inspected ProveIt sign-sequence construction and its small-cut theorem [7, 8]. We do not need the repository’s field arithmetic or stronger research claims. The resemblance to the Kanovei–Shelah definable hyperreal construction is real but limited: their actual index consists of fixed-length maps whose ranges are ultrafilters, not literally all well-orders of the reals [16]. We return to this distinction in [section 4.4](#).

[merge] *On source 08’s predecessor.* The “prior working manuscript on the continuum” that source 08 cites [2] is batch-76 manuscript 02, *Lexicographic Orders of Well-Orderings of the Continuum*, which is now the base source 02 of the repository’s reals report (placed in 2a04b60f2, written in 718bfa3b5 and 603f6ed31). Source 08 did not read that report. Its set-sized results re-prove the reals report’s general-alphabet theorems, which hold for every infinite linear alphabet, separable or not; they are recorded with pointers in [sections 6.1](#) and [6.4](#). The display (S08.1) is [note 19.17](#), and (S08.2) is [notes 6.21](#) and [6.23](#).

4.4 Source 08: the Kanovei–Shelah connection, carefully stated

Kanovei and Shelah’s definable nonstandard model uses an index A of maps

$$a : \mathfrak{c} \longrightarrow \mathcal{P}(\mathbb{N})$$

whose ranges are ultrafilters. They compare these maps lexicographically using the binary characteristic-function order on subsets of $\mathbb{N}$. Repetitions are allowed; the index is not a well-order [16]. It is therefore not literally the order of all bijective well-order enumerations of $\mathbb{R}$.

The relevant common idea is that ordinal-indexed codes can be linearly compared at their first unequal value even when no definable choice of one representative from each coded object is supplied. Our paired codes, terminal codes, and virtual relations explore that order-theoretic idea in the surreal setting. They do not provide a class ultrapower or a transfer theorem.

An extension to a class-sized ultrapower would need separate decisions about admissible predicates, function families, equivalence-class coding, truth or satisfaction, and the exact size of saturation. Neither a dense index order nor a surreal-sized skeleton settles these questions. In particular, the set-cut interpolation theorem is an order-theoretic theorem, not a claim of saturation for every language or every class-sized family of types.

[merge] The Kanovei–Shelah distinction is made in the same terms by all four sources (source 11 in [section 3](#), source 12 in [section 4.1](#), source 09 in [section 4.2](#), source 08 above) and by the reals report itself, which analyses the actual index (`lwo:eq:KS`, `lwo:sec:KS-topology`, `lwo:rk:sec:KS`). The uniform constructions it suggests are research questions here ([research question 34.10](#); source 09’s and source 08’s last directions).

5 Surreal order facts and foundational conventions

5.1 Signs, birthdays, and set cuts

A surreal number is represented by a sign sequence

$$s : \alpha \longrightarrow \{-, +\}$$

of set-ordinal length α , its birthday $\text{bd}(s)$. Numerical comparison at the first disagreement uses

$$- < \text{termination} < +.$$

This differs from dictionary order with every proper prefix smaller than every extension. A sign sequence can be on either side of an extension, depending on its next sign.

For an ordinal θ , let

$$\mathbf{No}_{<\theta} = \bigcup_{\alpha < \theta} \{-, +\}^\alpha.$$

It is a set. The full $\mathbf{No}$ is proper-class-sized, because all-plus sequences embed all ordinals. Every set of surreals has bounded birthdays by Replacement.

Definition 5.1. A class linear order L has the *set-cut property*, called η_{Ord} here, if for all sets $A, B \subseteq L$ with $A < B$ there is x with $A < x < B$. Empty sides are allowed. For a cardinal κ , the set-sized notation η_κ requires the same property when $|A \cup B| < \kappa$.

The surreal cut construction supplies this property for $\mathbf{No}$ [14, Section 2]. A separator can be chosen canonically as the unique simplest separator. This statement concerns *sets* of options; applying it to the whole class against the empty right side would be inconsistent.

Lemma 5.2 (Deleting a set). *If $S \subseteq \mathbf{No}$ is a set, then $\mathbf{No} \setminus S$ has the set-cut property. Every nonempty open interval of $\mathbf{No}$ is a proper class.*

Proof. Let $A < B$ be sets in $\mathbf{No} \setminus S$. First choose x with $A < x < B$, and put

$$S_- = \{s \in S : s \leq x\}, \quad S_+ = \{s \in S : x < s\}.$$

Then $A \cup S_- < B \cup S_+$. A separator of this new set cut avoids both parts of S . The argument allows empty sides. If an interval (a, b) were a set S , applying the result to $\{a\} < \{b\}$ in $\mathbf{No} \setminus S$ would be impossible. $\square$

Choosing both separators by simplicity makes the deletion argument uniform in the set parameters without global choice.

[merge] *Third route to the first clause of lemma 5.2 (source 12, Lemma 6.2).* Source 12 avoids a forbidden set by using that open intervals of $\mathbf{No}$ are proper classes.

Lemma 5.3 (Avoiding a forbidden set). *If $A < B$ are sets of surreals and D is a set, then some separator z of A, B lies outside D .*

Proof. Choose u between A and B , and then v between $\{u\}$ and B . Every point of (u, v) separates A, B , and that interval is a proper class. It cannot be contained in D . With a fixed global well-order, choose the first acceptable point, making the operation uniform. $\square$

[merge] Source 08 opens with its own toolkit, printed next. Its [proposition 5.4](#) restates `found:prop:proper` (Lean-proved in universe-relative form; [section 1.5](#)) and adds that neither the numerical order nor “birthday first” well-orders $\mathbf{No}$, which source 12 also proves ([section 13.1](#)). Its [theorem 5.6](#) sharpens the birthday bound of the cut theorem to a separator of *exactly* any prescribed birthday above the option birthdays; this is proved in Lean as `exists_cut_separator_of_birthday_lt`, by the same complete-lattice argument ([lemma 5.5](#)). Its [corollary 5.7](#) contains the first clause of [lemma 5.2](#) with a prescribed birthday, and its [definition 5.8](#) is [definition 5.1](#).

5.2 Source 08: surreal order and size, the minimal toolkit

We take a surreal number to be a sign sequence

$$s : \alpha \longrightarrow \{-, +\}, \quad \alpha \in \text{Ord}.$$

Its birthday is $\text{bd}(s) = \alpha$. To compare two sequences, extend them by a termination symbol 0 and compare at the first disagreement using $- < 0 < +$. This is the numerical order. The simplicity relation is proper initial-segment extension, and is not the numerical order.

Proposition 5.4. *Every birthday-bounded collection of surreals is a set, every set of surreals has bounded birthdays, and $\mathbf{No}$ is a proper class. Neither the numerical order nor “birthday first, then numerical order” is a well-order of $\mathbf{No}$.*

Proof. For a fixed ordinal λ , the sign sequences of length below λ form the set $\bigcup_{\alpha < \lambda} \{-, +\}^\alpha$. Replacement bounds the birthdays of any given set. Conversely, the all-plus sequences give a distinct surreal for every ordinal, so a set containing all surreals would bound all ordinals. The negative integers already descend numerically. For the second claim, let s_n have birthday ω , with n initial minus signs followed by plus signs forever. At coordinate n one has $s_{n+1}(n) = - < + = s_n(n)$, so $s_0 > s_1 > s_2 > \dots$ within a single birthday level. $\square$

5.2.1 A birthday-controlled separator theorem

We record a proof of the exact order-theoretic fact used later. It also explains the repository’s bounded-word implementation [8].

Lemma 5.5. *For every set ordinal β , the binary lexicographic order $\mathbb{B}_\beta = (\{0, 1\}^\beta, <_{\text{lex}})$ is a complete lattice.*

Proof. The least and greatest elements are the constant 0 and 1 words. Given a set A of words, form w recursively. At coordinate ξ , put $w(\xi) = 1$ exactly when some $a \in A$ agrees with the already constructed prefix $w \upharpoonright \xi$ and has $a(\xi) = 1$; otherwise put 0. If some $a \in A$ exceeded w , its first disagreement would contradict the rule for a 0 coordinate. If $v < w$, then at their first disagreement ξ we have $v(\xi) = 0$, $w(\xi) = 1$, and a witness $a \in A$ agreeing with w through that 1 coordinate. Thus $v < a$ and v is not an upper bound. This proves that $w = \sup A$, including $A = \emptyset$. Infima follow by reversing all bits. $\square$

Theorem 5.6 (Birthday-controlled cut filling). *Let A, B be sets of surreals with $A < B$, meaning $a < b$ for all $a \in A$, $b \in B$. If β is an ordinal strictly greater than every birthday in $A \cup B$, then a surreal z of birthday exactly β satisfies $A < z < B$. Either side may be empty.*

Proof. For a shorter word x , let $u_\beta(x)$ be the word of length β obtained by appending a plus to x and then filling all remaining coordinates with minuses. It is the least full-length word strictly above x . Dually, let $v_\beta(x)$ be x followed by a minus and then pluses; it is the greatest full-length word strictly below x . These assertions follow by inspecting the first differing sign: an extension above x must choose plus when x terminates, and the least possible continuation then uses only minuses.

If $x < y$ and both have length below β , then $u_\beta(x) \leq v_\beta(y)$. Indeed $u_\beta(x) \leq y$ by the same first-disagreement check; equality is impossible because their birthdays differ. Applying the dual greatest-word property gives the claim. By lemma 5.5, full-length sign words have a supremum, and

$$z = \sup\{u_\beta(a) : a \in A\}$$

satisfies $a < u_\beta(a) \leq z$ for every $a \in A$ and $z \leq v_\beta(b) < b$ for every $b \in B$. All words in this complete lattice have birthday exactly β . If A is empty, its supremum is the all-minus word. The vacuous case $\beta = 0$ causes no problem. $\square$

Corollary 5.7 (Fresh separators). *If $A < B$ are sets and S is a set of forbidden surreals, a separator $z \notin S$ exists. It can be required to have any specified birthday strictly exceeding all birthdays in $A \cup B \cup S$.*

Definition 5.8. A class linear order is an η_{Ord} -order if every pair of sets $A < B$ has a strict separator, including empty sides. For an infinite cardinal κ , an η_κ -order satisfies the same condition when $|A \cup B| < \kappa$.

The notation η_κ here uses a cardinal bound directly; it should not be confused with conventions in which the subscript is the index α in $\aleph_\alpha$. By [theorem 5.6](#), **No** is an η_{Ord} -order. Every interval is a proper class: if (a, b) were a set, a separator of $\{a\} \cup (a, b)$ and $\{b\}$ would be a new member of the same interval above all its members. No set is cofinal or cointial in **No**.

5.3 Which recursion principle is used?

Here GB has a ZF set part, elementary class comprehension, and class replacement; choice is stated separately. We use $\text{GBC} = \text{GB} + \text{GC}$, which includes set choice. Cardinality calculations and cardinal-valued cofinality statements throughout the article are read in ZFC, or in the set part of GBC. The explicitly definable sign code and the direct set-cut constructions retain their stated choice-free strength.

Ordinary recursion along Ord, with a *set* output at every stage and a rule using the set of earlier values and fixed class parameters, is available in GB. On each set ordinal perform the usual recursion with those parameters. Uniqueness makes the local solutions agree; elementary comprehension assembles their graph.

This is different from arbitrary elementary transfinite recursion, ETR, whose stages may themselves be classes. The latter supports iterated truth predicates and is stronger than GBC [\[22\]](#). The prefix searches, partial permutations, and back-and-forth maps used here have set-sized stages.

[merge] The other sources fix their foundational conventions in their own sections, printed whole next. They agree with source 11’s conventions, with one difference of base theory: when it analyses choice, source 12 works over $\text{GB} + \text{AC}$ ([section 1.3](#)).

5.4 Source 12: the surreal carrier and the foundational conventions

5.4.1 Sign sequences

We use the sign-sequence presentation

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each individual sequence is a set. Its length $\text{bd}(s) = \text{lh}(s)$ is its birthday. At the first difference, signs and termination are compared by

$$- < \text{end} < +.$$

This is the usual numerical order, not the ordinary “shorter comes first” dictionary convention. Simplicity is the separate relation of being a proper sign prefix. The standard surreal construction and its simplicity structure are treated by Gonshor and Ehrlich [\[12, 15\]](#).

We shall use the following standard feature, also obtainable directly from the construction: whenever A, B are *sets* of surreals with $A < B$, there is a surreal z with $A < z < B$. There is a unique such separator of minimum birthday. Moreover, if every member of $A \cup B$ has birthday less than an ordinal γ , a separator can be constructed with birthday at most γ . This is interpolation, not Dedekind completeness: no least upper bound is being asserted.

Every set of surreals has a birthday bound, by Replacement. Every nonempty numerical interval contains a proper class of surreals. To see the latter without arithmetic, start with $a < b$ and recursively choose x_α between $\{a\} \cup \{x_\xi : \xi < \alpha\}$ and $\{b\}$. The previous values form a set at each stage. Thus one obtains distinct points for all ordinals α .

5.4.2 Well-orders and set-likeness

A class well-order on a class X is a strict linear class relation R for which every nonempty subclass has an R -least element. In GBC this agrees with the usual formulations using nonempty set subsets or absence of descending ω -sequences; the relationship is discussed in [27, §3]. We use strict relations throughout.

It is *set-like* if

$$\text{Pred}_R(x) = \{y \in X : yRx\}$$

is a set for every x . A proper-class set-like well-order is isomorphic to Ord : map x to the ordinal type of its set of predecessors. The range is an initial segment of Ord , and cannot be a set ordinal when X is proper. This construction uses only set-sized order types.

Not all class well-orders are set-like. The ordered sums $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, and $\text{Ord} + \text{Ord} + 1$ are examples. Their notation means ordered sums of class orders, not new von Neumann ordinals.

5.4.3 Foundational levels

Our set-sized results are theorems of ZFC. Our global results are in GBC, meaning Gödel–Bernays set theory with global choice. When analyzing the necessity of choice we work over GB plus ordinary, set-level AC. The full collection of global well-orders is handled by formulas with class variables, not by pretending that a proper class is an element of another GBC class.

Kelley–Morse theory allows stronger comprehension, but its classes still have sets, not proper classes, as members. Moving from GBC to KM does not by itself create a powerclass object of all subclasses of $\mathbf{No}$. One can instead use a larger set-theoretic universe, a set model with a specified collection of classes, or an explicitly higher-order theory. These are distinct interpretations and will be kept distinct.

5.4.4 Why an ordinal-length recursion here need not invoke ETR

A recursion on Ord that produces one *set* at each stage and uses the set of previous values is available by the ordinary recursion theorem with class parameters in GB. In contrast, ETR permits a genuinely class-valued recursive stage, represented by a class of pairs of a stage and an arbitrary set. Even a truth hierarchy brings stronger principles into play [27, §2]. Our forth constructions, block lengths, and canonical enumeration of a class with set-sized initial segments are set-valued recursions. We do not invoke a universal satisfaction predicate or unrestricted class-valued recursion.

[merge] Like source 12, source 09 works over $\text{GB} + \text{AC}$ when it analyses choice. Its lemma 5.10 is a second route to the first clause of lemma 5.2: it adds the forbidden points inside the cut to the left side, where source 11 first splits the forbidden set at a separator. Its bound (1) is `hset:eq:cut-bound` of the birthday-cutoff report [4], and its definition 5.9 is definition 5.1.

5.5 Source 09: foundational conventions and the surreal order

5.5.1 Sets, classes, and the meaning of a family

All set-sized results are in ZFC. Global results are in GBC, here meaning Gödel–Bernays class theory with global choice. Classes have sets as members. A proper class cannot be an element of another class. An expression such as “the set of class functions E_i ” is therefore replaced by a *uniform family*: a class relation

$$F \subseteq I \times \text{Ord} \times \mathbf{No}$$

whose i th section is the graph of E_i . The index I may be a set or a class of sets. If I is a set, this is a uniformly set-indexed family, not a literal set of proper classes.

This distinction does not prevent quantification over individual class variables. For example, “for every two class bijections E, F , there is a least ordinal where they differ” is a legitimate GBC assertion. What is unavailable is a class having all such bijections as members. Rangel and Mariano explicitly distinguish the set-subsets of a class from a nonexistent class of all its subclasses [17, §1.1.1].

Global choice will be used in its standard consequence that a proper class can be enumerated by Ord with set-sized initial segments. One way to obtain this consequence is to well-order each rank layer using a global choice function, concatenate the layers, and collapse the resulting set-like well-order. Restricting such an order to any proper subclass again gives order type Ord.

5.5.2 Sign sequences and the cut property

Use the sign-sequence realization

$$\mathbf{No} = \bigcup_{\alpha \in \text{Ord}} \{-, +\}^\alpha.$$

Each displayed element is a set function. Its domain is its *birthday*, denoted $\text{bd}(x)$. Numerical order compares the first different sign using

$$- < \text{termination} < +.$$

It is different both from ordering by birthday and from the dictionary order on arbitrary surreal-valued words used later. The all-plus word of length α represents the ordinal α .

The standard simplicity/cut theorem says that, for sets $A, B \subseteq \mathbf{No}$ with $A < B$, there is a unique simplest surreal strictly between them. Here $A < B$ means $a < b$ for every $a \in A, b \in B$, with empty sides allowed. We write this simplest separator as $\{A \mid B\}$. A useful birthday bound is

$$\text{bd}\{A \mid B\} \leq \sup\{\text{bd}(x) + 1 : x \in A \cup B\}. \quad (1)$$

These are standard sign-sequence facts [12]; the inspected repository also contains explicit small-cut and birthday-bound statements on its sign carrier [9].

In particular, every set of surreal numbers has strict upper and lower bounds. No set is cofinal or cointial in $\mathbf{No}$, and no set is the whole of $\mathbf{No}$. These assertions do not authorize a cut constructor with arbitrary proper-class option families.

Definition 5.9. A class linear order D has the *set-cut property* if every pair of sets $A, B \subseteq D$ satisfying $A < B$ has an element $d \in D$ with $A < d < B$. Empty sides are allowed.

The surreal order has the set-cut property. In terminology used in the literature this is an absolute or class analogue of an η -order. We use the explicit definition rather than relying on potentially ambiguous subscripts such as η_{Ord} .

Lemma 5.10 (Avoiding a prescribed set). *If D has the set-cut property and $U \subseteq D$ is a set, then $D \setminus U$ also has the set-cut property.*

Proof. Let $A, B \subseteq D \setminus U$ be sets with $A < B$, and put

$$U_0 = \{u \in U : A < u < B\}.$$

Then $A \cup U_0 < B$. Choose x with $A \cup U_0 < x < B$. This x is in the original open cut. If it belonged to U , it would belong to U_0 , giving $x < x$. Thus $x \notin U$. $\square$

5.5.3 The precise recursion strength used here

We repeatedly construct a sequence of *sets* indexed by ordinals, where the next value is a uniquely specified set obtained from the set-sized previous history and fixed class parameters. This recursion is available in GBC: set recursion with class parameters constructs all set-length approximations, uniqueness makes them compatible, and elementary class comprehension joins them.

This is not the principle ETR, whose stages can themselves be proper classes. The latter is genuinely stronger than GBC and must not be silently invoked by saying “perform class recursion” [24, §2]. Every global recursion in the proofs below is explicitly of the former, set-valued kind.

[merge] Source 08’s foundational section, printed next except for its Theorem 8.2, which is recorded in [note 5.12](#). One sentence is corrected, as proposed by the review of the four archives in the repository’s Hilbert’s-tenth research tree (`Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_batch80_surreal.md`, commit `c338541e3`, with the patch `surreal_research1_wellfounded_scope.patch` there). Source 08 wrote: “For non-set-like relations, the model’s class collection can matter to the well-order assertion.” Under GBC, internal well-foundedness of a fixed class relation is equivalent to the absence of a set-coded descending ω -sequence ([27, Section 3]; source 11 makes the same distinction in [section 20](#)), so two class expansions with the same sets agree about it for a relation they share; what can change is which relations are available, and internal well-foundedness differs from external well-foundedness. The printed sentence is the patch’s text. No statement of source 08 depends on the original sentence.

5.6 Source 08: class foundations and set-like well-orders

From this section onward, the default theory is GBC: Gödel–Bernays class set theory with Global Choice. Sets are the objects that may be members; classes are collections of sets. A class function maps set arguments to set values. Its image of a set is a set by class replacement. Elementary class comprehension allows formulas with class parameters but only set quantifiers in the defining formula.

5.6.1 Why Global Choice is named explicitly

Global Choice supplies a set-like global well-order of the universe. Restricting it to any class gives a set-like well-order, and every proper class can then be enumerated by Ord. We use this to choose one baseline $b : \text{Ord} \rightarrow \mathbf{No}$, to enumerate class carriers during back-and-forth, and to select witnesses uniformly. Ordinary set Choice alone does not justify these global choices.

Many local results need less. Pairwise lexicographic comparison, the extension of one specified set prefix relative to a given b , and the set-cut argument use only class replacement, elementary comprehension, set Choice, and the displayed class parameters. Keeping b as an explicit parameter is therefore useful in weaker foundations.

Definition 5.11. A class well-order $\prec$ is *set-like* when, for every element x , its predecessor class $\{y : y \prec x\}$ is a set. By a class well-order here we mean a linear class relation such that every nonempty subclass available in the model has a least element.

For a set-like relation, it suffices to require well-foundedness on sets: a nonempty subclass has a witness x , and one may search the set of predecessors of x together with x for its least member. Over GBC, well-foundedness of any fixed class relation is equivalent to the absence of a set-coded descending ω -sequence. Keeping the sets and the relation fixed while enlarging the class collection therefore does not change its internal well-foundedness. What may change is which class relations are available; internal and external well-foundedness must also be distinguished. [merge] (This replaces one sentence of source 08; see the note before this section.)

Note 5.12 (Source 08, Theorem 8.2). *Every set-like well-order of a proper class has a unique order isomorphism from Ord. Conversely, every class bijection $e : \text{Ord} \rightarrow \mathbf{No}$ induces a set-like well-order of $\mathbf{No}$.* This is the last paragraph of the proof of [theorem 13.1](#) and source 09’s [proposition 13.6](#), proved by the same rank map: the predecessors of x form a well-ordered set of ordinal type $r(x)$; the range of r is downward closed, and it cannot be a set ordinal, since the inverse of r would then map a set ordinal onto a proper class, contradicting class replacement. Conversely the predecessors of $e(\alpha)$ form the set $e[\alpha]$.

Thus all set-like proper-class well-orders have the same abstract order type. The interesting lexicographic structure lies in their labels, not in distinct ordinal types. It is incorrect either to introduce a set ordinal equal to “the order type of all ordinals” or to quotient away the enumerations and expect the lexicographic information to survive.

5.6.2 The recursion principle actually used

All transfinite constructions below have *set-valued* stages and set-sized histories at each ordinal stage. In GBC, fix the class parameters and a rule assigning a unique set to each set history. The ordinary set recursion theorem with those parameters constructs a solution through each set ordinal. Uniqueness makes these initial solutions compatible, and elementary comprehension forms their union as a class. This is the set-valued recursion needed here.

It is not the principle of elementary transfinite recursion with *class-valued* stages. For example, constructing a new truth class at each stage is a substantially different task. The distinctions among GBC, GBC + ETR, and KM are genuine; the class-recursion and class forcing literature documents their strength [23, 25]. No equivalence with those stronger principles is claimed for the constructions in this paper.

5.7 Universality and uniqueness of the saturated class order

Proposition 5.13. *In ZFC, every set linear order embeds into $\mathbf{No}$. In GBC, every class linear order embeds uniformly into $\mathbf{No}$. Any two proper-class η_{Ord} orders are isomorphic in GBC.*

Proof. Enumerate the source in a cardinal-length well-order, or in type Ord for a proper class, using global choice. At stage α , the earlier images form a set. Separate the images of earlier points below the new point from those above it, and choose a separator. This extends an order embedding. Each partial map is a set.

For uniqueness, globally enumerate both class orders and alternately insert the first unused point from each enumeration. The appropriate set cut in the other order is fillable. Global choice selects a separator when none is distinguished. The union of the partial isomorphisms is a surjective class order isomorphism. $\square$

The same construction, started from a specified set-sized partial isomorphism, gives set homogeneity. These are the order-theoretic universality and homogeneity features of $\mathbf{No}$ [15]. The assertion that every class order embeds is made with global choice; it is not used as an unproved converse to global choice.

[merge] *Source 12's version (its Lemmas 6.5 and 6.6).* It exposes the size of the forth recursion, with a birthday bound for set orders, and proves the class isomorphism $\mathbf{No} \setminus A \cong \mathbf{No}$ that source 11 uses in [proposition 19.25](#).

Lemma 5.14 (Surreal universality, with the size of the recursion exposed). *In GBC, every class linear order admits a class order-embedding into $\mathbf{No}$. For a set order enumerated in type λ , one can arrange that the image of its ξ th enumerated element has birthday at most ξ , for each $\xi < \lambda$.*

Proof. Use a global-choice enumeration of the carrier, either by an ordinal or by Ord. When the next element is considered, its previously embedded predecessors and successors form two sets $A < B$. Send it to the simplest surreal separator. It is distinct from all previous images and preserves order. Every stage has set-sized history. Induction gives the birthday bound, because the earlier birthdays are less than the current stage. This is the standard forth argument behind surreal universality [15, 20]. $\square$

Lemma 5.15 (Deleting a set does not change the numerical class order). *In GBC, $\mathbf{No} \setminus A$ is order-isomorphic to $\mathbf{No}$ for every set A . Every nonempty open numerical interval is also order-isomorphic to $\mathbf{No}$.*

Proof. Each target has interpolation for arbitrary separated sets, by [lemma 5.3](#), adding interval bounds when needed. Each is a proper class and has an Ord-enumeration by global choice. Carry out a back-and-forth between the two enumerations. At each ordinal stage, extend the partial isomorphism to the next unused domain point and then to the next unused target point. The previous partial map is a set, and the required cut can be filled while avoiding its set-sized range. At the end the class map is onto and order-preserving. $\square$

[merge] *Source 09's route.* Source 09 proves universality into any class order with the set-cut property, uniqueness together with set-homogeneity, and the isomorphisms $\mathbf{No} \setminus U \cong \mathbf{No}$ and $(a, b) \cong \mathbf{No}$; its proofs are the forth and back-and-forth recursions of [proposition 5.13](#), stated for a general target.

5.8 Source 09: the classical universality mechanism, with proofs

The following facts are classical features of the surreal order; we include the arguments because the same mechanism will identify our lexicographic core. The class-universal and set-homogeneous viewpoint appears, for example, in [17].

Theorem 5.16 (Class universality). *In GBC, every class linear order embeds into $\mathbf{No}$. More generally, every class linear order embeds into any class linear order with the set-cut property.*

Proof. For a set-sized domain, choose a well-ordering and enumerate it by an ordinal. For a proper-class domain L , use global choice to fix a class bijection $a : \text{Ord} \rightarrow L$. At stage α , assume the values $h(a(\xi))$ for $\xi < \alpha$ have been selected. Form the two sets

$$A_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\xi) <_L a(\alpha)\}, \quad B_\alpha = \{h(a(\xi)) : \xi < \alpha, a(\alpha) <_L a(\xi)\}.$$

Earlier choices preserve order, so $A_\alpha < B_\alpha$. Choose a separator in the target, using its set-cut property and a fixed global well-order for a definite choice. The new value is different from every

earlier value and has exactly the required comparisons. At each stage the partial map is a set. The resulting class function is the embedding. $\square$

Theorem 5.17 (Uniqueness and set-homogeneity). *In GBC, any two class linear orders with the set-cut property are isomorphic. Moreover, an order isomorphism between set-sized suborders of these orders extends to a class order isomorphism.*

Proof. Neither order is a set: applying the set-cut property with the entire carrier on the left would otherwise be contradictory. Fix Ord-length enumerations of both carriers. Start with the given set partial isomorphism. At each ordinal stage, first put the least not-yet-mapped element of the first enumeration into the domain. Its required image is constrained by two sets of already mapped elements; the set-cut property supplies an image. Next put the least omitted element of the second enumeration into the range by the symmetric construction. Take unions at limit stages. Every initial partial map is a set, so the recursion is of the permitted kind. The final map is total and onto. $\square$

Corollary 5.18. *For every set $U \subseteq \mathbf{No}$, $\mathbf{No} \setminus U \cong \mathbf{No}$. Every nonempty open interval of $\mathbf{No}$ is likewise isomorphic to $\mathbf{No}$.*

Proof. Use lemma 5.10 and theorem 5.17. For an interval (a, b) , add a to every lower constraint set and b to every upper constraint set. $\square$

These isomorphisms concern numerical order alone and need not preserve simplicity or arithmetic.

[merge] Source 08’s Section 12 (“The skeleton is a surreal line”) contains two further statements of the classical mechanism and the skeleton theorem (note 19.17).

Note 5.19 (Source 08, Theorem 12.1, “Uniqueness of a class saturated line”). *In GBC, any two proper-class η_{Ord} -orders whose elements are sets are order-isomorphic.* This is the last clause of proposition 5.13 and theorem 5.17, by the same back-and-forth. Source 08 spells out that the separator chosen in the forward step cannot be a previous image, that a fixed set-like global well-order chooses each separator uniformly among sets, and that the least-unmatched rule handles every element: after enough stages all elements of a given bounded index have been included.

Note 5.20 (Source 08, Theorem 12.3, “Universality for class linear orders”). *Every class linear order L with set-valued elements admits a class order embedding into $\mathbf{No}$, and hence into $\mathcal{P}_s(b)$.* This is the second clause of proposition 5.13 and theorem 5.16, composed with note 19.17. Source 08 adds: “The input order need not be set-like. A point with a proper class of predecessors is harmless because only the previously *processed* elements, rather than all its predecessors, enter a stage of the construction. This explains why Ord + 1 and more complicated non-set-like class well-orders can embed into the skeleton even though each *enumeration represented by* a skeleton point induces a set-like well-order on $\mathbf{No}$.”

Part I

Sets of surreals

6 Well-order codes on a fixed set alphabet

Let X be an infinite linearly ordered set, and put $\lambda = |X|$. We work in ZFC throughout this section. A well-order on the whole carrier X has a unique increasing enumeration $f : \alpha \rightarrow X$, where

increasing refers to that well-order, not to the native order of X . Its domain satisfies $|\alpha| = \lambda$, and hence $\alpha < \lambda^+$. Write $\mathcal{W}(X)$ for these bijective enumeration codes, and introduce two related sets:

$$\mathcal{I}(X) = \bigcup_{\alpha < \lambda^+} \{f : \alpha \hookrightarrow X\}, \quad \mathcal{W}_\lambda(X) = \{f : \lambda \longrightarrow X : f \text{ is bijective}\}.$$

Thus $\mathcal{I}(X)$ codes well-orders of arbitrary subsets of X , including the empty subset, whereas $\mathcal{W}(X)$ always uses the whole carrier. All three sets inherit the lexicographic order defined earlier. A proper prefix is smaller than its extension in $\mathcal{I}(X)$. Two distinct members of $\mathcal{W}(X)$ cannot be proper prefixes of one another: the shorter enumeration would already have used every element of X .

The repository's real-well-ordering report already proves the exhaustive and cardinal-length statements for arbitrary infinite alphabets [1]. We record those results together with the injective-word variant, and include a common proof to keep the size distinctions explicit. Neither density nor completeness of the native order is needed.

Theorem 6.1 (Cardinality and exact ordinal spectrum). *Each of $\mathcal{W}_\lambda(X)$, $\mathcal{W}(X)$, and $\mathcal{I}(X)$ has cardinality 2^λ , contains a copy of every linear order of cardinality at most λ , and contains neither λ^+ nor $(\lambda^+)^*$. Consequently the ordinals embeddable in any of them are exactly the ordinals $\alpha < \lambda^+$.*

Proof. There are at most λ^+ permitted domains. For each domain there are at most $\lambda^\lambda = 2^\lambda$ functions, so $|\mathcal{I}(X)| \leq \lambda^+ \cdot 2^\lambda = 2^\lambda$. For the reverse inequality, partition X into pairs $\{a_\xi, b_\xi\}$, $\xi < \lambda$, with $a_\xi < b_\xi$. The lexicographic order on $\lambda \times \{0, 1\}$, taking ξ first, has order type $2 \cdot \lambda = \lambda$. For $\varepsilon \in 2^\lambda$, list the ξ -th pair as (a_ξ, b_ξ) if $\varepsilon(\xi) = 0$, and as (b_ξ, a_ξ) otherwise. The resulting map

$$2^\lambda \longrightarrow \mathcal{W}_\lambda(X)$$

is an order embedding when 2^λ has lexicographic order. This proves all three cardinality assertions.

Let L be any linear order of cardinality at most λ , and fix an injection $\iota : L \hookrightarrow \lambda$. Associate to $t \in L$ the binary function c_t whose value at $\iota(s)$ is 1 precisely when $s < t$, and whose remaining coordinates are 0. If $t < u$, then $c_t \leq c_u$ coordinatewise, and the inequality is strict at $\iota(t)$. Their first difference therefore has the correct direction. The preceding pair construction embeds L into $\mathcal{W}_\lambda(X)$.

It remains to exclude longer monotone sequences. Suppose that $(f_\xi)_{\xi < \lambda^+}$ is strictly increasing or strictly decreasing in $\mathcal{I}(X)$. We recursively construct eventual common prefixes p_α of length α , for $\alpha < \lambda^+$. At a limit stage, take the union of the earlier prefixes. Regularity of λ^+ ensures that a tail of the original sequence extends that union. At a successor step, at most one member of this tail equals p_α ; discard it and all earlier indices. The values at coordinate α are now monotone. A monotone sequence indexed by a tail of λ^+ , taking at most λ values, is eventually constant: the first occurrence indices of all its values have a supremum below λ^+ . Append this eventual value to p_α . Every constructed prefix is injective because it occurs in members of the original sequence. The union of the prefixes through λ^+ would therefore be an injection $\lambda^+ \hookrightarrow X$, a contradiction. Finally, an ordinal at least λ^+ contains λ^+ , whereas every smaller ordinal has cardinality at most λ . $\square$

The distinction between the models also affects their elementary order topology. The order $\mathcal{W}_\lambda(X)$ is dense. Indeed, let $f <_{\text{lex}} g$, with first difference at ξ . If the tail of f after ξ has an ascent, interchanging the two entries produces a point strictly between f and g . If the tail of g has a descent, the corresponding interchange works in the other direction. If neither exists, the first remaining tail is strictly decreasing and the second strictly increasing. Their common set of entries, after removing the two entries at ξ , is infinite and would be both well-ordered and reverse-well-ordered. This is impossible.

By contrast, $\mathcal{W}(X)$ has adjacent pairs. Choose $a < b$ in X , enumerate $X \setminus \{a, b\}$ as a prefix p , and compare $p \frown \langle a, b \rangle$ with $p \frown \langle b, a \rangle$. Any intervening enumeration would have to share p , after which only these two orders remain. Thus a cardinal-length convention changes the resulting linear order even though it does not change its cardinality or ordinal embedding spectrum.

Proposition 6.2 (Exactly where jumps occur). *Suppose $f <_{\text{lex}} g$ in $\mathcal{W}(X)$, with common prefix p up to their first difference. Put $Y = X \setminus \text{ran}(p)$, $a = f(\text{dom}(p))$, and $b = g(\text{dom}(p))$, so $a < b$. The pair is adjacent precisely when Y is finite, a, b are consecutive in its native order, the tail of f after a lists $Y \setminus \{a\}$ decreasingly, and the tail of g after b lists $Y \setminus \{b\}$ increasingly.*

Proof. If an unused element lies strictly between a and b , start an enumeration with that element after p and complete it; this lies between f and g . Likewise, an ascent after a , or a descent after b , can be interchanged to create an intervening enumeration. Adjacency thus forces the two remaining tails to be decreasing and increasing, respectively. Their common native suborder $Y \setminus \{a, b\}$ is then both well-ordered and reverse-well-ordered. It must be finite: an infinite well-order contains its first ω elements, a subset with no greatest element. This proves necessity. Conversely, an intervening enumeration must share p . There is no possible next entry strictly between a and b . Among enumerations beginning with a , the displayed decreasing continuation is the largest; among those beginning with b , the increasing continuation is the smallest. Neither branch admits an intervening point. $\square$

This description also identifies the finite components generated by adjacency. Write the domain of an enumeration uniquely as $\theta = \delta + n$, with δ a limit ordinal and $n < \omega$. Its component consists of the $n!$ permutations of the final n entries, keeping the prefix of length δ fixed. In particular, words of limit length have no immediate neighbor. A jump cannot cross from one such component to another, because its common-prefix remainder would be infinite.

Endpoints can be classified just as concretely. The least element of $\mathcal{W}(X)$, if one exists, must always choose the least unused native element: otherwise interchanging that entry with a later smaller one decreases the enumeration. Hence a least element exists exactly when the native order on X is a well-order, and is its increasing enumeration. The greatest-element statement is the reverse assertion. For $\mathcal{W}_\lambda(X)$, the same argument shows that a least element exists exactly when the native order type is the initial ordinal λ , not merely an arbitrary ordinal of cardinality λ . For example, an alphabet of native type $\lambda + 1$ has a least full enumeration, but no least λ -length permutation.

6.1 Pointers to the reals report

[merge] The theorems of this section hold for every infinite linearly ordered set and are theorems of the repository’s reals report, which proves them in that generality. Source 11 says so (section 3); its repository audit (11-raw-orders-repository-audit.md) adds that their application to birthday fragments “should be attributed as applications of existing repository results, not marketed as new general theorems”. All four sources re-prove them: source 08 from batch-76 manuscript 02 (now the reals report’s base source), sources 09 and 12 deliberately, and source 11 to keep the size distinctions explicit. They are printed once, here, with these pointers; no novelty is claimed for any of them.

Statement	Here	Reals report
Prefix-freeness of exhaustive enumerations	before theorem 6.1 ; lemma 6.3 ; note 6.11 ; note 6.18	<code>lwo:lem:prefix-free</code>

Statement	Here	Reals report
Cardinality 2^λ ; pair swaps embed $\mathbb{B}_\lambda$ in $\mathcal{W}_\lambda(X)$ and $\mathbb{B}_\theta$ in $\mathcal{W}(X)$ for $\theta < \lambda^+$	theorem 6.1 ; lemma 6.6 and theorem 6.8 ; note 6.12 ; notes 6.19 and 6.21	<code>lwo:prop:cardinality</code> , <code>lwo:lem:all-small-cubes</code>
Every order of cardinality $\leq \lambda$ embeds (cut coding), even in every fixed-type slice	theorem 6.1 ; lemma 6.5 ; note 6.12 ; note 6.20	<code>lwo:lem:cut-code</code> , <code>lwo:thm:ordinal-spectrum(a)</code> , <code>lwo:sp:thm:universality</code>
No monotone λ^+ -sequence; ordinal spectrum $\beta < \lambda^+$	theorem 6.1 ; lemma 6.7 ; lemma 6.13 and theorem 6.14 ; notes 6.22 and 6.23	<code>lwo:lem:eventual</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code>
Strict hierarchy of binary cubes	lemma 6.4	<code>lwo:thm:cube-strict</code>
Cube spectrum $\theta < \lambda^+$ and coding length $\ell_2(\mathcal{W}(X)) = \lambda^+$	theorem 6.8 ; notes 6.23 and 6.24	<code>lwo:thm:full-length</code>
Least and greatest enumerations	paragraph after proposition 6.2 ; note 6.16	<code>lwo:lem:extrema</code>
Adjacent pairs; finite blocks of $n!$ points	proposition 6.2 and the paragraph after it; note 6.17 ; notes 6.25 and 6.26	<code>lwo:thm:adjacency</code> , <code>lwo:thm:blocks</code>
Absence of λ^+ is not a sufficient embeddability criterion	remark 6.9 ; remark 6.15	the reals report's non-claims

The arguments are the reals report's: pair swaps, characteristic-function cut codes, stabilization of a common prefix along a long monotone sequence, and cylinder fusion for strict cubes (the reals report credits the strict cube hierarchy to Corollary `cor:strict` of the point-separating game report and prints its fusion proof as a second route). Two of the sources' routes are printed in full: source 12's Section 9, because the singular transition of [section 8](#) uses its lemmas, and source 09's stabilization lemma for a regular cardinal ([lemma 6.13](#)). Source 08's statements are recorded in notes ([section 6.4](#)).

6.2 Source 12's route: a general set-sized spectrum theorem

We temporarily forget the surreal structure. Let X be any infinite linearly ordered set, and write $|X| = \mu$, an initial cardinal. The spectra below depend only on the cardinality, not on density, separability, or field operations of X .

Lemma 6.3 (Prefix-freeness of exhaustive enumerations). *Two distinct exhaustive injective ordinal enumerations of X are never proper prefixes of one another. Hence first-difference lexicography is unambiguous on $\mathcal{W}(X)$.*

Proof. An exhaustive prefix has already used every element of X . An injective proper extension would have to use a new element, which is impossible. $\square$

For any ordinal γ , let $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$. For linear orders L, M , write $L \hookrightarrow M$ for an order-embedding and $L \simeq_e M$ for embeddings in both directions. Define

$$\ell_2(L) = \min\{\gamma : L \hookrightarrow \mathbb{B}_\gamma\}$$

when such a binary coding ordinal exists. This is an ordinal-valued rank, not a cardinal invariant.

Lemma 6.4 (Strict binary hierarchy). *For ordinals α, β ,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

In particular $\ell_2(\mathbb{B}_\alpha) = \alpha$.

Proof. Zero-padding proves the forward construction when $\alpha \leq \beta$. It suffices to rule out an embedding $f : \mathbb{B}_{\beta+1} \rightarrow \mathbb{B}_\beta$.

Recursively construct an input prefix s_ξ of length ξ and an output prefix t_ξ of length ξ , so that every input extending s_ξ is sent to an output extending t_ξ . At a successor stage, split the input cylinder into its two next-bit halves and split the output cylinder likewise. The two input halves have strictly ordered images. They cannot both have images meeting both output halves: an image point in the upper output half from the lower input half would exceed an image point in the lower output half from the upper input half. Thus one input half has image contained in a single output half. Choose those halves to continue.

At a limit take unions; the corresponding cylinders are nonempty and the containment persists. At stage β , two inputs remain, differing in their last bit, but their images have the same full β -long output. This contradicts injectivity. For $\alpha > \beta$, padding embeds $\mathbb{B}_{\beta+1}$ into $\mathbb{B}_\alpha$, giving the general impossibility. $\square$

Lemma 6.5 (Characteristic-cut coding). *Every linear order of cardinality at most μ embeds in $\mathbb{B}_\mu$.*

Proof. Enumerate its carrier by distinct coordinates d_ξ , padding unused coordinates if necessary. Send x to the bits $\mathbf{1}[d_\xi < x]$. For $x < y$, every differing coordinate changes from 0 to 1, and the coordinate representing x does change. Thus the least difference gives the correct order. $\square$

Lemma 6.6 (Pair orientation coding). $\mathbb{B}_\mu \hookrightarrow \mathcal{W}_\mu(X)$. *More generally, $\mathbb{B}_\gamma \hookrightarrow \mathcal{W}(X)$ for every $\gamma < \mu^+$.*

Proof. Partition X into μ unordered pairs, sort each pair by X 's order, and orient the pair according to its bit. The ordinal sum of μ two-element blocks has type $2 \cdot \mu = \mu$.

For a general $\gamma < \mu^+$, choose $|\gamma|$ disjoint pairs and retain a disjoint reserve of size μ . Such a choice is possible even when $|\gamma| = \mu$. List the pairs in type γ , then well-order all remaining labels in type μ . The resulting enumeration has length $2 \cdot \gamma + \mu < \mu^+$. Its comparison is decided by the first oriented pair at which the inputs differ. $\square$

Lemma 6.7 (No μ^+ -long monotone chain). *Neither μ^+ nor its reverse embeds into $\mathcal{W}(X)$.*

Proof. Suppose $(e_i)_{i < \mu^+}$ is strictly increasing. We recursively stabilize its entries on final segments of the index set. Assume that a final segment agrees on a prefix p of length $\delta < \mu^+$. No enumeration in a further infinite final segment can end there, by lemma 6.3. Their δ th entries form a nondecreasing μ^+ -sequence in a set of size μ .

Such a sequence is eventually constant: choose one first occurrence for each value in its range; their at most μ indices have supremum below μ^+ by regularity. Past that supremum no larger value can first appear. At a limit stage $\delta < \mu^+$, take the supremum of the earlier stabilization indices, still below μ^+ , and the union of the earlier prefixes.

This would produce an injective sequence of labels from X of length μ^+ : every set-sized earlier part is shared by a final segment of injective enumerations. That contradicts $|X| = \mu$. Replace nondecreasing by nonincreasing for the reverse order. $\square$

Theorem 6.8 (Cardinality and exact spectra). *For every infinite linearly ordered set X of cardinality μ ,*

$$\begin{aligned} |\mathcal{W}(X)| &= |\mathcal{W}_\mu(X)| = 2^\mu, \\ \beta &\hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+, \\ \beta &\hookrightarrow \mathcal{W}_\mu(X) \iff \beta < \mu^+, \\ \mathbb{B}_\gamma &\hookrightarrow \mathcal{W}(X) \iff \gamma < \mu^+, \\ \ell_2(\mathcal{W}(X)) &= \mu^+. \end{aligned}$$

The ordinal statements also hold for reversed ordinals, and every linear order of cardinality at most μ embeds even into $\mathcal{W}_\mu(X)$.

Proof. Pair coding gives 2^μ elements, while all well-orders are relations on X , giving the upper bound $2^{|X \times X|} = 2^\mu$. The lower ordinal bounds and universality follow from [lemmas 6.5 and 6.6](#); [lemma 6.7](#) gives the upper bounds. The same characteristic coding applies to reversed ordinals.

For the cube spectrum use [lemma 6.6](#). If $\gamma \geq \mu^+$, $\mathbb{B}_\gamma$ contains an increasing chain of type μ^+ : use the sequences that are 1 before a moving cut and 0 afterward. Thus it cannot embed into $\mathcal{W}(X)$.

Embed X into $\mathbb{B}_\mu$ using [lemma 6.5](#). Encode an exhaustive enumeration of length $\alpha < \mu^+$ by concatenating its μ -long entry codes, then pad with zeroes to length μ^+ . The unpadded length $\mu \cdot \alpha$ has cardinality at most μ , so is less than μ^+ . Distinct enumerations disagree before either ends, and their encoded first disagreement preserves the order. This proves $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\mu^+}$.

If it embedded in $\mathbb{B}_\rho$ for $\rho < \mu^+$, then the copy of $\mathbb{B}_{\rho+1}$ supplied by [lemma 6.6](#) would contradict [lemma 6.4](#). Hence the rank is exactly μ^+ . $\square$

Remark 6.9 (What the rank does not say). Although $\ell_2(\mathcal{W}(X)) = \mu^+$, the cube $\mathbb{B}_{\mu^+}$ does *not* embed back into $\mathcal{W}(X)$. An order can first admit a binary encoding at an ordinal without containing the full cube of that ordinal. Nor is the absence of μ^+ and its reverse claimed sufficient for an arbitrary order to embed in $\mathcal{W}(X)$.

6.3 Source 09's route

[merge] Source 09's Section 4 works for any linearly ordered set. Its first statements are recorded in notes; its stabilization lemma and its proof of the ordinal spectrum are a second route and are printed in full.

Note 6.10 (Source 09, Definition 4.1). $\mathcal{W}_\alpha(X) = \text{Bij}(\alpha, X)$ and $\mathcal{W}(X)$ is the union over all ordinals α ; if $|X| = \kappa$ is infinite, the possible lengths are exactly $\kappa \leq \alpha < \kappa^+$. This is the $\mathcal{W}(X)$ of [section 6](#) and of the reals report.

Note 6.11 (Source 09, Lemma 4.2, “Exhaustiveness removes the prefix ambiguity”). *Distinct members of $\mathcal{W}(X)$ have a first unequal coordinate strictly below both of their lengths*, so $e <_{\text{lex}} f$ at that coordinate is a strict linear order (for transitivity compare the two first-disagreement ordinals of $e < f$ and $f < g$). This is `two:lem:prefix-free` and [lemma 6.3](#), by the same argument.

Note 6.12 (Source 09, Theorem 4.3, “Cardinality and universality”). *For every infinite linearly ordered set X of cardinality κ , $|\mathcal{W}_\kappa(X)| = |\mathcal{W}(X)| = 2^\kappa$; the lexicographic cube 2^κ embeds into $\mathcal{W}_\kappa(X)$; consequently every linear order of size at most κ embeds into $\mathcal{W}_\kappa(X)$ and hence into $\mathcal{W}(X)$.* This is [theorem 6.1](#), `two:prop:cardinality` and `two:thm:ordinal-spectrum(a)`, with the same pair-orientation and characteristic-function codes. Source 09 stresses that $2 \cdot \kappa = \kappa$ is ordinal multiplication.

Second route (source 09): eventual stabilization at a regular cardinal.

Lemma 6.13 (Eventual stabilization). *Let μ be a regular infinite cardinal and let X have cardinality less than μ . Every nondecreasing map $f : \mu \rightarrow X$ is eventually constant.*

Proof. Some fiber must be unbounded: otherwise the union of fewer than μ bounded fibers would be bounded. Once a value in such a fiber is reached, monotonicity and the unboundedness of the fiber prevent any subsequent strict increase. $\square$

Theorem 6.14 (Exact ordinal spectrum). *For $|X| = \kappa$ infinite and any ordinal β ,*

$$\beta \hookrightarrow \mathcal{W}(X) \iff \beta < \kappa^+.$$

The same equivalence holds for β^ , the reverse order of β .*

Proof. The positive direction follows from [note 6.12](#). For the negative direction put $\mu = \kappa^+$ and suppose that $(e_\xi)_{\xi < \mu}$ is strictly increasing in $\mathcal{W}(X)$. We construct an injective common eventual prefix of length μ , which is impossible in the alphabet X .

Assume a prefix p of length $\delta < \mu$ is shared by all e_ξ beyond some index $t_\delta < \mu$. At limits this holds by taking the supremum of the earlier cutoffs; regularity ensures that the new cutoff remains below μ . If p exhausts X , every enumeration on this tail is exactly p , contrary to strict increase. Otherwise each tail enumeration has an entry at coordinate δ . As the words share all earlier coordinates, these entries are nondecreasing in ξ . By [lemma 6.13](#) they eventually have one common value. Add it to p . Injectivity of the enumerations makes this a new value, not used earlier in p .

Continuing for every $\delta < \mu$ gives μ distinct elements of X , a contradiction. An embedding of any $\beta \geq \mu$ would restrict to one of μ . Reversing the alphabet proves the negative statement for μ^* and its longer extensions. $\square$

Remark 6.15. The absence of κ^+ and its reverse is not asserted to characterize all linear orders embeddable in $\mathcal{W}(X)$. Indeed $\mathcal{W}(X)$ already contains the binary cube of cardinality 2^κ , so “size at most κ ” is a universality guarantee, not a necessary cardinality condition for an individual order to embed.

Note 6.16 (Source 09, Proposition 4.7). $\mathcal{W}(X)$ has a least element exactly when the alphabet order on X is a well-order, and a greatest element exactly when its reverse is. This is the paragraph on endpoints after [proposition 6.2](#) and `two:lem:extrema`, by the same argument (a least enumeration must take the least unused element at every position).

Note 6.17 (Source 09, Proposition 4.8, “Finite residuals are exactly the adjacency mechanism”). *If $e < f$ in $\mathcal{W}(X)$ have common prefix p before their first disagreement and $R = X \setminus \text{ran}(p)$, they are adjacent precisely when R is finite and their remaining words are consecutive permutations of R in finite lexicographic order.* This is [proposition 6.2](#) and `two:thm:adjacency`, by the same argument through [note 6.16](#).

Even for a dense alphabet X , the full space $\mathcal{W}(X)$ can therefore have adjacent points. This finite-tail phenomenon disappears for global enumerations: deleting a set prefix from **No** never leaves a finite remainder.

6.4 Source 08's set-sized sections

[merge] Source 08's Sections 3 ("All well-orderings of a fixed set"), 4 ("The exact ordinal and binary-cube spectra") and 5 ("Adjacent pairs and finite tails") re-prove the reals report's general theorems by the same arguments. They are recorded here with source 08's own remarks. Throughout, $(X, <)$ is an infinite linearly ordered set of cardinality κ , its initial ordinal.

Note 6.18 (Source 08, display (3) and Lemma 3.1, "No prefix ambiguity"). Source 08 defines

$$\mathcal{W}_\alpha(X) = \{e : \alpha \rightarrow X : e \text{ is a bijection}\}, \quad \mathcal{W}(X) = \bigcup_{\kappa \leq \alpha < \kappa^+} \mathcal{W}_\alpha(X) \quad (\text{S08.3})$$

and proves: *two distinct members e, f of $\mathcal{W}(X)$ disagree before either one ends; consequently*

$$e <_{\text{lex}} f \iff e(\delta) < f(\delta), \quad \delta = \min\{\xi < \min(\text{dom } e, \text{dom } f) : e(\xi) \neq f(\xi)\} \quad (\text{S08.4})$$

defines a linear order without an end-of-word convention. This is `lwo:lem:prefix-free`. Source 08 adds: "Sorting by length *before* reading entries gives a different order. For instance, choosing one word of each possible length would produce a chain of type κ^+ in that alternative." [Note 6.22](#) shows this is impossible in (S08.4).

Note 6.19 (Source 08, Lemma 3.2, "Pair coding"). *For every ordinal $\theta < \kappa^+$ there is an order embedding $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$; in particular $\mathbb{B}_\kappa \hookrightarrow \mathcal{W}_\kappa(X)$.* This is `lwo:lem:all-small-cubes`, `lwo:prop:cardinality` and [lemma 6.6](#), with the same blocks (a_ξ, c_ξ) and (c_ξ, a_ξ) followed by a fixed enumeration of the unused elements.

Note 6.20 (Source 08, Lemma 3.3). *Every linear order L of cardinality at most κ embeds into $\mathbb{B}_\kappa$, by the bits $1[d < x]$ at distinct coordinates.* This is `lwo:lem:cut-code` and [lemma 6.5](#).

Note 6.21 (Source 08, Theorem 3.4). *$\mathcal{W}(X)$ has cardinality 2^κ and contains every linear order of cardinality at most κ ; the embedding can already be made into $\mathcal{W}_\kappa(X)$.* This is `lwo:prop:cardinality` and `lwo:thm:ordinal-spectrum(a)`. Source 08 notes that neither the continuum hypothesis nor any regularity of κ occurs, so the theorem applies to arbitrary birthday cutoffs and to numerically scattered as well as dense sets of surreals.

Note 6.22 (Source 08, Theorem 4.1, "Prefix stabilization"). *There is no strictly increasing or strictly decreasing sequence of length κ^+ in $\mathcal{W}(X)$.* This is `lwo:thm:no-long`; the proof is the stabilization of [lemma 6.13](#) and [theorem 6.14](#): a monotone κ^+ -sequence with at most κ values has an unbounded fibre in the regular cardinal κ^+ and is constant from its first occurrence on.

Note 6.23 (Source 08, Corollary 4.2, "Exact spectra"). *For every ordinal β and set ordinal θ : $\beta \hookrightarrow \mathcal{W}(X)$, $\beta^{\text{op}} \hookrightarrow \mathcal{W}(X)$ and $\mathbb{B}_\theta \hookrightarrow \mathcal{W}(X)$ each hold exactly when the ordinal is below κ^+ .* This is `lwo:thm:ordinal-spectrum` and the cube clause of `lwo:thm:full-length`, as in [theorem 6.8](#); the chain $f_\eta = 1[\xi < \eta]$ of type κ^+ in $\mathbb{B}_{\kappa^+}$ excludes the larger cubes.

Note 6.24 (Source 08, Proposition 4.3, "A uniform binary upper coding"). *There is an order embedding $\mathcal{W}(X) \hookrightarrow \mathbb{B}_{\kappa^+}$, by concatenating cut codes of the entries and padding with zeros.* This is the upper bound of `lwo:thm:full-length`, as in [theorem 6.8](#). Source 08's remark: these results locate $\mathcal{W}(X)$ between a long hierarchy of binary cubes without identifying it with one; the absence of a κ^+ -chain is compatible with cardinality 2^κ and does not mean that all suborders are small.

Note 6.25 (Source 08, Theorem 5.1, "Adjacency classification"). *Let $e < f$ in $\mathcal{W}(X)$ first disagree at δ , with common prefix p , remaining alphabet $Y = X \setminus \text{ran } p$, $a = e(\delta)$ and $c = f(\delta)$. They are adjacent exactly when (i) Y is finite, (ii) $a < c$ are consecutive in Y , and (iii) after a the word e lists $Y \setminus \{a\}$ decreasingly and after c the word f lists $Y \setminus \{c\}$ increasingly.* This is [proposition 6.2](#) and `lwo:thm:adjacency`, by the same argument.

Note 6.26 (Source 08, Corollary 5.2, “Finite condensation blocks”). *If the length of e is $\lambda + n$ with λ zero or a limit and $n < \omega$, its finite-condensation block consists of the words with the same prefix of length λ and all $n!$ permutations of the remaining n values; a limit-length word has a singleton block.* This is the paragraph after [proposition 6.2](#) and `lwo:thm:blocks`. Source 08 adds: “The phrase ‘singleton block’ does not assert that the point is isolated: it means that it has no finite chain of jumps to another point.”

7 Birthday cutoffs and their set-sized orders

Use the sign presentation of surreal numbers: $x \in \mathbf{No}$ is a function from an ordinal $\text{bd}(x)$ to $\{-, +\}$. At the first difference, the comparison is $- < \text{termination} < +$. This presentation and the fundamental set-cut theorem are treated in Gonshor and in the modern account of Bagayoko and van der Hoeven [12, 14]. For an infinite cardinal κ , define

$$\mathbf{No}_{<\kappa} = \{x \in \mathbf{No} : \text{bd}(x) < \kappa\}.$$

This is a set. Counting sign sequences gives

$$|\mathbf{No}_{<\kappa}| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}.$$

Here $2^{<\kappa}$ denotes the supremum of 2^μ over cardinals $\mu < \kappa$; it is at least κ .

Corollary 7.1. *Put $\lambda = 2^{<\kappa}$. Then $\mathcal{W}(\mathbf{No}_{<\kappa})$ has cardinality 2^λ , is universal for linear orders of cardinality at most λ , and admits exactly the ordinals below λ^+ . The same conclusions hold for its cardinal-length permutation suborder and for the injective-word order $\mathcal{I}(\mathbf{No}_{<\kappa})$.*

The parameter controlling this spectrum is the cardinality λ of the alphabet. It need not equal the birthday cutoff κ . For $\kappa = \omega$, the alphabet consists of the dyadic rationals. Its well-ordering order has cardinality $2^{\aleph_0}$, contains every countable linear order, and does not contain ω_1 . For $\kappa = \omega_1$, the alphabet consists of the surreals with countable sign expansions and has cardinality $\mathfrak{c} = 2^{\aleph_0}$. Its well-ordering order consequently has cardinality $2^\mathfrak{c}$ and ordinal spectrum $\alpha < \mathfrak{c}^+$, without any use of the continuum hypothesis. When $2^{<\kappa} = \kappa$, the spectrum simplifies to $\alpha < \kappa^+$.

Lemma 7.2. *If κ is regular infinite, $\mathbf{No}_{<\kappa}$ fills every cut $A < B$ with $|A \cup B| < \kappa$. Such a cut can still be filled after excluding any prescribed subset of cardinality less than κ .*

Proof. The birthdays of the members of $A \cup B$ have supremum below κ , by regularity. The simplest surreal between them has birthday at most the supremum of their birthdays plus one, and hence still belongs to $\mathbf{No}_{<\kappa}$. For the exclusion statement, let S be the forbidden set, and let S_0 consist of its members that already lie strictly between A and B . Fill the cut $A \cup S_0 < B$. Its filler lies in the original interval but strictly above every forbidden point in that interval. $\square$

We call this property η_κ , always allowing either side of a cut to be empty. The cardinal parameter in this notation is explicit; conventions indexing eta orders by ordinal subscripts vary in the literature. This elementary form of saturation is all that the stage arguments below require.

Regularity is necessary for this particular cutoff conclusion. If κ is singular, a cofinal sequence of ordinals below κ , of length $\text{cf}(\kappa) < \kappa$, is unbounded in $\mathbf{No}_{<\kappa}$. Indeed, a surreal of birthday β is at most the ordinal β , the all-positive sign expansion of that length. Every candidate upper bound is therefore exceeded by some member of the cofinal sequence. The cut consisting of this sequence

on the left and the empty set on the right has no filler in $\mathbf{No}_{<\kappa}$. This failure concerns the bounded stage; it does not weaken the set-cut theorem for the full surreal class.

[merge] Source 09's Section 5 adds the stage sizes for every ordinal θ , not only for cardinals, the density and cofinality of $\mathbf{No}_{<\theta}$, and a table. The cardinal clause of its [proposition 7.3](#) is source 11's count above, and (2) is [corollary 7.1](#).

7.1 Source 09: birthday-bounded versions of the problem

Set

$$\mathbf{No}_{<\theta} = \{x \in \mathbf{No} : \text{bd}(x) < \theta\}, \quad \kappa_\theta = |\mathbf{No}_{<\theta}|.$$

Each $\mathbf{No}_{<\theta}$ is a set, and every set of surreal numbers is contained in one such $\mathbf{No}_{<\theta}$. The latter assertion follows by taking the supremum of successor birthdays.

Proposition 7.3. *For every ordinal θ ,*

$$|\mathbf{No}_{<\theta}| = \sum_{\alpha < \theta} 2^{|\alpha|}.$$

In particular, for an infinite cardinal τ , $|\mathbf{No}_{<\tau}| = 2^{<\tau} := \sup_{\lambda < \tau} 2^\lambda$. If θ is an infinite limit ordinal, $\mathbf{No}_{<\theta}$ is dense without endpoints and has cofinality and coinitality $\text{cf}(\theta)$.

Proof. There are exactly $2^{|\alpha|}$ sign words of length α , and words of different lengths are distinct. For an infinite cardinal τ , the supremum of these cardinalities is at least τ , so multiplying it by τ does not enlarge it.

For $x < y$ in an infinite limit stage, the simplest cut separator has birthday at most $\max(\text{bd}(x), \text{bd}(y)) + 1 < \theta$. Larger positive and negative ordinals of birthday below θ rule out endpoints. The ordinals below θ are cofinal, and their negatives are coinital. Conversely, a subset whose birthdays are bounded below θ is bounded in $\mathbf{No}_{<\theta}$ by an ordinal below θ . A cofinal subset must therefore have birthdays cofinal in θ , establishing the lower cofinality bound. $\square$

The fixed-set results now give the exact ordinal spectrum

$$\beta \hookrightarrow \mathcal{W}(\mathbf{No}_{<\theta}) \iff \beta < \kappa_\theta^+ \quad (\kappa_\theta \text{ infinite}). \quad (2)$$

This is a cardinal bound attached to the entire alphabet, not merely to the cofinality of its numerical order.

Stage	Alphabet size	Size of its well-ordering space	Ordinal spectrum
$\mathbf{No}_{<\omega}$	$\aleph_0$	$2^{\aleph_0}$	$\beta < \omega_1$
$\mathbf{No}_{<\omega_1}$	$2^{\aleph_0} = \mathfrak{c}$	$2^{\mathfrak{c}}$	$\beta < \mathfrak{c}^+$
$\mathbf{No}_{<\tau}$, τ strongly inaccessible	τ	2^τ	$\beta < \tau^+$

The second row does not identify $\mathfrak{c}^+$ with ω_2 without an additional continuum hypothesis. At finite stages, $|\mathbf{No}_{<n}| = 2^n - 1$, so the number of well-orders is $(2^n - 1)!$.

[merge] Source 08's Section 6 begins with the same count, as ([S08.5](#)), and then treats the regular case; its statements are recorded in the notes after this extract.

7.2 Source 08: birthday cutoffs and the regular-cardinal model

For an ordinal λ , write

$$\mathbf{No}_{<\lambda} = \{x \in \mathbf{No} : \text{bd}(x) < \lambda\}.$$

Its exact cardinality is

$$|\mathbf{No}_{<\lambda}| = \sum_{\alpha < \lambda} 2^{|\alpha|}. \quad (\text{S08.5})$$

This follows directly from the disjoint birthday levels in the sign-sequence representation. If this cardinal is infinite, all the results of [section 6.4](#) apply to $\mathcal{W}(\mathbf{No}_{<\lambda})$. No algebraic closure property of the cutoff is needed.

A particularly transparent case occurs at a regular infinite cardinal κ satisfying $2^{<\kappa} = \kappa$. This includes $\kappa = \omega$ and every strongly inaccessible cardinal; it also includes successor cardinals under appropriate cardinal arithmetic hypotheses.

Note 7.4 (Source 08, Proposition 6.1). *If κ is regular infinite and $2^{<\kappa} = \kappa$, then $|\mathbf{No}_{<\kappa}| = \kappa$ and the numerical order of $\mathbf{No}_{<\kappa}$ is an η_κ -order.* The second clause is [lemma 7.2](#) (which needs only regularity), proved by the same birthday bound; it is the classical saturation of $\mathbf{No}_{<\kappa}$ recorded in the birthday-cutoff report (`hset:rem:saturation`, through `hset:eq:cut-bound`) [4].

Note 7.5 (Source 08, Lemma 6.2). *Replace each minus by 00 and each plus by 11 and append the terminal block 01; the resulting binary word $d(s)$ of length $2 \cdot \text{bd}(s) + 2$ is prefix-free and $s < t$ exactly when $d(s) <_{\text{lex}} d(t)$.* This is source 12's [lemma 19.33](#), by the same comparison $00 < 01 < 11$.

Note 7.6 (Source 08, Theorem 6.3, “The initial-length cutoff order”). *If κ is regular infinite and $2^{<\kappa} = \kappa$, then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_\kappa, \quad |\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa; \quad (\text{S08.6})$$

moreover $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ is an η_κ -order, and its ordinal and reverse-ordinal spectra consist exactly of the ordinals below κ^+ . The equimorphism is case 2 of source 12's [theorem 8.3](#), proved the same way (pair coding one way, the unpadding code [note 7.5](#) with partial sums below κ by regularity the other way). The η_κ clause is source 11's [proposition 12.1](#) for $X = \mathbf{No}_{<\kappa}$; source 08's set-sized cut argument is the same common-coordinate search, with fresh values from [theorem 5.6](#) and a completion of length κ , and it refers to [note 19.13](#) for the detailed invariant. The spectra are [theorem 6.1](#).

Note 7.7 (Source 08, Remark 6.4). Equimorphism in (S08.6) is not isomorphism: the binary cube has endpoints and adjacent pairs, whereas $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ is dense without endpoints. Order embeddings need not preserve jumps or be continuous for the order topologies.

The regularity assumption has specific jobs: it bounds option birthdays and first-disagreement coordinates, and controls partial sums of code lengths. At a singular cutoff these steps may fail. The general cardinality and ordinal-spectrum theorems remain valid, but the η_κ and fixed-length equimorphism conclusions are not asserted without their hypotheses.

[merge] The singular case that this paragraph leaves open is settled for the coding length by source 12's dichotomy ([section 8](#)): at a singular strong limit the fixed-length order is $\simeq_e \mathbb{B}_{\kappa \cdot \kappa}$, not $\mathbb{B}_\kappa$, and it is $\simeq_e \mathbb{B}_{2^{<\kappa}}$ when $2^{<\kappa} > \kappa$ ([section 1.7](#), item (c)). Source 11's remark at the end of [section 7](#) gives the failure of η_κ at a singular cutoff.

[merge] The next section is source 12's Section 10. Its dichotomy is new against the repository; it answers in part the reals report's question “Other ground orders” ([section 1.7](#)). It uses source 12's self-delimiting code ([lemma 19.33](#), printed in [part IV](#)) and its lemmas of [section 6.2](#). Here S_κ of source 12 is printed $\mathbf{No}_{<\kappa}$, and its “binary coding rank” is the binary coding length ℓ_2 of the reals report.

8 Birthday cutoffs and the singular transition (source 12)

Let κ be an infinite initial cardinal and put

$$\mathbf{No}_{<\kappa} = \{s \in \mathbf{No} : \text{bd}(s) < \kappa\}, \quad \mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa} = \sup_{\lambda < \kappa} 2^{|\lambda|}.$$

The cardinality identity follows by counting all levels. The supremum is at least κ , and summing κ levels of size at most that supremum does not increase it. The reverse inequality is supplied by the levels themselves. We do not assume $\mathbf{No}_{<\kappa}$ is a field; none of the following proofs uses arithmetic closure.

8.1 A scheduler that preserves the first varying coordinate

Lemma 8.1 (Row scheduler). *Suppose a set order X of infinite cardinality κ has disjoint subsets $(X_i)_{i < \kappa}$, each containing a copy of X and each of cardinality κ . Suppose their complement is either empty or has cardinality κ . Then*

$$X_{\text{lex}}^\kappa \hookrightarrow \mathcal{W}_\kappa(X).$$

Proof. Fix embeddings $\phi_i : X \rightarrow X_i$ and a baseline enumeration of each X_i in length κ . For $x \in X$, transpose the label $\phi_i(x)$ with the first label of that baseline, giving an enumeration $r_{i,x}$ of X_i whose first value is $\phi_i(x)$. Its dependence on x is fixed and uniform. If the complement is nonempty, add a fixed filler row indexed before all variable rows. The row index order still has type $1 + \kappa = \kappa$.

Well-order row-stage pairs $(i, j) \in \kappa \times \kappa$ first by $\max(i, j)$ and then lexicographically within a level. This schedule has type κ : the predecessors of any pair have cardinality less than κ , since both coordinates there are bounded below κ , while the entire set of pairs has size κ . The first appearance of row i is $(i, 0)$, and these first appearances occur in increasing row order.

For an input $(x_i)_{i < \kappa}$, output the j th entry of r_{i,x_i} when pair (i, j) is scheduled. The output is an exhaustive bijection in type κ . If two inputs first differ in row i , every earlier row agrees, no later row has appeared before $(i, 0)$, and the first output discrepancy is $\phi_i(x_i)$ versus $\phi_i(y_i)$. Its sign is exactly the sign of the input comparison. $\square$

Lemma 8.2 (Surreal cone decomposition). *If $|\mathbf{No}_{<\kappa}| = \kappa$, then*

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \simeq_e (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa.$$

Proof. One embedding is the inclusion of bijective sequences among all sequences. For the other, partition the sign sequences with a minus into the cones with prefixes

$$p_i = +^i \frown -, \quad i < \kappa.$$

They are disjoint, and prefixing by p_i gives a numerical order isomorphism from $\mathbf{No}_{<\kappa}$ onto that cone. For $i, \alpha < \kappa$, $i + 1 + \alpha < \kappa$; this uses that κ is an initial cardinal and is valid even when it is singular. Removing the initial prefix also leaves a sign sequence of length less than κ . The complement consists of all-plus sequences, one for each length below κ , so has cardinality κ . Apply lemma 8.1. $\square$

Theorem 8.3 (Exact minimal-slice dichotomy). *For κ and $\mu = 2^{<\kappa}$ as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Consequently its binary coding rank is respectively $\kappa \cdot \kappa$ or μ . The exceptional case is exactly the case of a singular strong-limit cardinal κ .

Proof. We separate the three mechanisms.

Case 1: $\mu > \kappa$. By lemma 19.33, each member of $\mathbf{No}_{<\kappa}$ has a prefix-free numerical binary code of length less than κ . Pad each such code to length κ . Concatenating the entry codes of a μ -long enumeration produces a binary word of length $\kappa \cdot \mu = \mu$. The equality holds because μ is an initial cardinal strictly larger than κ : every proper partial product $\kappa \cdot \xi$, $\xi < \mu$, has cardinality below μ . Thus $\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \hookrightarrow \mathbb{B}_\mu$. Pair coding gives the reverse embedding.

Case 2: $\mu = \kappa$ and κ is regular. Use the unpadding, self-delimiting code $c(s)$. In a sequence of κ entries, the sum of code lengths before any stage $\xi < \kappa$ is less than κ , by regularity. The full sum is κ , since every length is positive. Consequently concatenation embeds even $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$ into $\mathbb{B}_\kappa$. Pair coding again supplies the reverse embedding for $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$.

Case 3: $\mu = \kappa$ and $\delta = \text{cf } \kappa < \kappa$. The padded single-entry code embeds $\mathbf{No}_{<\kappa}$ into $\mathbb{B}_\kappa$, whence

$$\mathcal{W}_\kappa(\mathbf{No}_{<\kappa}) \hookrightarrow (\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa \hookrightarrow \mathbb{B}_{\kappa \cdot \kappa}.$$

Choose increasing nonzero ordinals $(\lambda_j)_{j < \delta}$ cofinal in κ . Their ordinal sum is κ . As $\delta \cdot \kappa = \kappa$, partition the κ input coordinates into κ consecutive blocks of length δ . In each block, restrict its j th coordinate to sign sequences of exactly length λ_j . The lexicographic product of those coordinates is the binary cube of the sum of their lengths. Thus, over all the blocks, this restricted product is

$$\mathbb{B}_{\sum_{\xi < \kappa} \sum_{j < \delta} \lambda_j} = \mathbb{B}_{\kappa \cdot \kappa}.$$

It embeds into $(\mathbf{No}_{<\kappa})_{\text{lex}}^\kappa$, which embeds into $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ by lemma 8.2. This gives the claimed equimorphism. The rank assertions follow from lemma 6.4.

Finally, a singular strong limit satisfies $2^{<\kappa} = \kappa$. Conversely, suppose κ is singular and $2^{<\kappa} = \kappa$. If $2^\lambda = \kappa$ for some $\lambda < \kappa$, choose $\rho < \kappa$ with $\rho \geq \lambda, \text{cf } \kappa$. Then $\kappa \leq 2^\rho \leq 2^{<\kappa} = \kappa$, so $2^\rho = \kappa$, contradicting König's inequality $\text{cf}(2^\rho) > \rho$. Thus $2^\lambda < \kappa$ for every $\lambda < \kappa$. $\square$

Corollary 8.4 (A complete cube test for the minimal slice). *Let $\rho = \kappa \cdot \kappa$ in the exceptional case of theorem 8.3, and let $\rho = \mu$ otherwise. Then*

$$\mathbb{B}_\gamma \hookrightarrow \mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \iff \gamma \leq \rho.$$

In every case its ordinal embedding spectrum is still precisely $\{\beta : \beta < \mu^+\}$.

Example 8.5 (An unconditional singular example). Let $\kappa = \beth_\omega = \sup_{n < \omega} \beth_n$. It is a singular strong-limit cardinal of cofinality ω , independently of GCH. Thus

$$\ell_2(\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})) = \kappa \cdot \kappa > \kappa, \quad \ell_2(\mathcal{W}(\mathbf{No}_{<\kappa})) = \kappa^+.$$

All three displayed ordinals must be distinguished. The first two have the same cardinality, but are different coding lengths; the successor cardinal is larger still.

Example 8.6 (Regular cutoffs). For $\kappa = \omega$, $\mathbf{No}_{<\kappa}$ is the order of finite-birthday surreals, and the minimal slice is equimorphic with $\mathbb{B}_\omega$. For $\kappa = \omega_1$, its cardinality is $2^{\aleph_0}$, and the minimal slice is equimorphic with $\mathbb{B}_{2^{\aleph_0}}$, whether or not CH holds. At a strongly inaccessible κ , it is equimorphic with $\mathbb{B}_\kappa$.

8.2 Why the singular phenomenon is genuinely ordinal

A set of κ coordinates does not license arbitrary reordering of those coordinates in a lexicographic product. In the singular case, a short cofinal block of coordinate lengths can already consume ordinal length κ ; repeating that event κ times consumes $\kappa \cdot \kappa$. Replacing this ordinal sum by its cardinality would incorrectly collapse the rank back to κ . The strict-cube lemma shows that the lost information is mathematically detectable.

The row scheduler addresses a separate possible error: an arbitrary bijection $\kappa \times \kappa \rightarrow \kappa$ need not preserve the first varying row. Bounding by the maximum coordinate ensures that all earlier observable output is independent of later varying rows. This is why the product-to-permutation embedding is valid even at singular κ .

[merge] The last sentence of [corollary 8.4](#) is [corollary 7.1](#) for the minimal slice. The dichotomy computes binary coding length only; source 11's question on the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$ without $2^{<\kappa} = \kappa$ ([research question 34.7](#)) stays open.

[merge] Dated note, 3 October 2026 (batch 81): that question is now answered by sources 17, 15 and 18 in Part VI: point characters (q, q) with $q = \text{cf } \mu$, the exact gap spectrum and its multiplicities, and the saturation threshold $\eta_{\text{cf } \kappa}$ of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$, for every infinite cardinal κ ([sections 30](#) and [33](#)). Source 15 extends the dichotomy itself to the strata $\mu \cdot \gamma + n$ ([theorem 31.2](#)).

Part II

Set-length words

9 Set-length surreal words embed explicitly in the surreal line

Let $\mathcal{T}$ be the class of all set-length words $f : \alpha \rightarrow \mathbf{No}$, and let $\mathcal{I}$ be its injective subclass. The latter codes well-orders of set-sized subsets of $\mathbf{No}$. Both are ordered lexicographically with proper prefixes smaller. Their members are sets, so these are ordinary definable classes in $\mathbf{GB}$, not collections whose members are proper classes. They should not be confused with all global well-orders of the entire surreal class.

In particular, a word cannot exhaust $\mathbf{No}$: its range is a set by Replacement. Passing from fixed alphabets to $\mathcal{I}$ consequently changes which carrier is being well-ordered from one point of the lexicographic order to another. This is a mathematically natural extension, but it answers a different question from ordering all class well-orders on a single proper-class carrier. Making this distinction before forming any collection of codes is what prevents a hidden ascent from classes to collections of classes.

The following construction uses neither a global enumeration of $\mathbf{No}$ nor Global Choice. It uses only ordinal concatenation of sign expansions, whose general properties are discussed in [\[14, Section 3.2\]](#).

Theorem 9.1 (Explicit sign coding). *There is an explicitly defined order embedding $\mathcal{E} : \mathcal{T} \rightarrow \mathbf{No}$. It maps the empty word to 0, maps every nonempty word to a positive surreal, and restricts to an order embedding of $\mathcal{I}$. Singleton words give embeddings in the opposite direction.*

Proof. For a sign expansion s , let $D(s)$ duplicate each sign: replace $-$ by $(-, -)$, and $+$ by $(+, +)$. For $x \in \mathbf{No}$, put

$$C(x) = (+) \frown D(s_x) \frown (+, -).$$

The length here is the ordinal $1 + 2 \cdot \text{bd}(x) + 2$. If s_x, s_y first differ at a sign, their doubled blocks have the same comparison. If s_x is a proper prefix of s_y , the terminal block $(+, -)$ of $C(x)$ is compared with the next doubled block of $C(y)$. We have

$$(-, -) < (+, -) < (+, +).$$

This is precisely the comparison of a missing sign with a negative or positive continuation. Thus C preserves the surreal order. Moreover, two distinct values of C cannot be prefixes of one another: the same first differing block supplies an actual differing sign before either code ends.

For $f : \alpha \rightarrow \mathbf{No}$, concatenate these blocks in domain order:

$$\mathcal{E}(f) = \bigcap_{i < \alpha} C(f(i)).$$

Replacement and ordinal summation make this a set-length sign expansion. If two words first differ at i , their preceding blocks coincide and their blocks at i have a first differing sign. Prefix freedom prevents that difference from being displaced into a subsequent block. The comparison therefore agrees with the comparison of the original entries. If f is a proper prefix of g , the expansion $\mathcal{E}(g)$ continues $\mathcal{E}(f)$ with the initial $+$ of the next block, so $\mathcal{E}(f) < \mathcal{E}(g)$. This also proves the assertions about empty and nonempty words. Finally, $x \mapsto \langle x \rangle$ embeds $\mathbf{No}$ into both word classes. $\square$

Accordingly, $\mathcal{T}$, $\mathcal{I}$, and $\mathbf{No}$ are pairwise equimorphic as class linear orders. This conclusion asserts embeddings both ways, not isomorphism. Section 10 exhibits an obstruction to isomorphism that persists after the empty word is removed.

Proposition 9.2 (Birthday bounds). *The coding has the following bounds.*

1. If κ is regular infinite, $\alpha < \kappa$, and $f(i) \in \mathbf{No}_{<\kappa}$ for every $i < \alpha$, then $\mathcal{E}(f) \in \mathbf{No}_{<\kappa}$.
2. For any infinite cardinal κ , put $\lambda = 2^{<\kappa}$. If f is an arbitrary injective word over $\mathbf{No}_{<\kappa}$, then $\mathcal{E}(f) \in \mathbf{No}_{<\lambda^+}$.

Proof. The birthday of the image is

$$\text{bd}(\mathcal{E}(f)) = \sum_{i < \alpha} (1 + 2 \cdot \text{bd}(f(i)) + 2),$$

with ordinal addition and multiplication. For the first assertion, this sums fewer than κ ordinals below κ , and regularity keeps the sum below κ . For the second, injectivity gives $|\alpha| \leq \lambda$; each block has cardinality at most λ . The whole sign expansion has cardinality at most λ , so its ordinal length is below λ^+ . $\square$

The first bound need not hold at a singular cutoff. Choose distinct ordinals $\gamma_i < \kappa$, $i < \text{cf}(\kappa)$, cofinal in κ , and regard them as surreal entries. The word has length less than κ , but the lengths of its code blocks are unbounded in κ , so its encoded birthday is at least κ . Also, injectivity is essential to the unrestricted second bound: constant words over a one-element alphabet can have every ordinal length.

10 Local characters and set suprema of word orders

In this section $\mathcal{W}$ denotes either $\mathcal{T}$ or $\mathcal{I}$. Deleting the set of entries of an existing prefix causes no difficulty when choosing a new coordinate: apply Lemma 5.2. We say that a segment has no set cofinal subset rather than assigning it a cardinal cofinality that does not exist.

Theorem 10.1 (Exact local character). *Let $f \in \mathcal{W}$ have domain α .*

1. *If α is a nonzero limit ordinal, then $\text{cf}(\mathcal{W}_{<f}) = \text{cf}(\alpha)$.*
2. *If α is a successor ordinal, then $\mathcal{W}_{<f}$ has no set cofinal subset.*
3. *For every α , the segment $\mathcal{W}_{>f}$ has no set coinitial subset.*

The empty word is the minimum of $\mathcal{W}$.

Proof. Suppose first that α is a nonzero limit. Its proper prefixes $f \upharpoonright \beta$, $\beta < \alpha$, are cofinal below f . A word $g < f$ is either a proper prefix, or differs from f first at some coordinate ξ ; in the latter case it lies below every prefix long enough to include coordinate ξ . A cofinal sequence of indices in α therefore supplies a cofinal subset of size $\text{cf}(\alpha)$. Conversely, to each $g < f$ assign a prefix length strictly large enough that the corresponding prefix exceeds g . Fewer than $\text{cf}(\alpha)$ such lengths are bounded below α , so the corresponding words cannot be cofinal.

Next let $\alpha = \beta + 1$, write $p = f \upharpoonright \beta$, and take a set $A < f$. For each member a properly extending p , its next coordinate satisfies $a(\beta) < f(\beta)$. Choose an unused surreal d strictly above all these coordinates and strictly below $f(\beta)$. Then $h = p \frown \langle d \rangle$ is below f . It exceeds every member of A extending p ; the remaining members are at most p , so they also lie below h . Thus A is not cofinal.

Finally, take a set $B > f$. Collect $b(\alpha)$ for those $b \in B$ properly extending f , and choose an unused d below all these coordinates. The word $h = f \frown \langle d \rangle$ lies above f . It lies below the members of B extending f . Every other member of B first differs earlier with a larger coordinate, and consequently exceeds every extension of f , including h . Hence $f < h < B$. $\square$

Corollary 10.2. *Neither $\mathcal{T}$ nor $\mathcal{I}$, with or without its empty word, is order-isomorphic to $\mathbf{No}$. Each has nontrivial set suprema, but a set with no minimum cannot have an infimum in either word order.*

Proof. Take an injective word f of length ω . Its nonempty finite prefixes increase to f , and f is their supremum by the preceding theorem. No surreal can be the supremum of a set lying strictly below it: the surreal set-cut property would insert a smaller upper bound. This distinguishes the orders even after deleting their minimum. If f were the infimum of a set B not containing f , part (3) would produce h with $f < h < B$, contradicting the definition of infimum. $\square$

These local obstructions coexist with very rich intervals. Given $f <_{\text{lex}} g$ with a first differing coordinate, vary that coordinate through a surreal interval strictly between the two values and disjoint from the set of preceding entries. If f is a proper prefix of g , vary a new coordinate below the next entry of g instead. The surreal set-cut property provides such an interval after excluding the preceding entries. Every surreal interval (u, v) is order-isomorphic to $\mathbf{No}$, for example by

$$t \mapsto u + \frac{v - u}{2} \left(1 + \frac{t}{1 + |t|} \right).$$

Thus every open word interval contains a copy of $\mathbf{No}$, and Theorem 9.1 then places a copy of the entire word order inside it. Interval universality therefore does not imply that set cuts are filled.

11 The complete obstruction to filling a set cut

The preceding prefix example is the only nontrivial obstruction. Here and below, $A < B$ means that every member of A is strictly below every member of B ; either set may be empty.

Theorem 11.1 (Word-cut classification). *A set cut $A < B$ in $\mathcal{W}$ has no interpolant if and only if one of the following holds:*

1. *B contains the empty word;*
2. *B has a minimum b of nonzero limit length, and A is cofinal in $\mathcal{W}_{<b}$.*

If neither obstruction occurs, an interpolant can be chosen to have nonzero successor length.

Proof. Either displayed obstruction plainly prevents an interpolant. For the converse, perform the following common-prefix search. Start with p empty. Maintain these invariants: members of A not extending p lie below p ; members of B not extending p lie above every extension of p . Here extending allows equality.

If $p \in B$, stop. Otherwise, with $\delta = \text{dom}(p)$, form the sets of next coordinates

$$A(p) = \{a(\delta) : a \in A, p \text{ is a proper prefix of } a\}, \quad B(p) = \{b(\delta) : b \in B, p \text{ is a proper prefix of } b\}.$$

Every member of $A(p)$ is at most every member of $B(p)$, since the corresponding words satisfy $a < b$. If these coordinate sets are disjoint, all these inequalities are strict. Choose a surreal c strictly between them, outside $\text{ran}(p)$. Then $p \frown \langle c \rangle$ interpolates the original cut by the invariants, and the construction is complete.

If the two coordinate sets meet, their intersection consists of a unique element c : two distinct common elements would violate the preceding inequalities. Replace p by $p \frown \langle c \rangle$. The invariants persist because the discarded A -branches have smaller next coordinates and the discarded B -branches have larger ones. In the injective-word case, c is unused, since it is witnessed by an injective word extending the previous prefix. At a limit stage take the union of the prefixes. It is a set word, is injective when required, and satisfies the same invariants. A member of B cannot have become a proper prefix of this union: the search would already have stopped upon reaching that member.

This procedure requires only set recursion. Choose an ordinal strictly above the domains of all words in $A \cup B$. The search cannot continue taking common next coordinates up to that length, because there would be no words supplying those coordinates. It therefore terminates. The remaining possibility is termination at $p \in B$. The invariants show that $p = \min B$. If its length is zero, the first obstruction holds.

Termination cannot occur at a successor length $\delta + 1$. The last appended coordinate was chosen from the intersection, so some member of A extends this successor prefix. Such a word is at least p , contrary to $A < B$. Hence a nonzero terminal length α is a limit. For every $\beta < \alpha$, a member of A witnessed the common coordinate at stage β ; it extends $p \upharpoonright (\beta + 1)$ and lies below p . These witnesses exceed prefixes of arbitrarily large length. Since the proper prefixes are cofinal below p , A is cofinal below p . This is exactly the second obstruction. $\square$

Corollary 11.2 (Successor words). *The suborder of $\mathcal{W}$ consisting of words of nonzero successor length fills every set cut. In GBC, it is order-isomorphic to **No**.*

Proof. Neither obstruction in Theorem 11.1 can occur when the upper set consists of successor words. The construction returns another successor word, proving the set-cut property. Under Global Choice, enumerate each of the two proper classes by a set-like class well-order and perform back-and-forth. At every ordinal stage the previously matched elements form a set; their images specify a set cut, and an unused filler extends the partial order isomorphism. The union is an isomorphism. This is the class analogue of the usual eta-order back-and-forth, with its choice assumption made explicit; compare the class framework in [15]. $\square$

The distinction is precise. Deleting the empty word alone does not remove the prefix-supremum obstruction. Restricting to successor lengths does remove it. The explicit embeddings of Theorem 9.1 require no Global Choice, while the back-and-forth identification just given uses it.

[merge] Source 09’s class of set-length injective words is $\mathcal{I}$ (its $\mathcal{T}$). Its [proposition 11.3](#) obtains the equimorphism with **No** from class universality in GBC; source 11’s [theorem 9.1](#) gives explicit embeddings in ZF, and [corollary 10.2](#) the non-isomorphism.

11.1 Source 09: allowing the underlying set to vary

11.1.1 The set-word class

Every well-order on a set of surreals has a unique injective exhaustive enumeration of its range. Thus all such labeled set well-orders are represented by the genuine proper class

$$\mathcal{I} = \{p : p \text{ is an injective function } \alpha \rightarrow \mathbf{No} \text{ for some ordinal } \alpha\}.$$

Every member of $\mathcal{I}$ is a set. The collection is proper because it contains every singleton word (x) with $x \in \mathbf{No}$.

For $p, q \in \mathcal{I}$, compare at the first unequal entry when one exists; otherwise the proper prefix is smaller. This is a strict linear order. One can verify transitivity by padding words with a formal termination symbol below every surreal, but this symbol is *not itself a surreal number*. It is simply part of the definition of word order.

Proposition 11.3. *The dictionary order on $\mathcal{I}$ is dense, has the empty word as its least element, and has no greatest element. In GBC it is equimorphic with **No**, but it is not isomorphic to **No**, even after removing the empty word.*

Proof. If $p < q$ differ at coordinate δ , their common prefix has set range. By [lemma 5.10](#), choose a fresh surreal between the two entries at δ and append it to this prefix. The resulting word lies strictly between p and q . If p is a proper prefix of q , append to p a fresh value smaller than $q(\text{dom } p)$; this also gives an intermediate word. Every word has a proper injective extension, so there is no greatest word.

The singleton map $x \mapsto (x)$ embeds **No** into $\mathcal{I}$. Conversely, $\mathcal{I}$ is a class linear order and embeds into **No** by [theorem 5.16](#). The final nonisomorphism assertion follows from the next proposition. $\square$

Note 11.4 (Source 09, Proposition 6.2, “Limit prefixes create small one-sided character”). *If $p \in \mathcal{I}$ has nonzero limit length λ , its proper prefixes are cofinal in the open initial segment below p , the cofinality of that segment is exactly $\text{cf}(\lambda)$, and $p = \sup_{\mathcal{T}}\{p \upharpoonright \xi : \xi < \lambda\}$.* This is part (1) of [theorem 10.1](#) and the first step of the proof of [corollary 10.2](#), by the same argument.

In **No**, the set-cut property forbids a set cofinal immediately below any fixed point. Thus the difference in [note 11.4](#) is intrinsic, not just an artifact of the empty word. An embedding into **No** does not contradict this: the image need not preserve suprema in the ambient order.

11.1.2 Labeled objects versus order types

There is only one increasing exhaustive enumeration of a set already well-ordered by the numerical surreal order. In contrast, $\mathcal{W}(X)$ allows all well-orders on the labeled set X . If one instead quotients all set well-orders by abstract order isomorphism, only their ordinal types remain. Lexicographic comparison of their numerical entries does not descend to that quotient: two distinct two-letter words can have identical abstract order type and different positions in dictionary order. The distinctions between labels, inherited order, and order type must be retained.

[merge] Source 08’s Section 7 (“Varying set-sized subsets: a different class order”) uses the same class $\mathcal{I}(\mathbf{No}) = \mathcal{I}$ with proper prefixes first.

Note 11.5 (Source 08, Proposition 7.1). *The prefix-small lexicographic order on $\mathcal{I}(\mathbf{No})$ is linear; its empty word is a minimum, its nonempty part is dense without endpoints, and it contains a copy of $\mathbf{No}$ through one-entry words. Nevertheless its nonempty part is not an η_{Ord} -order: for an injective word p of length ω , the set of its nonempty finite prefixes and $\{p\}$ have no separator. This is [corollary 10.2](#) with the obstruction (2) of [theorem 11.1](#), by the same argument. Its last paragraph is printed next.*

More generally, for a word p of nonzero limit length λ , its proper prefixes are cofinal below it, and the left cofinality at p is $\text{cf}(\lambda)$. For the lower bound, a family of fewer than $\text{cf}(\lambda)$ words below p has its first disagreement or termination coordinates bounded below λ , and so is bounded by a proper prefix. Thus even very large numerical alphabets do not erase cofinalities created by termination conventions.

If “well-orders of subsets” is instead required to mean *numerically increasing* enumerations, it defines a further subclass of $\mathcal{I}(\mathbf{No})$. The singleton embedding of $\mathbf{No}$ still works, and the limit-prefix obstruction persists using an increasing p . We do not identify that subclass with the arbitrary-enumeration class. Likewise a numerically well-ordered *proper subclass* of $\mathbf{No}$ need not be set-like: a bounded increasing copy of Ord followed by its upper endpoint has type $\text{Ord} + 1$. Such copies also follow from the universality theorem in [section 5.8](#).

12 Saturation at a regular stage

For regular infinite κ , restrict to successor-length injective words of length below κ , with entries in $\mathbf{No}_{<\kappa}$. This order is η_κ . Indeed, a cut with fewer than κ words has all its word lengths bounded below κ . The common-prefix search in the preceding proof consequently stops below κ , and [Lemma 7.2](#) supplies the last coordinate while excluding the fewer than κ entries already used. [Proposition 9.2](#) embeds this order into $\mathbf{No}_{<\kappa}$; singleton words embed $\mathbf{No}_{<\kappa}$ back. When $2^{<\kappa} = \kappa$, both orders have cardinality κ , and the ordinary back-and-forth of length κ makes them isomorphic.

A related saturation statement applies to full permutations.

Proposition 12.1. *Let X be an η_κ -order of cardinality κ , where κ is regular infinite. Then $\mathcal{W}_\kappa(X)$ is η_κ . Its global and local cofinalities and coinitalities are all exactly κ .*

Proof. For a cut $A < B$ of total cardinality below κ , apply the common-coordinate search to its permutations. If both sides are nonempty, the first-disagreement indices of the pairs in $A \times B$ have supremum below κ . The search cannot continue sharing a coordinate past this bound; if either active side becomes empty, it stops sooner. The common prefix therefore has length below κ . The η_κ property chooses an unused coordinate strictly between the active coordinate sets. Complete the resulting prefix to a κ -permutation: its remaining entries and remaining positions both have cardinality κ , and its remaining interval of positions has order type κ . The completed permutation fills the cut.

Saturation excludes cofinal or coinital subsets of size below κ , both globally and on either side of a point. Conversely, [Theorem 6.1](#) excludes monotone sequences of length κ^+ . If one of these segments had no cofinal subset of size at most κ , recursively choosing a point above all earlier choices would produce such an increasing sequence. The decreasing argument is identical. Thus each of the stated characters equals κ . $\square$

In particular this proposition applies to $\mathbf{No}_{<\kappa}$ when $2^{<\kappa} = \kappa$. The resulting permutation order has cardinality 2^κ , so it is substantially larger than its alphabet while retaining the same local characters. It is not η_{κ^+} : that stronger cut property would construct an increasing κ^+ -sequence, contrary to Theorem 6.1.

Part III

Global choice and all class well-orders

13 Global choice and well-ordering the surreal class

The distinction between a set of surreal numbers and the entire surreal class changes the existence question. Here GB has the sets of ZF, without a choice axiom, and GC asserts the existence of a class function choosing an element of every nonempty set. A class well-order is a strict linear class relation for which every nonempty available subclass has a least element. Its underlying objects are still sets. The relation itself can be a proper class, and its predecessor classes need not be sets. These are the size distinctions emphasized in the repository's foundations discussion [3].

Throughout this section, $\mathfrak{W}_{\text{all}}$, $\mathfrak{W}_{\text{sl}}$, and $\mathcal{W}(X)$ when X is a proper class are metanotation for the indicated collections of class relations. We quantify over their members as class variables; we do not assert that those collections are GB classes. When X is a set, $\mathcal{W}(X)$ retains its ordinary set meaning from the preceding section. All constructions below are uniform formulas in the stated class parameters. In particular, restricting such a construction to an already given uniformly indexed family produces an ordinary class relation on the indexing domain.

Concretely, a family indexed by a class I is supplied by one class $\mathcal{R} \subseteq I \times \mathbf{No} \times \mathbf{No}$, whose slice at i defines the relation R_i . Writing this family does not make R_i a set or a member of I : the index i is a set, and the slice is a class defined with that parameter. This distinction permits uniform families of proper class relations while preserving the usual GB typing rules. No exhaustive indexing family is assumed in this section.

Theorem 13.1 (Existence and global choice). *Over GB, the following statements are equivalent:*

1. GC holds;
2. there is a class well-order of $\mathbf{No}$;
3. there is a set-like class well-order of $\mathbf{No}$;
4. there is a class bijection $\text{Ord} \rightarrow \mathbf{No}$.

Here set-like means that the predecessors of every element form a set.

Proof. We give the implication from an arbitrary well-order to global choice in detail. It also explains why set choice need not be assumed in the base theory. Let W be a class well-order of the sign-sequence presentation of $\mathbf{No}$. By ordinary set-valued transfinite recursion, with the fixed class parameter W , construct a set relation w_α well-ordering V_α , for every ordinal α .

Set $w_0 = \emptyset$. Suppose w_β has been constructed. There is a unique ordinal θ and a unique order isomorphism

$$e : \theta \longrightarrow (V_\beta, w_\beta).$$

For $A \subseteq V_\beta$, let s_A be the sign sequence of length θ whose value at ξ is $+$ exactly when $e(\xi) \in A$. The map $A \mapsto s_A$ is a bijection between $\mathcal{P}(V_\beta)$ and the set of all sign sequences of length θ . Define

$$A \, w_{\beta+1} \, B \iff s_A \, W \, s_B.$$

This is a set relation by separation with a class parameter, and it well-orders $V_{\beta+1} = \mathcal{P}(V_\beta)$. No selection of an enumeration is involved: the enumeration of an already specified set well-order is unique.

At a nonzero limit λ , order V_λ first by rank, and within rank γ by the restriction of $w_{\gamma+1}$:

$$x w_\lambda y \iff \text{rank}(x) < \text{rank}(y) \text{ or } (\text{rank}(x) = \text{rank}(y) = \gamma \text{ and } x w_{\gamma+1} y).$$

Since λ is a limit, every required $\gamma + 1$ is below λ . A nonempty subset of V_λ has a least occurring rank; the specified well-order on that rank then supplies its least member. Thus the recursion continues. Its stages are sets, even though their union over all ordinals is a class. This is precisely the ordinary GB recursion theorem, rather than a recursion whose individual values are proper classes.

For a nonempty set A , every element of A belongs to $V_{\text{rank}(A)}$. Choose the $w_{\text{rank}(A)}$ -least element of A . The recursion and this prescription define a single class choice function, establishing GC.

Conversely, global choice implies choice for sets. For each ordinal α , the set of well-orders of the birthday layer $S_\alpha = \{-, +\}^\alpha$ is nonempty. Apply the global choice function to this set, and order **No** first by birthday and then by the selected order within the birthday layer. This is a class well-order. Every predecessor class is contained in $\bigcup_{\beta \leq \alpha} S_\beta$ for some α , hence is a set. This proves the set-like version directly.

Finally, for any set-like well-order R of the proper class **No**, assign to x the ordinal order type of its predecessor set. These ranks form an initial subclass of Ord. That subclass cannot be a set: otherwise the inverse rank map, by class replacement on a set domain, would make **No** a set. It is therefore all of Ord, and the inverse rank map is the required bijection. Conversely, a class bijection from Ord transports its well-order to a set-like well-order of **No**. $\square$

The connection between surreal enumerations and global choice is discussed by Hamkins [18]; the general equivalences for global choice are collected in [19]. The explicit rank recursion above proves the assertion over the stated choiceless GB base. It does not assert that a global choice class is definable without parameters in the first-order language of set theory. Definability is an additional requirement.

[merge] *Second proof over GB + AC (source 12)*. Source 12 proves the equivalence by coding transitive closures on ordinals; set choice is used only to code each $\text{TC}(\{x\})$ on an ordinal. Its equivalence includes non-set-like well-orders, as source 11's does. Its subsection on birthday order follows its theorem.

13.1 Source 12's route over GB with set choice

Theorem 13.2 (Choice equivalence). *Over GB plus set-level AC, the following are equivalent:*

- (i) *global choice;*
- (ii) *a class bijection $\text{Ord} \rightarrow \mathbf{No}$;*
- (iii) *a set-like class well-order of $\mathbf{No}$;*
- (iv) *a class well-order of $\mathbf{No}$, not necessarily set-like.*

Proof. Global choice supplies a set-like global well-order of the universe, and hence of the proper class **No**. A proper set-like well-order has type Ord, giving (i) $\Rightarrow$ (iii) $\Rightarrow$ (ii). A bijection in (ii) pulls back ordinal order, giving (iii), and (iii) $\Rightarrow$ (iv) is immediate.

We prove the substantive reverse implication by coding sets. Suppose R well-orders **No**. Let x be a nonempty set and let $T_x = \text{TC}(\{x\})$; it contains x and all its elements. By set choice, there is

a least initial ordinal λ equinumerous with T_x . There is a fixed definable ordinal τ_λ large enough to code a binary relation on λ , together with one distinguished index, by a binary sequence of length τ_λ . For example, concatenate a $\lambda \cdot \lambda$ -long relation table and a λ -long singleton marker.

Consider the *set* of codes for well-founded extensional relations on λ whose Mostowski collapse has range T_x , and whose marked index collapses to x . This set is nonempty. Convert 0, 1 to $-$, $+$, view the codes as surreals, and take its R -least member. Decode it, and choose the element of x represented by the least ordinal index that belongs to the marked node in the coded membership relation.

The result is an element of x , uniquely determined by x and the class parameter R . Elementary class comprehension gives the graph of this choice operation on all nonempty sets. It is a global choice function. Notice that only a set of codes for each x was minimized: set-likeness of R was not needed. $\square$

Remark 13.3 (The explicit set-choice hypothesis). The converse proof uses set choice to code T_x on an ordinal. We do not claim the same equivalence over GB with no set-level choice. A purported stronger choice equivalence must address that missing coding step.

13.1.1 Birthday is not a canonical global well-order

Ordering first by birthday and then by numerical value does not work. For $n < \omega$, let s_n be the length- ω sign sequence consisting of n minuses followed by pluses forever. Then

$$\text{bd}(s_n) = \omega, \quad s_{n+1} < s_n.$$

The whole descending sequence lies in one birthday level. Simplicity itself is well-founded but is only a partial order; two incompatible sign sequences need not be comparable by simplicity.

With global choice one may choose a well-order on every birthday level and concatenate them in birthday order. The result is set-like because all earlier levels, and every initial part of the current level, form a set. The noncanonical choices have not disappeared: they are exactly the kind of information accounted for by [theorem 13.2](#).

Remark 13.4 ([[merge](#)]; source 12's question answered). Dated note, 2 October 2026. Source 12's [remark 13.3](#) declines the choiceless case, and its research question on the exact weak-choice strength ([remark 34.15](#)) asks which principle is equivalent to a class well-order of the sign-sequence surreals over GB without set choice. Source 11's [theorem 13.1](#) answers it: global choice itself. The “missing coding step” that [remark 13.3](#) names is avoided, not supplied: the well-orders w_α of V_α are built by set-valued recursion from the given well-order of the sign sequences of length $\theta = \text{otp}(V_\beta, w_\beta)$, a subset $A \subseteq V_\beta$ being coded by the sign sequence s_A of length exactly θ , so no ordinal coding of a transitive closure has to be chosen. The proof thus produces a uniformly defined sequence of well-orders of the sets V_α , in particular of the power sets of ordinals, which is the comparison the question suggests. The same mechanism, at the level of sets, is the ordinal-graph coding of the birthday-cutoff report (`hset:thm:choice`, whose range over ZF is the class of sets with well-orderable transitive closure) [[4](#)]; that theorem concerns a different statement and is unchanged. Hamkins's question whether surreal universality implies global choice [[21](#)] is not affected. (Corrected in place on 3 October 2026, following the Hilbert's-tenth programme's review [Computability/HilbertTenthProblem/Papers/research-wip/native-stream-queue/review_surreal_synthesis_0be9b9134.md](#) (commit 280583d2f): this note had called the sequence “coherent”. The orders w_α are uniformly defined, but a later one need not restrict to an earlier one. If W puts $-$ before $+$ among the sign sequences of length one and $+-$ before $--$ among those of length two, then w_2 puts $\emptyset$ before $\{\emptyset\}$ (codes $-$ and $+$), while w_3 puts $\{\emptyset\}$ before $\emptyset$ (codes $+-$ and $--$). The proof of [theorem 13.1](#) needs only the

uniform definition: it selects the $w_{\text{rank}(A)}$ -least element of each nonempty set A . Ordering all sets by rank and then, within rank γ , by $w_{\gamma+1}$, as at the limit stages, does give a class well-order whose restrictions to the sets V_α cohere; the proof does not use it.)

[merge] *Third proof over GB + AC, for set-like orders (source 09).* Source 09’s [proposition 13.6](#) is the last paragraph of the proof of [theorem 13.1](#) (and source 08’s [note 5.12](#)); its [proposition 13.7](#) recovers global choice from a class bijection $\text{Ord} \rightarrow \mathbf{No}$ through a definable class surjection of $\mathbf{No}$ onto V , using set choice only to prove surjectivity. By [theorem 13.1](#) that hypothesis is unnecessary ([remark 13.4](#)). Its last subsection leads to source 08’s relation codes ([section 17](#)).

13.2 Source 09: global well-orders and global enumerations

Definition 13.5. A class well-order $<$ on a class D is *set-like* when the predecessors of every $d \in D$ form a set. We take “class well-order” to mean that every nonempty subclass has a least element.

Proposition 13.6. *Set-like class well-orders on $\mathbf{No}$ correspond bijectively to class bijections $E : \text{Ord} \rightarrow \mathbf{No}$.*

Proof. Given E , transfer the ordinal order to its values. Every predecessor set is the image of a set ordinal, so this order is set-like and is a class well-order.

Conversely, for a set-like well-order, assign to x the ordinal order type of its set of predecessors. This assignment is an order isomorphism onto a downward-closed class of ordinals. Its range cannot be a set: otherwise its inverse, a class function on a set, would have set range $\mathbf{No}$. A proper downward-closed class of ordinals is all of Ord . Invert the assignment. $\square$

Global choice supplies at least one such bijection; fix one and call it b . None of these enumerations is increasing in the numerical order on $\mathbf{No}$. The numerical order is dense and has descending sequences, so it is not a well-order.

Proposition 13.7 (The baseline already entails global choice). *Over GB with ordinary choice for sets, the following are equivalent: global choice; a class bijection $b : \text{Ord} \rightarrow \mathbf{No}$; and a set-like class well-order of $\mathbf{No}$.*

Proof. Global choice gives the bijection as explained in [section 5.5](#), and [proposition 13.6](#) identifies the last two conditions without using global choice. It remains to recover global choice from b .

There is an elementary definable class surjection $d : \mathbf{No} \rightarrow V$, using only ordinary set choice to prove surjectivity. Here is the coding argument. A set code consists of a nonzero ordinal γ , a well-founded extensional relation $R \subseteq \gamma \times \gamma$, and a distinguished root $r < \gamma$. Its decoded value is the image of r under the Mostowski collapse of (γ, R) . Every set x has such a code: well-order a transitive set containing x , transport its membership relation to an ordinal, and mark the index corresponding to x .

These triples have a uniform injective encoding by sign sequences. Write γ plus signs, then one minus sign as a header. Append a binary word of length $\gamma \cdot \gamma$ recording R , with (u, v) assigned to position $\gamma \cdot v + u$, and then a word of length γ whose unique plus sign marks r . The first minus sign recovers γ ; the prescribed block lengths recover R and r . Decode valid codes by their set Mostowski collapses and send invalid codes to $\emptyset$. Everything in this decoder concerns set-sized relations, so it defines a class function by elementary comprehension, not a class truth predicate.

The composite $d \circ b : \text{Ord} \rightarrow V$ is a class surjection. Assign to each set x the least ordinal α with $d(b(\alpha)) = x$. These indices define a set-like global well-order of V : predecessors of x occur among the set of decoded values before its index. Taking the least member of each nonempty set in that well-order gives a global choice function. $\square$

Thus existence of the global baseline is not a choice-free substitute for global choice. The equivalence is a standard well-founded-coding argument, included to make the axiomatic cost explicit. It says nothing similar about merely well-ordering one birthday-bounded set.

13.2.1 Why non-set-like global well-orders are different

Not every class well-order is set-like. For example, $\text{Ord} + 1$ and $\text{Ord} + \text{Ord}$ are class well-orders; the final point in the first example, and every point in the second block of the second example, has a proper class of predecessors. Global choice permits transfer of these orders to the underlying class **No**.

Such orders have no exhaustive enumeration by ordinal positions alone. They are deliberately excluded from $\mathfrak{W}_{\text{sl}}$. Even two presentations of orders with a full first block of type Ord can agree throughout that block and differ later; there is then no ordinal position at which their full presentations first differ.

There is a different, fully explicit way to compare all class well-order *relations*, including non-set-like ones: fix a class bijection $c : \text{Ord} \rightarrow \mathbf{No} \times \mathbf{No}$ and compare the binary characteristic classes of the relations at their first differing c -coordinate. This is a legitimate comparison formula on class variables. However, it depends on the auxiliary coordinate enumeration and is not the exhaustive-enumeration lexicographic order analyzed here. Its collection of objects remains of the same higher logical level.

[merge] The relation-code comparison of source 09's last paragraph is source 08's [proposition 17.1](#), which computes its equimorphism type. Source 09's research direction on it ([section 34.3](#)) asks for more: a dense set-coded core and the representation of its cuts.

14 Lexicographic comparison without class ordinals

The set-sized enumeration method studied for the real numbers [1] suggests comparing the first unequal labels. Arbitrary class well-orders require a formulation that also permits a proper class of common earlier labels. No ordinal enumeration of that entire common prefix is needed.

For a strict class well-order R on a fixed class X , write

$$P_R(x) = \{y \in X : yRx\}.$$

A common labeled prefix of R, S is a subclass initial in both orders on which their restrictions agree. The adjective labeled matters: agreement here is equality of the actual elements and relations, rather than the existence of an abstract order isomorphism.

Lemma 14.1 (Unique first disagreement). *If $R \neq S$ are class well-orders of the same class X , there are unique distinct $a, b \in X$ such that*

$$P_R(a) = P_S(b) = C, \quad R \upharpoonright C = S \upharpoonright C.$$

The subclass C is their maximal common labeled prefix.

Proof. The class $D = \{x : P_R(x) \neq P_S(x)\}$ is nonempty, since predecessor classes determine a strict order. Let a be its R -least element and put $C = P_R(a)$. For every $y \in C$, $P_R(y) = P_S(y)$. In particular, ySa : otherwise aSy , which would imply aRy , contradicting yRa . Thus $C \subseteq P_S(a)$. Also, if zSy with $y \in C$, then zRy , so $z \in C$. Therefore C is S -initial as well as R -initial, and the restrictions of the orders to it agree.

The inclusion $C \subseteq P_S(a)$ is strict because $a \in D$. Let b be the S -least element outside C . The strict inclusion shows that bSa , hence $b \neq a$. Initiality and the minimality of b give $P_S(b) = C$.

For uniqueness, suppose a', b', C' satisfy the displayed conditions with $a' \neq b'$. Every element of C' has identical predecessor classes in the two orders. But $b'Sa'$, so the predecessor classes of a' differ. Consequently a' is exactly the R -least element of D ; hence $a' = a$, $C' = C$, and $b' = b$. Any common prefix properly extending C would have to include incompatible next labels, so C is maximal. $\square$

Fix now a native linear order $<_X$ on the labels. For distinct R, S , define $R <_{\text{lex}} S$ when the labels supplied by Lemma 14.1 satisfy $a <_X b$.

Theorem 14.2 (Raw lexicographic order). *This rule is a strict linear order on the class well-orders of X , understood as a definable relation on class variables. In particular it defines $\mathfrak{W}_{\text{all}}$ for $X = \mathbf{No}$ with its native numerical order.*

Proof. Existence and uniqueness of first disagreement give totality and asymmetry; the relation is irreflexive by definition. To verify transitivity, suppose $R <_{\text{lex}} S <_{\text{lex}} T$. Write (a, b) for the first unequal labels of R, S , and (c, d) for those of S, T . Their common prefixes are $C = P_S(b)$ and $D = P_S(c)$, respectively, and $a <_X b$, $c <_X d$.

If bSc , then $C \subsetneq D$. The orders S, T agree through the label b , so the first unequal labels of R, T are a, b , with common prefix C . Thus $R <_{\text{lex}} T$. If cSb , then $D \subsetneq C$; now R, S agree through c , and the first unequal labels of R, T are c, d , again giving the conclusion. If $b = c$, the prefixes coincide and the next labels of R, T are a, d . Since $a <_X b = c <_X d$, the conclusion follows in this case as well. $\square$

This proof compares labeled relations directly. It does not compare arbitrary class well-orders up to isomorphism, and makes no appeal to a collection of class ordinal types. The defining conditions use only set quantifiers with the given class relations as parameters: equality of predecessor classes means a universal statement about their set elements.

For example, the predicate that a, b are the first unequal labels can be written without quantifying over a common-prefix class:

$$\begin{aligned} a \neq b \quad \wedge \quad & \forall z (zRa \leftrightarrow zSb) \\ & \wedge \quad \forall x \forall y ((xRa \wedge yRa) \longrightarrow (xRy \leftrightarrow xSy)). \end{aligned}$$

Adding $a <_X b$ and existentially quantifying the two labels gives the lexicographic predicate. Consequently its restriction to the indexed family $\mathcal{R}$ described above is a GB class of pairs of indices. If the indexing contains duplicate orders, this relation orders the distinct relations, and becomes a strict linear order on indices whenever the indexing is injective. Passing instead to abstract isomorphism types would discard exactly the label information on which the construction depends.

Proposition 14.3 (Convex prefix cones). *Fix a well-ordered subclass (C, P) of X . The collection of all class well-orders of X extending (C, P) as an initial labeled segment is convex in the raw lexicographic order.*

Proof. Suppose R, S extend the prefix and $R <_{\text{lex}} T <_{\text{lex}} S$. Let D be the first-disagreement prefix of R, T . The subclasses C, D are initial in R , and therefore inclusion-comparable. If $C \subseteq D$, then T extends (C, P) , as required. If $D \subsetneq C$, the next R -label lies in C ; it is also the next S -label after D . Accordingly, T compares with R and S in the same direction, contradicting the strict inequalities. This reasoning permits C and D to be proper classes. $\square$

[merge] *Second route (source 12): the largest common labelled prefix.* Source 12's Section 3 proves lemma 14.1 and theorem 14.2 differently. Instead of the class D of points whose predecessor classes differ, it defines the largest common labelled prefix $C(R, S)$ by an elementary formula and proves directly that it is initial in both orders and contains every common labelled initial segment. Source 11's shipped check compares its predecessor rule with finite lexicographic order on all 266,272 pairs of permutations of at most six points (`data/11-raw-orders-finite_checks_results.txt`); source 12's checks the common-prefix formula on 15,017 finite pairs (`data/12-singular-finite_checks.json`).

14.1 Source 12's route: the largest common labelled prefix

The central observation is that comparing two *labelled* well-orders is easier than comparing arbitrary abstract class order types. Fix a class X equipped with an ambient strict linear order $<$, and let R, S be class well-orders of the same carrier X .

Definition 14.4 (Common labelled prefix). Define $C(R, S) \subseteq X$ by putting $x \in C(R, S)$ if

$$\begin{aligned} \text{Pred}_R(x) &= \text{Pred}_S(x), \\ \forall u, v \in \text{Pred}_R(x), \quad uRv &\iff uSv. \end{aligned}$$

Equality of the predecessor classes is expanded as a quantification over sets $u \in X$. Thus this is an elementary formula with class parameters R, S ; it is not a quantification over all possible common prefixes.

Lemma 14.5 (Maximal-prefix lemma). *$C(R, S)$ is an initial segment of both well-orders, their restrictions to it agree, and it contains every common labelled initial segment. If $R \neq S$, it is proper, and the least elements*

$$a = \min_R(X \setminus C(R, S)), \quad b = \min_S(X \setminus C(R, S))$$

exist and are distinct.

Proof. Suppose $x \in C(R, S)$ and yRx . Equality of the predecessor classes puts y before x in S as well. For zRy , transitivity gives zRx ; agreement of R, S below x then gives zSy . Conversely, zSy implies zSx , hence zRx , and agreement below x yields zRy . The orders agree below y because they agree below x . Therefore $y \in C(R, S)$. The symmetric argument proves initiality in S .

If $u, v \in C(R, S)$ and uRv , equality of the predecessor classes of v gives uSv . Thus the restrictions agree. If D is any common labelled initial segment and $x \in D$, both predecessor classes of x lie in D and agree, as do the restricted relations. Hence $D \subseteq C(R, S)$. If $C(R, S) = X$, equality of every predecessor class implies $R = S$.

For $R \neq S$, class well-foundedness supplies a, b . By initiality and minimality, $\text{Pred}_R(a) = C(R, S)$ and $\text{Pred}_S(b) = C(R, S)$. If $a = b$, these predecessor classes agree and the two relations agree on them. The defining formula would put a in $C(R, S)$, a contradiction. $\square$

Definition 14.6 (Full lexicographic comparison). For distinct R, S , put $R <_{\text{lex}} S$ precisely when $a < b$, with a, b from lemma 14.5. Identical relations are equal in this order.

Theorem 14.7 (Full class lexicographic order). *In GBC, the preceding comparison is a strict linear ordering of the predicate of all class well-orders of a fixed linearly ordered class X . Its definition uses only elementary class comprehension. In particular, ETR and the comparability of abstract class well-order types are not required.*

Proof. Irreflexivity and trichotomy follow from the maximal-prefix lemma and trichotomy in X . For transitivity suppose $R <_{\text{lex}} S <_{\text{lex}} T$. The classes $C(R, S)$ and $C(S, T)$ are initial segments of the same well-order S , so they are comparable by inclusion.

If $C(R, S)$ is a proper subset of $C(S, T)$, the first S -element b after $C(R, S)$ belongs to $C(S, T)$. Thus T agrees with S through that element. The first disagreement of R, T is exactly the earlier R, S disagreement, with the same two labels $a < b$. Consequently $R <_{\text{lex}} T$. The opposite proper-inclusion case uses the earlier S, T disagreement and is identical in form.

If the two common prefixes are equal to C , let a, b, c be the first labels after C in R, S, T . The hypotheses give $a < b < c$. Hence R, T also first disagree after C , and $R <_{\text{lex}} T$. Every construction of C , and every assertion about its next elements, expands into set quantifiers with the given class relations as parameters. $\square$

Remark 14.8 (Why this does not prove class order-type comparability). In general one asks whether two arbitrary class well-orders admit an initial-segment isomorphism. ETR is a sufficient hypothesis for that principle [27, Theorem 6]. Here no such isomorphism is constructed. Only their *identical labelled* portions are compared. As soon as the labels disagree, the fixed ambient order decides the answer. Confusing these two tasks would impose an unnecessary stronger axiom.

Example 14.9 (A disagreement after every ordinary ordinal). Fix $a < b$ in $\mathbf{No}$. Well-order $\mathbf{No} \setminus \{a, b\}$ in type Ord. Append first a, b , and in a second order append b, a . The orders agree throughout their initial Ord-block and differ only after it. No ordinary ordinal indexes that first disagreement, but [theorem 14.7](#) compares them immediately: their common labelled prefix is $\mathbf{No} \setminus \{a, b\}$.

Corollary 14.10 (Agreement with enumeration lexicography). *For set-sized exhaustive ordinal enumerations, and for class bijections $\text{Ord} \rightarrow X$, the common-prefix definition agrees with first-difference lexicography.*

Proof. Before the first unequal entry, exactly the same labels occur in exactly the same order. Their ranges form the common labelled initial segment. No later label can belong to that segment by initiality and the distinct next labels. In the Ord case the disagreement is a nonempty class of ordinals; restricting it below any one witness gives its least member. $\square$

[merge] Source 12's [remark 14.8](#) is source 11's remark after [theorem 14.2](#) and in [section 20](#): comparability of abstract class well-orders by initial-segment isomorphisms follows from ETR [27], and neither labelled comparison needs it. Its [example 14.9](#) is source 11's [example 16.3](#), and its [corollary 14.10](#) is the corresponding sentence after [definition 19.1](#).

15 The canonical initial set-like part

Every class well-order of $\mathbf{No}$ has a canonical initial segment of type Ord, even when its entire order is not set-like. This gives a structural description of all well-orders using ordinary class functions.

Theorem 15.1 (Canonical initial part). *For $R \in \mathfrak{W}_{\text{all}}$, let*

$$I_R = \{x \in \mathbf{No} : P_R(x) \text{ is a set}\}.$$

Then I_R is a proper initial subclass of the domain or is the entire domain; in either case it is a proper class. Its induced order is set-like and has a unique order enumeration

$$F_R : \text{Ord} \longrightarrow I_R.$$

The order R is the concatenation of this enumeration with its induced well-order on $\mathbf{No} \setminus I_R$.

Proof. If $x \in I_R$ and yRx , then $P_R(y) \subseteq P_R(x)$, so separation makes $P_R(y)$ a set. Thus I_R is initial. Suppose it were a set. Since **No** is proper, there would be an R -least element x outside it. Every predecessor of x would then belong to I_R , and initiality would place every member of I_R before x . Hence $P_R(x) = I_R$, a contradiction.

The induced order on I_R has the same predecessor sets as R . The ordinal-rank map therefore identifies it with an initial subclass of **Ord**. As in Theorem 13.1, a set-sized range would make I_R a set by replacement. Its range is consequently all of **Ord**. The inverse map is unique, and initiality gives the asserted concatenation. $\square$

Example 15.2 (Two proper blocks). Under GBC, fix $e : \text{Ord} \rightarrow \mathbf{No}$ bijective. The disjoint classes $A = \{e(2\alpha) : \alpha \in \text{Ord}\}$ and $B = \{e(2\alpha + 1) : \alpha \in \text{Ord}\}$ partition **No**, where the product is ordinal multiplication $2 \cdot \alpha$. Order each block by its displayed ordinal index and place all of A before all of B . The resulting well-order has class type $\text{Ord} + \text{Ord}$. Its canonical initial part is exactly A : a point of B has the whole proper class A among its predecessors. Hence an **Ord**-enumeration of the canonical part is not, in general, an enumeration of the entire well-order. Both blocks consist of ordinary surreal labels; their size affects their positions in the chosen well-order, not the status of individual surreal numbers as sets.

Let $\mathfrak{J}$ denote the metacollection of injective class functions $F : \text{Ord} \rightarrow \mathbf{No}$, with lexicographic comparison at their first ordinal of disagreement. Such a first disagreement exists for distinct functions: after choosing any witness, minimize within the set of earlier ordinals. Put $Y_F = \mathbf{No} \setminus \text{ran}(F)$.

Proposition 15.3 (Convex fibers). *Assume GBC. Every $F \in \mathfrak{J}$ occurs as F_R for some $R \in \mathfrak{W}_{\text{all}}$. The fiber over F is a convex copy of $\mathcal{W}(Y_F)$. Distinct fibers are ordered by the first unequal values of their functions F .*

Proof. The range of an injection from **Ord** is proper, since a set could not contain an injective image of all ordinals. Choose a well-order Q of Y_F , using the restriction of any fixed global well-order, and put the sequence F before Q . Each $F(\alpha)$ has exactly the set $F''\alpha$ of predecessors. Each residual point, however, has the proper class $\text{ran}(F)$ among its predecessors. Thus the resulting order has canonical function exactly F .

All orders with that function extend the same initial labeled order on $\text{ran}(F)$, and conversely every extension has that canonical function. Proposition 14.3 proves convexity. Restriction to Y_F and concatenation with F are inverse operations; first disagreement inside the residual orders proves that they preserve lexicographic comparison. Finally, if $F \neq G$, their first ordinal disagreement occurs before either residual order begins. It therefore determines the comparison of every member of one fiber with every member of the other. $\square$

We may summarize this result as the lexicographic-sum description

$$\mathfrak{W}_{\text{all}} \cong \sum_{F \in (\mathfrak{J}, <_{\text{lex}})} \mathcal{W}(Y_F).$$

This is uniform metanotation for the preceding theorem, not the assertion that its indexing hyperclass or all its fibers have been collected into a GB class. The set-like fragment $\mathfrak{W}_{\text{sl}}$ consists exactly of the fibers with $Y_F = \emptyset$, or equivalently the exhaustive enumerations. For finite Y_F of size m , its fiber is a convex block of $m!$ orders, with the usual lexicographic order on permutations.

Remark 15.4 (Singleton fibers and iteration). A singleton fiber need not be set-like. If F omits exactly one surreal z , its unique completion puts z after the entire **Ord**-sequence; it has class order

type $\text{Ord} + 1$. Conversely, every exhaustive F gives a singleton set-like fiber. Extracting the initial part once also does not justify transfinite iteration of this operation through class-sized tails. Such an iteration would require a separate specification of its indexing order, its objects, and its recursion principle; none is asserted here.

16 All jumps arise from finite residual permutations

The full order has jumps despite the extreme density of the native surreal line. A proper common prefix can already have used every label between the two labels at which comparison is decided.

Lemma 16.1 (Extreme enumerations). *For a class Z with a native linear order, a class well-order is lexicographically greatest among all well-orders of Z exactly when it is the native reverse order. It is lexicographically least exactly when it is the native forward order.*

Proof. If a well-order places y before z with $y < z$, transpose these two labels throughout the order. The result is another class well-order. Its first changed label is z in place of y , so it is lexicographically greater. A greatest order must therefore be strictly decreasing in the native order, which determines the entire relation. Conversely, in the reverse native order the next label at any first disagreement is the greatest remaining label, so every different well-order is smaller. The least-order assertion follows by reversing the native order. The argument includes empty and singleton domains. $\square$

The lemma is an existence criterion as well as a characterization: a greatest enumeration exists only when the native reverse order itself is well-founded. It is therefore not permissible to use “list the remaining labels in decreasing order” as an unconditional completion rule. For example, on a copy of the natural numbers with its usual native order, the increasing enumeration is least, but there is no greatest well-order. Finite residual domains have both extrema, and the adjacency theorem shows that precisely such domains can support a jump after a common prefix.

Theorem 16.2 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be class well-orders of X . Let C be their first-disagreement prefix, let $a < b$ be their respective next labels, and put $Y = X \setminus C$. They are adjacent if and only if all three conditions hold:*

1. *there is no $y \in Y$ with $a < y < b$;*
2. *the R -tail on $Y \setminus \{a\}$ is the native reverse order;*
3. *the S -tail on $Y \setminus \{b\}$ is the native forward order.*

These conditions force Y to be finite. Thus adjacency is exactly ordinary finite permutation adjacency after a possibly proper class common prefix.

Proof. Suppose first that the pair is adjacent. A label $y \in Y$ strictly between a, b could be placed immediately after C , followed by the restriction of R to the other remaining labels. This would give a well-order strictly between R, S ; thus condition 1 holds.

Any lexicographically greater replacement of the R -tail after a would also give an intermediate order: its next label after C would still be $a < b$. Hence this tail is greatest, and Lemma 16.1 gives condition 2. A lexicographically smaller replacement of the S -tail after b similarly gives condition 3. These modifications use restrictions and finite transpositions of existing class relations; they require no new choice principle.

The overlap $Z = Y \setminus \{a, b\}$ is now well-ordered both forward and backward in the native order. If Z were not finite, repeatedly take its least element after deleting the finitely many already selected. Ordinary recursion on ω and replacement produce a set of strictly increasing selected elements.

That set has no greatest member, contradicting reverse well-ordering. Thus Z , and hence Y , is finite. This argument does not assume that Z was a set in advance.

Conversely, suppose the three conditions hold and let $R <_{\text{lex}} T <_{\text{lex}} S$. Convexity of the prefix cone forces T to extend C . Let $t \in Y$ be its next label. A label below a makes $T < R$, and a label above b makes $T > S$; condition 1 excludes intermediate labels. If $t = a$, its remaining tail cannot exceed the greatest tail of R , so $T \leq R$. If $t = b$, its tail cannot precede the least tail of S , so $T \geq S$. Every possibility contradicts strict betweenness. $\square$

Example 16.3 (An adjacent pair of type $\text{Ord} + 2$). Assume GBC, choose surreal labels $a < b$, and give $C = \mathbf{No} \setminus \{a, b\}$ a set-like well-order P . Consider

$$R = P + (a, b), \quad S = P + (b, a).$$

Each has class order type $\text{Ord} + 2$, and R, S are adjacent. They have the same canonical initial enumeration, and constitute its entire two-point fiber. The notation $\text{Ord} + 2$ describes a class concatenation; it is not a set ordinal.

For $X = \mathbf{No}$, every adjacent pair has a cofinite, hence proper class, common prefix. Neither endpoint can be set-like. Moreover, if two set-like orders are given, their common prefix is a set. The deletion property of the surreal line leaves an unused label strictly between their next labels, and that set prefix can be completed to a set-like well-order. For completeness, this extension does not require an additional recursion principle. Retain the prescribed set prefix, put the new intermediate label next, and then append the restriction of a fixed set-like global well-order of $\mathbf{No}$ to the unused labels. A predecessor class in the appended part is contained in the union of the prescribed set prefix and a predecessor set of the fixed order, together with the one inserted label. It is therefore a set. Every label occurs once, and the concatenation is a well-order, so the completion belongs to $\mathfrak{W}_{\text{sl}}$. Thus the induced order $\mathfrak{W}_{\text{sl}}$ is dense, while $\mathfrak{W}_{\text{all}}$ has the finite terminal blocks classified above. Allowing all well-orders changes the local order structure in a concrete and unavoidable way.

[merge] *Second route (source 12): residual decomposition and adjacency.* Source 12's Section 8 proves [theorem 16.2](#) in the form “the residual class is finite and the residual permutations are consecutive”; the two criteria are equivalent, since a tail that is both well-ordered and reverse well-ordered is finite. Its [proposition 16.5](#) is the description of one prefix cone given by [propositions 14.3](#) and [15.3](#). Its [corollary 16.7](#) adds that interior points of the finite blocks with $n \geq 3$ are isolated in the order topology and that the set-like suborder is not dense in the full order, and its last paragraph adds the order-reversing involution induced by negation.

16.1 Source 12's route: unrestricted class well-orders and finite-tail geometry

The full order $\mathfrak{W}_{\text{all}}(\mathbf{No})$ contains $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ as an ordered subcollection, but it is not dense. The effect is not a failure of density of the surreal alphabet; it is exhaustion of almost all labels by a proper-class prefix.

Definition 16.4 (A prefix cylinder in the full order). Let $C \subsetneq X$ carry a fixed well-order P . Consider all class well-orders of X that have (C, P) as a labelled initial segment. Write this cylinder as $[P]_X$ and put $D = X \setminus C$.

Proposition 16.5 (Residual-order decomposition). *$[P]_X$ is convex and is order-isomorphic to the lexicographic order of all well-orders of the residual carrier D , with its inherited ambient order. If D is a set, this is the ordinary set order $\mathcal{W}(D)$, with the same notation extended to finite carriers. If $|D| = n < \omega$, the cylinder has exactly $n!$ elements.*

Proof. Restriction to D and ordered concatenation with P are inverse operations. Concatenation of two class well-orders is a class well-order: a nonempty subclass has its minimum in the first part if it meets that part, and otherwise in the second. The maximal common prefix after P is exactly the common prefix of the tails, so comparisons are preserved.

If a third order differs before completing P , its first discrepancy with every extension of P is the same and places it on the same side of all of them. This proves convexity. The last assertion counts permutations of a finite carrier. $\square$

Theorem 16.6 (Exact adjacency criterion). *Let $R <_{\text{lex}} S$ be two class well-orders of X , and let $C = C(R, S)$, $D = X \setminus C$. They are adjacent in the full order if and only if D is finite and their residual permutations are consecutive in the ordinary finite lexicographic order.*

Proof. Let $a < b$ be their first residual labels. If some $d \in D$ satisfies $a < d < b$, begin the residual order with d and well-order the remaining labels by a restriction of R . This gives an order strictly between. So an adjacent pair can have no such d .

If the tail of R after a has two labels u before v with $u < v$ in the ambient order, interchange the labels u, v at their two positions. The new well-order is larger than R , and still smaller than S because its first residual label remains a . Thus adjacency forces the R tail to be numerically decreasing. Dually, the tail of S after b must be numerically increasing.

On $D \setminus \{a, b\}$, both the ambient numerical order and its reverse are therefore well-orders. Such a carrier is finite: an infinite well-order contains an increasing ω -sequence, contradicting well-foundedness of its reverse. Hence D is finite.

Conversely, if D is finite, convexity in [proposition 16.5](#) says that every order between R, S must be another permutation of D after C . There is none exactly when the two permutations are consecutive. $\square$

Corollary 16.7 (Proper-class prefixes are responsible). *Adjacent pairs in $\mathfrak{W}_{\text{all}}(\mathbf{No})$ share a proper-class prefix. There are convex blocks of every factorial size $n!$. For $n \geq 3$, interior elements of such blocks are isolated in the order topology. The set-like suborder is not dense in the full order.*

Proof. A set prefix cannot exhaust all but finitely many surreals. Conversely, well-order $\mathbf{No} \setminus D$ in type Ord for any finite D , and append all its permutations. For $|D| \geq 3$, choose two nonconsecutive permutations in the resulting finite convex block. Their nonempty open interval contains only orders with a proper-class predecessor segment, so it contains no set-like order. $\square$

The full order still has no endpoints and no set-indexed cofinal or cointial family, by the first-entry argument of [note 19.28](#). It contains every class linear order, by [note 19.21](#). But neither global density nor interpolation for all set-indexed cuts survives: an adjacent pair already gives a two-sided cut that cannot be filled.

Pointwise surreal negation gives an order-reversing involution on both global orders: transport a well-order by the bijection $x \mapsto -x$. This preserves enumeration type and set-likeness while reversing the ambient comparison of the first unequal labels.

[merge] Source 08's Section 14 proposes two alternatives to the labelled comparison of [section 14](#): a fixed larger index order, and relation-code lexicography.

17 Non-set-like well-orders and alternative coding (source 08)

The restriction to set-like well-orders is substantive, not cosmetic. Choose distinct surreals $a < c$, and enumerate $\mathbf{No} \setminus \{a, c\}$ by Ord. Append a, c at the end in one well-order and c, a in another.

These are legitimate class well-orders of type $\text{Ord} + 2$, but they are not set-like: the final points have a proper class of predecessors.

The two enumerations agree at every *set ordinal* position. Their first disagreement occurs only after a proper-class initial segment, at the first appended point. Thus (S08.7), whose index is Ord , does not compare them. Treating “ Ord ” as one more set ordinal would conceal precisely the error.

17.1 A fixed larger index order

One can fix a class well-order J , such as $\text{Ord} + 2$, and compare class bijections $J \rightarrow \mathbf{No}$ at their first difference, using the order on J and the numerical order on $\mathbf{No}$. For the indicated concrete J this is entirely rigorous. For a general J , the existence of least members for the disagreement subclasses must be part of the specified class well-order property.

What does not follow is a canonical single comparison for all bijections having all possible non-set-like class well-order domains. Unlike set ordinals, the domains do not automatically come with a common set-ordinal coordinate system. A comparison theorem for all such domains, or a recursive alignment of their proper-class initial segments, needs an explicit formulation and an axiomatic audit. We do not assume an unproved class analogue of every set well-order theorem.

17.2 Relation-code lexicography works, but changes the question

There is an alternative that does not align enumeration domains. Fix a class bijection

$$q : \text{Ord} \longrightarrow \mathbf{No} \times \mathbf{No}.$$

Encode each strict class relation R on $\mathbf{No}$ by $\chi_R(\xi) = 1$ iff $q(\xi) \in R$. Compare these characteristic functions lexicographically. This gives a comparison of *all* class well-order relations, set-like or not. It depends on q , and is not generally the numerical lexicographic comparison of their exhaustive enumerations.

Proposition 17.1. *For every fixed q , the virtual order of all class well-order relations under relation-code comparison is equimorphic with $\mathbb{B}_{\text{Ord}}$. The lower embedding can be chosen to use only set-like well-order relations.*

Proof. The upper embedding is the characteristic-function code itself. For the lower embedding, partition $\mathbf{No}$ into disjoint two-element blocks using a baseline b . For each block $\{a, c\}$, let m be the smaller of the two positions where q lists (a, c) and (c, a) . The positions m are distinct and form a proper subclass of Ord ; they have a unique increasing enumeration by Ord . Order the blocks by this enumeration.

Within each block, choose its orientation so that the relation value at its first listed directed pair is the prescribed input bit. Order all elements in earlier blocks before all elements in later blocks. This is a set-like well-order, consisting of an Ord -sequence of two-element blocks. Reversing a block changes only the two directed relations internal to that block, not any relations between blocks. At the first differing input bit, its associated m is therefore the first changed coordinate of the characteristic function. The code preserves order. All selections and enumerations are uniform relative to q, b . $\square$

This supplies two coherent meanings of “lexicographic ordering of all well-orders”: numerical enumeration comparison on the set-like part, and relation-code comparison on all class well-orders. They have uniform embeddings into one another through the binary class order, but their definitions, canonicity properties, and possible topological invariants should not be conflated.

[merge] Dated note, 2 October 2026. Source 08 says (its Section 14.1) that a canonical single comparison for all bijections from all non-set-like class well-order domains does not follow without an explicit alignment of proper-class initial segments, and asks (its question 11) for a comparison principle for arbitrary class well-orders and its recursion requirements. Sources 11 and 12 supply one without aligning domains: compare the two relations on their largest common labelled prefix, then compare the next labels ([theorems 14.2](#) and [14.7](#)); its definition uses elementary comprehension only, not ETR and not comparability of abstract class order types. For the pair of type $\text{Ord} + 2$ displayed at the start of this section, the comparison is decided at the first appended point ([example 16.3](#), [example 14.9](#)). Source 08’s warning stands for the alignment of abstract domains, and relation-code lexicography remains a different order ([section 18](#) makes the same point on three points).

18 Numerically well-ordered subclasses and other interpretations (source 12)

The positive surreal ordinals form a numerically increasing class copy of Ord , but this is not a well-order of all of $\mathbf{No}$. More generally, [lemma 5.14](#) implies that every class well-order is isomorphic to a numerically well-ordered subclass of $\mathbf{No}$. That subclass need not be set-like in its inherited numerical order.

For an explicit example, the class

$$\left\{ -\frac{1}{\alpha + 1} : \alpha \in \text{Ord} \right\} \cup \{0\}$$

has numerical order type $\text{Ord} + 1$. The arithmetic here is surreal arithmetic on the embedded ordinal values; the reciprocal reverses positive order and the minus sign reverses it back. The final point has a proper class of predecessors. Putting a positive ordinal block after this negative block gives a numerical copy of $\text{Ord} + \text{Ord}$.

For a set $A \subseteq \mathbf{No}$, numerical well-ordering has the usual ordinal order type. Its birthday bound places it inside some $\mathbf{No}_{<\kappa}$. The class of all such set-sized well-ordered subsets is a legitimate class of sets. Its members are not exhaustive enumerations of a fixed carrier, however, so one must specify what happens when one enumeration is a prefix of another. The “shorter first” convention and the surreal $- < \mathbf{end} < +$ convention are different choices; neither can be silently borrowed from the prefix-free exhaustive setting.

Finally, ordering binary relation tables rather than enumerations is another legitimate but different construction. If one fixes a set-like well-order of $\mathbf{No} \times \mathbf{No}$, compares the characteristic functions of relations at the first unequal table entry, and restricts to well-orders, one gets a table-dependent order. A first changed relation bit is not generally the first changed enumerated label. Even on three points the two lexicographic orders can differ. The present article uses labelled initial segments, not relation-table lexicography.

[merge] Source 12’s shipped check records a three-point example on which relation-table lexicography and enumeration lexicography differ (`data/12-singular-finite_checks.json`, key `relation_table_vs_enumeration_example`).

Part IV

Set-like global well-orders

19 The lexicographic structure of global enumerations

The set-like case has a particularly strong structure theorem. Although the full collection of global enumerations is a higher-order object, one ordinary class of set codes already meets every cut specified by set-indexed families. That class, with its induced order, is isomorphic to the surreal line. The distinction between a set-indexed family and an ordinal-indexed class family is essential: an explicit omitted-range construction shows exactly where the interpolation theorem stops.

Throughout this section we work in GBC and fix a class bijection

$$e_0 : \text{Ord} \longrightarrow \mathbf{No}.$$

The existence of such a baseline is supplied by global choice; the relevant equivalences are discussed in [19]. We use the surreal set-cut property, its preservation under deleting a set (Lemma 5.2), and the universality and uniqueness statements of Proposition 5.13. These are order-theoretic properties; the field operations on $\mathbf{No}$ play no role in the main construction.

19.1 Class parameters and uniform families

Definition 19.1. For this section, $\mathfrak{J}$ denotes the collection of class injections $F : \text{Ord} \rightarrow \mathbf{No}$, and $\mathfrak{W}_{\text{sl}}$ denotes the subcollection of class bijections. For distinct $F, G \in \mathfrak{J}$, put

$$d(F, G) = \min\{\alpha \in \text{Ord} : F(\alpha) \neq G(\alpha)\}, \quad F <_{\text{lex}} G \iff F(d(F, G)) < G(d(F, G)).$$

A *uniformly set-indexed family* $(F_i)_{i \in I}$ is given by a set I and one class relation whose i th row is the graph of F_i .

These notations are interpreted relationally in two-sorted class theory. An individual F is a class, not a possible element of a GB class. Thus “for all $F \in \mathfrak{W}_{\text{sl}}$ ” abbreviates quantification over class functions with the stated property. Likewise, a uniformly indexed family is a single class of triples (i, α, x) , not a literal set containing proper classes. The first-difference definition is legitimate: choose any coordinate of disagreement and minimize within the set of preceding ordinals. The usual first-difference proof gives a strict linear comparison on these objects.

Uniformity matters whenever a set bound is taken. For example, if two uniformly set-indexed families have different rows across every pair, Class Replacement makes their cross first-difference ordinals a set. No principle of choosing an arbitrary family of classes is being assumed. Finite lists of given class parameters are automatically uniform, and the families of set-coded objects constructed below have a fixed evaluation formula, so their set-indexed subfamilies are uniform as well.

[merge] Sources 09 and 08 define the same comparison of global enumerations (source 09 in section 19.3; source 08 in its Section 9, “The virtual order of global enumerations”).

Note 19.2 (Source 08, display (7) and Proposition 9.1). For two different global enumerations e, f ,

$$\delta(e, f) = \min\{\alpha \in \text{Ord} : e(\alpha) \neq f(\alpha)\}, \quad e <_{\text{lex}} f \iff e(\delta(e, f)) < f(\delta(e, f)), \quad (\text{S08.7})$$

and *this comparison satisfies trichotomy and transitivity; it defines a linear comparison in the two-sorted sense*. This is definition 19.1 and the paragraph after it, by the same argument. Source 08’s Section 9.1 makes source 11’s point about uniform families: a family indexed by J is one class

relation $T \subseteq J \times \text{Ord} \times \mathbf{No}$; for a set J , class replacement bounds all cross first-disagreement coordinates; “merely saying ‘choose one class for every member of a set’ would not supply the uniform relation, and could hide an additional class-choice principle”; with repeated rows one works with the comparison preorder or an injective subfamily.

19.2 A class of bounded-support codes

Definition 19.3. A *normalized bounded-support code* is a set permutation $\sigma : \lambda \rightarrow \lambda$ of an ordinal λ satisfying

$$\lambda = \sup\{\xi + 1 : \xi < \lambda \text{ and } \sigma(\xi) \neq \xi\}.$$

The empty permutation is allowed. Extend σ to a class permutation $\hat{\sigma}$ of Ord by the identity outside λ , and evaluate the code as

$$E_\sigma = e_0 \circ \hat{\sigma}.$$

Write $\mathcal{P}_{\text{bd}} = \mathcal{P}_{\text{bd}}(e_0)$ for the ordinary class of these codes, ordered by the lexicographic comparison of their evaluations.

Normalization removes the otherwise harmless duplication obtained by adding fixed points to the end of a code. Every class permutation with bounded support has exactly one normalized code: restrict it to the supremum of one plus its moved coordinates. Its support is invariant under the permutation, so this restriction is a permutation of the indicated ordinal. Consequently distinct normalized codes evaluate to distinct enumerations, and the induced order on $\mathcal{P}_{\text{bd}}$ is a genuine class linear order. The class is proper, since the transpositions of 0 with arbitrary nonzero ordinals have distinct codes.

Here and below, identifying a code with its evaluated enumeration is a notational convenience. It does not convert the evaluated proper-class graph into a set. In particular, the assertion that $\mathcal{P}_{\text{bd}}$ is an ordinary class refers to the codes, while assertions comparing its members with arbitrary members of $\mathfrak{W}_{\text{sl}}$ use the common evaluation relation.

Lemma 19.4 (Completion of an arbitrary set prefix). *Every injective set function $p : \delta \rightarrow \mathbf{No}$, with δ an ordinal, is the initial segment of an enumeration represented in $\mathcal{P}_{\text{bd}}$. The completion can be chosen uniformly in p .*

Proof. Let $q = e_0^{-1} \circ p$. This is a set injection from δ into Ord . Choose an infinite initial cardinal γ strictly larger than δ and than $\sup \text{ran}(q)$. Both complements

$$\gamma \setminus \delta \quad \text{and} \quad \gamma \setminus \text{ran}(q)$$

have cardinality γ , since fewer than γ elements were removed. A bijection between the complements therefore extends q to a permutation of γ . Extend that permutation by the identity above γ , and normalize its code. Its evaluation agrees with p at every coordinate below δ . To make the construction uniform, fix the least permissible γ and then the least complement bijection under a fixed global well-order. These choices select sets; they do not require a choice principle for families of proper classes. $\square$

The lemma is stronger than merely extending p to some global well-order. Even an arbitrarily long set-sized prefix can be realized by changing the baseline only on a bounded set of coordinates. The bound need not be small: it must contain the baseline indices of all values prescribed by p .

Source 12’s prefix extension. [merge] Source 12’s Section 6 (“Set-like global well-orders: interpolation and universality”) proves the results of this section by the same arguments, except for its prefix extension, printed next, which is a different construction: it appends the restriction of the baseline order to the complement of the prefix, so its completion need not have bounded support. Source 12’s interpolation theorem, local characters and local universality are recorded in notes below; its avoidance lemma and its universality lemmas are printed in [section 5](#).

We now restrict to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, the class-bijection predicate. All statements about set- or class-indexed collections of its members mean uniform families, unless a larger ambient universe is explicitly specified.

Lemma 19.5 (Extending a set prefix). *Every injective set sequence $p : \alpha \rightarrow \mathbf{No}$ extends to a class bijection $\text{Ord} \rightarrow \mathbf{No}$. The extension can be chosen uniformly from p and one fixed baseline bijection B .*

Proof. Let $A = \text{ran}(p)$. Restrict the baseline order to $\mathbf{No} \setminus A$. It is a proper-class set-like well-order, so it has its canonical increasing enumeration in type Ord . Append that enumeration to p . The class order $\alpha + \text{Ord}$ is canonically isomorphic to Ord . The construction is set-valued recursion with B as a parameter. $\square$

[merge] *Source 09’s coding of the core.* Source 09 codes the core by fixed-point-free support permutations ([section 1.3](#)). Its prefix completion [lemma 19.6](#) has the proof of [lemma 19.4](#), and its [remark 19.7](#) warns why a larger cardinal is needed.

19.3 Source 09’s coding: canonical support codes and prefix completion

19.3.1 Lexicographic comparison of global enumerations

If $E, F : \text{Ord} \rightarrow \mathbf{No}$ are distinct, choose an ordinal γ where they differ. Their disagreement class intersected with $\gamma + 1$ is a nonempty set and has a least member. Thus

$$\Delta(E, F) = \min\{\alpha : E(\alpha) \neq F(\alpha)\}$$

is a set ordinal. Define

$$E <_{\text{lex}} F \iff E(\Delta(E, F)) < F(\Delta(E, F)).$$

The same first-disagreement argument as before proves trichotomy and transitivity. This proof quantifies over individual classes, not over a new class whose members would be enumerations.

19.3.2 Canonical set codes for eventual perturbations

A *support code* is a set function $\sigma : S \rightarrow S$, where $S \subseteq \text{Ord}$ is a set, σ is bijective, and $\sigma(\alpha) \neq \alpha$ for every $\alpha \in S$. The empty code is allowed. Extend it by the identity outside S :

$$\hat{\sigma}(\alpha) = \begin{cases} \sigma(\alpha), & \alpha \in S, \\ \alpha, & \alpha \notin S. \end{cases}$$

The code determines a global enumeration

$$E_{\sigma} = b \circ \hat{\sigma}. \tag{3}$$

Let $\mathcal{P}_s(b)$ be the class of these codes, ordered by the corresponding E_σ . Because fixed points are omitted, distinct codes yield distinct enumerations. The carrier of $\mathcal{P}_s(b)$ genuinely consists of sets.

A class enumeration E occurs this way exactly when it agrees with b at every sufficiently large ordinal. Indeed any set of ordinals is bounded. Conversely, if E equals b beyond a bound, then $b^{-1} \circ E$ has set support, and that support is invariant under the permutation and its inverse. This gives its unique support code.

Lemma 19.6 (Bounded-support prefix completion). *Every injective set word $p : \alpha \rightarrow \mathbf{No}$ is a prefix of some E_σ with $\sigma \in \mathcal{P}_s(b)$. More precisely, choose any infinite initial cardinal ν strictly above α and every ordinal in $b^{-1}[\text{ran } p]$. Then σ can be chosen with support contained in ν .*

Proof. Let $r = b^{-1} \circ p : \alpha \rightarrow \nu$. Since $\alpha < \nu$, both $\nu \setminus \alpha$ and $\nu \setminus \text{ran } r$ have cardinality ν . Choose a bijection between them. Together with r , this gives a permutation $\pi : \nu \rightarrow \nu$. Extend π by the identity outside ν and remove fixed points from its set code. Then $b \circ \pi \upharpoonright \alpha = p$. $\square$

Remark 19.7 (Why extra unused positions matter). A partial injection need not extend to a permutation on the union of its domain and range. For example, $n \mapsto n + 1$ on ω cannot extend to a permutation of ω while retaining every prescribed value. The proof avoids this defect by moving to a larger cardinal ν , where both complements have the same cardinality. Merely saying “complete the partial permutation” would leave a genuine gap.

For each ordinal γ , permutations of γ extended by the identity form a set-sized suborder of the core. Increasing γ gives compatible embeddings. Therefore

$$\mathcal{P}_s(b) = \bigcup_{\gamma \in \text{Ord}} \{\text{support codes of permutations of } \gamma\}.$$

Unlike naive unions of $\mathcal{W}(\mathbf{No}_{<\theta})$, this is a coherent direct system: the same global enumeration is retained when the support bound is increased.

[merge] Source 08’s Section 10 (“Set-supported permutations and prefix extension”) uses source 09’s support codes, writes $\mathcal{P}_s$ for their class and $\mathcal{P}_s(b)$ for the class ordered by

$$s <_b t \iff b \circ \widehat{s} <_{\text{lex}} b \circ \widehat{t}, \tag{S08.8}$$

an ordinary class relation since codes are sets, and calls $O_b = \{b \circ \widehat{s} : s \in \mathcal{P}_s\}$ the virtual orbit of b .

Note 19.8 (Source 08, Lemma 10.1, “Prefix extension inside one orbit”). *Every injective set word $p : \alpha \rightarrow \mathbf{No}$ is an initial segment of $b \circ \widehat{s}$ for some $s \in \mathcal{P}_s$.* This is lemma 19.4 and lemma 19.6, by the same proof through an initial cardinal θ above α and $\sup \text{ran}(b^{-1} \circ p)$, where both complements have cardinality θ . Source 08 notes that set choice supplies the complement bijection; that a support below any such θ suffices; that an arbitrary partial permutation of a fixed infinite set need not extend to a permutation of that set; and that under global choice the extension can be selected uniformly as the least valid code in a fixed set-like global well-order, “a choice of *sets*, not an invocation of choice among arbitrary classes”.

Note 19.9 (Source 08, Corollary 10.2). *There is a uniform order embedding $j_b : \mathbf{No} \hookrightarrow \mathcal{P}_s(b)$ for which $b \circ \widehat{j_b(x)}$ has first entry x ; its image is cofinal and cointial in $\mathfrak{W}_{\text{sl}}$ in the uniform-family sense.* The code is the transposition of 0 and $b^{-1}(x)$. This is proposition 19.20 (source 11’s E_x) and the first part of source 12’s note 19.28. Source 08: the images of the positive and of the negative ordinals are cofinal and cointial, so there are uniform class-indexed cofinal families although there are no set-indexed ones; “cofinality must not be confused with exhaustive parametrization”.

19.4 Interpolation of every uniformly set-indexed cut

Theorem 19.10 (Prefix interpolation). *Let $(A_i)_{i \in I}$ and $(B_j)_{j \in J}$ be uniformly set-indexed families in $\mathfrak{J}$, with*

$$A_i <_{\text{lex}} B_j \quad (i \in I, j \in J).$$

Either index set may be empty. There is an injective set prefix p such that every member of $\mathfrak{J}$ extending p is strictly above every A_i and strictly below every B_j . In particular, a member of $\mathcal{P}_{\text{bd}}$ lies between the two families.

Proof. If both index sets are nonempty, Class Replacement gives a set of ordinals $d(A_i, B_j)$. Choose β so that

$$d(A_i, B_j) < \beta \quad (i \in I, j \in J).$$

If one side is empty, put $\beta = 0$. We build an injective prefix p , keeping track of the rows that still agree with it. Call these rows *active*. The invariant is that every discarded A_i is below every extension of p , and every discarded B_j is above every extension of p .

Suppose p has length α and the construction has not stopped. Let

$$L = \{A_i(\alpha) : A_i \text{ is active}\}, \quad R = \{B_j(\alpha) : B_j \text{ is active}\}.$$

These are sets. Since active cross pairs agree before α , the hypothesis implies $\ell \leq r$ whenever $\ell \in L$ and $r \in R$. Thus $L \cap R$ has at most one member: two distinct common values, taken in reverse order as a left and a right value, would contradict this inequality.

If $L \cap R$ is empty, all cross inequalities are strict. By Lemma 5.2, choose

$$z \in \mathbf{No} \setminus \text{ran}(p), \quad L < z < R.$$

This notation includes an empty left or right side; if both are empty, choose any fresh value. Append z and stop. Every previously active left row is now strictly below every extension of the new prefix, and every previously active right row is strictly above it. The invariant handles the already discarded rows, proving the desired conclusion.

If $L \cap R = \{z\}$, append z and retain as active exactly the rows whose next value is z . The new value is fresh because each surviving row is injective. A discarded left value is strictly below z , since $z \in R$; a discarded right value is strictly above z , since $z \in L$. Hence the invariant is preserved. At a limit stage take the union of the preceding prefixes; its length is a set ordinal and it remains injective.

The procedure cannot continue with both sides active through a prefix of length β . Such active rows would agree below β , contrary to the bound on every cross first difference. Therefore it stops no later than the next step, giving a set prefix. Lemma 19.4 supplies a bounded-support completion. Notice that surjectivity of the input rows was never used; this is why the theorem was stated for $\mathfrak{J}$. $\square$

All recursion histories in this proof are sets. Even when the input rows are proper classes, their restrictions to a set ordinal are sets, and their next values form sets because the index sets are sets. The proof therefore uses ordinary set-valued transfinite recursion with class parameters, not ETR. The same observation applies to the back-and-forth argument below.

Note 19.11 (Source 12, Theorem 6.3, “Set-indexed interpolation”). *If I, J are sets and $(E_i)_{i \in I}$, $(F_j)_{j \in J}$ are uniform families in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} F_j$ for all i, j , there is $G \in \mathfrak{W}_{\text{sl}}(\mathbf{No})$ with $E_i <_{\text{lex}} G <_{\text{lex}} F_j$; empty sides are allowed, so $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ is dense without endpoints in the indexed-family sense. This is theorem 19.10, by the same forced-prefix construction; source 12 completes the prefix by lemma 19.5, so its interpolant need not lie in the bounded-support core.*

[merge] Sources 09 and 08 prove the same theorem with their support codes (source 09’s Section 9, “The interpolation theorem”, and source 08’s Section 11, “The set-cut interpolation theorem”).

Note 19.12 (Source 09, Theorem 9.1, “Uniform set-indexed interpolation”). *For uniformly set-indexed families $(E_i) <_{\text{lex}} (F_j)$ of global enumerations (I, J possibly empty) and every fixed baseline b , there is $\sigma \in \mathcal{P}_s(b)$ with $E_i <_{\text{lex}} E_\sigma <_{\text{lex}} F_j$.* This is [theorem 19.10](#), by the same construction; source 09 bounds the forced prefix by $\lambda + 1$, $\lambda = \sup\{\Delta(E_i, F_j) + 1\}$, and notes that the theorem “permits constraints from outside the chosen core, and the interpolant can be required to come from *any* specified baseline’s eventual-agreement class”.

Note 19.13 (Source 08, Theorem 11.1, “Orbit interpolation”). *For fixed b and uniform families $(l_i)_{i \in I}, (r_j)_{j \in J}$ of global enumerations, I, J sets, with $l_i <_{\text{lex}} r_j$, there is $s \in \mathcal{P}_s$ with*

$$l_i <_{\text{lex}} b \circ \hat{s} <_{\text{lex}} r_j \quad (i \in I, j \in J); \quad (\text{S08.9})$$

either family may be empty. This is [theorem 19.10](#), by the same active-row invariant (its display (S08.10) is the inequality $a \leq b$ between active next entries). Source 08 stresses that the separator lies in one fixed orbit “regardless of the supports, or lack of supports, of the input enumerations”.

Note 19.14 (Source 08, Corollary 11.2). *The virtual order $\mathfrak{W}_{\text{sl}}$ is dense without endpoints and has no uniformly set-indexed cofinal or coinital family; every orbit O_b is dense in $\mathfrak{W}_{\text{sl}}$ and realizes every uniformly set-sized separated cut; the class order $\mathcal{P}_s(b)$ is an η_{Ord} -order.* This is [corollary 19.15](#) and [proposition 19.27](#), by specializing the interpolation theorem.

Corollary 19.15. *The class order $\mathcal{P}_{\text{bd}}$ is an η_{Ord} order and is order-isomorphic to $(\mathbf{No}, <)$. Moreover, it meets every strict cut in $\mathfrak{W}_{\text{sl}}$ specified by uniformly set-indexed families.*

Proof. A set of codes is uniformly evaluable, so Theorem [19.10](#) applies to every separated pair of sets in $\mathcal{P}_{\text{bd}}$. Empty sides give the absence of endpoints, and singleton sides give density. We already know that $\mathcal{P}_{\text{bd}}$ is a proper class. Proposition [5.13](#) therefore identifies it with $\mathbf{No}$.

Explicitly, global choice enumerates each of the two classes by ordinals. Alternately extend a set-sized partial isomorphism to cover the next element of one enumeration and then the next element of the other. The lower and upper images at either step are sets, so the cut property supplies the new point. The union of this ordinary set-valued recursion is an isomorphism. The last assertion is the direct specialization of Theorem [19.10](#) to bijective input rows. $\square$

This is an isomorphism of the class of bounded-support codes with the surreal line. It is not an isomorphism between $\mathbf{No}$ and the full higher-order collection $\mathfrak{W}_{\text{sl}}$. Different baselines produce classes with the same abstract order type, but the argument does not single out a canonical isomorphism.

Note 19.16 (Source 09, Corollary 9.2, “The surreal core theorem”). *In GBC, $\mathcal{P}_s(b) \cong \mathbf{No}$. In particular $\mathcal{P}_s(b)$ is dense without endpoints, every set-sized partial order isomorphism between its suborders extends to an automorphism, and every class linear order embeds into it.* This is [corollary 19.15](#) (with the coding of [section 1.3](#)), with homogeneity from [theorem 5.17](#). Source 09 adds that no set family is cofinal or coinital among all global enumerations, or immediately on either side of a fixed global enumeration, while proper-class cofinal and coinital families exist, with first entries running through cofinal positive and coinital negative ordinals.

Note 19.17 (Source 08, Theorem 12.2, “Surreal-sized skeleton”). *For every fixed global enumeration b , $\mathcal{P}_s(b) \cong \mathbf{No}$; the code order is a proper class, is dense in $\mathfrak{W}_{\text{sl}}$ through its represented enumerations, and realizes every uniform set cut of $\mathfrak{W}_{\text{sl}}$.* This is [corollary 19.15](#), by the same uniqueness theorem ([note 5.19](#)); properness comes from [note 19.9](#). Source 08 adds: “The isomorphism is an isomorphism of *linear orders*, not of fields, of simplicity trees, of birthday filtrations, or of permutation groups.

No canonical isomorphism independent of Global Choice and b is asserted. One could transport a field structure along a selected isomorphism, but it would be additional, choice-dependent structure, not an arithmetic operation naturally defined by lexicographic comparison of enumerations.”

19.4.1 Eventual agreement and interleaved orbits (source 08)

Two enumerations e, f differ on a set of coordinates exactly when $e = b \circ \hat{s}$, $f = b \circ \hat{t}$ for some common orbit, equivalently when $e^{-1} \circ f$ has set support. Any set of ordinal coordinates is bounded, so this is also eventual equality: there is an ordinal α such that $e(\xi) = f(\xi)$ for every $\xi \geq \alpha$. The equivalence between the support conditions follows from injectivity of the baseline enumeration and class replacement.

Distinct orbits are disjoint as collections of enumerations, but every one is dense in $\mathfrak{W}_{\text{sl}}$. Hence these equivalence classes are not convex, and the quotient by eventual equality does not inherit a linear order by simply comparing representatives. This is a concrete obstruction, not merely a size warning.

Corollary 19.18 (The class of realizations of a set cut). *Under the hypotheses of Theorem 19.10, the ordinary class*

$$K = \{p \in \mathcal{P}_{\text{bd}} : (\forall i \in I \ A_i <_{\text{lex}} p) \text{ and } (\forall j \in J \ p <_{\text{lex}} B_j)\}$$

is itself order-isomorphic to $\mathbf{No}$.

Proof. It is an ordinary class by comprehension with the two uniform input relations as parameters. Suppose $C, D \subseteq K$ are sets with $C < D$. Add the evaluated codes of C to the lower input family and those of D to the upper input family. All cross inequalities remain strict: members of K already lie between the original sides. Theorem 19.10 supplies a member of K between C and D . Thus K has the set-cut property, including empty sides. It cannot be a set, since applying that property with lower side all of K would give a member above itself. The uniqueness statement of Proposition 5.13 now gives the isomorphism. $\square$

Thus the interpolation theorem produces a full proper-class order of available choices, even when the input constraints are numerous and occur at arbitrarily complicated set-ordinal coordinates. In particular, no set of codes can be order-dense in $\mathfrak{W}_{\text{sl}}$. Given such a set, choose a point strictly above all its evaluations, and then a second, larger point. The open interval between the two contains no code from the proposed dense set. The same argument applies within K : a set cannot be dense even in the class of realizations of one specified set cut. The ordinary class $\mathcal{P}_{\text{bd}}$ is therefore a natural scale for density here, while an exhaustive class of set codes will be ruled out by a separate diagonal argument.

Note 19.19 (Source 09, Corollary 9.3, “Dense copies in every interval”). *For global enumerations $E < F$, the class $\{\sigma \in \mathcal{P}_s(b) : E < E_\sigma < F\}$ is isomorphic to $\mathbf{No}$; the core meets every open interval of the global order and supplies a copy of every class linear order inside it.* This is [corollary 19.18](#) with $I = \{E\}$ and $J = \{F\}$, by the same argument.

19.5 Universality, cylinders, and local cofinality

Proposition 19.20. *The surreal line embeds explicitly into $\mathcal{P}_{\text{bd}}$. Consequently every ordinary class linear order embeds into $\mathcal{P}_{\text{bd}}$ and hence into $\mathfrak{W}_{\text{sl}}$, in the uniform evaluation sense.*

Proof. For $x \in \mathbf{No}$, transpose the values $e_0(0)$ and x in e_0 , and denote the resulting enumeration by E_x . Its coordinate permutation has support contained in $\{0, e_0^{-1}(x)\}$, so it has a code in $\mathcal{P}_{\text{bd}}$, and

$$E_x(0) = x, \quad x < y \iff E_x <_{\text{lex}} E_y.$$

For the general assertion, compose this embedding with the class-order universality of $\mathbf{No}$ from Proposition 5.13. $\square$

Thus all set ordinals, the class order Ord , its reverse, and class orders such as $\text{Ord} + \text{Ord}$ occur as suborders of the lexicographic comparison. This statement concerns the order *between* enumerations. Each individual enumeration in $\mathfrak{W}_{\text{sl}}$ still presents a well-order of type Ord ; its lexicographic environment need not be well-ordered.

Note 19.21 (Source 12, Theorem 6.7, “Local class universality and cylinders”). *Every class linear order has a uniform order-embedding into $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ and into every nonempty open interval of $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; for a set-sized injective prefix p , the cylinder $[p]$ is convex and isomorphic to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ by uniform transformations.* This is proposition 19.20 and proposition 19.25, by the same constructions ($x \mapsto E_x$ composed with lemma 5.14; a cylinder through $\mathbf{No} \setminus \text{ran}(p) \cong \mathbf{No}$, lemma 5.15). Source 12 lists the examples “all set ordinals, their reverses, $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, all set-sized lexicographic powers of $\mathbf{No}$ formed as class orders” and adds that “every class linear order” still means an order whose elements are sets; it does not include an unrestricted higher collection such as all class well-orders themselves.

Note 19.22 (Source 09, Theorem 10.2). *Every set ordinal, every reverse set ordinal and every class linear order embeds as a uniformly class-indexed family of global enumerations; all these embeddings can be taken in $\mathcal{P}_s(b)$, or within any specified nonempty global open interval.* This is proposition 19.20 with corollary 19.18. Source 09 lists Ord , Ord^* , $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, set-sized lexicographic powers of $\mathbf{No}$ and every class linear order on a class of set codes, and stresses that “the class order *being embedded* need not be set-like, although each well-order represented by a target global enumeration is set-like. These are two entirely different orders.” Its direct embedding $x \mapsto T_x \circ b$ (exchange $b(0)$ and x) is source 11’s E_x ; it “does not by itself give the surjectivity of the isomorphism”.

Note 19.23 (Source 08, Corollary 12.4). *The skeleton contains copies of every ordinal and its reverse, every class well-order, $\text{Ord} + 1$, $\text{Ord} + \text{Ord}$, $\mathbf{No} \times_{\text{lex}} \mathbf{No}$, and $(\mathbf{No}^\alpha, <_{\text{lex}})$ for every set ordinal α ; every injectively parametrized uniform class family of global enumerations, with its induced lexicographic order, also embeds into $\mathcal{P}_s(b)$.* The first part is proposition 19.20; the last clause is source 09’s proposition 19.24, printed next. Source 08: this concerns a uniform class-indexed family, not all class enumerations simultaneously, and replacing the set ordinal α by Ord in $\mathbf{No}^\alpha$ changes the size level of the elements and is not covered.

Proposition 19.24 (Representable suborders do not exceed surreal universality). *Every uniformly class-indexed family of global enumerations, modulo equal rows, has a class-coded linear order that embeds into $\mathbf{No}$.*

Proof. For a family with class index I , choose the least representative of each equal-row class in a fixed set-like global well-order of I . Equality of rows is the first-order condition $\forall \alpha E_i(\alpha) = E_j(\alpha)$, with the family as a class parameter. The representatives form a class of sets. Their first-disagreement order is a class linear order, to which theorem 5.16 applies. $\square$

This does not say that the full higher-level totality $\mathfrak{W}_{\text{sl}}$ embeds into $\mathbf{No}$; it says that every *uniformly class-coded portion* does.

Proposition 19.25 (Prefix cylinders). *For an injective set prefix p , let*

$$C_p = \{E \in \mathfrak{W}_{\text{sl}} : E \upharpoonright \text{dom}(p) = p\}.$$

The cylinder C_p is convex and order-isomorphic to $\mathfrak{W}_{\text{sl}}$, with the isomorphism interpreted as a uniform transformation of class parameters. Every nonempty open interval in $\mathfrak{W}_{\text{sl}}$, and every interval specified by a uniformly set-indexed strict cut, contains a whole such cylinder.

Proof. An enumeration that first departs from p at a coordinate of p lies on the same side of all its extensions. This proves convexity. Put $S = \text{ran}(p)$ and $\delta = \text{dom}(p)$. By Lemma 5.2 and Proposition 5.13, fix an order-isomorphism $h : \mathbf{No} \rightarrow \mathbf{No} \setminus S$. The ordinal tail $[\delta, \text{Ord})$ has order type Ord , with its canonical increasing parametrization $\xi \mapsto \delta + \xi$. Given $E \in \mathfrak{W}_{\text{sl}}$, prepend p to the enumeration $h \circ E$ on this tail. The result is a bijection onto $\mathbf{No}$, since its prefix covers S and its tail covers exactly the complement. Conversely, restrict an element of C_p to its tail and apply h^{-1} . These transformations are inverse and preserve first-difference comparisons. The final assertion follows from Theorem 19.10. $\square$

In particular, every such interval contains a copy of the entire comparison structure $\mathfrak{W}_{\text{sl}}$, not merely a copy of an arbitrarily large set order. This self-similarity does not invoke a class whose members are the cylinders; for each set prefix the displayed construction gives the required transformation, using the indicated class parameters.

Note 19.26 (Source 09, Proposition 10.4, “Cylinder self-similarity”). *Each cylinder $[p]$ (the global enumerations extending an injective set word p ; convex, and nonempty by lemma 19.6) is isomorphic to the full global comparison order by an explicit invertible transformation of class enumerations preserving lexicographic order and having exactly $[p]$ as its image. This is proposition 19.25, by the same construction $H_E \upharpoonright \alpha = p$, $H_E(\alpha + \xi) = h(E(\xi))$ with $h : \mathbf{No} \cong \mathbf{No} \setminus \text{ran } p$ (corollary 5.18). Source 09’s paragraph on two baselines follows; source 08 makes the same observation in its subsection printed after corollary 19.15.*

For two baselines b, c , their eventual-agreement collections are either identical or disjoint. They are identical exactly when b, c agree beyond some set ordinal. Otherwise no global enumeration can be eventually equal to both. Nevertheless, *each* such collection has a genuine set-coded presentation isomorphic to $\mathbf{No}$ and is dense in the entire global comparison order. Eventual agreement is therefore not a convex relation, and its quotient does not inherit a lexicographic linear order by comparing arbitrary representatives.

Proposition 19.27 (Approximants at every point). *For each $F \in \mathfrak{J}$ there are uniformly ordinal-indexed families (a_α) and (b_α) of codes in $\mathcal{P}_{\text{bd}}$ such that, whenever $\alpha < \beta$,*

$$a_\alpha <_{\text{lex}} a_\beta <_{\text{lex}} F <_{\text{lex}} b_\beta <_{\text{lex}} b_\alpha.$$

The lower family is cofinal among the global enumerations below F , and the upper family is coinital among those above F . Neither side admits a uniformly set-indexed cofinal or coinital family, respectively.

Proof. Choose a fresh value below $F(\alpha)$, append it to $F \upharpoonright \alpha$, and use Lemma 19.4 to obtain a_α . Choose a fresh value above $F(\alpha)$ to obtain b_α . Lemma 5.2 supplies the fresh values, and the fixed global well-order makes all choices uniform. For $\alpha < \beta$, the comparison between a_α and a_β is decided at α , giving the asserted inequality; the other comparisons are identical arguments.

If $E <_{\text{lex}} F$, choose $\alpha > d(E, F)$. Then a_α agrees with F at the decisive coordinate, so $E <_{\text{lex}} a_\alpha <_{\text{lex}} F$. This proves cofinality, and the upper assertion is symmetric. Finally, given any

uniformly set-indexed family below F , Theorem 19.10 applied with upper side $\{F\}$ supplies a larger lower approximant in $\mathcal{P}_{\text{bd}}$. Hence the original family was not cofinal. The dual argument proves the coinitality assertion. $\square$

For $F \in \mathfrak{W}_{\text{sl}}$, these are the local cofinality and coinitality statements at an actual point of the order. It is appropriate to say that ordinal-length class families suffice and set-sized families do not; assigning an ordinary set cardinal to these proper-class cofinalities would obscure the distinction.

Note 19.28 (Source 12, Proposition 6.4, “Cofinality and local character”). *No set-indexed uniform family is cofinal or coinital in $\mathfrak{W}_{\text{sl}}(\mathbf{No})$; there are increasing cofinal and decreasing coinital families indexed by Ord; at every point neither side has a set-indexed cofinal, respectively coinital, family, but both have Ord-indexed ones.* This is proposition 19.27 together with the remark that first entries of a set-indexed family are bounded, by the same argument; the cofinal family is $(E_\alpha)_{\alpha \in \text{Ord}}$ for the positive ordinals α .

Corollary 19.29 (Set saturation is not Dedekind completeness). *A nonempty uniformly set-indexed family in $\mathfrak{W}_{\text{sl}}$ with no greatest member has no supremum in $\mathfrak{W}_{\text{sl}}$. Dually, a family with no least member has no infimum. Both statements also hold in the ordinary class order $\mathcal{P}_{\text{bd}}$.*

Proof. Suppose that E is the supremum of a family (A_i) with no greatest member. Every A_i is then strictly below E . By Theorem 19.10, there is a bounded-support enumeration p with $A_i < p < E$ for every i . This makes p a smaller upper bound, a contradiction. The infimum argument reverses the inequalities, and the same bounded-support interpolant proves the assertions internal to $\mathcal{P}_{\text{bd}}$. $\square$

For example, take the enumerations E_n from the transposition construction, so that $E_n(0) = n$ for $n \in \mathbb{N}$. They form an increasing sequence of bounded-support codes. The enumeration whose first value is the surreal ordinal ω is an upper bound, but there is no least upper bound in either $\mathcal{P}_{\text{bd}}$ or $\mathfrak{W}_{\text{sl}}$. The obstruction is not a failure to find bounds: Theorem 19.10, with one input side empty, bounds every uniformly set-indexed family above and below. Rather, whenever a proposed bound is strictly beyond every member, there remains room to move the bound closer. This distinction will also matter when the full order of Dedekind cuts is embedded below. An order embedding preserves comparisons, but does not automatically preserve the role of a point as the least upper bound of a specified family.

[merge] Source 08’s Section 11.2, “Saturation is not Dedekind completeness”, proves corollary 19.29 by the same argument (a strictly smaller upper bound from note 19.13) and adds that the same assertion holds in $\mathcal{P}_s(b)$ and in $\mathbf{No}$: “Set-cut saturation supplies abundant *strict* separators; it is not the Dedekind completeness familiar from the real line.”

[merge] Bounded reflection is proved only by source 09.

19.6 Source 09: bounded reflection

19.6.1 Every set family reflects into one bounded permutation stage

Theorem 19.30 (Bounded reflection). *Let $(E_i)_{i \in I}$ be a uniformly set-indexed family of pairwise distinct global enumerations. For any baseline b , there are a set ordinal λ , an infinite initial cardinal ν , and support codes $\sigma_i \subseteq \nu \times \nu$ such that*

$$E_{\sigma_i} \upharpoonright \lambda = E_i \upharpoonright \lambda, \quad E_i < E_j \iff E_{\sigma_i} < E_{\sigma_j}.$$

The family order also embeds into $\mathcal{W}_\kappa(\mathbf{No}_{<\theta})$ for some birthday bound θ , where $\kappa = |\mathbf{No}_{<\theta}|$.

Proof. Choose $\lambda \geq 1$ strictly greater than every first-disagreement ordinal of a pair of distinct rows. This is possible by Replacement because $I \times I$ is a set. The set

$$U = \bigcup_{i \in I} \text{ran}(E_i \upharpoonright \lambda)$$

is a set of surreals. Choose one infinite initial cardinal ν strictly above λ and every ordinal in $b^{-1}[U]$. Apply lemma 19.6 to every row prefix at this same bound. All comparisons were resolved before λ , so they are preserved. Choice for this set family of completions suffices.

For the birthday version, take θ large enough that $U \subseteq \mathbf{No}_{<\theta}$ and its cardinality κ is strictly larger than the ordinal λ . Each prefix then removes fewer than κ letters from $\mathbf{No}_{<\theta}$. Enumerate the remainder in type κ and append it. The total ordinal length is $\lambda + \kappa = \kappa$. The resulting exhaustive words preserve all first disagreements. $\square$

Thus no counterexample involving a *uniform set family* needs unboundedly many coordinates in an essential way. But the bound depends on the family, and there is no single set bound working for every family. For each κ , the global core contains κ^+ , whereas theorem 6.14 rules it out of any fixed alphabet of size κ .

[merge] The failure of finite support is proved only by source 08; it shows why the core must allow set support.

19.7 Why finite support is insufficient (source 08)

Let F_b be the suborder of $\mathcal{P}_s(b)$ consisting of finite support codes. The explicit transpositions in note 19.9 show that F_b already contains a copy of $\mathbf{No}$ and hence every class linear order. Nevertheless it does not have the saturation or ambient density of the full set-supported skeleton.

Proposition 19.31. *The order F_b is not an η_{Ord} -order, and its represented enumerations are not dense in $\mathfrak{W}_{\text{sl}}$.*

Proof. Put $p(n) = b(n+1)$ for $n < \omega$. This injective prefix differs from b at every natural-number coordinate. Every finite injective prefix extends to a finite perturbation of b : after applying b^{-1} , its partial injection extends to a permutation of the finite union of its domain and range, because the two remaining finite complements have equal size.

For each n , choose fresh values $a_n < p(n) < c_n$ outside $\text{ran}(p \upharpoonright n)$, and extend $(p \upharpoonright n) \cap (a_n)$ and $(p \upharpoonright n) \cap (c_n)$ to finite perturbations l_n, r_n of b . These choices can be made uniformly. For all n, m one has $l_n < r_m$: at the earlier of their altered coordinates one sees either $a_n < p(n)$ or $p(m) < c_m$, and when $n = m$ one sees $a_n < c_n$.

A separator of these two countable families must agree with p at every finite coordinate. Otherwise at its first disagreement k , the rows l_m, r_m for $m > k$ both agree with p there, forcing the candidate below a left row or above a right row. Such a separator moves every natural-number coordinate of b and cannot have finite support. This proves failure of set-cut saturation.

Finally choose $u < v$ in $\mathfrak{W}_{\text{sl}}$ with common prefix p of length ω and distinct fresh entries immediately afterward, using note 19.8. Every enumeration between them must share that prefix, so none is a finite perturbation of b . Hence F_b is not dense in $\mathfrak{W}_{\text{sl}}$. $\square$

Thus containing every ordinary class order is weaker than being a surreal line or being a dense skeleton of the virtual ambient order. Set support, not merely finite support, is what permits arbitrary set-length prefixes to be imposed.

19.8 Binary class words, cuts, and diagonalization

Theorem 19.32. *The lexicographic orders $\mathfrak{W}_{\text{sl}}$ and $\{0, 1\}^{\text{Ord}}$ are order-bi-embeddable, where a binary word of length Ord means a class characteristic function. Both embeddings have uniform relational definitions in GBC.*

Proof. Partition the coordinate ordinals into successive pairs $2 \cdot \xi, 2 \cdot \xi + 1$. Here the multiplication is ordinal multiplication with 2 on the left; every ordinal has a unique such pair position. Sort the corresponding baseline values as

$$u_\xi = \min\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}, \quad v_\xi = \max\{e_0(2 \cdot \xi), e_0(2 \cdot \xi + 1)\}.$$

For a class $D \subseteq \text{Ord}$, put these two values in the order (u_ξ, v_ξ) if $\xi \notin D$, and in the order (v_ξ, u_ξ) if $\xi \in D$. Denote the resulting enumeration by T_D . Each pair retains exactly its original two values, so T_D is a bijection onto $\mathbf{No}$. At the least differing bit of two characteristic functions, 0 chooses the smaller first value and 1 the larger one. Thus $D \mapsto T_D$ preserves and reflects lexicographic comparison.

For the converse, fix an ordinal enumeration $(p_\xi)_{\xi \in \text{Ord}}$ of the set codes in $\mathcal{P}_{\text{bd}}$. Associate with $E \in \mathfrak{W}_{\text{sl}}$ the class

$$D_E = \{\xi \in \text{Ord} : p_\xi <_{\text{lex}} E\}.$$

If $E <_{\text{lex}} F$, then $D_E \subseteq D_F$, and the inclusion is strict: Corollary 19.15 supplies a code strictly between E and F . At the least ordinal belonging to $D_F \setminus D_E$, the characteristic functions therefore have values 0 and 1, respectively. This proves the second order embedding. Both constructions use elementary class comprehension with the input word or enumeration as a class parameter. $\square$

Bi-embeddability must not be replaced by isomorphism. Binary class words have a least and a greatest word and have adjacent pairs: after a set prefix, the word with next bit 0 and all subsequent bits 1 immediately precedes the word with next bit 1 and all subsequent bits 0. By contrast, $\mathfrak{W}_{\text{sl}}$ is dense and has no endpoints.

[merge] *Second route (source 12).* Source 12's Section 7 embeds $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ into binary class words by concatenating self-delimiting codes of the entries, instead of source 11's characteristic function of a dense class; the other embedding is source 11's pair swap.

19.9 Source 12's route to the binary-class representation

Let $\mathbb{B}_{\text{Ord}}$ denote the predicate of class subsets of Ord, regarded as binary functions on Ord and compared at their least disagreement. This is again notation for a predicate and relation on classes, not a powerclass object.

Lemma 19.33 (A self-delimiting numerical code). *Replace every minus in a sign sequence by 00, every plus by 11, and append the terminal block 01. The resulting binary ordinal word $c(s)$ has length $2 \cdot \text{bd}(s) + 2$. Distinct codewords are prefix-incomparable, and their first-difference comparison agrees with numerical comparison of the surreals.*

Proof. The three two-bit blocks have order $00 < 01 < 11$, exactly matching $- < \text{end} < +$. At a sign disagreement the first different block decides. When one sign sequence ends, its terminal block differs from the next sign block of the other. Hence neither complete codeword can be a prefix of another, and the numerical comparison is preserved. $\square$

Theorem 19.34 (Binary-class equimorphism). *In the sense of uniform class transformations,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}) \simeq_c \mathbb{B}_{\text{Ord}}.$$

Proof. For $E : \text{Ord} \rightarrow \mathbf{No}$, concatenate $c(E(\alpha))$ over all ordinals. The cumulative length before each stage is a set ordinal. All blocks are nonempty, so the cumulative lengths are unbounded in Ord and the result is a binary class on all of Ord . If E, F first disagree at α , their earlier block lengths and bits agree, and the code of that entry decides the same inequality by [lemma 19.33](#).

For the converse, use the disjoint pairs $a_\alpha < b_\alpha$ in the proof of [theorem 19.38](#). For a binary class A , list the pair in ascending order if its bit is 0, and descending order if its bit is 1. Put pair blocks in ordinal order. The first different bit gives the first different pair orientation and the correct lexical inequality. All labels occur exactly once. $\square$

This is not an isomorphism assertion. The binary-class order has least and greatest elements and adjacent pairs, such as a zero followed by all ones versus a one followed by all zeros. By [note 19.11](#), $\mathfrak{W}_{\text{sl}}(\mathbf{No})$ has no endpoints or adjacent pairs. The distinction between equimorphism and isomorphism is already visible at this level.

[merge] Source 08’s Section 13 (“Binary class sequences and the size obstruction”) defines an embedding between virtual collections as a *uniform operator*: “a formula, with displayed class parameters, assigning a unique output class to every input class, with class existence proved in the base theory. It is not a new class whose members are ordered pairs of proper classes.”

Note 19.35 (Source 08, Theorem 13.1, “Virtual binary equimorphism”). *Relative to a baseline $b : \text{Ord} \rightarrow \mathbf{No}$, there are uniform order embeddings*

$$\mathbb{B}_{\text{Ord}} \hookrightarrow \mathfrak{W}_{\text{sl}} \hookrightarrow \mathbb{B}_{\text{Ord}}; \quad (\text{S08.11})$$

thus these virtual orders are equimorphic in the uniform-operator sense; they are not order-isomorphic. This is [theorem 19.32](#). The first embedding is source 11’s pair swap; the second is source 12’s concatenation ([theorem 19.34](#)), with the code of [note 7.5](#) and block endpoints $D_0 = 0$, $D_{\xi+1} = D_\xi + (2 \cdot \text{bd}(e(\xi)) + 2)$, $D_\lambda = \sup_{\xi < \lambda} D_\xi$ defined by set-valued recursion. Source 08 adds that the jump “ $p \frown 0$ followed by constant 1” below “ $p \frown 1$ followed by constant 0” in $\mathbb{B}_{\text{Ord}}$ ceases to be adjacent in the image, because $\mathfrak{W}_{\text{sl}}$ is dense: “equimorphism does not preserve order topology”.

Corollary 19.36. *The higher-order order of all class initial segments of $\mathbf{No}$, ordered by inclusion, embeds into $\mathfrak{W}_{\text{sl}}$. In particular, this holds for any specified convention of Dedekind cuts. Conversely, each $E \in \mathfrak{W}_{\text{sl}}$ determines the initial segment $\{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}$, and these initial segments separate distinct enumerations.*

Proof. For a class initial segment $C \subseteq \mathbf{No}$, form $e_0^{-1}[C] \subseteq \text{Ord}$ and apply the pair-swap embedding of Theorem [19.32](#). Proper inclusion of initial segments gives proper inclusion of their inverse images, which implies the correct lex comparison of their characteristic functions. The converse is the dense-class construction in the second half of that theorem. $\square$

This corollary does not form a GB class of all Dedekind cuts of $\mathbf{No}$. Each individual cut is an allowed class parameter; the rule associates an enumeration to it and compares any two such results. An encompassing object belongs to a higher type or to an explicitly larger ambient universe. Nor does the corollary assert that the natural inclusion of $\mathcal{P}_{\text{bd}}$ in $\mathfrak{W}_{\text{sl}}$ is a Dedekind completion: an order embedding of cuts need not preserve their intended suprema or their original positions relative to the embedded line.

There is a useful description of the natural cuts produced by an enumeration outside the bounded-support family. If $E \in \mathfrak{W}_{\text{sl}}$ has no code in $\mathcal{P}_{\text{bd}}$, then

$$L_E = \{p \in \mathcal{P}_{\text{bd}} : p <_{\text{lex}} E\}, \quad U_E = \{p \in \mathcal{P}_{\text{bd}} : E <_{\text{lex}} p\}$$

partition $\mathcal{P}_{\text{bd}}$ into two nonempty classes. Proposition 19.27 shows that L_E has no set-sized cofinal subset and U_E has no set-sized coinital subset, while ordinal-indexed class families suffice on both sides. This cut is unfilled in $\mathcal{P}_{\text{bd}}$: a putative filling code and E would be separated by another bounded-support code. Transporting the cut through an isomorphism $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ therefore exhibits a surreal class cut with neither side controlled by a cofinal or coinital set. This is a description of individual additional enumerations relative to the fixed dense class, rather than a classification of all surreal class cuts.

Theorem 19.37 (No exhaustive uniform class of codes). *There is no ordinary class of set codes with a single class evaluation relation whose evaluated rows include every member of $\mathfrak{W}_{\text{sl}}$.*

Proof. If such a coding existed, global choice would enumerate its codes by ordinals, with repetitions if the class of codes were a set. Write the resulting row relation as $\text{Eval}(\xi, \alpha, x)$. Use the sorted pairs $u_\xi < v_\xi$ from Theorem 19.32, and define the class

$$D = \{\xi : \text{Eval}(\xi, 2 \cdot \xi, u_\xi)\}.$$

The associated pair-swap enumeration T_D differs from row ξ on the pair $(2 \cdot \xi, u_\xi)$. If that pair belongs to the old row, it does not belong to the graph of T_D , whose value there is v_ξ . If it does not belong to the old row, it belongs to the new graph, whose value there is u_ξ . Thus T_D is a member of $\mathfrak{W}_{\text{sl}}$ omitted by the proposed exhaustive list. The argument even permits invalid or nonfunctional rows in the evaluator. $\square$

The theorem explains why class universality of $\mathbf{No}$ cannot simply be applied to the full collection $\mathfrak{W}_{\text{sl}}$. It also pinpoints the limitation of coding only bounded modifications, or any other one uniformly evaluable family of set codes: such a family may be order-dense and may realize all set-indexed cuts, while still omitting global enumerations by diagonalization.

[merge] *Strengthening (source 12)*. Source 12's Section 5 proves that no uniform family of class well-orders indexed by a class of sets contains every class well-order of $\mathbf{No}$, and that the missing one can be chosen set-like; its corollary 19.39 is theorem 19.37. Its subsection on definable classes explains why enumerating formulas does not evade the theorem.

19.10 Source 12's strengthening: why the collection of all global well-orders is not a class

The formal obstacle is not that individual class relations are illegal. They are legitimate class variables. The obstacle is attempting to make them the elements of an ordinary class of the same theory.

A useful legal replacement is a *uniform family*. For example, $F \subseteq I \times \text{Ord} \times \mathbf{No}$ is a class of triples of sets, and its section F_i can be a class bijection for every $i \in I$. Similarly a class of triples (i, x, y) can code a family of class order relations. The index class I may be proper. These are single classes, not functions whose values are class objects.

Theorem 19.38 (No exhaustive class-coded family). *In GBC, no uniform family of class well-orders of $\mathbf{No}$ indexed by a class of sets contains every class well-order of $\mathbf{No}$. In fact, for every such family there is a set-like class well-order not among its sections.*

Proof. Global choice permits the family to be reindexed by an ordinal or by Ord . Extend an ordinal-indexed family to an Ord -indexed one by repeating an arbitrary default well-order; an empty family is trivial. Write the resulting relations as R_α .

Fix a bijection $B : \text{Ord} \rightarrow \mathbf{No}$ and partition its range into the pairs

$$\{B(2 \cdot \alpha), B(2 \cdot \alpha + 1)\} = \{a_\alpha, b_\alpha\}, \quad a_\alpha < b_\alpha.$$

Here multiplication is ordinal multiplication. The blocks of length two partition Ord . Define a new enumeration E by placing the two labels of pair α in the order *opposite* to their relative order in R_α . Keep the pair blocks in increasing α order.

This is a class bijection $\text{Ord} \rightarrow \mathbf{No}$, defined elementarily from the family and B . Its induced well-order disagrees with R_α on the relative order of a_α, b_α . Thus it is not any section of the family. $\square$

Corollary 19.39 (No uniform set-code solution). *There is no class D of set codes together with a uniform class decoder whose decoded relations are exactly all class well-orders of $\mathbf{No}$.*

This diagonal theorem applies equally to an attempted exhaustive coding of all class subsets of Ord through the pair construction. It does not say that an externally specified set model has too many classes to form an external set. It says that the corresponding exhaustive decoder is not available *internally at the same size level*.

19.10.1 Definable classes and the truth-predicate trap

Formula codes and set parameters are sets, but the semantic map “formula plus parameters $\mapsto$ its extension in the universe” is not a uniformly available truth predicate for the universe. For each fixed formula one may define a class. One may also work externally with all definable classes of a chosen model. Neither maneuver supplies an internal universal decoder defeating [theorem 19.38](#). The same caution applies to quantifying over all possible Lean universe levels: universe polymorphism is not an internal type of all universes.

[merge] Sources 09 and 08 prove the diagonal theorem for families of global enumerations (source 09’s Section 12, “Why all global well-orders cannot be uniformly coded by sets”).

Note 19.40 (Source 09, Theorem 12.1, “Uniform no-coding theorem”). *There is no class of set indices I and no uniform class relation $F \subseteq I \times \text{Ord} \times \mathbf{No}$ whose rows are exactly, or even include, all global enumerations of $\mathbf{No}$.* This is [theorem 19.37](#), by the same diagonal class $D = \{\alpha : (c(\alpha), 2 \cdot \alpha, a_\alpha^-) \in F\}$ along a class surjection $c : \text{Ord} \rightarrow I$, valid with nonfunctional rows. Source 09’s preceding paragraph defines the switches E_A that list the pair $\{a_\alpha^-, a_\alpha^+\} = \{b(2 \cdot \alpha), b(2 \cdot \alpha + 1)\}$ increasingly for $\alpha \notin A$ and decreasingly for $\alpha \in A$; they embed the binary-class comparison into the global one, “the proper-class counterpart of the pair construction” of [note 6.12](#), as in the first half of [theorem 19.32](#).

Note 19.41 (Source 08, Theorem 13.2, “No class-uniform exhaustive list”). *If a class relation $T \subseteq \text{Ord} \times \text{Ord} \times \mathbf{No}$ uniformly gives a global enumeration T_α for each ordinal α , there is a global enumeration e^* different from every T_α ; consequently no uniform family indexed by an ordinary set or class of sets exhausts $\mathfrak{W}_{\text{sl}}$.* This is [theorem 19.37](#), by the same switch at the even coordinate $2 \cdot \alpha$; families indexed by sets or proper classes are reindexed by Ord . Source 08 adds: “the class of set-supported perturbation codes does not parametrize the full collection. The isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ is compatible with the much larger size of $\mathfrak{W}_{\text{sl}}$: each orbit is a dense copy of the surreal line, not all global enumerations”, and “an assertion about an embedding of the entire virtual order into a class must first specify a higher-type notion of mapping; it cannot be inferred from ordinary class universality.”

19.10.2 What the theorem does and does not say

The theorem rules out a uniform master list by set parameters, including one indexed by a proper class of such parameters. It does not rule out large useful coded subfamilies such as $\mathcal{P}_s(b)$; indeed the latter is dense and class-universal.

It also does not mean that class functions cannot be discussed in GBC, nor that stronger foundations are inconsistent. They can be discussed one or several at a time, in uniform families, or as elements of a higher-level collection in a suitably stratified theory. What fails is compression of *all* of them into one same-level, set-indexed evaluation relation.

Enumerating formula strings does not evade this issue. A putative uniform evaluation of all formulas over the set universe would require an appropriate truth predicate; it is not obtained from syntax alone. Definitions with additional class parameters also need not remain in the original parameter collection. One must distinguish a metatheoretic list of formulas, individual definable classes, and an internal uniform family of their extensions.

19.11 Omitted ranges and the transition to raw well-orders

Theorem 19.42 (An omitted-range cut). *Fix $r \in \mathbf{No}$ and a class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r\}$. There are increasing and decreasing ordinal-indexed families (a_α) and (b_α) in $\mathcal{P}_{\text{bd}}$ such that*

$$a_\alpha <_{\text{lex}} b_\beta \quad (\alpha, \beta \in \text{Ord}),$$

but no member of $\mathfrak{W}_{\text{sl}}$ lies strictly between both entire families. The raw well-order that lists F and then r has type $\text{Ord} + 1$ and is the unique member of $\mathfrak{W}_{\text{all}}$ between these families.

Proof. Use the approximants from Proposition 19.27. They satisfy $a_\alpha <_{\text{lex}} F <_{\text{lex}} b_\beta$ for every pair of indices. If a global bijection E lay between both families and differed from F first at δ , an approximant with index greater than δ would place E on the wrong side. Therefore E would equal F at every ordinal, which contradicts its surjectivity.

Let R be the raw order obtained by appending r after the enumeration F . Its set-like initial part is precisely F : every point listed by F has set-sized predecessors, whereas r has a proper class of predecessors. Its comparisons with the approximants are decided at their prescribed ordinal first differences, so it lies between both families. Conversely, the same first-difference argument forces the core F_S of any raw order S between the families to equal F . By the core decomposition of Theorem 15.1, its residual order must be the unique order on the singleton $\{r\}$. Thus $S = R$. $\square$

The lower and upper families here are indexed by all ordinals, not by sets. Accordingly, the missing interpolant does not contradict Theorem 19.10. It gives a concrete boundary between the set-cut saturation of the set-like comparison and the additional points provided by non-set-like global well-orders.

[merge] *Source 09's route.* Source 09's Section 11 gives a concrete instance of theorem 19.42, with the nonexhaustive word $\xi \mapsto \xi$ and the fresh values -1 and $\alpha + 1$; its remark 19.44 is the general mechanism, which is source 11's theorem for an arbitrary omitted range.

19.12 Source 09's route: a proper-class cut with no interpolant

The interpolation theorem has an exact and visible size limitation. The following construction uses only ordinal-valued prefixes and the single nonordinal value -1 .

Theorem 19.43 (An explicit unfilled (Ord, Ord) configuration). *For any baseline b , there are uniformly class-indexed families $(L_\alpha)_{\alpha \in \text{Ord}}$ and $(U_\alpha)_{\alpha \in \text{Ord}}$ lying in $\mathcal{P}_s(b)$ such that*

$$\alpha < \beta \implies L_\alpha < L_\beta < U_\beta < U_\alpha, \quad L_\alpha < U_\beta \quad \text{for all } \alpha, \beta,$$

but there is no global enumeration E with $L_\alpha < E < U_\beta$ for every α, β .

Proof. Let $g : \text{Ord} \rightarrow \mathbf{No}$ be $g(\xi) = \xi$. This is an injective class word but not an exhaustive enumeration of $\mathbf{No}$. For each α , use [lemma 19.6](#) to choose core completions of the prefixes

$$(g \upharpoonright \alpha) \frown (-1), \quad (g \upharpoonright \alpha) \frown (\alpha + 1),$$

calling them L_α and U_α , respectively. A fixed global well-order on the set support codes makes these choices uniform. The appended letters are fresh. For $\alpha < \beta$, comparison at coordinate α gives $L_\alpha < L_\beta$ and $U_\beta < U_\alpha$. Every lower word is below every upper word, by comparison at the smaller divergence coordinate.

Suppose a global enumeration E lay between all these bounds. If $E(\delta) \neq \delta$ for some ordinal, take the least such δ . If $E(\delta) < \delta$, the lower word $L_{\delta+1}$ agrees with g through coordinate δ and therefore lies above E , a contradiction. If $E(\delta) > \delta$, the upper word $U_{\delta+1}$ lies below E , also a contradiction. Hence $E(\delta) = \delta$ for every ordinal δ , making $E = g$. But g omits, for example, -1 , and is not surjective onto $\mathbf{No}$. $\square$

Here the notation (Ord, Ord) describes an increasing lower and a decreasing upper cofinal presentation, not a pair of set cardinal cofinalities. More explicitly, compare every global enumeration to the nonexhaustive word g . Every enumeration differs from g at a set ordinal and lies on exactly one side. The lower family in the theorem is cofinal on the lower side, and the upper family is coinital on the upper side. Neither side has an endpoint. The missing point would have to be the nonexhaustive branch g itself.

Every set subfamily of these constraints has an interpolant by [note 19.12](#). The failure is therefore genuinely at the proper-class level, not a missed set-sized limit. In particular, the space of all global enumerations is not a class-cut completion of its surreal core.

Remark 19.44 (A general missing-branch mechanism). The same proof works for any injective class word $g : \text{Ord} \rightarrow \mathbf{No}$ which is not surjective. At each set stage choose fresh values below and above $g(\alpha)$ outside the earlier range, and complete the two prefixes. Any interpolant would agree with g at every coordinate and would therefore fail surjectivity. The surreal set-cut property guarantees the required fresh values.

Example 19.45 (Two omitted points produce adjacent raw orders). Let $r < s$ and choose $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{r, s\}$ bijectively. The approximants to F have exactly two raw interpolants: append r, s in that order, or append s, r . Both raw orders have type $\text{Ord} + 2$, and the first is lexicographically smaller. They are adjacent in $\mathfrak{W}_{\text{all}}$. Indeed, a raw order between them cannot have a different core, since an ordinal first difference would put it on the same side of both. With core F fixed, its residual order is a well-order of a two-element set, and these are exactly the two displayed possibilities. Thus the density theorem for $\mathfrak{W}_{\text{sl}}$ does not extend to all raw well-orders.

Corollary 19.46 (Exactly which raw set cuts admit bounded completions). *Let $(A_i)_{i \in I} <_{\text{lex}} (B_j)_{j \in J}$ be a uniformly set-indexed strict cut in $\mathfrak{W}_{\text{all}}$. A member of $\mathcal{P}_{\text{bd}}$ lies between the two sides if and only if*

$$F_{A_i} \neq F_{B_j} \quad (i \in I, j \in J),$$

where F_R is the canonical ordinal enumeration of the set-like initial part of a raw well-order R . The condition is vacuous when a side is empty.

Proof. If the cross cores are different, their comparisons are decided at ordinal first differences and have the same direction as the given raw comparisons. Apply Theorem 19.10 to these uniform families in $\mathfrak{J}$. Every comparison of the resulting completion with a raw input is already decided within its core, giving the required raw inequalities.

Conversely, suppose a cross pair shares a core F . This core cannot be surjective: a surjective core determines the entire raw order, contradicting the strict comparison of the pair. Any $E \in \mathfrak{W}_{\text{sl}}$ therefore differs from F at an ordinal coordinate. That first difference places E on the same side of both raw orders, so E cannot lie between them. This excludes a bounded-support completion in particular. $\square$

The criterion isolates the obstruction without appealing merely to size. Distinct ordinal cores can be separated by one bounded change of the baseline, even for a whole set-indexed cut. Two opposite sides with the same core require information in the residual non-set-like order, beyond every ordinal coordinate available to a global set-like enumeration.

Part V

Foundations, models and formalization

20 Formalizing the subject without changing its meaning

20.1 A bounded development in ordinary set theory

For any ordinal θ , use the set $X = \mathbf{No}_{<\theta}$. The well-order relations on X form a subset of $\mathcal{P}(X \times X)$. Their lexicographic comparison can be defined either by unique ordinal enumeration or by the predecessor formula. This is ordinary ZFC mathematics. No inaccessible cardinal, class theory, or new logical sort is required.

This approach is particularly good for proving the general alphabet theorem, investigating cardinal spectra, and checking how a proposed construction changes birthdays. One should distinguish $\mathcal{W}(X)$, which permits every possible ordinal type of cardinality $|X|$, from its minimal-length stratum. Their cardinalities agree, but their jumps and finer order structure need not.

A family of results for every θ does not automatically give a coherent global map. To take a union, the maps must agree on overlaps. The repository's bounded-surreal nonembedding example makes this restriction concrete.

20.2 Second-order class theory

In GB or GBC, the formula

$$\text{ClassWO}(R) \wedge \text{ClassWO}(S) \implies R = S \vee \text{Lex}(R, S) \vee \text{Lex}(S, R)$$

is meaningful with class variables. One can prove trichotomy, transitivity, existence theorems, and uniform-family embedding theorems in this language. It is unnecessary to form a class whose members are R, S .

When a family is indexed by a class I of sets, one class relation $T \subseteq I \times \mathbf{No} \times \mathbf{No}$ supplies its rows. If the rows are pairwise distinct, substituting them into the elementary predecessor formula produces a class linear order on I . With repeated rows, one instead has the corresponding comparison on codes modulo equal evaluations. This is the correct format for most class-sized examples and applications.

In contrast, an expression such as

$$\{R : R \text{ is a class well-order of } \mathbf{No}\}$$

does not define an ordinary GB class of members. Our fraktur notation abbreviates a domain of class quantification. The diagonal theorem shows why a single uniform set-code replacement cannot recover the full domain either.

20.3 Why KM does not erase the extra level

KM strengthens comprehension by allowing class quantifiers, but its classes still have *sets* as members. It therefore does not turn the totality of class well-orders into an ordinary class. A genuine object containing arbitrary classes as members requires another sort, or a containing universe in which those classes become sets.

This distinction matters when one wants an arbitrary topology of higher-order open collections, a measure on all global well-orders, or a power object of their totality. Such notions require an explicit ambient theory. Hyperclass forcing provides one example of higher-level foundational practice [28]; it is not a reason to silently ignore the level distinction in bare KM.

There is also a separate recursion issue. Extracting the canonical initial Ord-part of one raw well-order is an elementary class construction. Iteratively deleting proper-class tails through an arbitrary transfinite sequence would have class-valued stages and needs its own analysis. General comparability of abstract class well-orders by initial-segment isomorphisms is another stronger topic; ETR implies it [27]. None of that machinery is needed for the labeled comparison proved in this article.

20.4 An explicit containing-universe model

A useful concrete model assumes a strongly inaccessible cardinal κ in an ambient universe and takes

$$U = V_\kappa, \quad \mathcal{C} = \mathcal{P}(V_\kappa).$$

This is a sufficient setting for interpreting a full class universe. The internal surreal class becomes the ambient set

$$\mathbf{No}^U = \mathbf{No}_{<\kappa}, \quad |\mathbf{No}^U| = \kappa.$$

All its ambient well-orders form a set of cardinality 2^κ . Those that are internally set-like have ambient order type κ ; unrestricted well-orders may have any ambient ordinal type of cardinality κ .

To see the set-like assertion, every initial segment of a κ -enumeration has fewer than κ members, with ranks bounded below κ ; regularity puts it in U . Conversely, an ordinal type strictly exceeding κ has an initial segment of cardinality κ , which cannot be a set of U . A smaller type cannot enumerate the entire carrier.

The bounded-support class becomes an ambient order of cardinality κ , while the full κ -permutation order has cardinality 2^κ . Thus a κ -sized dense suborder can coexist with a larger order of all enumerations. The exact ambient ordinal cutoff remains κ^+ .

Every ordinal *of* U is below κ . There is no contradiction between internally embedding all ordinals and having an ambient successor-cardinal barrier. Inaccessibility is an additional assumption for this full universe example; it is not required for an individual birthday fragment.

20.5 A strict nonembedding result

The containing-universe interpretation makes a further comparison decidable by ordinary set-order invariants.

Theorem 20.1 (Nonembedding in a full universe). *In the preceding full interpretation $U = V_\kappa$, $\mathcal{C} = \mathcal{P}(V_\kappa)$, with ambient ZFC and κ strongly inaccessible,*

$$\mathfrak{W}_{\text{sl}}(\mathbf{No}^U) \hookrightarrow \mathfrak{W}_{\text{all}}(\mathbf{No}^U), \quad \mathfrak{W}_{\text{all}}(\mathbf{No}^U) \not\hookrightarrow \mathfrak{W}_{\text{sl}}(\mathbf{No}^U).$$

These are assertions about the actual ambient sets of internal class well-orders and arbitrary ambient order embeddings.

Proof. The first embedding is inclusion. Put $X = \mathbf{No}^U$. The target of the proposed reverse embedding is the dense order of all bijections $\kappa \rightarrow X$. Its bounded-support suborder $D = \mathcal{P}_{\text{bd}}^U$ is dense and has ambient cardinality κ . For the upper bound, codes with domain $\lambda < \kappa$ are subsets of $\lambda \times \lambda$; strong inaccessibility bounds their number below κ , and there are κ possible domains. Transpositions give the matching lower bound.

Fix labels $a < b$ in X . There are 2^κ bijections $F : \kappa \rightarrow X \setminus \{a, b\}$, by the fixed-alphabet cardinality theorem. Each determines a convex adjacent pair

$$R_F = F + (a, b) <_{\text{lex}} S_F = F + (b, a)$$

in the unrestricted order. Distinct F 's determine distinct, disjoint convex fibers; one entire pair precedes the other.

If $\Phi : \mathfrak{W}_{\text{all}}(X) \hookrightarrow \mathfrak{W}_{\text{sl}}(X)$ existed, the intervals

$$(\Phi(R_F), \Phi(S_F))$$

would be 2^κ pairwise disjoint nonempty open intervals in the target. Each contains a point of D . Choosing the least such point under a fixed well-order of D would inject 2^κ into κ , a contradiction. $\square$

Thus equal cardinality 2^κ and equal ordinal embedding cutoff κ^+ do not imply equimorphism. The useful invariant here is the size of a family of disjoint open intervals relative to a dense suborder. The repository already uses related separability and cellularity arguments for the real alphabet [1, Remark `lwo:sp:rem:conditional`]; the argument above supplies the precise surreal-universe version.

There is also a direct class-theoretic obstruction, provided the inverse evaluation relation can be formed by impredicative comprehension.

Theorem 20.2 (No definable uniform embedding in $\text{KM} + \text{GC}$). *For every fixed formula, with fixed set and class parameters, that defines a total transformation $R \mapsto \Phi(R)$ from $\mathfrak{W}_{\text{all}}(\mathbf{No})$ to $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, the theory $\text{KM} + \text{GC}$ proves that this transformation is not an order embedding. The transformation is specified relationally: the defining formula assigns a unique output class graph to each input class well-order. This is a schema for fixed formulas, not quantification over a third sort of arbitrary transformations.*

Proof. Suppose such an order embedding were defined. Fix $a < b$ and a global well-order of the set codes in $\mathcal{P}_{\text{bd}}$. For each class bijection $F : \text{Ord} \rightarrow \mathbf{No} \setminus \{a, b\}$, form the adjacent raw pair R_F, S_F from the preceding proof. Density of $\mathcal{P}_{\text{bd}}$ in $\mathfrak{W}_{\text{sl}}$ supplies codes strictly between $\Phi(R_F)$ and $\Phi(S_F)$. Let q_F be the globally least such code. Comprehension in KM makes this a legitimate least-element

definition, even if the formula specifying Φ contains class quantifiers. Disjointness of the image intervals implies

$$q_F = q_G \implies F = G.$$

Now use KM comprehension to form the ordinary class relation

$$\text{Eval}(q, \alpha, x) \iff \exists \text{ class } F [F : \text{Ord} \xrightarrow{\sim} \mathbf{No} \setminus \{a, b\} \wedge q = q_F \wedge F(\alpha) = x].$$

Its rows are empty or the graphs of unique such bijections, and every such bijection occurs. Compose all rows with one fixed class bijection $\mathbf{No} \setminus \{a, b\} \rightarrow \mathbf{No}$. The resulting single class evaluator, with the ordinary set codes $q \in \mathcal{P}_{\text{bd}}$, contains every global enumeration of $\mathbf{No}$. This contradicts Theorem 19.37. $\square$

The inverse evaluator is the decisive extra comprehension step: its definition quantifies over an arbitrary class F . The preceding proof does not claim that elementary GB comprehension suffices. Nor does it exclude arbitrary external maps between the available-class collections of a nonfull model. In particular, for an externally countable model these collections can both be countable, and its externally dense set-like order can admit an external copy of its unrestricted order. Such a map need not be representable by any uniform definition in the model. This is why the stated interpretation of an embedding matters.

20.6 Internal and external well-foundedness

For a model $(M, \mathcal{X})$, an internally well-founded class relation need not be externally well-founded. An external descending sequence might not be represented by a set in M . Thus comparing the internal theory with an external collection of all relations requires care.

Over GBC, well-foundedness of a class relation can be characterized by the absence of a set-coded descending ω -sequence [27, Section 3]. Keeping the sets and an existing relation fixed while changing only the available classes does not arbitrarily change that relation's internal well-foundedness in suitable models. What can change is which class relations are available at all. External correctness can be ensured by an appropriate well-founded full universe interpretation, as above.

20.7 A practical Lean architecture

The repository already has a universe-indexed sign carrier [7]. Its type structure can be displayed schematically as:

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : forall i, signAt i != 0 <-> i < birthday
```

This is mathematical pseudocode using ASCII logical notation, not a new executable Lean declaration. The key point is the level separation: the full carrier is one universe above the types indexing its small sign lengths.

The following existing declarations were inspected:

Module or declaration	Role in the proposed development
<code>SignSequence.ordinalOrderEmbedding</code>	The all-plus ordinal embedding
<code>birthdays_bounded</code> , <code>small_bounded</code>	Control of small parameter families and bounded fragments
<code>SignSequence.not_small</code>	The carrier cannot be treated as small at its ordinal level
<code>SignSequenceCutOperation.cut</code> and <code>cut_realizes</code>	Actual separators for small cut data
<code>SizeObstructions.noUnrestrictedCuts</code>	The obstruction to unrestricted cut filling

These are source-level observations, not results of a fresh build. The current repository contains later arithmetic modules as well; historical proposal notes should not be mistaken for the present implementation status.

A sensible formalization proceeds in four parts.

1. **Set carriers.** For $X : \text{Type } u$, represent well-orders as relations on X together with their laws. Prove the predecessor-based comparison and its equivalence with ordinal enumeration. This avoids transporting across ordinal representatives at every comparison.
2. **Ordinal words.** Define words with small ordinal domains, an injective subtype, ordinal concatenation, and the sign code. Formalize prefix-freeness and first disagreement before proving the cardinal spectrum.
3. **Relative set-likeness.** If X lives in $\text{Type } (u+1)$, express the condition that each predecessor subtype is `Small.{u}`. A well-order of this larger X has its unrestricted order type at the larger ordinal universe. A same-level enumeration must not be inferred without the relative smallness hypothesis.
4. **Uniform families.** Keep an explicit index type and an evaluation relation. The bounded-support codes are sets of ordinal pairs at the selected level. Formulate the interpolation theorem only for small index families and the no-code theorem at its correct larger level.

Lean universes and Mathlib ordinals provide precisely the needed separation [29, 30]. Universe polymorphism is a schema of well-typed constructions, not a term denoting the union of all universes. This article supplies no new Lean file and makes no machine-verification claim.

[merge] Source 12's Section 11 identifies both global orders in the full-class model $(V_\kappa, \mathcal{P}(V_\kappa))$ of [theorem 20.1](#). Its clause (iv) gives a second proof of that theorem's non-embedding: the full order $\mathcal{W}(\mathbf{No}_{<\kappa})$ has binary coding length κ^+ ([theorem 6.8](#)), whereas the set-like order $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ has coding length κ ([theorem 8.3](#), regular case), and coding length cannot decrease under embeddings ([lemma 6.4](#)). Source 11's proof counts 2^κ disjoint intervals against a dense suborder of size κ .

21 A full-class model that connects the two size levels (source 12)

Assume in an ambient ZFC universe that κ is strongly inaccessible. Consider the two-sorted structure

$$\mathcal{M} = (V_\kappa, \mathcal{P}(V_\kappa)),$$

with all external subclasses of V_κ available as its classes. This is a familiar full-class interpretation of strong class set theory: regularity and the strong-limit property ensure the required closure of set operations and class Replacement. It satisfies KM and global choice. We use this as an explicit relative model, not as a claim that inaccessible cardinals are necessary for the earlier GBC theorems.

Theorem 21.1 (Exact external identification). *For the full-class interpretation above:*

- (i) *the internal surreal carrier is externally $\mathbf{No}_{<\kappa}$, and its internal ordinals are the external ordinals below κ ;*
- (ii) *the internal set-like global surreal well-orders, viewed as an external set order, are exactly $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$;*
- (iii) *all internal global surreal well-orders, viewed externally, form exactly $\mathcal{W}(\mathbf{No}_{<\kappa})$;*
- (iv) *both external orders have cardinality 2^κ and ordinal spectrum $\beta < \kappa^+$, but their binary coding ranks are respectively κ and κ^+ . Therefore the full order does not embed into the set-like order.*

Proof. A sign sequence belongs to V_κ precisely when its ordinal domain has length less than κ . Inaccessibility gives $|\mathbf{No}_{<\kappa}| = \kappa$.

Every external relation on $\mathbf{No}_{<\kappa}$ is a permitted internal class: each ordered pair of its labels is an element of V_κ . External well-foundedness implies internal well-foundedness. Conversely, if a relation is externally ill-founded, ambient choice supplies a descending ω -sequence. Its countably many labels have ranks bounded below κ by regularity, so the sequence is an element of V_κ and witnesses internal ill-foundedness. Thus the well-orders agree.

An internal set has external cardinality less than κ . Conversely, a subset of $\mathbf{No}_{<\kappa}$ of cardinality less than κ has bounded ranks and is an element of V_κ . Therefore internal set-likeness means that every proper predecessor segment has cardinality less than κ . An external well-order of κ labels has that property exactly when its order type is κ , proving (ii). Arbitrary external well-orders on those labels have types exactly the ordinals of cardinality κ , proving (iii). Comparisons agree by [theorem 14.7](#). Finally apply [theorems 6.8](#) and [8.3](#) and the monotonicity of coding rank under embeddings. $\square$

21.1 Why this does not contradict surreal universality

Internally, each class linear order has elements that are sets of V_κ and hence externally at most κ elements. The external collection of all class well-orders has 2^κ elements and is not an internal class carrier. Thus it is outside the scope of the internal class-universality theorem.

If $\lambda > \kappa$ is another inaccessible cardinal, the entire external set $\mathcal{W}(\mathbf{No}_{<\kappa})$ can be studied as an ordinary set in V_λ . It then embeds into the larger surreal carrier there by set-order universality. Its labels have moved up one foundational level; no same-level universal object has been created.

The singular cutoff $\beth_\omega$ in [section 8](#) must not be used as though $V_{\beth_\omega}$ were this model. That rank lacks the regularity used in Replacement and in the well-foundedness argument. The singular theorem is an ordinary ZFC theorem about a bounded set, not a full-class model construction at a singular height.

21.2 Henkin classes, countable models, and relativity

If the class collection is some $\mathcal{X} \subsetneq \mathcal{P}(M)$, only those well-orders whose relations lie in $\mathcal{X}$ are admitted. External cardinality and saturation can then differ from the full-class calculation. A countable model may have an externally countable collection of admitted classes and external countable cofinal families, while internally having no *coded set-indexed* cofinal family. The missing external families are simply not internal objects.

For nonstandard or non- β class models, one must additionally separate internal and external well-foundedness. Our exact identification in [theorem 21.1](#) used transitivity, full classes, and regularity; it was not asserted for arbitrary models. The general subtleties of class well-foundedness are discussed in [\[27, §3\]](#).

[merge] Clause (ii) of [theorem 21.1](#) is source 11's statement that the internally set-like well-orders have ambient type κ , and clause (iii) its statement that unrestricted ones have every ambient type of cardinality κ ([section 20](#)). Source 12's remark that the singular cutoff $\beth_\omega$ must not be used as though $V_{\beth_\omega}$ were such a model is the counterpart of source 11's remark that inaccessibility is needed for the full universe example but not for an individual birthday fragment.

[merge] Source 09's inaccessible model is the same setting. Its clause (ii) is the bounded form of [corollary 19.15](#), its clause (iii) of [theorem 19.10](#), and its clause (iv) the bounded form of [theorem 19.43](#); its $\mathcal{E}_\kappa$ is $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$, the set-like order of [theorem 21.1\(ii\)](#).

22 A concrete universe enlargement: all objects become sets (source 09)

A useful consistency-safe visualization of the size jump is available in ordinary set theory with a strongly inaccessible cardinal. This assumption is for the illustration, not for the earlier ZFC or GBC results.

Fix a strongly inaccessible cardinal κ , put $X = \mathbf{No}_{<\kappa}$, and choose a bijection $b : \kappa \rightarrow X$. Externally, X is a set of cardinality κ . In the smaller universe V_κ , it plays the role of the entire surreal class. Every element of X and every set-length prefix of length below κ belong to V_κ .

Define the external set

$$\mathcal{E}_\kappa = \text{Bij}(\kappa, X)$$

and let $\mathcal{P}_s^{<\kappa}(b)$ contain its enumerations eventually equal to b below κ , equivalently the permutations of κ with bounded support. Regularity makes bounded support equivalent here to support of cardinality less than κ .

Theorem 22.1 (Inaccessible set-sized model). *In this setting:*

- (i) $|\mathcal{P}_s^{<\kappa}(b)| = \kappa$ and $|\mathcal{E}_\kappa| = 2^\kappa$.
- (ii) $\mathcal{P}_s^{<\kappa}(b) \cong X$, as linear orders, and the core is order-dense in $\mathcal{E}_\kappa$.
- (iii) Every separated pair of families in $\mathcal{E}_\kappa$ of combined size less than κ has an interpolant in the core.
- (iv) There is a separated (κ, κ) -configuration in the core with no interpolant even in $\mathcal{E}_\kappa$.

Proof. By [proposition 7.3](#), $|X| = 2^{<\kappa} = \kappa$. There are fewer than κ permutations on any fixed $\alpha < \kappa$, because κ is strong limit. Their union over κ stages has size at most κ , and the swaps of position zero with other positions give the reverse bound. The cardinality of $\mathcal{E}_\kappa$ is 2^κ by [note 6.12](#).

For fewer than κ lower and upper constraints, regularity bounds all cross first-disagreement coordinates below κ . The forced-prefix construction in [note 19.12](#) therefore has length below κ . Its chosen separators remain in X : the option families and previously used values have size below κ , their birthdays are bounded below κ , and (1) supplies a separator of birthday below κ . Prefix completion can be carried out at a larger cardinal still below κ . This proves (iii), and in particular density and the $< \kappa$ -cut property of the core.

The same birthday argument gives the $< \kappa$ -cut property of X . A back-and-forth of length κ , with partial maps of size less than κ at every stage, gives the isomorphism in (ii). Finally repeat [theorem 19.43](#) for $\alpha < \kappa$ and the nonexhaustive word $\alpha \mapsto \alpha$ into X . Both -1 and each required successor ordinal belong to X , and the same comparison argument proves (iv). $\square$

This model explains the apparently conflicting assertions that every class-coded portion embeds into the surreal line and yet all enumerations are larger. A class of set codes over V_κ has external

size at most κ ; the set $\mathcal{E}_\kappa$ has external size 2^κ . It is an object of the larger universe, not such a class of codes in the smaller universe. External arbitrary families of κ or more class objects are not automatically small-universe sets.

Even externally, $\mathcal{E}_\kappa$ only represents well-orders of length κ . All well-orders on the external set X also include lengths α with $\kappa < \alpha < \kappa^+$. Thus a universe jump must record not just the carrier but which ordinal lengths are admitted.

[merge] Source 08's Section 15 discusses the same frameworks and the same full-class model; its (S08.13) is [theorem 21.1\(ii\)](#).

23 Universes, class models, and paradox avoidance (source 08)

The preceding proofs can be studied in several foundations, provided the meaning of “all” is kept explicit.

23.1 ZFC with definable classes

In ordinary ZFC, **No** is given by a formula defining sign sequences. Statements about a specified finite list of definable class functions can be translated into statements about their formulas. Set-sized cutoffs and all theorems about $\mathcal{W}(X)$ are ordinary set theorems. Global conclusions can be formulated as schemes relative to specified definable enumerations or in a theory supplying a definable global well-order.

However, “take the set of formulas defining class well-orders and interpret them all in V ” is not a free replacement for a class collection. Formula codes are sets, but uniform unrestricted truth for all formulas of the universe is a separate issue. One cannot silently insert a full satisfaction predicate for V into the base theory. Uniformity and definability claims must state their truth resources.

23.2 Two-sorted GBC and stronger class theories

In a model $(M, \mathcal{C})$ of GBC, the virtual collection $\mathfrak{W}_{\text{sl}}$ means the bijections whose graphs belong to $\mathcal{C}$. The cut families in [note 19.13](#) are represented by one relation in $\mathcal{C}$, indexed by an M -set. The skeleton consists of the M -set support codes, interpreted relative to $b \in \mathcal{C}$. All class unions and selections asserted by the proofs are made inside this model.

Passing from GBC to KM strengthens class comprehension, but it still does not make proper classes eligible as members of ordinary classes. Stronger comprehension is not an additional membership sort. A literal object containing every class enumeration belongs to an explicitly higher-order theory or to an external metatheory.

The literature on class recursion gives useful warnings here: even natural uniform truth or forcing relations may require principles stronger than GBC [23, 25]. Our elementary lexicographic operators do not themselves supply such truth relations.

23.3 A concrete full-class model at an inaccessible cardinal

Let κ be strongly inaccessible in an external set universe, and consider the full-class model

$$(V_\kappa, \mathcal{P}(V_\kappa)). \tag{S08.12}$$

The second occurrence of $\mathcal{P}_s$ here is the ordinary power-set notation, not the permutation-code class. Internally, Ord^{V_κ} is a proper class; externally it is the set κ . Likewise the internal surreal class is externally $\mathbf{No}_{<\kappa}$ and has cardinality κ .

The full collection of internal global enumerations is, externally,

$$\{e : \kappa \longrightarrow \mathbf{No}_{<\kappa} : e \text{ is a bijection}\}, \quad (\text{S08.13})$$

a set of cardinality 2^κ . An internal set-supported perturbation has support of size below κ , equivalently bounded support in κ . There are exactly κ canonical codes: at each bound there are fewer than κ possibilities by strong inaccessibility, and there are κ bounds; the first-entry embedding gives the matching lower bound. Thus externally the skeleton has size κ and the full enumeration order size 2^κ .

This is a direct set-theoretic model of the size distinction. The internal isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ becomes an external isomorphism between two κ -sized η_κ -orders. Internally “every set cut” means externally every cut specified by fewer than κ elements, not every external set cut.

Also, (S08.13) is the *fixed-length* order $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$. It is not the external order of all well-orderings of the set $\mathbf{No}_{<\kappa}$, which also allows lengths $\kappa+1$, $\kappa+\omega$, and all other ordinals of cardinality κ . Those longer enumerations may have internally proper-class initial segments and are not internal set-like well-orders. This is an important bridge between section 7.2 and the global theory.

Strong inaccessibility is used here to obtain a convenient full model and its size calculation. It is not an additional hypothesis of the GBC theorems: it describes one external way to realize their set/class distinction.

23.4 Restricted classes and external viewpoints

For a general class model, enlarging $\mathcal{C}$ may add new global enumerations and new binary class words while leaving the sets of M unchanged. Thus the virtual ambient orders depend on the chosen class collection. The explicit formulas for pair coding, prefix extension, and diagonalization still work with their stated parameters in each model, but one must not identify an internal “all” with an external one.

Similarly, an internal well-foundedness assertion in a nonstandard or nontransitive model need not mean external well-foundedness. Concrete full-class models such as (S08.12) avoid some of these ambiguities; unrestricted absoluteness claims do not follow from our proofs.

23.5 What prevents the paradoxes

The safeguards are structural. A surreal is a set sign sequence, not a sequence of length Ord. Its birthday is a set ordinal. A support code is a set, while its identity extension is a class function. A uniform family is one class relation with set-valued indices, not a set containing proper classes. The full collection of class enumerations is virtual or lives at a higher type. Finally, “every set cut is realized” never means that a separator is required between two arbitrary proper classes.

For instance, a surreal above every ordinal would contradict proposition 5.4; the left options in that purported cut are a proper class, so theorem 5.6 does not apply. Likewise a list of every set-supported code could be a class enumeration, but an exhaustive class-uniform list of all global enumerations is ruled out by note 19.41. Neither obstruction is bypassed by changing notation.

[merge] The three remaining sections are the other sources’ formalization proposals. The modules they name are proposals and do not exist; every existing declaration they cite exists at the current revision, and none of them states a result of this report (section 1.5).

24 A formalization architecture with explicit size boundaries (source 12)

24.1 Start with a universe-polymorphic set theorem

The most efficient verified starting point is independent of surreal arithmetic. For a type X with a linear order, represent a well-order on X by its binary relation and a proof of well-foundedness and linearity. Define the common prefix by equality of predecessor sets and equality of the restricted relations, exactly as in [section 14.1](#). Prove maximality, distinct next labels, and transitivity of comparison. This gives a generic theorem reusable for reals, surreal cutoffs, and other ordered alphabets.

For a fixed ordinal domain, represent an enumeration as an equivalence between an ordinal's carrier and X . For all well-orders of a small X , using relations on X often avoids unnecessarily quantifying over every ordinal in a larger universe. The ordinal enumeration can be recovered later from the well-order type. This is a representation choice, not an identification of all ordinals with a small type.

24.2 Use the existing sign-sequence foundations selectively

The inspected repository file

```
Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean
```

contains, among others,

```
isPrefix_of_minimum_birthday
existsUnique_prefix_of_ordConnected
existsUnique_minimum_birthday_of_ordConnected
existsUnique_simplest_separator
```

[10]. Its comments explicitly distinguish uniqueness of a simplest separator from existence of separators for arbitrary small cuts. The last theorem assumes a separator exists. It must not be imported as though it independently established all cut-filling instances.

A suitable interface for the order-theoretic development needs a birthday map, a numerical linear order, sign concatenation and truncation, the first-difference comparison, and small-cut existence with a birthday bound. Field operations can be omitted except in illustrative examples.

24.3 Suggested proof modules

Module	Principal obligations
CommonPrefix	The elementary definition; initiality; maximality; next-label inequality; generic full lexical linearity.
ExhaustiveLex	Prefix-freeness; equivalence with ordinal first-difference comparison; transport between relations and enumerations.
BinaryCubes	Cylinder fusion; strict hierarchy; characteristic-cut embeddings; precise ordinal concatenation.
PermutationCoding	Pair orientation, product scheduler, transpositions, and exhaustive output.

Module	Principal obligations
BirthdayCutoffs	Size of $\mathbf{No}_{<\kappa}$; prefix-free numerical code; cone partition; regular and singular sums.
ClassFamilies	A formal set/class model, sections of class relations, baseline enumeration, and diagonal non-enumerability.
UniverseModels	A chosen inaccessible-rank interpretation; internal/external well-foundedness and identification of the two orders.

The finite common-prefix theorem should be proved directly for relations, not by first developing all of ordinal arithmetic. The scheduler proof should record both surjectivity and the first-row-appearance invariant; a bare cardinal pairing equivalence is insufficient. The singular theorem must use ordinal sums, never a cardinal cast that silently loses order.

24.4 Universe accounting

Schematically, a sign-sequence type built from small ordinal lengths has one universe level for its ordinal index and potentially a higher level for the total carrier. One should retain explicit levels rather than promise a concrete declaration before checking the chosen ordinal API. A bounded carrier with a specified small birthday bound is the cleanest first implementation.

Three operations must remain visibly different: taking a subtype of one fixed surreal carrier; lifting an existing carrier to a larger universe; and constructing all surreals of the larger universe. The third adds new birthdays and is not merely a type lift. Likewise, a type of relations on a fixed carrier can exist at a higher level without becoming a same-level class of all class relations.

For global theorems, two safe routes are available. One may formalize GB or GBC models and express theorems about their class predicates. Or one may formalize the bounded analogue inside a hierarchy of explicit universes and separately prove its interpretation in a set/class model. Neither route requires an inconsistent type of all types.

24.5 Axioms, kernels, and honest validation

A full formalization should record the uses of classical logic, set choice, global choice, and any ambient model-existence assumptions. The common-prefix comparison should not acquire an ETR assumption merely because abstract class order comparison sometimes uses one. The inaccessible-rank example is conditional on that ambient cardinal, whereas the bounded singular theorem needs no inaccessible cardinal.

The present package includes exact executable tests of finite shadows: sign coding, pair coding, the scheduler, the common-prefix formula, and finite adjacency. These are useful implementation checks and counterexample detectors. They are not kernel-checked proofs of any infinite or class theorem, and no supplied file is advertised as such.

[merge] Source 09's frameworks and Lean dependency plan.

25 Formalization strategies and paradox avoidance (source 09)

25.1 Four suitable frameworks

Birthday-bounded ZFC. For the most immediately formalizable results, fix $\mathbf{No}_{<\theta}$ and a length bound. Everything is then a set: relations, permutations, words, embeddings, and counterexamples.

The exact ordinal spectrum and finite residual adjacency results belong here. No inaccessible cardinal is needed.

Two-sorted GBC. Use set variables for surreal numbers, support codes, ordinals, and bounded prefixes; use class variables for global enumerations. Treat $E <_{\text{lex}} F$ as a formula on class variables. State family theorems with an explicit uniform ternary relation. The core is a genuine class linear order. This framework proves all the global results above without a class-of-classes object.

A larger universe or explicit higher sorts. To make the entire collection of smaller-universe global enumerations an object, put it in a larger universe, or use separate sorts for sets, classes, and collections of classes. Membership and comprehension must be stratified. The inaccessible model in [section 22](#) demonstrates one controlled interpretation. A higher-sort theory requires its own axioms; merely writing a new symbol for all classes does not specify one.

Universe-polymorphic dependent type theory. A type-theoretic formalization can construct a carrier relative to a universe, then form its function types at an appropriate larger level. It must not infer that those functions are indexed by the original small ordinals or that every large subtype is small. Universe polymorphism handles repeated constructions; it does not identify the universe levels.

25.2 Kelley–Morse does not create classes of classes

KM strengthens class comprehension, but its ordinary two-sorted interpretation still has sets as the members of classes. It does not turn proper classes into elements of other classes. Similarly, adding ETR resolves certain class-valued recursion problems but does not invalidate [note 19.40](#). These are separate issues: recursion strength, comprehension strength, and object level.

For non-set-like class well-orders, recursion along predecessors may need class-valued histories. One should therefore not reuse our set-valued recursion justification without checking the new hypotheses. The literature on ETR supplies the appropriate distinctions [\[24, 26\]](#).

25.3 Concrete reuse of the ProveIt sign-sequence foundation

At the pinned snapshot, the foundational module declares

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff : ...
```

The ASCII arrow above is only a typographical transcription. The actual module supplies a linear order via zero-extended signs and contains the following relevant named declarations [\[7\]](#):

Declaration	Role in this project
<code>lt_iff</code>	First-disagreement characterization of sign order.
<code>ordinalOrderEmbedding</code>	Increasing embedding of small ordinals.
<code>birthdays_bounded</code>	Birthday bounds for explicitly small families.
<code>small_bounded</code>	Smallness of bounded sign fragments.
<code>not_small</code>	The entire carrier does not fit in the lower universe.

The companion cut-operation file contains `cut_realizes` and `birthday_cut_le_iSup`, together with the separator theorem `exists_separator_of_small_sets` [9]. Its documentation explicitly distinguishes this order/cut development from supplying field operations. The present proofs need only the order/cut development.

25.4 A proposed Lean dependency plan

A practical implementation should proceed in the following dependency order, keeping the formal claims smaller than the whole manuscript at first.

1. **Fixed-alphabet enumeration order.** Represent an enumeration by its ordinal domain and an equivalence onto a fixed small alphabet. Prove prefix-freeness before defining variable-domain first disagreement. Then prove pair-switch coding, cardinality bounds, and the κ^+ stabilization obstruction.
2. **Small injective prefixes.** Prove the extension lemma for a partial injection into a sufficiently large ordinal. Keep the equal cardinality of both complements as an explicit lemma. This part is independent of surreals.
3. **Abstract small-cut alphabets.** Parameterize the avoidance and forced-prefix interpolation lemmas by a linear order with separators for explicitly small option families. Instantiate with the existing sign carrier only afterwards.
4. **Coded support core.** Use a small support, a permutation on it, and a no-fixed-point condition, or use bounded permutations followed by a proved equivalence relation and canonical normalization. Supply a baseline equivalence as an explicit parameter.
5. **Large-family interface.** Represent rows by a single function or relation, retain its universe levels, and prove bounded reflection for small index types. State separately which large index orders possess an ordinal enumeration with small initial segments.
6. **Back-and-forth and the diagonal.** Formalize the set-valued recursion invariant. Only after its smallness obligations are discharged should the abstract core isomorphism and no-coding statements be instantiated.

In a universe-indexed version, a proposed baseline has the shape $\text{Ordinal}.\{u\} \simeq \text{SignSequence}.\{u\}$. Its existence is not inferred here merely because both types live in $\text{Type}(u + 1)$; equal universe level does not imply equipotence. Likewise, the type of *all* functions between these carriers can be formed in the ambient type theory without becoming a small-universe class of set codes. This is exactly the distinction exhibited by [theorem 22.1](#).

25.5 A checklist of invalid shortcuts

The following shortcuts would change the mathematical problem or introduce an unjustified assumption:

- treating **No** as a set in its original universe, or treating all global enumerations as members of an ordinary class;
- using an arbitrary class option family in the surreal cut constructor, or replacing set-indexed interpolation by class-indexed interpolation;
- assuming every class well-order has type `Ord`, or confusing the well-order represented by an enumeration with the lexicographic order of enumerations;

- completing a partial injection without checking complement cardinalities, or using a set of individually given class witnesses without a uniform family relation;
- transferring a proof of equimorphism to isomorphism, or assuming an order embedding preserves ambient suprema;
- attributing the new global core and diagonal theorems to existing Lean declarations merely because the requisite sign-cut primitives are present.

[merge] Source 08’s Lean plan. The declarations of its table exist; source 08 correctly identifies `exists_cut_separator_of_birthday_lt` as the formal landing point of [theorem 5.6](#) and [corollary 5.7](#) ([section 1.5](#)).

26 A Lean formalization plan tied to ProveIt (source 08)

The repository was inspected at commit

ae7c1e6aae3dc58c9c638b35646865ad6c10387d.

The relevant project is `Algebra/SurrealNumbers`. Its README explicitly distinguishes research drafts from machine-checked results and describes an axiom audit [6]. This manuscript uses that distinction rather than treating all repository prose as a formal theorem.

26.1 Inspected declarations

The source `Surreal/Foundations/SignSequence.lean` defines a universe-indexed carrier of the form

```
structure SignSequence : Type (u + 1) where
  birthday : Ordinal.{u}
  signAt : Ordinal.{u} -> SignType
  signAt_ne_zero_iff :
    forall i, Iff (Not (signAt i = 0)) (i < birthday)
```

This display is an ASCII rendering of the inspected declaration, not a new compiled file. Its zero extension uses the ordered values minus, termination, and plus. The source gives the numerical linear order via Mathlib’s lexicographic function `order` and proves `lt_iff`, `ordinalOrderEmbedding`, `birthdays_bounded`, `small_strict_upper_bounds`, `not_small`, and `small_bounded` [7].

Two declarations in `SignSequenceCut.lean` are particularly relevant: `exists_cut_separator_of_birthday_lt` and `small_cut_fillers`. The construction uses full-length binary words, uses their complete lexicographic order, and produces a separator of exactly a prescribed birthday above all option birthdays [8]. This is the formal landing point for [theorem 5.6](#) and [corollary 5.7](#); it is stronger than merely having a dense linear order.

Inspected ingredient	Role in a formalization
<code>SignSequence.lt_iff</code>	Express every comparison by its least differing coordinate
<code>birthdays_bounded</code>	Bound options indexed by a lower-universe small type
<code>not_small</code>	Exclude collapsing the whole carrier into the birthday universe
<code>small_bounded</code>	Represent a fixed birthday fragment by a genuinely small type
<code>exists_cut_separator_of_birthday_lt</code>	Choose a separator outside a small forbidden family by increasing its birthday
<code>small_cut_fillers</code>	Discharge the ordered-cut interface used by the abstract arguments

26.2 Smallness is not the same as having a type

In Lean, `SignSequence.{u}` and `Ordinal.{u}` live in `Type (u + 1)`. They are not small relative to universe `u`. A `Set SignSequence` is a predicate on the whole larger carrier; it need not be a small family of surreals. Thus a formal cut theorem must carry a condition such as `Small.{u} I` on its index type, not merely declare its options to be a `Set`.

The same applies to permutations. A predicate-defined support in the larger universe is not automatically an admissible set support at the smaller level. The orbit should use explicit small support codes or an equivalent `Small.{u}` hypothesis. Otherwise the crucial prefix bound and the intended distinction between skeleton and full enumeration space are lost.

One safe interface keeps an equivalence

```
b : Equiv Ordinal.{u} SignSequence.{u}
```

as an explicit parameter. The present manuscript does not claim that the repository already constructs this equivalence with all the required relative-smallness properties. An implementation must either prove that baseline theorem in its universe framework or retain the baseline as a stated assumption.

26.3 Suggested module boundaries

The following are proposed modules, not existing files or completed formalizations.

1. `WellOrders/Exhaustive.lean`: exhaustive enumerations of a fixed carrier, no-prefix ambiguity, first-disagreement order, and relations versus enumerations.
2. `WellOrders/PairCoding.lean`: binary pair blocks, small-order universality, cardinality, prefix stabilization, and the exact ordinal spectrum.
3. `WellOrders/FiniteTails.lean`: cylinder extrema, adjacency, and finite-condensation blocks.
4. `WellOrders/PrefixExtension.lean`: small partial injections, choice of a larger support bound, complement bijections, and canonical moved-point codes.
5. `WellOrders/SetCuts.lean`: the active-row invariant, the bound on cross-disagreement coordinates, stopping, and strict separation before completion.
6. `WellOrders/BoundedPermutations.lean`: the code carrier, its small-cut interface, and the relative-universe form of the surreal-skeleton theorem.
7. `WellOrders/ClassBinaryCoding.lean`: terminal block codes, ordinal concatenation, and uniform diagonalization in the chosen higher-type representation.

The first three modules can be developed without surreal arithmetic or class metatheory. The next three isolate the delicate smallness and extension arguments. The final module should be delayed until the meaning of a uniform operator on larger function types is fixed.

26.4 Two complementary implementation routes

A *cutoff-first* implementation proves theorems for $\mathbf{No}_{<\kappa}$ and $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$ with explicit regularity and cardinal hypotheses. All carriers are ordinary types and the formal statements closely match [note 7.6](#). This is the simplest route to substantial checked mathematics.

A *universe-relative* implementation abstracts over a large ordered carrier, a lower-universe small-cut operation, and a baseline equivalence. It formalizes the local orbit argument relative to that notion of smallness. Care is needed when proving back-and-forth: an arbitrary well-order of a large type may have initial segments too large for the small-cut operation. One must use an

enumeration whose proper initial segments are small, rather than invoke an unrestricted large-type choice order.

A third option is a deep formalization of a two-sorted class theory or of a specified model $(M, \mathcal{C})$. This is the best route for comparing axiomatic strengths and internal/external semantics, but is not necessary to certify the set-sized combinatorics.

26.5 Trust boundary

The cited repository files were read, not rebuilt in this work. No new Lean module accompanies this manuscript, and no new kernel-checking claim is made. A future implementation should pin the repository and Mathlib revisions, compile in dependency order, reject unfinished proofs, and inspect the resulting axiom dependencies. Finite tests of codes, even exhaustive ones, cannot certify the class interpolation or back-and-forth arguments.

[merge] The four plans agree on a first milestone that is independent of surreal arithmetic: the predecessor or common-prefix comparison of well-orders on a small carrier, prefix-freeness and the successor-cardinal obstruction, then bounded-support prefix completion and small-family interpolation, reusing the existing small-cut interface. This is a proposal recorded for the collection's formalization work; no such module exists.

Part VI

Set-sized strata: topology, cut spectra and block types

[merge] (3 October 2026.) This Part opens the batch-81 addition of six further manuscripts, sources 13–18 (section 1.8); its notation is fixed in section 1.3.1. It prints everything the six sources prove about set-sized enumeration orders: the density and cellularity of every stratum $\mathcal{W}_\alpha(X)$ of every infinite linear alphabet (section 27), conditional incompleteness and global characters (section 28), the cut spectrum of the alphabet $\mathbf{No}_{<\kappa}$ (section 29), the characters, fusion theorem, exact gap spectrum and gap multiplicities of the minimal slice $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ (section 30), the coding lengths, cube spectra and density of the longer strata $\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$ (section 31), and the six sources' set-sized baselines and warnings (section 32). Section 33 records which questions of this report and of the reals report the Part answers.

The bases are chosen by generality, as in Parts I–V: source 16 for density, cellularity and incompleteness (every infinite alphabet, every stratum), source 17 for the alphabet and the minimal slice (every infinite cardinal cutoff, with no hypothesis on $2^{<\kappa}$), source 15 for the longer strata (the only source treating them). Source 18, written before sources 13–17 had arrived and without knowledge of them, proves most of the minimal-slice results a third time; its density, characters and interpolation proofs differ in detail and are printed as routes, its gap-spectrum proof repeats source 17's and is recorded in notes, and its general permutation principle and exact gap multiplicities are new and printed in full. The whole Part is set theory in ZFC; no class is involved.

Throughout the Part, κ is an infinite cardinal used as a birthday cutoff, $\mu = 2^{<\kappa} = |\mathbf{No}_{<\kappa}|$, $r = \text{cf } \kappa$ and $q = \text{cf } \mu$; a general alphabet X is an infinite linearly ordered set of cardinality μ . Source 16 wrote κ for $|X|$ and θ for the cutoff, and source 18 wrote ν, τ for r, q ; their text is printed in the convention above (section 1.3.1).

27 Density and cellularity at every fixed type (source 16)

[merge] Source 16's Section 5, printed whole. Its κ (the cardinality of X) is printed μ , its cutoff θ is printed κ , and its d, c are printed d, c , cell (section 1.3.1); its equations keep their numbers as (S16.n). Source 18 proves the same theorem independently, with a different lemma and a different lower-bound family (section 27.1); the reals report proves the case $X = \mathbb{R}$ (note 27.5).

For a set linear order L with its *order topology*, let $d(L)$ be the least cardinality of a dense subset. Let $\text{cell}(L)$ be the supremum of the cardinalities of families of pairwise disjoint nonempty open sets. Always $\text{cell}(L) \leq d(L) \leq |L|$. A prefix cylinder is the collection of enumerations extending one fixed injective prefix. Cylinders are convex, but need not themselves be open. Our lower bounds use actual open intervals lying inside cylinders; our upper bounds use a family that meets every nonempty open set.

Lemma 27.1 (No jump between distinct short-prefix branches). *If $e < f$ in $\mathcal{W}_\alpha(X)$ first differ at a coordinate $\delta < \mu$, then they are not adjacent.*

Proof. Write $p = e \restriction \delta = f \restriction \delta$ and $a = e(\delta) < b = f(\delta)$. If there is no point between e and f , then e must be the maximum of its $p \frown a$ cylinder, and f the minimum of its $p \frown b$ cylinder. Any ascent in the remaining suffix of e can be swapped to produce a larger member of its cylinder. Thus that suffix is numerically strictly decreasing. Similarly, the suffix of f must be strictly increasing. Their common remaining label set is $X \setminus (\text{ran}(p) \cup \{a, b\})$, of cardinality μ . Its numerical order would be both a well-order and a reverse well-order. An infinite linear order cannot have both properties. This contradiction proves the claim. $\square$

Theorem 27.2 (Exact density and cellularity). *Let $|X| = \mu$ be infinite and $|\alpha| = \mu$. Then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha = \mu, \mu + 1, \text{ or } \mu + 2, \\ 2^\mu, & \text{all other } \alpha. \end{cases} \quad (\text{S16.3})$$

Moreover $d(\mathcal{W}(X)) = \text{cell}(\mathcal{W}(X)) = 2^\mu$. No regularity assumption is needed.

Proof. For any $\delta < \mu$ there are $\mu^{|\delta|}$ injective prefixes of length δ , and each extends to an admissible enumeration. One way to see the lower count is to choose, for each coordinate, its label in a separate μ -sized cell of a partition of X ; the choices are automatically injective. Every such cylinder contains at least three points and hence a nonempty ambient open interval. Distinct cylinders at one level are disjoint and convex. Consequently

$$\text{cell}(\mathcal{W}_\alpha(X)) \geq \sup_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Suppose $\alpha = \mu + n$, $n \leq 2$. Group enumerations by their common prefix of length μ . A nonempty group has exactly $n!$ points, coming from the permutations of the omitted n labels. Thus every group has at most two points. By Lemma 27.1, distinct groups have dense quotient order: if two groups were consecutive, the last point of the first and first point of the second would be adjacent and differ before μ .

It follows that every nonempty open set contains a cylinder specified before μ , except possibly a singleton consisting of an isolated global endpoint. Indeed, select a group strictly between the relevant boundary groups, choose a member of it, and bound its first disagreement with the boundaries by a coordinate below μ . The resulting prefix cylinder lies inside the open set. Select

one completion for each injective prefix of length less than μ and add any isolated global endpoints. This gives a dense set of size at most

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu}.$$

Together with the lower bound this proves the first line of (S16.3), including singular μ .

If $\alpha = \mu + n$ for finite $n \geq 3$, fix $F \subseteq X$ of size n . There are 2^μ bijections $p : \mu \rightarrow X \setminus F$. The completion cylinder for each is a finite convex chain of $n! \geq 6$ points. An interior point has both immediate neighbors in that chain; the interval between those neighbors is its singleton, an open set in the entire order. Different prefixes give 2^μ disjoint open singletons.

In every remaining case $\alpha \geq \mu + \omega$. Let τ be the order type of the interval $[\mu, \alpha)$, so $\alpha = \mu + \tau$ and $|\tau|$ is infinite. Choose $Y \subseteq X$ with $|Y| = |\tau|$ and $|X \setminus Y| = \mu$. Each of the 2^μ bijections $p : \mu \rightarrow X \setminus Y$ gives a convex cylinder of completions with residual order $\mathcal{W}_\tau(Y)$. Every such cylinder has at least three points, so supplies a nonempty ambient open interval. These intervals are pairwise disjoint. The cardinal upper bound $|\mathcal{W}_\alpha(X)| \leq 2^\mu$ finishes the proof for all strata.

For $\mathcal{W}(X)$ use the same construction with a fixed residual set of three labels. The resulting six-point cylinders are convex even when all enumeration lengths are admitted. Thus $\text{cell}(\mathcal{W}(X)) \geq 2^\mu$, and the cardinality bound gives equality throughout. $\square$

Remark 27.3 (Why two terminal labels differ from three). There can be 2^μ adjacent pairs in $\mathcal{W}_{\mu+2}(X)$ while its density is only $\mu^{<\mu}$. An adjacent pair does not contain an open interval. Three or more terminal labels give a permutation block with an interior point, and that point is isolated. This is why the topological transition occurs at $\mu + 3$, rather than $\mu + 1$ or $\mu + 2$. If $\mu^{<\mu} = 2^\mu$, the two displayed values coincide; the structural distinction still holds.

Corollary 27.4 (Birthday-stage topology). *For every infinite cardinal κ , put $\mu = 2^{<\kappa}$. Formula (S16.3) applies to $\mathbf{No}_{<\kappa}$, whose cardinality is μ . If κ is regular and $2^{<\kappa} = \kappa$, the first three strata have density and cellularity κ , whereas all later strata and the full order have density and cellularity 2^κ .*

Proof. Only the last simplification needs comment. Regularity and $2^{<\kappa} = \kappa$ imply $\kappa^{<\kappa} = \kappa$: for $\lambda < \kappa$, every function $\lambda \rightarrow \kappa$ has bounded range, and for $\rho < \kappa$ one has $\rho^\lambda \leq 2^{\rho \cdot \lambda} \leq \kappa$. Taking the union of κ bounds proves the assertion. $\square$

Note 27.5 ([merge]; the reals report's case). For $X = \mathbb{R}$, the reals report proves $d(\mathcal{W}_\alpha) = \text{cell}(\mathcal{W}_\alpha) = 2^{<\mathfrak{c}}$ for $\alpha \in \{\mathfrak{c}, \mathfrak{c} + 1, \mathfrak{c} + 2\}$ and $2^\mathfrak{c}$ for $\mathfrak{c} + 3 \leq \alpha < \mathfrak{c}^+$ (lwo:prop:all-strata-density), with the adjacent pairs at $\mathfrak{c} + 2$ and the isolated points from $\mathfrak{c} + 3$ on (lwo:prop:finite-topology) and the minimal slice in lwo:prop:fixed-topology. Since $\mathfrak{c}^\lambda = 2^\lambda$ for infinite $\lambda < \mathfrak{c}$, one has $\mathfrak{c}^{<\mathfrak{c}} = 2^{<\mathfrak{c}}$, and theorem 27.2 with $X = \mathbb{R}$ is exactly that result, by the same two mechanisms (short-prefix cylinders, and the finite terminal permutation block). Source 16 does not cite these statements; it cites the reals report only for the set baseline (note 32.1). New in source 16 is the generality: every infinite linear alphabet, of any cardinality including singular ones and $\aleph_0$, with no density or saturation assumption on X .

Note 27.6 ([merge]; the birthday-cutoff instances, and why three formulas agree). For $X = \mathbf{No}_{<\kappa}$, corollary 27.4 gives $d = \text{cell} = \mu^{<\mu}$ on $\mathcal{W}_\mu, \mathcal{W}_{\mu+1}, \mathcal{W}_{\mu+2}$ and 2^μ on every longer stratum. Three other sources compute instances: source 17 for the minimal slice ($d = w = \text{cell} = \mu^{<\mu}$, theorem 30.2), source 18 for every stratum of every alphabet (theorems 27.8 and 27.9, printed next), and source 15 for the block types $\mathcal{W}_{\mu \cdot \gamma + n}$, with the value $2^{<\mu}$ when $\rho_\kappa = \mu$, $\gamma = 1$ and $n \leq 2$, and 2^μ otherwise (corollary 31.7). The formulas agree. If $\mu > \kappa$ then $\mu \leq 2^\kappa$, so for $\lambda < \mu$ one has $\mu^\lambda \leq 2^{\kappa \cdot \lambda} = 2^{\max(\kappa, \lambda)} \leq 2^{<\mu}$, whence $\mu^{<\mu} = 2^{<\mu}$. If $\mu = \kappa$ is regular then $2^{<\kappa} = \kappa$ and $\kappa^{<\kappa} = \kappa$ (proof of

corollary 27.4). If $\mu = \kappa$ is singular, κ is a strong limit, $\rho_\kappa = \kappa \cdot \kappa$ falls under source 15's second case, and $\kappa^{<\kappa} = 2^\kappa$ (corollary 30.26). In every case the four sources give the same cardinal.

27.1 Source 18's route: exact cardinal invariants of bounded enumeration orders

[merge] Source 18's Section 4 up to its residual-alphabet subsection, printed whole as a second route to theorem 27.2. Source 18 presents these theorems as advances; within batch 81 source 16 proves them too, and source 18 could not know it (its pin precedes the arrival of sources 13–17). Source 18's status table says that its proof review “corrected the arbitrary-alphabet density construction to include possible isolated global endpoints”; source 16's proof treats the same endpoints (“except possibly a singleton consisting of an isolated global endpoint”). Its remaining subsections are printed in section 30.8, its residual-alphabet lemmas in notes 29.8 and 29.9.

Throughout this section, X is an infinite linearly ordered set of cardinality μ , identified with its initial ordinal when it is used as an index. If $|\alpha| = \mu$, write

$$\mathcal{W}_\alpha(X) = \{e : \alpha \longrightarrow X : e \text{ is bijective}\},$$

ordered by the first unequal coordinate. A proper-prefix comparison never occurs between two exhaustive enumerations of the same carrier. We use $d(L)$ for the least cardinality of a topologically dense subset of the order topology, and $\text{cell}(L)$ for the supremum of the cardinalities of pairwise disjoint families of nonempty open sets. The lower-bound families constructed below attain the asserted cellularities, so no convention concerning whether the supremum is attained affects the answers. All assertions in this section are theorems of ZFC.

27.1.1 Prefix cylinders and the minimal enumeration type

For an injective prefix $p : \delta \rightarrow X$, $\delta \leq \alpha$, its cylinder is

$$C_p = \{e \in \mathcal{W}_\alpha(X) : e \restriction \delta = p\}.$$

Every cylinder is convex. If $\delta < \mu$, the cylinder is nonempty: both the unused alphabet and the remaining index set have cardinality μ , so a bijection between them completes the prefix. Such a cylinder has at least three elements, obtained, for example, by permuting three unused labels in a fixed completion. It therefore contains a nonempty open interval of the ambient enumeration order. The entire cylinder need not be open; the proof uses only the open interval it contains.

Lemma 27.7 (The dense order of cofinite prefixes). *For $n < \omega$, let*

$$Q_n(X) = \{p : \mu \rightarrow X : p \text{ is injective and } |X \setminus \text{ran } p| = n\}.$$

With first-difference comparison, $Q_n(X)$ is a dense linear order. Consequently $\mathcal{W}_\mu(X) = Q_0(X)$ is dense.

Proof. Take $p < p'$ and let $\delta < \mu$ be their first disagreement. If there are $\delta < \beta < \gamma < \mu$ with $p(\beta) < p(\gamma)$, interchanging these two entries gives $p < s < p'$, with $s \in Q_n(X)$. Similarly, a descent $p'(\beta) > p'(\gamma)$ after δ gives $p < s < p'$ by an interchange in p' . If neither operation is available, the tail of p after δ is strictly decreasing and the tail of p' after δ is strictly increasing. Their ranges have an infinite common subset: they omit from X only their common prefix, of size less than μ , two first-disagreement labels, and two finite sets of omitted labels. On that common subset the numerical order would be both a well-order and a reverse well-order. This is impossible for an infinite set. Indeed, an infinite well-order contains an initial suborder of type ω , which has no largest element. Thus one of the interchanges is available. $\square$

Theorem 27.8 (Exact density and cellularity at the minimal type). *For every infinite linearly ordered alphabet X of cardinality μ ,*

$$d(\mathcal{W}_\mu(X)) = \text{cell}(\mathcal{W}_\mu(X)) = \mu^{<\mu}.$$

No regularity assumption on μ , and no density assumption on X , is needed.

Proof. There are $\mu^{<\mu}$ injective prefixes of length less than μ . For the upper bound, choose one completion for each such prefix. Given $e < f$, density supplies g strictly between them. Choose $\delta < \mu$ larger than both first-disagreement coordinates of (e, g) and (g, f) . Then $C_{g|\delta} \subseteq (e, f)$, so the selected completions are dense. Since a dense linear order with at least two points has no isolated endpoint, meeting all nonempty open intervals also gives topological density. This proves $d \leq \mu^{<\mu}$, and a dense set meets every member of a disjoint nonempty open family, giving $c \leq d$.

For the lower bound, choose distinct labels $a_\delta \in X$ for $\delta < \mu$. For each δ , take every injective prefix of length $1 + \delta$ whose first label is a_δ . There are $\mu^{|\delta|}$ such prefixes. To justify this count for infinite δ , functions $\delta \rightarrow \mu$ can be encoded as injective maps $\delta \rightarrow \delta \times \mu \cong \mu$ by recording the coordinate alongside its value; the reverse inequality is immediate. The finite cases are equally direct. Prefixes in different chosen levels have different first labels, while two distinct prefixes within a level have equal lengths and disagree. Their cylinders are therefore pairwise disjoint. Choose one nonempty open interval in each cylinder. The resulting disjoint family has cardinality

$$\sum_{\delta < \mu} \mu^{|\delta|} = \mu^{<\mu},$$

because $1 + \delta < \mu$ for every $\delta < \mu$. This proves the lower bound and also proves that the stated cellularity is attained. $\square$

27.1.2 A sharp transition at the third terminal coordinate

Theorem 27.9 (Density and cellularity for every fixed type). *If $|X| = \mu$ is infinite and $\mu \leq \alpha < \mu^+$, then*

$$d(\mathcal{W}_\alpha(X)) = \text{cell}(\mathcal{W}_\alpha(X)) = \begin{cases} \mu^{<\mu}, & \alpha \in \{\mu, \mu + 1, \mu + 2\}, \\ 2^\mu, & \mu + 3 \leq \alpha < \mu^+. \end{cases}$$

These formulas are valid for arbitrary numerical orders on X .

Proof. First take $\alpha = \mu + n$ with $n \leq 2$. Restriction to the first μ coordinates exhibits $\mathcal{W}_{\mu+n}(X)$ as a lexicographic sum over $Q_n(X)$ of finite fibers of size $n!$; the omitted n labels are the only possible final labels. Each fiber has at most two points. By Lemma 27.7, the quotient $Q_n(X)$ is dense. Thus every nonempty open interval in $\mathcal{W}_{\mu+n}(X)$ contains a point whose μ -prefix lies strictly between the prefixes of its two bounds: if the bounds had the same prefix, their fiber of at most two points would have empty open interval. A prefix of this intermediate point of length less than μ , taken beyond both relevant first disagreements, has its entire cylinder inside the interval. Choose one completion of every prefix of length less than μ , and also include the global least and greatest points whenever they exist. The resulting family is topologically dense: it meets every nonempty open interval, while the added endpoints handle a possible isolated endpoint in a two-point terminal fiber. Adding at most two points does not change its cardinality bound $\mu^{<\mu}$. The antichain of cylinders from the preceding proof gives the matching lower bound. This argument explicitly deals with the possible jumps when $n = 2$; it does not assume that every cylinder is open, or that every interval contains a point separated from both bounds merely by choosing an arbitrary intermediate point.

Now suppose $\alpha \geq \mu + 3$, and write $\alpha = \mu + \beta$, where β is the order type of its tail after the first μ positions. Choose $T \subseteq X$ with $|T| = |\beta|$ and $|X \setminus T| = \mu$. If $|\beta| = \mu$, use a partition of X into two μ -element sets; otherwise remove a set of size $|\beta|$. There are 2^μ bijections $p : \mu \rightarrow X \setminus T$. This standard count follows either from the cardinality theorem for well-order enumerations or directly by orienting μ disjoint pairs of labels. For each such p , the cylinder C_p consists of all continuations that biject β onto T . Since $\beta \geq 3$, it contains at least three points and hence a nonempty open interval. The cylinders are pairwise disjoint because the prefixes are distinct and have the same length. We obtain 2^μ disjoint nonempty open intervals. On the other hand, $|\mathcal{W}_\alpha(X)| \leq \mu^\mu = 2^\mu$, so no density or cellularity can exceed 2^μ . $\square$

Remark 27.10. For $\mu^{<\mu} < 2^\mu$, the values for $\mu + 2$ and $\mu + 3$ differ. Thus merely adding one terminal coordinate can change both density and cellularity. For singular strong-limit μ , both expressions are already 2^μ , so the geometric distinction remains but the cardinal invariants coincide. The formula is a new consequence beyond the baseline's finite-residual adjacency classification; it is not a new claim about the already-established ordinal embedding spectrum.

28 Countable incompleteness and global characters (source 16)

[merge] Source 16's Section 6 up to its subsection on regular saturated stages, which is printed in [section 30.4](#).

Conditional Dedekind completeness means that every nonempty bounded-above set has a supremum, equivalently every nonempty bounded-below set has an infimum. It should not be confused with filling all small strict cuts, with order density, or with the completeness of a separately adjoined Dedekind completion.

Theorem 28.1 (Every infinite stratum is incomplete). *For every infinite linearly ordered set X and every ordinal α with $|\alpha| = |X|$, $\mathcal{W}_\alpha(X)$ is not conditionally Dedekind complete. A countable bounded monotone family witnesses this failure. The same conclusion holds for $\mathcal{W}(X)$.*

Proof. Every infinite linear order contains an increasing or decreasing ω -sequence. For completeness, choose a countable set of distinct points and color each pair of its indices according to the numerical comparison; the infinite Ramsey theorem gives such a sequence. First assume $a_0 < a_1 < a_2 < \dots$ in X , and put $p(n) = a_{2n}$.

For each $n \geq 0$ choose U_n extending the finite prefix

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n+1}),$$

and for $n \geq 1$ choose L_n extending

$$(a_0, a_2, \dots, a_{2n-2}, a_{2n-1}).$$

All these prefixes extend to the prescribed type α . First disagreement gives $L_n < L_{n+1}$, $U_{n+1} < U_n$, and $L_n < U_m$ for all n, m . Hence (U_n) is bounded below.

Let g be any lower bound. If it first differs from the infinite prefix p at a finite coordinate j , it must differ downward: an upward difference would give $g > U_{j+1}$. Consequently $g < L_n$ for every $n > j$, so it is not the greatest lower bound.

Otherwise g agrees with all of p . Its remaining suffix contains all odd labels a_{2n+1} . That suffix cannot be numerically strictly decreasing: the positions of these increasing labels would form an infinite descending sequence of ordinals. It therefore has an ascent. Swap the two labels of such an

ascent. The new enumeration has the same type, is strictly above g , and still agrees with p at every finite coordinate. It remains below every U_n . Again g is not greatest. If no completion of p with the specified type exists, this second case simply never occurs.

Thus (U_n) has no infimum. Reversing the numerical alphabet treats a decreasing ω -sequence and yields the dual failure of a supremum. The argument for the full order is identical: swapping an ascent preserves whatever ordinal type the candidate bound happens to have. $\square$

Proposition 28.2 (Global characters from the first label). *If X has no greatest element, then*

$$\text{cf}(\mathcal{W}_\alpha(X)) = \text{cf}(\mathcal{W}(X)) = \text{cf}(X)$$

for every admissible infinite α . There is a dual statement for coinitality when X has no least element. In particular, for every infinite cardinal cutoff κ , all the fixed strata and the full order on $\mathbf{No}_{<\kappa}$ have global cofinality and coinitality $\text{cf}(\kappa)$.

Proof. Complete a cofinal family of possible first labels to enumerations. It is cofinal in the lexicographic order. Conversely, the first labels of a cofinal family of enumerations must be cofinal in X , or a larger first label would put an enumeration above the entire family. For $\mathbf{No}_{<\kappa}$, the ordinals below κ are numerically cofinal and the negatives coinital. Any family with bounded birthdays is bounded numerically, so the least possible size is $\text{cf}(\kappa)$. $\square$

Note 28.3 ([merge]; global characters in sources 15, 17 and 18). For the minimal slice of $\mathbf{No}_{<\kappa}$, the global part of [proposition 28.2](#) ($\text{cf} = \text{ci} = r$) is also proved by source 17 ([theorem 30.2](#), with the same value for every cylinder in [lemma 30.1](#)), by source 15 ([note 30.7](#)) and by source 18 ([theorem 30.24](#), also for $\mathcal{W}_{\mu+n}$, and [note 30.13](#) for cylinders), each by the same first-label projection. Source 16's proposition is the general form: it holds for every stratum and for the full order $\mathcal{W}(X)$ of any alphabet without a greatest element. Source 16's subsection "Regular saturated stages", the last part of its Section 6, is printed in [section 30.4](#) with the other gap spectra.

29 The cut spectrum of the alphabet $\mathbf{No}_{<\kappa}$ (source 17)

[merge] Source 17's Section 4, printed whole. Its $\Sigma(L)$ is printed $\text{GapChar}(L)$ (the reals report's symbol), and its displayed equations keep their numbers as (S17. n). At regular uncountable κ its main theorem is already in the repository ([note 29.7](#)). Sources 15 and 18 prove the alphabet lemmas again; their statements follow in [notes 29.8](#) to [29.10](#).

For the remainder of the set-sized analysis, fix an infinite cardinal κ , regarded as an initial ordinal, and write

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad q = \text{cf } \mu. \quad (\text{S17.3})$$

Here $\mathbf{No}_{<\kappa}$ consists of all sign sequences of ordinal length below κ , ordered by first difference with $- < \text{termination} < +$. We write Reg for the infinite regular cardinals. No arithmetic closure of X is assumed. In particular, no field hypothesis is being inferred from the mere fact that κ is a cardinal.

Definition 29.1 (Characters and genuine gaps). For a point x of a dense order L without endpoints, its left and right characters are $\text{cf}(L_{<x})$ and $\text{ci}(L_{>x})$. A *gap* is a partition $L = A \sqcup B$ into nonempty initial and final parts, with $A < B$, such that A has no greatest element and B no least element. Its character is $(\text{cf } A, \text{ci } B)$; $\text{GapChar}(L)$ denotes the set of these pairs. Thus a split immediately before or after an actual point is not a gap. Cofinalities of orders with a greatest element would be 1, but do not occur in $\text{GapChar}(L)$.

29.1 Size, intervals, and deletion

Lemma 29.2. *The order X has cardinality μ . Every nonempty open interval and every nonempty open ray in X has cardinality μ . Consequently, deletion of fewer than μ labels leaves an order-dense, cofinal and coinital suborder, with μ labels still in each nonempty interval and ray.*

Proof. There are $2^{|\alpha|}$ sign sequences of length α , so $|X| = \sum_{\alpha < \kappa} 2^{|\alpha|} = 2^{<\kappa}$; the supremum on the right is at least κ and absorbs the number of summands. Given $a < b$, choose a sign sequence z strictly between them, of length below κ . A sufficiently long extension of z fixes its comparisons with a, b before any subsequent signs. Hence an entire sign cone lies inside (a, b) . If its fixed prefix has length $\delta < \kappa$, then all extensions of lengths $\delta + \gamma$, $\gamma < \kappa$, remain below κ : an infinite initial ordinal is closed under sums of two smaller ordinals. This cone has μ elements. The same construction applies to rays. Removing fewer than μ elements cannot empty a set of cardinality μ and leaves its cardinality unchanged, even when μ is singular. $\square$

Lemma 29.3. *The cofinality and coinitality of X are r , and every point of X has left and right character r . The same assertions hold in $Y = X \setminus Z$ whenever $|Z| < \mu$.*

Proof. The ordinal surreals below κ are cofinal, and a cofinal sequence of ordinals of length r gives an upper bound r . A family of fewer than r sign sequences has bounded birthdays below κ and is bounded above and below by ordinal surreals, proving the reverse bound.

At x , the sequences

$$x \frown - \frown +^\gamma \quad \text{and} \quad x \frown + \frown -^\gamma \quad (\gamma < \kappa)$$

are respectively cofinal below and coinital above x ; a cofinal choice of γ 's of length r suffices. Conversely, fewer than r points have bounded birthdays. A longer sign extension of x lies between x and all the points on the appropriate side. Hence the character is not smaller than r .

For Y , replace each member of a cofinal or coinital sequence by a remaining label in an appropriate interval. Such labels exist by Lemma 29.2. Any smaller sequence cofinal in Y would also be cofinal in the relevant interval or ray of X , by density. $\square$

29.2 The alphabet's own gaps

The alphabet is not Dedekind complete. Even at a regular uncountable cutoff, its gaps need not have two equal characters. Distinguishing this fact from its uniform *point* characters is essential.

Theorem 29.4 (Cut spectrum of the birthday alphabet).

$$\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

If $|Z| < \mu$, the same formula holds for $X \setminus Z$.

Proof. Let (A, B) be a gap of X . Define a sign sequence s of length κ by set recursion: at stage α , the already defined prefix $s \upharpoonright \alpha$ is in X , so put $s(\alpha) = +$ if that prefix belongs to A , and $s(\alpha) = -$ if it belongs to B . Put

$$I_+ = \{\alpha < \kappa : s(\alpha) = +\}, \quad I_- = \{\alpha < \kappa : s(\alpha) = -\}.$$

The nodes $s \upharpoonright \alpha$ for $\alpha \in I_+$ increase and are cofinal in A ; the nodes for $\alpha \in I_-$ decrease and are coinital in B . To verify cofinality, compare any $x \in A$ with the path s . If x is a path prefix it is one of the displayed lower nodes. Otherwise the first disagreement cannot place x to the right of

the chosen path: at a minus choice the corresponding node is in B . It therefore places x below a lower node. The dual argument handles B .

Neither index set has a greatest element, since the cut is a gap. At least one is unbounded in κ and has cofinality r . An unbounded other side also has cofinality r ; a bounded other side has an infinite regular cofinality $\theta < \kappa$. This proves the upper inclusion. Conversely, the path $+^\theta \frown -^\kappa$, with the evident length κ interpretation, has lower and upper index cofinalities (θ, r) . Reversing signs gives (r, θ) . A path with both signs cofinal in κ gives (r, r) . In each case the induced partition has no endpoint, as the associated node sequences have no last member.

Now let $Y = X \setminus Z$ be as stated. Density of Y implies that a gap of Y corresponds either to a gap of X or to a single removed point of X : there cannot be two X -points between the two sides, because a Y -point would lie between them. At an X -gap, density preserves both cofinalities by choosing intermediate Y -points. At a removed point both characters are r by [Lemma 29.3](#). Conversely, every X -gap survives restriction to Y , with the same characters. This proves the deletion assertion. $\square$

Example 29.5 (A warning about equal-character guesses). For $\kappa > \omega$, the path which first has ω plus signs and then only minus signs determines an (ω, r) gap. It lies at a boundary of a limit sign cone. The cone itself contains many surreal numbers; none is its missing boundary point. Thus uniform point character r does not mean that every gap has character (r, r) .

Lemma 29.6. *One has $\mu \geq \kappa$ and $q \geq r$.*

Proof. The first assertion follows from the count in [Lemma 29.2](#). If the increasing cardinal sequence 2^ν , $\nu < \kappa$, never reaches its supremum μ , a purported cofinal family in μ of size below r is bounded by some 2^ν : choose an index bound for each of its members and then use $\text{cf } \kappa = r$ to bound all those indices. Thus $\text{cf } \mu \geq r$. If the supremum is attained, the powers are constantly μ for all sufficiently large cardinals $\nu < \kappa$. König's inequality $\text{cf}(2^\nu) > \nu$ then gives $\text{cf } \mu \geq \kappa$. For completeness, that inequality follows from $\rho^{\text{cf } \rho} > \rho$ for infinite cardinals ρ , applied to $\rho = 2^\nu$: if $\text{cf } \rho \leq \nu$, then $\rho^{\text{cf } \rho} \leq (2^\nu)^\nu = 2^\nu$, a contradiction. $\square$

Note 29.7 ([merge]; the repository's case of [theorem 29.4](#)). At a regular uncountable cutoff κ , [theorem 29.4](#) is the case $M = N$ of the cutoff gap decomposition of the report `surreal-fields-a-cross-universes` (`univ:gs:lem:boundary`), quoted in `birthday-cutoffs-and-hereditary-sets` (`hset:rem:cutoffgaps`): with no fresh signs, the gap pairs of $\mathbf{No}_{<\kappa}$ are (θ, κ) and (κ, θ) for every infinite regular $\theta \leq \kappa$, realized by the same sign paths $+^\theta \frown -^\kappa$. Source 17 does not cite these reports. New in source 17 are the singular cutoffs, where the bounded side ranges over the regular $\theta < \kappa$ while the unbounded side has character $r = \text{cf } \kappa < \kappa$; the countable cutoff $\kappa = \omega$, where only (ω, ω) occurs; and the deletion clause.

Note 29.8 (Source 18, Lemmas 4.5 and 5.3, and source 15, Lemma 9.1). Source 18's Lemma 4.5 ("Large intervals and unchanged cut sides after deletion"): *every nonempty numerical open interval in $\mathbf{No}_{<\kappa}$ has cardinality μ ; if $U \subseteq \mathbf{No}_{<\kappa}$ has cardinality less than μ , then $\mathbf{No}_{<\kappa} \setminus U$ meets every such interval; in particular (1) $\text{cf}(\mathbf{No}_{<\kappa}) = \text{ci}(\mathbf{No}_{<\kappa}) = r$, (2) immediately below or above any point the cofinality or coinitality is r , and (3) every nonprincipal side of a cut retains its cofinality or coinitality after deletion of U , as do the two open sides at a point, even a deleted one* (an endpoint retained in the residual alphabet still gives a principal side of character 1). Its Lemma 5.3 ("Small deletions preserve order density") repeats this for $R = X \setminus U$ and adds that *every proper gap of R has character in $\text{GapChar}(X) \cup \{(r, r)\}$* . Source 15's Lemma 9.1 ("Full-size numerical intervals"): *every nonempty open interval of $\mathbf{No}_{<\kappa}$, and every nonempty open ray, contains a sign-prefix cone order-isomorphic to $\mathbf{No}_{<\kappa}$; in particular each has cardinality μ* . These are [lemmas 29.2](#) and [29.3](#) and the deletion clause of [theorem 29.4](#), by the same sign-cone argument. Source 15 makes the

cone explicit in three cases (neither endpoint a prefix of the other: take $a \frown +$; a a proper prefix of b : take $b \frown -$; b a proper prefix of a : take $a \frown +$), and source 18 obtains the cone from the birthday-controlled separator theorem (section 5.2) and adds an explicit approach to a point from below by $x \frown - \frown +^\gamma$, γ cofinal in κ , and from above by the sign-reversed words.

Note 29.9 (Source 15, Lemma 9.3, and source 18, Lemma 4.6). Both sources state more than lemma 29.6: $q \geq r$; if the supremum $2^{<\kappa}$ is not attained then $q = r$, and if it is attained, $2^\lambda = \mu$ for some cardinal $\lambda < \kappa$, then $q \geq \kappa$. The proofs are source 17's: cofinal exponents give $q \leq r$ in the unattained case, a shorter cofinal family would have bounded exponent witnesses, and in the attained case König's inequality $\text{cf}(2^\lambda) > \lambda$ for all λ between the attaining exponent and κ gives $q \geq \kappa$.

Note 29.10 (Source 18, Lemma 5.2, "Alphabet gap spectrum"). For $X = \mathbf{No}_{<\kappa}$, $\text{GapChar}(X) = \{(r, r)\} \cup \{(\theta, r), (r, \theta) : \theta < \kappa \text{ an infinite regular cardinal}\}$; every point has left and right character r , and the global cofinality and coinitality are also r . This is theorem 29.4 with lemma 29.3, by the same recursion $s(\alpha) = +$ if and only if $s \restriction \alpha \in A$. Source 18 adds the warning that the lemma "corrects a tempting but false simplification: unfilled alphabet cuts need not all have character (r, r) "; at $\kappa = \omega_1$ the cut below all positive infinite elements, represented by $+^\omega -^{\omega_1}$, has character (ω, ω_1) , and "the asymmetric alphabet gaps will, however, already be covered by the longer-prefix construction in P ". This is the point of example 29.5.

30 The minimal slice $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ (source 17; routes of sources 15, 16 and 18)

[merge] Source 17's Sections 5 and 6 are printed in their order, with two insertions: source 18's general permutation principle (section 30.3), stated before source 17's exact gap spectrum because that theorem and source 16's are instances of it, and source 18's multiplicities (section 30.5). The second and third routes to the characters and the saturation threshold (sources 15 and 18) follow source 17's saturation corollary. Source 17 writes $[p]$ for the cylinder of an injective prefix p ; sources 16 and 18 write C_p for the same set. All three call a cylinder meeting both sides of a cut *straddling*.

30.1 Exact local and topological invariants of the minimal slice (source 17)

Put $P = \mathcal{W}_\mu(X)$. For an injective prefix $p : \alpha \rightarrow X$ with $\alpha < \mu$, let

$$[p] = \{f \in P : f \restriction \alpha = p\}, \quad Y_p = X \setminus \text{ran } p.$$

The set Y_p has cardinality μ . Every such prefix has a completion: the remaining ordinal interval $[\alpha, \mu)$ has order type μ , so a bijection from it to Y_p suffices. This elementary extension fact would be false for arbitrary prescribed prefixes of length μ .

Lemma 30.1 (The cylinder base). *Every $[p]$ is nonempty, convex, and clopen in the order topology of P . The cylinders form a base. No cylinder has an endpoint. Moreover,*

$$\text{cf}[p] = \text{ci}[p] = r.$$

Proof. Convexity is the first-difference property. Given $f \in [p]$, change its entry at coordinate α to a fresh smaller or larger label and then complete. Lemma 29.2 provides such labels after deletion of $f[\alpha]$. This produces bounds on either side of f within $[p]$, proving openness and absence of

endpoints. All length- α cylinders partition P , so the complement is also open. For $a < f < b$, a prefix of f extending beyond its first differences with a, b has its entire cylinder inside (a, b) . This proves the base assertion.

The first unused coordinate maps the cylinder onto Y_p in lexicographic blocks. Choosing one completion for cofinally many first unused labels shows $\text{cf}[p] \leq \text{cf } Y_p = r$. Conversely, the first unused entries of a cofinal family in $[p]$ must be cofinal in Y_p , since otherwise a larger entry starts a cylinder above the entire family. Cointinality is dual. $\square$

Theorem 30.2 (Characters and size invariants). *Every point $f \in P$ has character (q, q) , and*

$$\text{cf } P = \text{ci } P = r, \quad \text{d}(P) = \text{w}(P) = \text{cell}(P) = \mu^{<\mu}. \quad (\text{S17.4})$$

Here d is density, w is topological weight, and cell is cellularity as the supremum of cardinalities of pairwise disjoint families of nonempty open sets. In particular, P is zero-dimensional. The formula for cell does not assert that its supremum is always attained by one family.

Proof. For each coordinate $\delta < \mu$, choose $f_\delta^- < f < f_\delta^+$ whose first difference from f is exactly δ . Such choices are possible by the preceding lemma. Along a cofinal set of coordinates of order type q , the lower choices are increasing and cofinal below f , and the upper choices are decreasing and cointial above it. Conversely, for a family of fewer than q elements below f , the first-difference coordinates are bounded below μ . A lower approximation differing later lies above them all. Thus neither point character is below q . The global cofinalities are [Lemma 30.1](#) at the root.

The number of prefixes is at most $\sum_{\alpha < \mu} \mu^{|\alpha|} = \mu^{<\mu}$, giving that upper bound for weight, density, and cellularity. At each fixed length α , the cylinders are disjoint nonempty open sets. For $1 \leq \lambda < \mu$, the number of injective λ -words over X is μ^λ : a fixed injection $j : \lambda \times X \rightarrow X$ sends any function h to the injection $\xi \mapsto j(\xi, h(\xi))$. Hence both density and cellularity are at least μ^λ for every such λ . Taking suprema gives the reverse bound. Finally $d \leq w$ and $c \leq w$ complete the equalities. The clopen base gives zero-dimensionality. $\square$

Remark 30.3. The point character is controlled by $\text{cf } \mu$, whereas the cofinality of a whole cylinder is controlled by $\text{cf } \kappa$. These can differ. Confusing the two would lose the central feature of the cut classification below. Hereditary density also equals $\mu^{<\mu}$, since every subspace has weight at most $\text{w}(P)$ and P itself attains this density.

Note 30.4 (Source 17, Proposition 5.4, “Set-alphabet background, with a direct proof”). *The order P has cardinality 2^μ . Every linear order of cardinality at most μ embeds into every nonempty open interval of P . An ordinal β embeds in P if and only if $\beta < \mu^+$; the same bound holds for reverse well-ordered suborders.* Source 17 calls these “already part of the source’s theory”. They are the set baseline ([theorem 6.1](#); reals report `lwo:prop:cardinality`, `lwo:thm:ordinal-spectrum`, `lwo:sp:thm:universality`) applied inside a cylinder, whose residual alphabet has size μ , together with [lemma 30.1](#); the proof is the pair-orientation, initial-segment and coordinate-stabilization argument of [section 6](#).

30.1.1 Fusion with an exhaustiveness requirement

Theorem 30.5 (A dense-open fusion theorem). *The intersection of fewer than q open sets is open. The intersection of at most q dense open subsets of P is dense. In particular, when $q = \omega$, P is a Baire space.*

Proof. At a point in fewer than q open sets, choose a cylinder neighborhood for each. Their prefix lengths have supremum below μ , so the longest union prefix gives a cylinder inside the intersection.

Let $(U_i)_{i < q}$ be dense open sets and start with any nonempty cylinder. Write $X = \bigcup_{i < q} X_i$, where the X_i are increasing and each has cardinality below μ . Recursively refine the current prefix to a cylinder contained in U_i , using density and the cylinder base, and then append every member of X_i not yet used, in a set well-order of length below μ . At stages below q , the accumulated length is still below μ , including at limit stages. At the end the union has length exactly μ and range X : its length is at most μ , and it has used μ distinct labels. It is therefore an element of P , belongs to every U_i , and extends the starting prefix. Fewer dense open sets can be padded by copies of P . $\square$

The coverage step is essential. Bare prefix fusion of length μ can produce an injective word omitting a label, which is not a point of P . The theorem does not claim a complete metrizable or a general isomorphism classification.

Note 30.6 ([merge]; repository analogues and the regular size-matched case). For regular κ with $2^{<\kappa} = \kappa$ one has $\mu = r = q = \kappa$, and the character part of [theorem 30.2](#) is source 11's [proposition 12.1](#) (every local and global character κ , and η_κ). For $X = \mathbb{R}$ the reals report proves the analogues: point characters $\text{cf } \mathfrak{c}$ and global characters ω (`lwo:prop:fixed-topology`), weight $2^{<\mathfrak{c}}$ (`lwo:sp:prop:minimalweight`), and that fewer than $\text{cf } \mathfrak{c}$ open sets have open intersection (`lwo:rk:cor:Pspace`). Sources 15, 17 and 18 do not cite these. New in sources 15, 17 and 18 is the alphabet $\mathbf{No}_{<\kappa}$ at every infinite cardinal κ , in particular the separation of the point character $q = \text{cf } \mu$ from the global character $r = \text{cf } \kappa$; new in source 17 is the second half of [theorem 30.5](#), where every fusion sequence must also exhaust the alphabet.

Note 30.7 (Source 15, Theorem 9.2, “Local and global cofinalities”). Source 15 writes $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ for P , and $\text{cf}^-(f)$, $\text{ci}^+(f)$ for the cofinality of $L_{<f}$ and the coinitality of $L_{>f}$. For every $f \in L$, $\text{cf}^-(f) = \text{ci}^+(f) = \text{cf } \mu$; the global cofinality and coinitality are $\text{cf } \kappa$; consequently the ordinary local topological character of every point of L is $\text{cf } \mu$. This is the character part of [theorem 30.2](#), by the same argument (a swap of the entry at η with a later occurrence of a fresh smaller or larger label gives $f_\eta^\mp$ with first difference exactly η ; cofinally many η suffice, fewer than $\text{cf } \mu$ first differences are bounded). Source 15 adds the local base of size $\text{cf } \mu$ formed by pairs from the two sides, and the examples: “at the cutoff $\kappa = \omega_1$ the alphabet cardinal is $\mu = 2^{\aleph_0}$. The global cofinality is always ω_1 , whereas the local character is $\text{cf}(2^{\aleph_0})$. Under cardinal arithmetic making these cardinals different, the distinction survives at every point of the permutation order. At the singular strong-limit cutoff $\kappa = \beth_\omega$, every point has countable local character, even though the density and cellularity of the minimal slice are 2^κ .”

30.2 The complete nonprincipal cut spectrum of the minimal slice (source 17)

30.2.1 Three ways that a cut can occur

For a gap (A, B) of P , say that a cylinder $[p]$ *straddles* the cut when it meets both A and B . There is at most one straddling cylinder at any fixed length, because the cylinders at that length are disjoint convex blocks.

Lemma 30.8 (Cut trichotomy). *Every gap of P is of exactly one of the following structural forms, using the first stage at which the construction stops.*

1. **A coordinate cut.** Some straddling $[p]$, of length $\alpha < \mu$, has no straddling child. Its children are partitioned by a proper cut of the residual alphabet Y_p .
2. **A limit-prefix boundary.** At a nonzero limit $\delta < \mu$, all shorter prefixes on a coherent path straddle, but the union cylinder $[p]$ lies wholly on one side. The cut is exactly the left or right boundary of $[p]$.

3. An omitted-range branch. *A coherent prefix straddles at every length below μ . Their union is an injection $f : \mu \rightarrow X$ which is not surjective, and the gap is comparison with f .*

Proof. Begin at the empty prefix, whose cylinder straddles. At a successor stage, continue into the unique straddling child when it exists; otherwise every child belongs wholly to one side, giving case 1. At a limit $\delta < \mu$, the union is an injective prefix and has a nonempty cylinder. If that cylinder does not straddle, it is wholly on one side. Every word strictly below it differs at an earlier coordinate, and is below a lower member of an earlier straddling cylinder, so belongs to A . The dual argument holds above the cylinder. This proves exactly the boundary assertion in case 2.

If the process continues through all lengths below μ , let f be the union injection. For any $g \in P$ different from f , its first difference determines its cut side: a later straddling cylinder gives witnesses which force the orientation. If f were surjective it would itself belong to P . Whichever side contains it would then have an endpoint, contrary to the gap hypothesis. Thus f is not surjective, and all words in P are classified by their comparison with it. These alternatives cover the set recursion through μ , and are disjoint by the first stopping stage. $\square$

Lemma 30.9 (Characters in the three cases). *In case 1, the gap character belongs to $\text{GapChar}(X) \cup \{(r, r)\}$. In case 2 it is respectively (cf δ, r) or $(r, \text{cf } \delta)$. In case 3 it is (q, q) .*

Proof. In case 1 partition the residual alphabet into nonempty parts $U < V$, according as a child lies on the lower or upper side. If U has no largest label, the lower character is cf U ; choose one point in cofinally many children. If U has largest label u , the last lower child is cofinal on the lower side and has cofinality r . The upper side has the dual description. When neither side has an endpoint, Theorem 29.4 applies to Y_p . When one has an endpoint, the other has character r by Lemma 29.3; the endpoint side is replaced by an entire child of character r .

At the left boundary of a limit cylinder $[p]$, lower approximations obtained by a smaller fresh entry at coordinates cofinal in δ are cofinal below the cylinder. No family of fewer than cf δ can be cofinal, since its first disagreements with p would be bounded below δ . The upper side has $[p]$ as a coinital subset, and the cylinder has coinitality r . The right boundary is dual.

For an omitted-range branch, approximations from either side at cofinally many coordinates give upper bounds q for both characters. Fewer than q disagreements are bounded, so a later approximation is closer to the branch. This proves equality. $\square$

30.3 A general permutation-order principle (source 18)

The tree argument is not specific to surreal numbers. It separates the contribution of the alphabet from the new cuts introduced by long prefixes and omitted branches.

Theorem 30.10 (Gap spectrum under permutation expansion). *Let D be a dense linear order without endpoints of infinite cardinality μ . Assume that every nonempty open interval has cardinality μ , every point has the same two-sided character ν , and both global cofinalities are also ν . Let $\text{Perm}_\mu(D)$ be the lexicographic order on bijections $\mu \rightarrow D$, and put $q = \text{cf } \mu$. Then*

$$\begin{aligned} \text{GapChar}(\text{Perm}_\mu(D)) = & \text{GapChar}(D) \cup \{(\nu, \nu), (q, q)\} \\ & \cup \{(\theta, \nu), (\nu, \theta) : \theta < \mu \text{ is an infinite regular cardinal}\}. \end{aligned}$$

Proof. Small deletions leave an order dense in D , so the residual-cut analysis and the straddling-cylinder trichotomy apply verbatim. These give the upper bound. All the explicitly added pairs are attained by the same prefix-boundary and omitted-branch constructions. Finally, each gap of D lifts to a gap of the permutation order by placing a permutation on the left or right according to

its first digit. With neither digit side principal, the two characters are unchanged. This gives the reverse inclusion. $\square$

[merge] *The steps the proof leaves to the reader.* Source 18 writes “apply verbatim”; the two facts it uses are written out here. Put $P = \text{Perm}_\mu(D)$.

Small deletions add only (ν, ν) . Let $U \subseteq D$ with $|U| < \mu$ and $R = D \setminus U$. Every interval of D has μ points, so R meets every interval of D ; in particular R is dense, has no endpoints, and has global cofinalities ν and point characters (ν, ν) , because cofinal and cointial families transfer between D and R through points of the other order. Let (A, B) be a gap of R . Two points of D cannot lie between A and B , since a point of R would lie between them. If no point of D lies between them, the downward closure of A and the upward closure of B in D form a gap of D with the same characters. If one point x of D lies between them, then $x \in U$, A is cofinal in $D_{<x}$ and B cointial in $D_{>x}$, and the character is the point character (ν, ν) of x . Hence $\text{GapChar}(R) \subseteq \text{GapChar}(D) \cup \{(\nu, \nu)\}$.

The three cases. Let (A, B) be a gap of P . The prefixes p of length below μ whose cylinders C_p meet both sides form a chain under extension, as in lemma 30.8. If there is a longest one, p , the residual alphabet $R_p = D \setminus \text{ran } p$ (with $|\text{ran } p| < \mu$) is cut into the digits whose child cylinders lie in A and those whose child cylinders lie in B ; both parts are nonempty and, R_p being dense, they do not both have endpoints. If neither has one, the character of (A, B) is the character of this gap of R_p , which lies in $\text{GapChar}(D) \cup \{(\nu, \nu)\}$ by the first step: $A \cap C_p$ is cofinal in A and is the lexicographic sum of the lower child cylinders. If the lower part has a last digit u , A is cofinally the child cylinder $C_{p \frown u}$, whose cofinality is that of its residual alphabet, ν , while B approaches u from above in R_p , with cointiality ν ; the dual case is the same, so the character is (ν, ν) . If the straddling lengths have a limit supremum $\delta < \mu$, the union p has a nonempty cylinder lying on one side, the gap is a boundary of C_p , and its character is $(\text{cf } \delta, \nu)$ or $(\nu, \text{cf } \delta)$: the approximations with a smaller (larger) unused digit at coordinates cofinal in δ exist because rays of D contain intervals of size μ , and C_p has global characters ν . If the lengths are cofinal in μ , the union is a nonsurjective injection $\mu \rightarrow D$ and the character is (q, q) , exactly as in case 3 of lemma 30.9. This proves the upper bound.

Attainment. For a regular $\theta < \mu$, the two boundaries of the cylinder of an injective prefix of length θ have characters (θ, ν) and (ν, θ) ; the lower boundary of a one-digit cylinder $C_{\langle x \rangle}$ has character (ν, ν) , since its lower side approaches the digit x from below and $C_{\langle x \rangle}$ has cointiality ν ; a bijection of μ onto $D \setminus \{x\}$ gives (q, q) ; and a gap of D lifts at the first digit, as source 18 says. $\square$

Theorem 30.11 (Exact gap spectrum). *For every infinite cardinal κ , with the parameters in (S17.3),*

$$\text{GapChar}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \{(r, r), (q, q)\} \cup \{(\theta, r), (r, \theta) : \theta \in \text{Reg}, \theta < \mu\}. \quad (\text{S17.5})$$

Proof. The upper inclusion follows from the two preceding lemmas and $\kappa \leq \mu$: alphabet gap pairs involving a regular cardinal below κ already belong to the displayed set.

For (r, r) , fix a label $x \in X$ and place on the lower side every word whose first entry is at most x , and on the upper side every word whose first entry exceeds x . The last lower child has cofinality r , and the upper first-entry ray has cointiality r . Neither side has an endpoint.

For any infinite regular $\theta < \mu$, choose an injective prefix of length θ . Its two boundary cuts have characters (θ, r) and (r, θ) by the limit-boundary calculation. The fresh smaller and larger labels used there exist at every coordinate, so both sides of both cuts are nonempty.

Finally choose a bijection $f : \mu \rightarrow X \setminus \{x\}$ for some label x . Each proper prefix is completable in P , but f itself is not in P . Its comparison cut is a gap of character (q, q) . $\square$

Note 30.12 ([merge]; [theorem 30.11](#) as an instance). $D = \mathbf{No}_{<\kappa}$ satisfies the hypotheses of [theorem 30.10](#) with $\nu = r$: [lemmas 29.2](#) and [29.3](#) give the interval size and the uniform point and global characters. With $\text{GapChar}(\mathbf{No}_{<\kappa})$ from [theorem 29.4](#), whose asymmetric pairs $(\theta < \kappa)$ are among those with $\theta < \mu$, the principle returns exactly ([S17.5](#)). Source 17’s proof above is the direct one.

Note 30.13 (Source 18, Theorem 5.1, “Exact gap spectrum”, and Lemma 5.4). Source 18’s Section 5 “settles the cut-spectrum question for the minimal-length permutation order at every infinite cardinal birthday cutoff”. It writes P for the bijections $\mu \rightarrow \mathbf{No}_{<\kappa}$, defines a *proper Dedekind gap* as a partition $P = A \sqcup B$ into nonempty sets with $A < B$, A without greatest and B without least element, of character (cf A , ci B), and writes $\text{GapChar}(P)$ for the set of characters (“this convention excludes principal cuts and the two external ends”). Its Theorem 5.1 is ([S17.5](#)) together with: *every displayed pair is attained; (1) every point of P has left and right character q ; (2) the global cofinality and coinitality are r ; (3) the principal cuts have character $(1, q)$ or $(q, 1)$; (4) if empty sides are assigned cofinality 0, the external ends have character $(0, r)$ and $(r, 0)$* . Its Lemma 5.4: *every cylinder C_p has cofinality and coinitality r and no endpoints; if p has nonzero limit length $\delta < \mu$, the lower boundary of C_p is a proper gap of character (cf δ, r) and the upper boundary one of character $(r, \text{cf } \delta)$* . These are [lemma 30.1](#), case 2 of [lemma 30.9](#) and [theorems 30.2](#) and [30.11](#), and source 18’s proof is source 17’s: the straddling prefixes form a chain under extension, and its three cases (a longest straddling prefix, a limit supremum $\delta < \mu$, lengths cofinal in μ) are the three cases of [lemma 30.8](#), in the same order, with the same character computations. Source 18 attains (r, r) by the boundary of a successor-length cylinder $C_{\langle x \rangle}$ (the exterior side approaches the digit x) rather than by source 17’s first-entry split; “every infinite regular $\theta < \mu$ can be used as δ ” in Lemma 5.4, so the asymmetric pairs are attained “independently of which of them already occur as gaps in the alphabet”.

30.4 Source 16’s route: regular saturated stages over any saturated alphabet

[merge] The last subsection of source 16’s Section 6. It is printed here, after source 17’s spectrum, because it computes the same spectrum for the regular size-matched case and generalizes it to every regular saturated alphabet of the same size ([note 30.15](#)).

Suppose μ is regular infinite and X is an η_μ order of size μ . The earlier report proves that $\mathcal{W}_\mu(X)$ is itself η_μ , with every local and global cofinality and coinitality μ ([proposition 12.1](#)). Here is the mechanism. If one side is empty, choose a fresh first label beyond all first labels on the other side and complete it. For two nonempty families of fewer than μ lower and upper enumerations, bound their cross first disagreements below μ . Follow their forced common coordinates until the active next labels have strictly separated lower and upper sets. Choose a fresh interpolating label using saturation, and complete the resulting prefix. Fewer than μ used labels can be avoided by a further small cut. The procedure must stop before the bound is passed. Small cut filling gives the lower character bound; [Note 32.1](#) gives the upper bound.

Theorem 30.14 (Complete gap-character spectrum). *Under the preceding hypotheses, the cofinality pairs of proper unfilled Dedekind cuts in $\mathcal{W}_\mu(X)$ are exactly*

$$(\mu, \lambda) \quad \text{and} \quad (\lambda, \mu), \quad \omega \leq \lambda \leq \mu, \quad \lambda \text{ regular.} \quad (\text{S16.4})$$

Both sides of a cut here are nonempty, and neither has an endpoint.

Proof. The two characters are infinite regular cardinals. Neither exceeds μ , because a larger character would produce a monotone sequence of type μ^+ , contrary to [Note 32.1](#). They cannot both be below μ : cofinal and coinital sets of those sizes would form a strict small cut, and saturation would insert a point in the supposed unfilled cut. This proves necessity.

For an infinite regular $\lambda < \mu$, fix an injective prefix p of length λ . Its completion cylinder C_p is convex and has no endpoints; its cofinality and coinitality are μ , by the same saturation argument. The cut just above the entire cylinder has lower character μ . Enumerations above the cylinder first differ upward from p at some coordinate below λ . Choose such upward deviations at a cofinal sequence of coordinates in λ ; their completions form a coinital family of size λ on that upper side. Fewer than λ deviation coordinates are bounded, and a later upward deviation lies below all of them. Thus the upper character is exactly λ . The cylinder has no maximum and the upper side has no minimum, so this cut is unfilled. The lower edge gives (λ, μ) .

For (μ, μ) fix $a \in X$ and split the order according to first label $\leq a$ or $> a$. The lower side is cofinally the cylinder with first label a , of character μ . The upper side has coinitality μ , inherited from the native segment above a . Both sides have no endpoint. This realizes the final case. $\square$

For regular κ satisfying $2^{<\kappa} = \kappa$, these conclusions apply to $\mathbf{No}_{<\kappa}$. They answer the earlier report's regular-stage density, cellularity, and cut-character questions under precisely those hypotheses. Singular stages and regular stages of larger cardinality require separate analysis; the theorem does not erase the singular coding phenomenon in (S16.2).

Note 30.15 ([merge]; source 16's theorem as an instance). An η_μ order X of regular cardinality μ satisfies the hypotheses of [theorem 30.10](#) with $\nu = \text{cf } \mu = \mu$: it is dense without endpoints, its intervals have size μ , and every point and both ends have character μ , by saturation and $|X| = \mu$. An η_μ order has no gap with both characters below μ , so the principle gives the pairs (μ, θ) , (θ, μ) with $\theta \leq \mu$ regular, which is (S16.4). For $X = \mathbf{No}_{<\kappa}$ with κ regular and $2^{<\kappa} = \kappa$, [theorem 30.14](#) and [example 30.20](#) coincide. Source 17 is more general in the cutoff (any κ , no hypothesis on $2^{<\kappa}$), source 16 in the alphabet (any regular saturated alphabet). For $X = \mathbb{R}$ the reals report's `lwo:rk:thm:Pgaps` is the corresponding spectrum, $\{(\omega, \theta), (\theta, \omega) : \theta < \mathfrak{c} \text{ regular}\} \cup \{(\text{cf } \mathfrak{c}, \text{cf } \mathfrak{c})\}$; it is source 17's formula with $r = \omega$, $q = \text{cf } \mathfrak{c}$, $\mu = \mathfrak{c}$ and the alphabet spectrum $\text{GapChar}(\mathbb{R}) = \emptyset$ (the formula's (r, r) comes from a principal digit split, and (ω, ω) is the case $\theta = \omega$). Neither source 16 nor source 17 cites it.

30.5 How many gaps have each character? (source 18)

[merge] “The same classification” and “the three cases” below are the straddling-cylinder classification of the gaps of P : a longest straddling prefix, a limit supremum of straddling lengths below μ , and lengths cofinal in μ , which are cases 1, 2 and 3 of [lemma 30.8](#) ([note 30.13](#)). Source 18's d_* is the density $\mu^{<\mu}$ of P ([theorem 30.2](#)); its prefix variable q is printed p' , since $q = \text{cf } \mu$ here. The theorem is the minimal-slice analogue of the reals report's count of $2^{\mathfrak{c}}$ gaps of $\mathcal{W}$ (`lwo:sp:thm:gapcount`), which source 18 does not cite.

The same classification gives exact multiplicities, not just a list of the possible character pairs. Write

$$d_* = \mu^{<\mu} = \sup_{\lambda < \mu} \mu^{|\lambda|}.$$

This is also the cardinality of the tree of injective prefixes of length less than μ .

Theorem 30.16 (Exact multiplicities). *The minimal permutation order P has exactly 2^μ proper Dedekind gaps. For a pair (a, b) in the spectrum of [Note 30.13](#), let $G_{a,b}$ be the set of proper gaps of that character. Then*

$$|G_{a,b}| = \begin{cases} 2^\mu, & (a, b) = (q, q), \\ d_*, & (a, b) \neq (q, q). \end{cases}$$

In particular, the Dedekind completion of P , with or without its two external ends, has cardinality 2^μ .

Proof. First, every full nonsurjective injection $f : \mu \rightarrow X$ gives a proper gap of character (q, q) . Fix $x_* \in X$ and consider injections into $X \setminus \{x_*\}$. There are 2^μ such injections: for a lower bound partition this set into μ disjoint pairs and choose one entry of each pair according to a binary word; the upper bound is $\mu^\mu = 2^\mu$.

Distinct injections give distinct gaps. If $f < g$ first differ at α , choose an unused digit strictly between $f(\alpha)$ and $g(\alpha)$ and extend the common prefix followed by that digit to a bijection. This is possible because the intervening interval has cardinality μ , while fewer than μ digits have been used. The resulting member of P lies strictly between f and g . There are therefore at least 2^μ gaps of character (q, q) .

For an upper bound on all gaps, the tree of short prefixes has cardinality $d_* \leq 2^\mu$, an alphabet of size μ has at most 2^μ cuts, and there are at most $\mu^\mu = 2^\mu$ full injections. The three cases of the straddling-cylinder classification therefore give at most 2^μ gaps altogether. This proves both the total number and the multiplicity of (q, q) .

Next fix an infinite regular $\theta < \mu$. There are exactly d_* prefixes whose lengths have cofinality θ . For the nontrivial lower bound, take any infinite cardinal λ with $\theta \leq \lambda < \mu$ and put

$$\delta = \lambda \cdot \theta.$$

Then $\delta < \mu$, $|\delta| = \lambda$, and $\text{cf } \delta = \theta$. There are μ^λ injective words of this length. One way to see the lower bound is to partition X into λ disjoint sets, each of cardinality μ , and choose the digit at each coordinate from its assigned set. Taking the supremum over λ gives d_* . The opposite bound is the cardinality of the whole prefix tree.

Distinct prefixes give distinct lower-boundary cuts. For incomparable prefixes, an unused digit strictly between their first disagreeing digits yields a permutation between their cylinders; the interval has size μ and fewer than μ digits have been used. If p' properly extends p , choose at the first additional coordinate an unused digit smaller than the corresponding digit of p' . This gives a point of C_p below all of $C_{p'}$; a larger choice gives a point above all of $C_{p'}$. Thus the lower boundaries, and also the upper boundaries, are distinct in this case as well. Consequently there are at least d_* gaps of each character (θ, r) and (r, θ) .

It remains to bound the number of gaps of any character other than (q, q) . Full-length omitted branches contribute none of these. Limit-cylinder boundaries contribute at most d_* gaps. For the maximal-prefix case distinguish two possibilities.

If $q = r$, the character being counted is not (r, r) . In the alphabet sign-path description, a gap not of character (r, r) has one sign set bounded below κ , so its path is eventually constant. There are at most $\sum_{\delta < \kappa} 2^{|\delta|} = \mu$ such paths. Deleting a small set only adds (r, r) gaps, and a principal digit cut likewise lifts to that character. Thus each maximal prefix contributes at most μ gaps of the character under consideration, and all short prefixes together contribute at most $d_* \cdot \mu = d_*$.

If $q \neq r$, then $\mu > \kappa$: equality would give $\text{cf } \mu = \text{cf } \kappa$. Every alphabet cut is represented by a length- κ sign word, so there are at most 2^κ of them. Cuts created by deletions and retained points add at most μ . Since $\kappa < \mu$,

$$2^\kappa \leq \mu^\kappa \leq d_*.$$

Again all maximal prefixes together contribute at most d_* gaps.

Every allowed character other than (q, q) has the form (θ, r) or (r, θ) for a regular $\theta < \mu$. The only apparent exception would be (r, r) with $r = \mu$, but then $\kappa = \mu$ is regular and $q = r$, so that pair is precisely the already excluded (q, q) . The prefix-boundary lower bound therefore applies to every remaining character, completing the multiplicity calculation.

Finally $|P| = 2^\mu$, and its completion adds one point for each proper gap (and optionally its two ends). The completion cardinality is consequently 2^μ . $\square$

30.5.1 Consequences and distinctions

The exact formula makes several distinctions transparent.

1. If $\mu = \kappa$ is regular, then $r = q = \kappa$ and the spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta < \kappa \text{ infinite regular}\}.$$

In particular, the order's η_κ property does not prohibit asymmetric gaps with one small side.

2. At a singular strong-limit cutoff, $\mu = \kappa$ and $r = q = \text{cf } \kappa$. The asymmetric pairs range over every infinite regular $\theta < \kappa$, including regular cardinals much larger than $\text{cf } \kappa$.
3. If $\kappa = \omega_1$ and $2^{\aleph_0} = \aleph_2$, the proper gap spectrum consists exactly of

$$(\omega, \omega_1), \quad (\omega_1, \omega), \quad (\omega_1, \omega_1), \quad (\omega_2, \omega_2).$$

Every point has character (ω_2, ω_2) , while both global cofinalities are ω_1 . Thus local character, global character, and gap character cannot be conflated.

4. If $\kappa = \omega$, then $\mu = r = q = \omega$ and every proper gap has character (ω, ω) .

The set of possible character pairs depends only on μ , r , and q ; its proof retains substantially more information. Every gap has a precise structural cause: a split between next digits, a boundary of a limit-length cylinder, or a nonsurjective branch of full length. Coinciding character pairs need not have the same structural cause. Nor does equality of these spectra by itself establish order isomorphism.

Natural further problems include classifying the order of the gaps of each character, studying which of the three mechanisms can be recognized intrinsically from the pure linear order, and extending the analysis to strata of nonminimal enumeration length. In the last problem a prefix can leave a small residual alphabet near the end of the domain, so the uniform large-interval argument used here no longer applies without a separate analysis of the ordinal tail.

30.6 Saturation and illustrative cases (source 17)

Definition 30.17. For an infinite regular cardinal τ , an order is an η_τ order when every pair of separated sets $A < B$, each of size below τ , has a strict interpolant. Empty sides are allowed.

Corollary 30.18 (Exact saturation threshold). *The order P is an η_r order, and is not an η_{r+} order.*

Proof. Suppose $A < B$ have sizes below r and no strict interpolant. If one side is empty, the other would be cofinal or coinital in P , contradicting (S17.4). If both are nonempty, let

$$D = \{x : \exists a \in A (x \leq a)\}, \quad U = \{x : \exists b \in B (b \leq x)\}.$$

They are disjoint, and no strict interpolant means that they cover P . If D has a greatest element x , then B is coinital immediately above x , contradicting right point character $q \geq r$. If U has a least element, the dual contradiction uses A . If neither has an endpoint, (D, U) is a gap with both characters below r , forbidden by (S17.5). An (r, r) gap exists; cofinal and coinital witnesses of size r show failure of η_{r+} . $\square$

Example 30.19 (The countable birthday cutoff). At $\kappa = \omega$, X is the order of finite sign sequences, $\mu = r = q = \omega$, and P has countable weight and density, character (ω, ω) at every point, and only (ω, ω) gaps. It is a Baire, zero-dimensional order. These facts do not identify it with the real line, whose order topology has a completely different connectedness structure.

Example 30.20 (Regular size-matched cutoffs). If κ is regular and $2^{<\kappa} = \kappa$, then $\mu = r = q = \kappa$. The gap spectrum is

$$\{(\kappa, \kappa)\} \cup \{(\theta, \kappa), (\kappa, \theta) : \theta \in \text{Reg}, \theta < \kappa\}.$$

The order is η_κ but not $\eta_{\kappa+}$. The small character on one side of a limit-prefix boundary does not violate η_κ , because the other side requires κ witnesses.

Example 30.21 (A singular strong-limit cutoff). At $\kappa = \beth_\omega$, one has $\mu = \kappa$ and $r = q = \omega$. Every point has countable character, but for every infinite regular $\theta < \kappa$ there are gaps of character (θ, ω) and (ω, θ) . Weight, density and cellularity are $\kappa^{<\kappa} = 2^\kappa$. This complements rather than replaces the source's $\kappa \cdot \kappa$ binary-coding-length theorem (theorem 8.3).

[merge] (3 October 2026.) Source 17 wrote “are $\kappa^{<\kappa}$, with no extra cardinal-arithmetic simplification asserted”. At a singular strong limit κ one has $2^\kappa \leq \kappa^{\text{cf } \kappa} \leq \kappa^{<\kappa} \leq 2^\kappa$, so the value is 2^κ ; source 18 proves this identity in corollary 30.26, and source 15 obtains the same value 2^κ from its block-type theorem (corollary 31.7).

Remark 30.22 (What has not been classified). The preceding formulas concern the minimal type μ . They do not extend by replacing μ with an arbitrary ordinal α of cardinality μ : a prefix of length below α can already use μ labels, leaving a radically smaller residual alphabet. Nor do equal values of these invariants imply order-isomorphism. Both issues remain in the research program at the end of this article.

30.7 Source 15's route to the saturation threshold

[merge] Source 15's Section 9, “Exact local characters at every cardinal cutoff”. Its Lemma 9.1 is recorded in note 29.8, its Theorem 9.2 in note 30.7 and its Lemma 9.3 in note 29.9; its saturation theorem is proved directly, by a prefix search, not from the gap spectrum as in corollary 30.18, and is printed in full. Its $\delta = \text{cf } \kappa$ is printed r . Its first sentence describes this report at source 15's pin; the point characters it mentions are now theorem 30.2.

The repository leaves the point characters of $\mathcal{W}_{2^{<\kappa}}(\mathbf{No}_{<\kappa})$ open when $2^{<\kappa}$ need not equal κ . They can be computed exactly. The local invariant is controlled by the enumeration length, whereas the global invariant is controlled by the numerical alphabet.

Throughout this section let

$$X = \mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}, \quad r = \text{cf } \kappa, \quad L = \mathcal{W}_\mu(X),$$

where κ is an infinite initial cardinal. For $f \in L$, write $\text{cf}^-(f)$ for the cofinality of the open initial segment $L_{<f}$, and $\text{ci}^+(f)$ for the coinitality of the open final segment $L_{>f}$.

We use the convention that an order is η_ν if every cut $A < B$ with $|A| + |B| < \nu$ has a strict interpolant, including the cases where one side is empty. This does not assert that an arbitrary cut with larger sides is filled.

Theorem 30.23 (Exact saturation threshold for the minimal slice). *The order $L = \mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ is $\eta_{\text{cf } \kappa}$. For every infinite cardinal $\nu > \text{cf } \kappa$, it is not η_ν . Equivalently, among infinite cardinal cut-size parameters,*

$$L \text{ is } \eta_\nu \iff \nu \leq \text{cf } \kappa.$$

In particular, at every regular κ , the minimal slice is η_κ without the additional assumption $2^{<\kappa} = \kappa$.

Proof. Let $A < B$ be subsets of L with $|A| + |B| < r$. Any family of fewer than r surreal labels has birthdays bounded below κ . Consequently, for any strict numerical cut formed by such labels, there are two points $u < v$ of X strictly between its sides. This follows from the usual surreal cut construction applied twice; all resulting birthdays stay below κ . The same statement holds with an empty side. By [Note 29.8](#), the interval (u, v) has cardinality μ . Thus a numerical cut of this size admits a separator outside any prescribed set of fewer than μ labels.

If one of A, B is empty, choose the first entry strictly above all first entries on the left or below all first entries on the right, and complete to a permutation. Suppose both are nonempty. For each $(a, b) \in A \times B$, let $d(a, b) < \mu$ be its first-difference coordinate. By [Note 29.9](#),

$$\beta = \sup\{d(a, b) + 1 : a \in A, b \in B\} < \mu.$$

Build an injective prefix p , keeping as active those rows of A and B that still agree with p . At a stage η , every active left value at coordinate η is at most every active right value. If a value occurs on both sides, it is unique; append that value to p and keep the rows taking it. Every discarded left row is already below all extensions of the new prefix, and every discarded right row is already above them. The appended value is fresh because it belongs to active injective rows.

If the active value sets have no common value, they form a strict numerical cut. Choose a separator outside the range of p using the preceding full-size-interval argument, append it, and stop. If an active side becomes empty, use the same argument for the resulting one-sided cut. At a limit stage take the union of the prefixes and retain the rows that extend it. The search cannot still have active rows on both sides at stage β : such a pair would agree at every coordinate below β , contradicting the definition of β . Therefore it stops with a prefix of length less than μ .

That prefix has used fewer than μ labels. Its remaining labels and remaining positions both have cardinality μ , and an exhaustive completion has order type μ . Every such completion lies strictly above A and below B , proving η_r .

Finally, [Note 30.7](#) gives a global cofinal subset of size r . Together with an empty right side it is a cut of total size below every cardinal $\nu > r$, but it has no upper interpolant. Hence no such η_ν property holds. $\square$

These results settle the point-character part, and the maximal uniform cut-size threshold, of the earlier minimal-slice question. They do not compute its full two-sided unrealized-cut spectrum. In particular, saturation below $\text{cf } \kappa$ and failure above that threshold do not specify which individual larger pairs of regular cofinalities occur as gaps.

[merge] (3 October 2026.) The “full two-sided unrealized-cut spectrum” that source 15 leaves open is source 17’s [theorem 30.11](#) (and source 18’s [note 30.13](#)); its multiplicities are [theorem 30.16](#).

30.8 Source 18’s route and extension: characters, interpolation and two regimes

[merge] The rest of source 18’s Section 4, printed whole. Its point characters and interpolation theorem are source 17’s [theorem 30.2](#) and [corollary 30.18](#) and source 15’s [theorem 30.23](#), proved again with a different interpolation argument (bounded dropout coordinates); it extends both to the types $\mu + n$ and to the finite-cofinite prefix orders Q_n ([lemma 27.7](#)), and it exhibits a bounded (r, r) obstruction explicitly. Its corollary on two regimes merges with source 17’s examples [examples 30.20](#) and [30.21](#); its second part supplies the value $\kappa^{<\kappa} = 2^\kappa$ used in [example 30.21](#).

30.8.1 Exact point characters and global cofinalities

For a point e in a linear order, its left character means the cofinality of $(-\infty, e)$, and its right character means the coinitality of (e, ∞) . An immediate predecessor or successor therefore gives

character 1 on the respective side.

Theorem 30.24 (Complete local and global characters at the minimal type). *Every point of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ has left and right character $q = \text{cf}(\mu)$, whereas*

$$\text{cf}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \text{ci}(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = r.$$

More generally, $Q_n(\mathbf{No}_{<\kappa})$ has point characters (q, q) for every finite n . Accordingly, in $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$:

- *if $n = 0$ or 1 , every point has characters (q, q) ;*
- *if $n \geq 2$, the first and last points of each finite $n!$ -point fiber have characters $(q, 1)$ and $(1, q)$ respectively;*
- *all interior points of a finite fiber have characters $(1, 1)$.*

Interior points exist exactly when $n \geq 3$. The global cofinality and coinitality remain r for every finite n .

Proof. Fix $p \in Q_n(\mathbf{No}_{<\kappa})$ and $\delta < \mu$. There are μ labels numerically below $p(\delta)$, and equally many above it. The labels used before δ , together with the finitely many labels omitted from p , exclude fewer than μ labels. Choose a smaller available label which therefore occurs later in the μ -prefix, and transpose its occurrence with position δ . This gives $p_\delta^- < p$ with first difference exactly δ . An analogous transposition gives $p_\delta^+ > p$.

As δ runs along an increasing cofinal sequence in μ , the points p_δ^- increase cofinally toward p , while p_δ^+ decrease coinitially toward it. This proves both upper bounds q . Conversely, fewer than q points below p have first-disagreement coordinates bounded below μ . Changing p at a later coordinate gives a point above all of them and below p . They cannot be cofinal. The dual argument gives the right lower bound. This proves the character statement for Q_n .

The map $e \mapsto e(0)$ is onto $\mathbf{No}_{<\kappa}$ and respects comparisons whenever the first coordinates differ. A family cofinal in the enumeration order must therefore have first coordinates cofinal in $\mathbf{No}_{<\kappa}$; conversely, choose one enumeration over every member of a numerical cofinal sequence. This proves the global cofinality formula, and the coinitality formula is dual. Finally, $\mathcal{W}_{\mu+n}(\mathbf{No}_{<\kappa})$ is the lexicographic sum over Q_n of finite ordered fibers of size $n!$. Immediate neighbors inside a fiber give character 1, while approach to the outside of a fiber is controlled by the character q of the quotient point. $\square$

30.8.2 Sharp interpolation at every infinite cardinal cutoff

We say that L is η_r if for all sets $A, B \subseteq L$ with $|A \cup B| < r$ and $a < b$ for every $a \in A, b \in B$, there is $z \in L$ satisfying $A < z < B$. Empty sides are allowed. The sharpness result below also uses a bounded cut with both sides nonempty, so it does not depend on the empty-side convention.

Theorem 30.25 (Exact guaranteed interpolation). *For every infinite initial cardinal κ , with notation as above,*

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \text{ is } \eta_r, \quad r = \text{cf}(\kappa).$$

It has a bounded unfilled cut of character (r, r) , so it is not η_{r+} . The same interpolation and sharpness assertions hold for $\mathcal{W}_{\mu+1}(\mathbf{No}_{<\kappa})$. For $n \geq 2$, adjacent points in the terminal fibers already preclude η_ω .

Proof of interpolation. First $\mathbf{No}_{<\kappa}$ is η_r . A family of fewer than r numerical options has birthdays bounded below κ , so the set-cut theorem supplies a separator with birthday less than κ . Applying it twice supplies a nonempty interval of possible separators, still between the prescribed options. By [Note 29.8](#), such an interval has μ labels.

Let $A < B$ be sets of input enumerations with $|A \cup B| < r$. We construct an injective common prefix p until a completion separating the inputs becomes possible. At a prefix of length $\delta < \mu$, the active inputs are

$$A_p = \{a \in A : a \upharpoonright \delta = p\}, \quad B_p = \{b \in B : b \upharpoonright \delta = p\}.$$

The invariant is that all discarded lower inputs lie below every completion of p , while all discarded upper inputs lie above every completion of p . Let L and U be the sets of next entries of the active lower and upper inputs. Then every $l \in L$ is at most every $u \in U$.

If $L \cap U = \emptyset$, the two next-entry sets are strictly separated (with empty sides permitted). Choose an interval of numerical separators between them. It has μ labels, whereas $\text{ran } p$ has size less than μ . Choose an unused separator z in it. Every completion of $p \frown (z)$ lies strictly between all inputs, and an exhaustive completion of the required type exists by cardinality. The construction stops.

Otherwise, $L \cap U$ consists of exactly one label z : two different common labels would contradict $L \leq U$. This label is unused, since an active input is injective. Append it to p . Inputs with a smaller lower entry or a larger upper entry are discarded in the correct direction, maintaining the invariant. At limits take the union of the prefixes.

Suppose the construction never stopped before μ . The union is an injective word p of length μ . Every input whose μ -prefix differs from p drops out at its first disagreement. There are fewer than $r \leq q = \text{cf}(\mu)$ inputs, so these dropout coordinates are bounded below μ . At most one input can have μ -prefix exactly p : for type μ the whole enumeration is determined, and for type $\mu + 1$ its single omitted label determines the last entry. Since $A < B$, that one input cannot lie on both sides. Beyond the bound on the dropout coordinates, at least one active side is therefore empty, forcing the stopping case. This contradiction proves that the construction stops at a coordinate below μ . $\square$

Proof of the bounded obstruction. Choose a sign word s of length exactly κ such that both its plus positions and minus positions are cofinal in κ , and whose first two signs are $+, -$. For example, use the parity of the finite tail of each ordinal coordinate. Then $0 < s < 1$, while $s \notin \mathbf{No}_{<\kappa}$. Prefixes $s \upharpoonright \gamma$ at cofinally many positions with $s(\gamma) = +$ lie below s and are cofinal below its cut in $\mathbf{No}_{<\kappa}$; prefixes at cofinally many minus positions are coinital above it. Choose such two cofinal collections indexed by r .

Every $x \in \mathbf{No}_{<\kappa}$ has its comparison with s decided before κ . A sufficiently long chosen prefix therefore passes x on the appropriate side. No element of $\mathbf{No}_{<\kappa}$ separates the two prefix families. Neither side can have a smaller cofinal family: fewer than r birthdays are bounded below κ , and a longer appropriate prefix passes every label in that family. Thus this is a bounded numerical gap of character (r, r) .

Pull the cut back under the first-coordinate map $e \mapsto e(0)$. Choose arbitrary completions for the two numerical cofinal families. A hypothetical separating enumeration would need its first entry to lie in the missing numerical cut, which is impossible. Conversely those completions are cofinal and coinital on their respective sides, and the first-coordinate projection proves the matching lower bounds. Enumerations starting with 0 and 1 bound the whole configuration. This supplies the claimed bounded (r, r) obstruction. $\square$

30.8.3 Regular and singular consequences

Corollary 30.26 (Two especially transparent regimes). *1. If κ is regular and $2^{<\kappa} = \kappa$, then*

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = \kappa,$$

and every local character and global cofinality or coinitality is κ . The order is η_κ and has a bounded unfilled (κ, κ) -cut.

2. If κ is a singular strong-limit cardinal, then

$$|\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})| = 2^\kappa, \quad \text{d} = \text{cell} = 2^\kappa,$$

while every point character and both global cofinalities are $\text{cf}(\kappa)$. Its sharp interpolation threshold is $\eta_{\text{cf}(\kappa)}$.

Proof. In the regular case, $\kappa^{<\kappa} = \kappa$. For a fixed $\lambda < \kappa$, every function $\lambda \rightarrow \kappa$ has bounded range, and the possibilities for each range bound are at most $2^{<\kappa} = \kappa$; summing over κ range bounds gives $\kappa^\lambda = \kappa$ for nonzero $\lambda < \kappa$.

In the singular strong-limit case, $2^{<\kappa} = \kappa$. Let $r = \text{cf}(\kappa)$ and take a cofinal sequence $(\kappa_\xi)_{\xi < r}$ below κ . A subset of κ is determined by its intersections with the κ_ξ . Each intersection has fewer than κ possibilities by the strong-limit hypothesis, so

$$2^\kappa \leq \kappa^r \leq \kappa^{<\kappa} \leq \kappa^\kappa = 2^\kappa.$$

Thus $\kappa^{<\kappa} = 2^\kappa$. The other conclusions now follow from the preceding theorems and the baseline cardinality theorem. $\square$

What these results settle, and what they do not. The baseline explicitly left density, cellularity and much of the singular-cutoff interpolation analysis as continuation questions. The preceding theorems determine density and cellularity for every fixed enumeration type and every infinite alphabet, and give exact local/global characters and sharp guaranteed interpolation at every infinite initial birthday cutoff for the minimal slice. They do not identify the order up to isomorphism. In particular, equality of these invariants is not an isomorphism theorem, and the old exact ordinal spectrum remains logically separate from the new topological information. The full spectrum of unfilled cut characters requires an additional analysis of boundaries of transfinite prefix cylinders.

31 Longer strata: every cardinal-block type (source 15)

[merge] Source 15's Section 8, printed whole: the only batch-81 treatment of strata longer than the minimal one. Its r_κ is printed ρ_κ (by [theorem 8.3](#), $\rho_\kappa = \ell_2(\mathcal{W}_\mu(\mathbf{No}_{<\kappa}))$), so that r keeps the meaning $\text{cf} \kappa$; its $D_{\sigma, m}$ is printed $\mathbb{B}_\sigma \times_{\text{lex}} m$, so that no letter clashes with its core $\mathcal{D}$ ([section 1.3.1](#)). The section extends source 12's [theorem 8.3](#) ($\gamma = 1$, $n = 0$) along the targets of [research question 34.11](#). Its entry code ([lemma 31.1](#)) is source 12's [lemma 19.33](#) and source 16's mechanism in (S16.2); its cube clause is [lemma 6.4](#) (reals report `lwo:thm:cube-strict`); its row scheduler ([lemma 31.3](#)) is source 12's [lemma 8.1](#) run blockwise along γ ; its finite-deletion lemma ([lemma 31.4](#)) is [lemma 8.2](#) with a finite set removed; its finite-capacity lemma is, by its own account, the finite-fiber fusion argument of the reals report. Source 15 states its finite-deletion lemma for singular strong-limit κ ; the hypothesis is not used in its proof.

The earlier minimal-slice theorem (section 8) extends to all enumeration types whose remainder on division by the alphabet cardinal is finite. This includes the explicitly proposed next cases $\mu \cdot 2$ and $\mu \cdot \gamma + n$. It does not classify an additional infinite remainder smaller than μ . All products and sums of ordinals in this section are ordinal operations.

Let κ be an infinite initial cardinal, and put

$$X = \mathbf{No}_{<\kappa}, \quad \mu = |X| = 2^{<\kappa}, \quad \rho_\kappa = \begin{cases} \kappa \cdot \kappa, & \kappa \text{ is a singular strong-limit cardinal,} \\ \mu, & \text{otherwise.} \end{cases}$$

For an ordinal α with $|\alpha| = \mu$, write $\mathcal{W}_\alpha(X)$ for the bijections $\alpha \rightarrow X$, compared by their first unequal entry in the numerical order of X . For a positive integer m , write

$$\mathbb{B}_\sigma \times_{\text{lex}} m = \mathbb{B}_\sigma \times_{\text{lex}} \{0, \dots, m-1\}, \quad \mathbb{B}_\sigma = (2^\sigma, <_{\text{lex}}),$$

the finite chain $m = \{0, \dots, m-1\}$ being the last coordinate.

Lemma 31.1 (An explicit entry code and the strict cube hierarchy). *Replace each minus sign by 00, each plus by 11, and terminate with 01. The resulting code $c(s)$ is prefix-free and preserves numerical surreal order. Its length is $2 \cdot \text{bd}(s) + 2$, less than κ for $s \in \mathbf{No}_{<\kappa}$. Furthermore,*

$$\mathbb{B}_\alpha \hookrightarrow \mathbb{B}_\beta \iff \alpha \leq \beta.$$

Proof. The three blocks $00 < 01 < 11$ encode precisely minus, termination, and plus. Distinct completed codes cannot be proper prefixes: at the first termination of one of them the other either terminates too or has a different block. Padding a completed entry therefore does not alter its comparison with another entry.

For the cube assertion, extension by zeros gives the embedding when $\alpha \leq \beta$. Suppose $\alpha > \beta$ and an embedding in the other direction exists. Successively read target bits, retaining at each step a full source child cylinder whose image lies on one side of the next target-bit split. Such a child exists: if the lower child does not lie wholly below the split, every point of the upper child lies above it. At limits take unions of the chosen prefixes. After β steps, all points extending one source prefix of length β have the same whole target word. There are at least two source extensions since $\beta < \alpha$, contradicting injectivity. This is the same cylinder fusion mechanism used in the prior binary-length calculations. $\square$

When $\mu = \kappa$ is singular, κ is a strong limit. Otherwise $2^\rho = \kappa$ for some $\rho < \kappa$ would also give $2^\lambda = \kappa$ at $\lambda = \max(\rho, \text{cf } \kappa) < \kappa$, contradicting König's inequality $\text{cf}(2^\lambda) > \lambda$. Thus the three cases used below are exhaustive.

Theorem 31.2 (Cardinal blocks and a finite terminal permutation). *For every $0 < \gamma < \mu^+$ and every $n < \omega$,*

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e (\mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!).$$

In particular, if κ is a singular strong limit, then

$$\mathcal{W}_{\kappa \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} \times_{\text{lex}} \{0, \dots, n! - 1\}.$$

The statement concerns equimorphism; it does not identify these permutation orders with the generally nondense binary products.

We give the scheduler detail before proving the theorem, since a cardinality-preserving reindexing alone would not preserve lexicographic order.

Lemma 31.3 (Blockwise row scheduling). *Let Y be a linearly ordered set of infinite cardinality κ , let $0 < \gamma < \kappa^+$, and put $\lambda = \kappa \cdot \gamma$. Suppose that Y contains pairwise disjoint subsets $(Y_i)_{i < \lambda}$, each of cardinality κ , with an order embedding $\phi_i : X \rightarrow Y_i$, and that $Z = Y \setminus \bigcup_{i < \lambda} Y_i$ is either empty or has cardinality κ . Then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(Y).$$

Proof. For each $i < \lambda$, fix a bijection $b_i : \kappa \rightarrow Y_i$. For $x \in X$, transpose $\phi_i(x)$ with $b_i(0)$ in that enumeration; denote the result by $b_{i,x}$. Its first value is $\phi_i(x)$, and for equal x the whole row is the same.

Split the variable rows into the consecutive blocks $\{\kappa \cdot \beta + \xi : \xi < \kappa\}$, $\beta < \gamma$. Within one block schedule row-column pairs $(\xi, j) \in \kappa^2$ first by $\max(\xi, j)$ and then lexicographically inside each level. Every pair has fewer than κ predecessors: all earlier pairs lie in a square bounded below κ , and such a square has cardinality less than κ , even when κ is singular. The schedule therefore has order type κ . The first occurrence of row ξ is $(\xi, 0)$, and these first occurrences have increasing row indices.

If $Z \neq \emptyset$, include its fixed κ -long enumeration as a filler row before the variable rows in the first block. The row-index order is $1 + \kappa = \kappa$, so the same construction applies. The filler row has no input parameter. Every later block uses only its κ variable rows.

For $x = (x_i)_{i < \lambda}$, output the entries of b_{i,x_i} by these successive block schedules. Each block has length κ and there are γ blocks, so the output has length exactly λ . The disjoint rows and the filler exhaust Y exactly once.

If inputs $x <_{\text{lex}} x'$ first differ at row $i = \kappa \cdot \beta + \xi$, earlier blocks agree completely. In block β , every output before the first entry of row i comes from a lower row or from the fixed filler. Those outputs agree. The first unequal output is therefore $\phi_i(x_i) < \phi_i(x'_i)$ at the first entry of row i . This proves both injectivity and order preservation. No regularity of γ or κ was used. $\square$

Lemma 31.4 (Finite deletion does not obstruct the singular scheduler). *If κ is a singular strong limit, $X = \mathbf{No}_{<\kappa}$, $F \subseteq X$ is finite, and $\lambda = \kappa \cdot \gamma$ for $0 < \gamma < \kappa^+$, then*

$$X_{\text{lex}}^\lambda \hookrightarrow \mathcal{W}_\lambda(X \setminus F).$$

Proof. For $\xi < \kappa$, let C_ξ be the sign-sequence cone with prefix $+\xi \frown -$. The cones are disjoint, and prefixing gives an order isomorphism $X \rightarrow C_\xi$. Indeed, $\xi + 1 + \eta < \kappa$ for $\xi, \eta < \kappa$, because κ is an initial cardinal; deletion of the prefix gives the inverse. Each cone has cardinality κ . At most finitely many cones meet F . Select κ unaffected cones and index them by λ , which has cardinality κ .

The complement of those selected cones in $X \setminus F$ has cardinality κ : it contains all but finitely many of the all-plus sequences, one for each length below κ , and is a subset of X . The preceding lemma now applies, with that complement as the filler. In particular, no unexplained bijection of a grid with λ is being used in place of a lexicographic scheduler. $\square$

Proof of Theorem 31.2. Write $\lambda = \mu \cdot \gamma$ and $\sigma = \rho_\kappa \cdot \gamma$. We first construct the upper embedding. Use the repository's prefix-free, numerical-order-preserving binary code $c(s)$ for each $s \in X$, with positive length less than κ .

If $\mu > \kappa$, pad each individual entry code to length κ . Prefix-freeness ensures that this preserves all entry comparisons. A block of μ entries then consumes ordinal length $\kappa \cdot \mu = \mu$. Applying this separately to the γ successive blocks gives a prefix code of length $\mu \cdot \gamma = \sigma$.

If $\mu = \kappa$ and κ is regular, concatenate the unpadded codes in each κ -entry block. Each proper partial sum of their lengths is below κ by regularity, while the full sum is κ because all code lengths are positive. Each block therefore consumes exactly κ , again giving length σ .

In the remaining case $\mu = \kappa$ is singular, it is a strong limit. Pad each entry code to length κ . Each block now consumes $\kappa \cdot \kappa$, and the complete λ -entry prefix has code length $\kappa \cdot \kappa \cdot \gamma = \sigma$.

In all cases, a prefix of length λ from a bijection $\lambda + n \rightarrow X$ has a binary code of length exactly σ . The remaining n labels form a finite set determined by that prefix. Append the rank of their terminal permutation in its increasing lexicographic list of $n!$ permutations. Different prefixes compare inside the binary code; equal prefixes have the same remaining labels and compare by the finite rank. This gives $\mathcal{W}_{\lambda+n}(X) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n!)$.

For the reverse embedding in either nonexceptional case, reserve a fixed n -element set $F \subseteq X$ and partition $X \setminus F$ into pairs indexed by λ . Such a partition exists because $|\lambda| = \mu$. List each pair in its increasing or decreasing orientation according to the corresponding bit of a member of $\mathbb{B}_\lambda$. Since λ is a nonzero limit ordinal, $2 \cdot \lambda = \lambda$. Append the requested one of the $n!$ permutations of F . Here $\sigma = \lambda$, giving the required embedding.

It remains to obtain the longer cube in the singular strong-limit case. Let $\delta = \text{cf } \kappa < \kappa$, and choose nonzero ordinals $(\tau_j)_{j < \delta}$ cofinal in κ , with $\sum_{j < \delta} \tau_j = \kappa$. The identities

$$\delta \cdot \kappa = \kappa, \quad \delta \cdot \lambda = \delta \cdot (\kappa \cdot \gamma) = \lambda$$

allow the λ input coordinates of X^λ to be grouped into λ consecutive blocks of δ coordinates. In each block, restrict its j th coordinate to surreals whose sign sequences have length exactly τ_j . That coordinate is a copy of $\mathbb{B}_{\tau_j}$, so the entire restricted product is

$$\mathbb{B}_{\sum_{i < \lambda} \sum_{j < \delta} \tau_j} = \mathbb{B}_{\kappa \cdot \lambda} = \mathbb{B}_{\kappa \cdot \kappa \cdot \gamma} = \mathbb{B}_\sigma.$$

By Lemma 31.4, this product embeds into $\mathcal{W}_\lambda(X \setminus F)$. Append any of the $n!$ permutations of F , obtaining $(\mathbb{B}_\sigma \times_{\text{lex}} n!) \hookrightarrow \mathcal{W}_{\lambda+n}(X)$. $\square$

31.1 Exact coding lengths, embedded cubes, and comparisons

The next capacity argument is the finite-fiber fusion argument already used in the real-alphabet report. It is included to make the numerical consequences of the extension explicit.

Lemma 31.5 (Finite fibers over a binary cube). *Let σ, τ be nonzero limit ordinals and let m, n be positive integers. Then*

$$(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\tau \times_{\text{lex}} n) \iff \sigma < \tau \text{ or } (\sigma = \tau \text{ and } m \leq n).$$

Moreover,

$$\ell_2(\mathbb{B}_\sigma \times_{\text{lex}} m) = \sigma + \lceil \log_2 m \rceil, \quad \mathbb{B}_\theta \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} m) \iff \theta \leq \sigma + \lceil \log_2 m \rceil.$$

Proof. For equal binary lengths, suppose $f : (\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow (\mathbb{B}_\sigma \times_{\text{lex}} n)$. Recursively choose a source binary prefix s_ξ and target binary prefix t_ξ , both of length ξ , such that

$$f(C_{s_\xi} \times \{0, \dots, m-1\}) \subseteq C_{t_\xi} \times \{0, \dots, n-1\}.$$

At a successor stage the target's next bit cuts the image into two ordered parts. Of the source cylinder's two full children, at least one lies wholly on one side of that cut: if its lower child is not wholly on the lower side, its upper child is wholly on the upper side. Keep that child and the corresponding target bit. At a limit stage take unions of prefixes. The invariant persists, and the source cylinder remains nonempty. At stage σ , all m points over one source binary sequence lie in one target fiber of size n . Thus $m \leq n$. The converse is immediate.

Coding the finite chain of size m by $\lceil \log_2 m \rceil$ bits gives $(\mathbb{B}_\sigma \times_{\text{lex}} m) \hookrightarrow \mathbb{B}_{\sigma + \lceil \log_2 m \rceil}$. No cube shorter than σ can contain $(\mathbb{B}_\sigma \times_{\text{lex}} m)$, since it already contains $\mathbb{B}_\sigma$ and the binary cube hierarchy

is strict. A target $\mathbb{B}_{\sigma+j} = (\mathbb{B}_\sigma \times_{\text{lex}} 2^j)$ with $j < \omega$ can contain it only if $m \leq 2^j$, by the same fusion argument. This proves the coding-length formula.

Likewise, $\mathbb{B}_{\sigma+j}$ embeds into $(\mathbb{B}_\sigma \times_{\text{lex}} m)$ exactly when $2^j \leq m$. Once the first forbidden finite extension has been excluded, every larger cube is excluded because it contains that extension. This proves the complete cube spectrum.

Finally, if $\sigma < \tau$, then $\sigma + \lceil \log_2 m \rceil < \tau$ because τ is a limit. Code into that shorter cube and then into the target. If $\tau < \sigma$, the target codes into a cube of length $\tau + \lceil \log_2 n \rceil < \sigma$, whereas the source contains $\mathbb{B}_\sigma$; strictness excludes an embedding. This proves the comparison criterion. $\square$

Corollary 31.6 (The complete invariant on these block types). *Under the hypotheses of Theorem 31.2, put $\sigma = \rho_\kappa \cdot \gamma$. Then*

$$\ell_2(\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})) = \sigma + \lceil \log_2(n!) \rceil,$$

and

$$\mathbb{B}_\theta \hookrightarrow \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \iff \theta \leq \sigma + \lceil \log_2(n!) \rceil.$$

For $0 < \gamma, \eta < \mu^+$ and $m, n < \omega$,

$$\mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa}) \hookrightarrow \mathcal{W}_{\mu \cdot \eta + m}(\mathbf{No}_{<\kappa}) \iff \gamma < \eta \text{ or } (\gamma = \eta \text{ and } n! \leq m!).$$

Proof. Apply the finite-capacity lemma and use strict monotonicity in the right argument of $\gamma \mapsto \rho_\kappa \cdot \gamma$ for a fixed nonzero left factor. Here each $\rho_\kappa \cdot \gamma$ is a nonzero limit ordinal. $\square$

For an unconditional singular example, take $\kappa = \beth_\omega$. The types $\kappa \cdot 2$, $\kappa \cdot 2 + 3$, and κ^2 have binary coding lengths respectively

$$\kappa^2 \cdot 2, \quad \kappa^2 \cdot 2 + 3, \quad \kappa^3.$$

The type $\kappa \cdot 2 + 3$ contains the cube of length $\kappa^2 \cdot 2 + 2$ but not the cube of its own coding length. These distinctions are invisible to cardinality and to the ordinal embedding spectrum, which remain 2^κ and $\{\beta : \beta < \kappa^+\}$ throughout.

31.2 Density and cellularity, including the singular minimal slice

Write $d(L)$ for the least cardinality of a dense subset in the order topology, and $\text{cell}(L)$ for the supremum of cardinalities of families of pairwise disjoint nonempty open sets. An additional consequence of the coding theorem is an exact answer for these two invariants on the same block types. In particular, the singular minimal slice has the largest possible cellularity, a fact not visible from its unchanged ordinal embedding spectrum.

Corollary 31.7 (Density and cellularity of the block types). *For $L = \mathcal{W}_{\mu \cdot \gamma + n}(\mathbf{No}_{<\kappa})$ as above,*

$$d(L) = \text{cell}(L) = \begin{cases} 2^{<\mu}, & \rho_\kappa = \mu, \gamma = 1, n \leq 2, \\ 2^\mu, & \text{otherwise.} \end{cases}$$

Consequently

$$d(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = c(\mathcal{W}_\mu(\mathbf{No}_{<\kappa})) = \begin{cases} 2^\kappa, & \kappa \text{ is a singular strong limit,} \\ 2^{<\mu}, & \text{otherwise.} \end{cases}$$

Proof. First consider $\rho_\kappa = \mu$, $\gamma = 1$, and $n \leq 2$. The finite factor $n!$ has size at most two. There is an embedding $f : L \rightarrow (\mathbb{B}_\mu \times_{\text{lex}} n!)$. No point of L is isolated. Indeed, L has no endpoints because the native surreal alphabet has neither a least nor a greatest element; swapping the first entry with a smaller or larger label moves either way without changing the enumeration type. The adjacency criterion for exhaustive permutations applies inside a fixed type: if the unused remainder at a first difference is infinite, an ascent in the lower tail or a descent in the upper tail gives an intermediate permutation of the same type. Adjacency is therefore confined to the finite terminal permutation component. For $n \leq 2$ each such component has at most two points, so no point has both an immediate predecessor and an immediate successor. Thus no point is isolated, and every nonempty open set contains at least five points.

For every binary prefix s of length less than μ whose cylinder meets the first-coordinate projection of $f[L]$, choose one element $x_s \in L$ whose image has that prefix. There are at most $\sum_{\xi < \mu} 2^{|\xi|} = 2^{<\mu}$ such choices. These choices are dense. To see this, take five increasingly ordered points $x_0 < \dots < x_4$ in any given nonempty open interval. Each binary fiber has at most two points, so the binary first coordinate of $f(x_2)$ is strictly between those of $f(x_0)$ and $f(x_4)$. Choose a prefix long enough to distinguish it from both outer coordinates. Its whole binary cylinder lies strictly between them, and the chosen x_s lies in the original open interval. Hence $d(L) \leq 2^{<\mu}$.

Conversely L contains $\mathbb{B}_\mu$. For each $\xi < \mu$, its $2^{|\xi|}$ cylinders of length ξ each contain three ordered points. The ambient open intervals in L between the images of the outer two points are nonempty, since they contain the image of the middle point, and they are pairwise disjoint. Therefore $\text{cell}(L) \geq 2^{<\mu}$, proving equality in the first case since always $\text{cell}(L) \leq d(L)$.

In every remaining case, put $\sigma = \rho_\kappa \cdot \gamma$. If $\sigma > \mu$, the embedded cube $\mathbb{B}_\sigma$ has 2^μ distinct cylinders of prefix length μ , each with at least three points because σ is a limit ordinal greater than μ . The preceding interval construction gives $\text{cell}(L) \geq 2^\mu$. If $\sigma = \mu$ but $n \geq 3$, the embedded order $(\mathbb{B}_\mu \times_{\text{lex}} n!)$ has a fiber with at least three points over each of its 2^μ binary sequences. The intervals between the images of its first and last fiber points are again pairwise disjoint and nonempty. Thus $\text{cell}(L) \geq 2^\mu$ here as well. Finally $d(L) \leq |L| = 2^\mu$, giving the claimed equalities. $\square$

31.3 The boundary of the result

Write a general admissible type as $\alpha = \mu \cdot \gamma + \zeta$, $\zeta < \mu$. The theorem settles the case that ζ is finite. When ζ is infinite, the preceding long prefix leaves a residual numerical suborder of cardinality $|\zeta|$; that residual order varies with the prefix. Neither the size nor the coding behavior of this smaller alphabet can simply be replaced by those of $\mathbf{No}_{<\kappa}$. An exact treatment must control the possible residual suborders and the code lengths available uniformly for their permutations. Thus the result gives a precise affirmative continuation of the proposed cardinal-block program while leaving the smaller infinite-tail classification explicitly open.

Remark 31.8 ([merge]; the reals report's display at $X = \mathbf{No}_{<\kappa}$). The reals report asks under which hypotheses on X the display $\mathcal{W}_{\lambda+n}(X) \simeq_e \mathbb{B}_\lambda \times_{\text{lex}} n!$ still holds (its question “Other ground orders”, `lwo:sec:research`); its batch-80 note records that [theorem 8.3](#) gives the case $n = 0$, $\lambda = \mu$ for $X = \mathbf{No}_{<\kappa}$, except at a singular strong limit. For $\lambda = \mu \cdot \gamma$, $0 < \gamma < \mu^+$, and every $n < \omega$, [theorem 31.2](#) gives $\mathcal{W}_{\lambda+n}(\mathbf{No}_{<\kappa}) \simeq_e \mathbb{B}_{\rho_\kappa \cdot \gamma} \times_{\text{lex}} n!$, and by [lemma 31.5](#) two such products with nonzero limit lengths are equimorphic only when the lengths are equal. So the display holds whenever κ is not a singular strong limit ($\rho_\kappa = \mu$). At a singular strong limit ($\mu = \kappa$, $\rho_\kappa = \kappa \cdot \kappa$) it holds exactly when $\kappa \cdot \kappa \cdot \gamma = \kappa \cdot \gamma$, that is, when $\gamma = \kappa^\omega \cdot \delta$ for some ordinal δ (then $\kappa \cdot \gamma = \gamma$, by Cantor normal form to base κ); it fails for $\gamma = 1, 2, \kappa, \dots$. The display is therefore not simply false at singular strong limits. Strata λ that are not multiples of μ (the first is $\mu + \omega$), cutoffs that are not cardinals, and other alphabets remain open, as source 15 says in [section 31.3](#).

32 Set-sized baselines and warnings in sources 13–18

[merge] Every batch-81 source restates the set-sized baseline of the reals report and of Part I (section 6 and theorem 6.1): cardinality 2^μ , universality for orders of size at most μ , and the exact ordinal spectrum $\beta < \mu^+$. They are recorded here once, with the sources’ own provenance statements, followed by the set-level warnings of sources 13, 14 and 17. None of these statements is new.

Note 32.1 (Source 16, Theorem 4.1, “Basic set-alphabet results”). Source 16’s Section 4 fixes “an infinite linearly ordered set X of cardinality μ ” and records that exhaustive enumerations are prefix-incomparable and that an injective prefix of length less than μ extends to a bijection of every ordinal type α with $|\alpha| = \mu$. *The orders $\mathcal{W}(X)$ and $\mathcal{W}_\mu(X)$ have cardinality 2^μ . Every linear order of cardinality at most μ embeds into $\mathcal{W}_\mu(X)$. For every ordinal β , $\beta \hookrightarrow \mathcal{W}(X)$ if and only if $\beta < \mu^+$, and the same statement holds with $\mathcal{W}_\mu(X)$ in place of $\mathcal{W}(X)$, and for the reverse of β .* Its proof is the pair-orientation, initial-segment and coordinate-stabilization proof of section 6. Source 16’s status line: “Theorem 4.1 is background from the ProveIt reals and surreal reports; it is reproved here to make the later arguments self-contained. In particular, ‘all ordinals embed’ is false at every fixed set stage.”

Birthday cutoffs and a singular distinction (source 16’s Subsection 4.1). For $\mathbf{No}_{<\kappa}$, κ an infinite cardinal, substitute $\mu = 2^{<\kappa}$ for the alphabet cardinal in Note 32.1. The ordinal embedding cutoff is μ^+ , not automatically κ^+ . The earlier report also proves a sharper binary-coding distinction:

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} 2^{\kappa \cdot \kappa}_{\text{lex}}, & \kappa \text{ is a singular strong-limit cardinal,} \\ 2^\mu_{\text{lex}}, & \text{otherwise.} \end{cases} \quad (\text{S16.2})$$

Here $\kappa \cdot \kappa$ is ordinal multiplication. This statement is equimorphism, meaning embeddings in both directions, not isomorphism.

The upper coding mechanism replaces a negative sign by 00, a positive sign by 11, and termination by 01. These codes are prefix-free and preserve numerical comparison. At a regular stage, a sum of fewer than κ code lengths remains below κ . At a singular strong-limit stage it can already reach κ before all κ coordinates have been processed; padding every entry to length κ gives total length $\kappa \cdot \kappa$. The matching lower bound in the report uses a cofinal sequence of sign lengths and an order-preserving row scheduler (theorem 8.3). We do not reclassify the binary coding ranks of all longer strata here. The topological classification below, by contrast, covers every admissible enumeration length and every infinite alphabet.

Note 32.2 (Source 18, Theorem 3.1, “Set-sized universality and the exact ordinal spectrum”). Source 18’s Section 3 defines $\mathcal{W}(X)$ and $\mathcal{W}_\alpha(X)$ as here, notes that two exhaustive enumerations are never proper prefixes of one another, so that no termination convention is needed, and that “labels are essential: two two-point well-orders can be isomorphic as abstract orders and still have different first numerical labels”. *Let X be an infinite linear order of cardinality μ . Then $|\mathcal{W}(X)| = |\mathcal{W}_\mu(X)| = 2^\mu$. Every linear order of cardinality at most μ embeds into $\mathcal{W}_\mu(X)$. An ordinal β embeds into $\mathcal{W}(X)$ if and only if $\beta < \mu^+$. The same exact spectrum holds for reverse ordinals and for every nonempty stratum $\mathcal{W}_\alpha(X)$.* The proof is that of section 6; source 18 adds that the stabilization argument “uses the regularity of the successor cardinal μ^+ , not regularity of μ ”, and that cardinality 2^μ is a poor guide, since such an order contains no increasing sequence of type μ^+ . Its subsection “What the existing binary-rank result adds” restates theorem 8.3 (binary coding length $\kappa \cdot \kappa$ at a singular strong-limit cutoff, μ otherwise), calls it “background rather than a dependency of our new cut-spectrum proof”, and adds: “Replacing these ordinal sums by cardinalities would erase a

genuine distinction. Our results below add topological information not determined by this coding length.”

Note 32.3 (Source 15, Lemma 3.2 and Theorem 3.3, “Set-sized baseline, from the prior reports”). Source 15’s Lemma 3.2 (“Exhaustiveness removes prefix ambiguity”): *distinct exhaustive bijective enumerations of the same set have a first unequal coordinate in both domains; neither is a proper prefix of the other*. Its Theorem 3.3 is the baseline of [note 32.1](#) for every nonempty fixed-type slice, proved as in [section 6](#), with the fixed type $\alpha = \lambda + n$ handled by reserving n final labels and orienting pairs indexed by λ . Source 15 says: “This proof uses no special property of real or surreal arithmetic. It is reproduced to fix the dependence of the later theorems; its content is already in the audited reports. If X is finite of size n , the whole order is the finite chain of its $n!$ permutations. The empty carrier has one empty well-order.” Its subsections on birthday cutoffs and on set-length words, which close its Section 3, follow. Its Lemma 3.1 (fresh separators) belongs to the class layer and is recorded with the other batch-81 restatements of [lemma 5.2](#).

32.1 Source 15’s birthday cutoffs and set-length words

[merge] The last two subsections of source 15’s Section 3; its X_κ is printed $\mathbf{No}_{<\kappa}$, and its forward reference to its Section 8 points to [section 31](#).

32.1.1 Birthday cutoffs

For an infinite initial cardinal κ , let

$$X_\kappa = \mathbf{No}_{<\kappa}, \quad \mu = |X_\kappa| = 2^{<\kappa}.$$

Counting sign levels proves the cardinality formula: the supremum of their cardinalities is at least κ , and the union of κ levels does not exceed it. The preceding theorem gives

$$|\mathcal{W}(X_\kappa)| = 2^{2^{<\kappa}}, \quad \beta \hookrightarrow \mathcal{W}(X_\kappa) \iff \beta < (2^{<\kappa})^+.$$

At regular κ , a cut using fewer than κ labels has birthdays bounded below κ , and the usual cut construction stays in the cutoff. Singular κ need not have this property. The fixed-type calculation in [Section 31](#) makes the distinction visible even when the cardinalities and ordinal spectra coincide.

32.1.2 Why all set-length words are a different order

If X varies, the class of all pairs (X, R) with X a set of surreals and R a well-order is a legitimate class of sets. Its ordinal enumerations are injective set-length words. Proper prefixes now occur. For example (0) is a prefix of $(0, 1)$. One may put shorter words first, shorter words last, or use a termination symbol placed inside the alphabet. These comparisons agree on exhaustive enumerations of a fixed carrier but need not agree on variable carriers.

The prior report gives an explicit self-delimiting sign code embedding all set-length surreal words, with shorter-first order, into $\mathbf{No}$. Its code also exhibits why this order need not be isomorphic to $\mathbf{No}$: a word of nonzero limit length is the supremum of its shorter prefixes, whereas a surreal point has no set-sized cofinal family below it. Restricting to successor-length words removes that particular obstruction and yields a class order isomorphic to $\mathbf{No}$ in GBC (this report’s [part II](#), source 11’s Sections 9–11 there). An embedding theorem therefore must not be strengthened to an isomorphism theorem merely because both carriers are proper classes.

32.2 Source 13's fixed-set section

[merge] Source 13's Section 4, "The fixed-set theory and birthday fragments", printed whole; its theorem, a restatement credited by source 13 itself, is printed as a note, and its X_κ is printed $\mathbf{No}_{<\kappa}$.

The set-sized layer is important both in its own right and as a safe formalization starting point. We record its already-established results with their provenance rather than repeating the earlier report in full.

Note 32.4 (Source 13, Theorem 4.1, "Fixed-alphabet results from the repository"). Let X be an infinite linearly ordered set, $|X| = \mu$, and compare exhaustive enumerations using its label order. Then

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \hookrightarrow \mathcal{W}(X) \iff \beta < \mu^+,$$

and the same spectrum holds for reversed ordinals. Every linear order of cardinality at most μ embeds in $\mathcal{W}(X)$.

Provenance and proof mechanisms (source 13). These are the general-alphabet results of the repository's real well-ordering report, carried into Section 6 of the surreal report ([1]; section 6). The cardinal lower bound codes independent binary choices by pair reversals in an enumeration; the upper bound counts well-order relations on X . Universality uses binary codes for initial segments of a small order. The ordinal upper bound is a prefix-stabilization obstruction, stronger than a cardinality bound. We invoke that established written theorem here; this paragraph is not a replacement proof of its stabilization argument.

For an infinite cardinal κ , let

$$\mathbf{No}_{<\kappa}, \quad \mu = 2^{<\kappa}.$$

Counting signs gives $|\mathbf{No}_{<\kappa}| = \mu$: there are $2^{|\alpha|}$ sign sequences of length α , and the union over $\alpha < \kappa$ has that supremum cardinality. Thus the ordinal spectrum of $\mathcal{W}(\mathbf{No}_{<\kappa})$ is exactly the ordinals below $(2^{<\kappa})^+$. This does not say that only ordinals below the birthday cutoff embed.

32.2.1 A warning about the minimal-length layer

Write $\mathbb{B}_\theta = (\{0, 1\}^\theta, <_{\text{lex}})$, and write $L \simeq_e M$ for mutual order embeddability. The merged surreal report proves the following sharper distinction (theorem 8.3, Theorem 8.3 of the report at its pin):

$$\mathcal{W}_\mu(X_\kappa) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases} \quad \mu = 2^{<\kappa}.$$

The multiplication in $\kappa \cdot \kappa$ is *ordinal* multiplication. At a singular strong limit, $\mu = \kappa$, but the lengths κ and $\kappa \cdot \kappa$ need not have the same lexicographic effect. This is an existing result, not an advance of the present manuscript.

There is no automatic passage from equimorphism to isomorphism, from a minimal-length stratum to all strata, or from independently chosen cutoff embeddings to a coherent global map. The last passage requires agreement on overlaps. These distinctions become especially visible in the product obstruction below.

32.2.2 Set-sized approximations to the new cut theorem

For an arbitrary set alphabet, prefix cylinders can terminate by exhaustion and can have extrema. The crossing arguments of source 13's Section 8 (its reconstruction of cuts from prefix cylinders) still provide a useful tree description, but its clean all-ordinal formulation depends on two genuinely global facts: every set prefix has a completion, and after deleting its range there are fresh labels

on both sides of any remaining set cut. A bounded alphabet need not have either property at its boundary. Therefore the new global theorem is not asserted by simply replacing **No** by every birthday fragment in its statement.

Note 32.5 (Source 14, Theorem 2.1, “Set-carrier baseline, inherited”). *In ZFC, if $|X| = \mu$ is infinite, then $|\mathcal{W}(X)| = 2^\mu$ and $\beta \hookrightarrow \mathcal{W}(X)$ if and only if $\beta < \mu^+$, for ordinals β . Every linear order of cardinality at most μ embeds in $\mathcal{W}(X)$. Adjacent pairs are exactly the consecutive permutations of a finite residual alphabet after a common prefix.* This is [section 6](#) with source 11’s adjacency theorem (reals report `two:thm:adjacency`). Source 14 prints it as “inherited from the repository, not a new result of this continuation” and gives no proof: “The complete proof is in the predecessor report; we do not substitute counting for that argument.” Its set-level remark follows.

[merge] The rest of source 14’s Subsection 2.2, “Fixed carriers and the inherited spectra”.

The size assertion is plausible for two different reasons: a relation on X has at most 2^μ possibilities, while independently choosing the orientation of μ disjoint pairs gives 2^μ distinct well-orders. The ordinal upper bound is a lexicographic obstruction, not a cardinality bound: 2^μ can be much larger than μ^+ . The complete proof is in the predecessor report; we do not substitute counting for that argument.

For an infinite cardinal κ , sign sequences give

$$|\mathbf{No}_{<\kappa}| = 2^{<\kappa}.$$

Thus [Note 32.5](#) applies with $\mu = 2^{<\kappa}$. The repository also proves a more refined minimal-slice dichotomy: with $\mathbb{B}_\delta = (\{0, 1\}^\delta, <_{\text{lex}})$,

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \kappa \text{ is singular strong limit,} \\ \mathbb{B}_\mu, & \text{otherwise.} \end{cases}$$

Here $\cdot$ is *ordinal* multiplication and $\simeq_e$ means mutual order embeddability, not isomorphism. This inherited result should not be confused with the support-cardinal theorem proved below: birthday bounds restrict the alphabet, whereas support bounds restrict how much of a fixed global enumeration is changed.

32.3 Source 17’s labelled well-orders of a fixed set

[merge] Source 17’s Subsection 2.1, from its map of interpretations. Let $(X, <)$ be a set with a specified linear order. A well-ordering R of X has a unique exhaustive enumeration $e_R : \alpha_R \rightarrow X$, where α_R is its ordinal type. For two distinct well-orders, neither exhaustive enumeration is a proper prefix of the other: after listing every label once there is no fresh label to append. Therefore a first unequal coordinate exists, and we define

$$R <_{\text{lex}} S \iff e_R(\delta) < e_S(\delta), \quad \delta = \min\{\xi : e_R(\xi) \neq e_S(\xi)\}.$$

Write $\mathcal{W}(X)$ for all such well-orders, and $\mathcal{W}_\alpha(X)$ for the suborder with enumeration type exactly α . Equivalently, $\mathcal{W}_\alpha(X)$ consists of bijections $\alpha \rightarrow X$. The notation $\mathcal{W}_\mu(X)$ below always uses the initial ordinal of the cardinal $\mu = |X|$.

The source establishes, for infinite $|X| = \mu$,

$$|\mathcal{W}(X)| = 2^\mu, \quad \beta \text{ embeds in } \mathcal{W}(X) \iff \beta < \mu^+,$$

and universality for linear orders of cardinality at most μ ([theorem 6.1](#); [1]). These are labelled-order results. Quotienting well-orders by abstract order-isomorphism removes exactly the labels on which the comparison depends.

33 What this Part answers

[merge] Dated notes, 3 October 2026. Each item names the question, where it is asked, and the result that answers it. Where the question index of [section 34](#) (dated 2 October 2026) says otherwise, these notes supersede it.

- (a) *Source 11's question on the cut spectrum and point characters of the minimal slice without $2^{<\kappa} = \kappa$* ([research question 34.7](#)): **answered**. Point characters (q, q) , global characters r , density, weight and cellularity $\mu^{<\mu}$ ([theorem 30.2](#)), the exact gap spectrum ([theorem 30.11](#)), its multiplicities ([theorem 30.16](#)), and the saturation threshold η_r ([corollary 30.18](#) and [theorems 30.23](#) and [30.25](#)).
- (b) *Source 12's question on the invariants of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$* ([research question 34.13](#)): answered except for the isomorphism type and homogeneity, which no source determines (source 17's [remark 30.22](#); source 18 says “equality of these invariants is not an isomorphism theorem”).
- (c) *Source 08's question 2, invariants at regular cutoffs* ([section 34.4](#)): **answered**. Density and cellularity of every stratum ([theorem 27.2](#) and [corollary 27.4](#)), point characters and cut cofinalities ([theorem 30.14](#) and [example 30.20](#)). This completes item (d) of [section 1.7](#).
- (d) *Source 08's question 3, which small cuts fail at singular cutoffs*: answered in character terms by [theorems 29.4](#) and [30.11](#) (source 17 does not claim this question); its coding half was already answered by [theorem 8.3](#) (item (c) of [section 1.7](#)).
- (e) *Source 12's question on all fixed enumeration types* ([research question 34.11](#)) and *source 08's question 4*: answered for $\alpha = \mu \cdot \gamma + n$, $0 < \gamma < \mu^+$, $n < \omega$, the first targets the question suggests ([theorem 31.2](#) and [corollary 31.6](#)); open for an infinite remainder $\zeta < \mu$ ([section 31.3](#)). The density of every stratum is [theorem 27.2](#).
- (f) *Source 09's direction 7, what survives away from inaccessible truncations* ([section 34.3](#)): answered for the minimal slice; it is η_r and not η_{r+} , $r = \text{cf } \kappa$ ([corollary 30.18](#) and [theorem 30.23](#)).
- (g) *The reals report's question “Other ground orders”* ([lwo:sec:research](#); item (e) of [section 1.7](#)): answered further for $X = \mathbf{No}_{<\kappa}$, every stratum $\lambda = \mu \cdot \gamma$ and every n ([remark 31.8](#)); open for strata that are not of this form, for noncardinal cutoffs and for other alphabets.
- (h) *The reals report's questions on other alphabets* ([lwo:rk:q:alphabets](#), [lwo:sp:q:alphabets](#)): the density and cellularity of every stratum and the incompleteness of every stratum are settled for every infinite linear alphabet ([theorems 27.2](#) and [28.1](#)); the gap spectrum is settled for the minimal slice of every alphabet satisfying the hypotheses of [theorem 30.10](#), which bears on the “general cut-spine transfer theorem” that [lwo:sp:q:alphabets](#) asks for; the gap spectrum of the full order $\mathcal{W}(X)$ stays open.

Source 16 also says that its regular-stage results “answer the earlier report's regular-stage density, cellularity, and cut-character questions under precisely those hypotheses” (end of [section 30.4](#)), and source 15 that its block theorem gives “a precise affirmative continuation of the proposed cardinal-block program” ([section 31.3](#)); items (c) and (e) record the same.

Questions, conclusions and status

34 Research questions

[merge] The four sources' questions are printed per source in the following subsections, as proposed continuations, not as a census of open problems. The index below groups the duplicates, credits every source that asks them, and records the status after the merge (dated 2 October 2026). Questions of source 09 and source 08 are numbered by the order of their subsections and paragraphs in sections 34.3 and 34.4.

Topic	Asked by	Status
Weak-choice strength of a well-order of No	12: remark 34.15	answered by theorem 13.1
Comparison of non-set-like class well-orders	08: question 11	answered by theorems 14.2 and 14.7
Coding lengths at singular cutoffs	08: question 3	answered in part by theorem 8.3
Invariants, cut spectra and isomorphism type at cutoffs	11: research question 34.7 ; 12: research question 34.13 ; 08: questions 1, 2	open; regular case partly by proposition 12.1
Fixed noninitial enumeration types	12: research question 34.11 ; 08: question 4	open (reals report: lwo:sec:research)
Noncardinal cutoffs; away from inaccessible truncations	12: research question 34.12 ; 09: direction 7	open
Represented cuts, exhaustive branches and higher-size cuts	11: research question 34.1 ; 09: direction 3; 08: question 8	open
Full virtual lexicographic products	08: question 7	open
Weakest comprehension and minimal axioms	11: research question 34.2 ; 12: research question 34.16 ; 09: direction 2; 08: question 10	open
Iterated residual decomposition; proper-class blocks	11: research question 34.3 ; 12: research question 34.14	open
Automorphisms; recovering prefix structure; baselines	11: research questions 34.4 and 34.5 ; 12: research question 34.17 ; 09: direction 4; 08: questions 5, 6	open
Birthday and support bounds	11: research question 34.6 ; 09: direction 6; 08: question 13	open
Dependence on the class realization	11: research question 34.8 ; 12: research question 34.18 ; 08: question 12	open
A first verified milestone	11: research question 34.9 ; 09: direction 1; 08: question 13	proposals
Uniform constructions indexed by codes (Kanovei–Shelah)	11: research question 34.10 ; 09: direction 9; 08: question 15	open
Relation-code comparisons	09: direction 5; 08: question 9	open; equimorphism type by proposition 17.1
Termination conventions for set words	09: direction 8; 08: question 14	open
Compatibility between birthday cutoffs	12: research question 34.19	open

34.1 Source 11's research questions and topics

The questions below are proposed extensions of the present development. Except where a source is explicitly named, they are not asserted to be established open problems in the literature. Several ask for a stronger classification than the proved results supply.

34.1.1 A precise description of the represented cuts

Fix the dense ordinary class $\mathcal{P}_{\text{bd}} \cong \mathbf{No}$. Every set-like global enumeration determines a lower cut of $\mathcal{P}_{\text{bd}}$, and so does every injective nonexhaustive Ord-word. The omitted-range example shows that the first collection does not fill every cut generated by the second.

Research question 34.1. Characterize, using the prefix tree and its numerical labels, exactly which class cuts of $\mathcal{P}_{\text{bd}}$ arise from injective Ord-words, and which of these words are exhaustive.

A useful first target is a necessary-and-sufficient compatibility condition on the intersections of a cut with prefix cylinders. The challenge is to separate missing labels, impossible repeated labels, and cuts that escape every coherent branch. This should first be stated for a fixed uniform class cut, before quantifying over an entire hyperclass of cuts.

34.1.2 The embedding relationship between the two global orders

The set-like fragment and binary class words are equimorphic, while the unrestricted order has additional convex fibers. Theorem 20.1 rules out a reverse embedding in the full universe interpretation. Theorem 20.2 also rules out a definable uniform reverse embedding in $\mathbf{KM} + \mathbf{GC}$.

Research question 34.2. What is the weakest comprehension scheme over GBC that proves the uniform nonembedding result? Can the inverse-evaluation step be replaced by an elementary argument, or is a genuine strengthening necessary?

A precise project should specify the syntactic complexity of the proposed transformation and of the comprehension used to invert its code assignment. In the bounded setting one can instead compare the density and cellularity of all well-orders of $\mathbf{No}_{<\kappa}$ with their minimum-length stratum, especially when κ is not inaccessible. The full-universe case is already settled above; the remaining questions concern strength and wider hypotheses.

34.1.3 Iterated residual decomposition

One extraction of the canonical initial Ord-part requires only elementary class comprehension. A residual proper class has another such part, suggesting a hierarchy of successive blocks.

Research question 34.3. What is the exact recursion strength needed to organize iterated residual decompositions uniformly, and what class order types can occur as their block-index orders?

An initial project could analyze finitely many blocks and specified set-length constructions, recording where the recursion state becomes a class. The general case should be compared with known ETR results on class well-order comparability, rather than assumed to follow from the one-step theorem.

34.1.4 Recovering the prefix structure from the bare order

The definitions exhibit a labeled tree of prefixes. An automorphism of the bare lexicographic order need not obviously preserve its displayed labels or cylinders.

Research question 34.4. Which prefix cones, canonical-core fibers, and finite permutation blocks are definable from the linear order alone? What extra predicates make the reconstruction unique?

Adjacency directly detects finite stretches, but a singleton fiber can be exhaustive or can omit one label. Thus local block size alone cannot recover set-likeness. An analysis of the automorphism group may clarify what the order remembers about the original enumeration.

34.1.5 The bounded-support choice of baseline

Every baseline enumeration gives an ordinary class $\mathcal{P}_{\text{bd}}$ abstractly isomorphic to $\mathbf{No}$. If two baselines differ on a proper class of positions, their bounded-support subfamilies can be disjoint.

Research question 34.5. How do the two dense classes sit inside the same $\mathfrak{W}_{\text{sl}}(\mathbf{No})$, and when does an automorphism of that higher-order order carry one onto the other?

The abstract isomorphism of the two dense orders does not automatically extend to all global enumerations. The represented-cut problem is likely relevant. A restricted first case is a baseline change obtained by an order automorphism of the numerical surreal line.

34.1.6 Sharper birthday bounds and minimal codes

The explicit code gives a usable bound

$$\text{bd}(E(f)) = \sum_{i < \text{dom } f} (1 + 2 \cdot \text{bd}(f(i)) + 2).$$

It is designed for a transparent proof, not asserted to be optimal.

Research question 34.6. Which birthday bounds are optimal for uniform embeddings of injective surreal words? Can shorter prefix-free coefficient codes preserve useful simplicity relations?

Regular cutoffs permit same-stage compression; singular cutoffs show why a cardinal bound alone is insufficient. A sharpened result should specify whether it preserves only numerical order, prefix extension, simplicity, or all of these.

34.1.7 Saturation of more general bounded alphabets

If X is an η_κ order of cardinality κ , its minimal-length permutation order inherits η_κ . Birthday fragments often have cardinality $2^{<\kappa}$, which may exceed κ .

Research question 34.7. Determine the exact cut spectrum and point characters of $\mathcal{W}_\lambda(\mathbf{No}_{<\kappa})$, where $\lambda = 2^{<\kappa}$, without assuming $2^{<\kappa} = \kappa$.

The issue combines the size of a used prefix with the number and birthdays of possible numerical separators. The real-alphabet formulas in the precursor report cannot settle this case by substitution.

34.1.8 Class extensions and available global orders

Two models may have the same sets and hence the same individual surreal numbers, but different available classes and different global well-orders.

Research question 34.8. Which features of the lexicographic theory are invariant under extensions of the class part, and how does the represented-cut collection change?

Uniform formulas such as first divergence remain the same for the existing relations. Existence of additional enumerations changes the larger totality. This offers a way to distinguish properties of the labeled surreal order from properties of the surrounding class theory.

34.1.9 A verified first milestone for ProveIt

Research question 34.9. Can the explicit word compression and the direct predecessor comparison be formalized first, with a small, reusable interface independent of surreal arithmetic?

These are concrete, finite-dependency targets: ordinal concatenation, numerical first difference, injectivity, and well-order predecessor sets. The next milestone is the general successor-cardinal obstruction, followed by bounded-support completion and the small-family interpolation theorem. This ordering keeps the central algorithms separate from a formalization of an entire higher-order set theory.

34.1.10 Uniform mathematical constructions indexed by well-orders

The Kanovei–Shelah method suggests using a definable linear order of codes to index constructions without selecting one privileged code. The surreal setting raises additional smallness conditions.

Research question 34.10. Which finite-support or bounded-support constructions can be indexed by a uniform class family of global well-orders, and which require a genuine higher-order index object?

A first investigation should specify its support sets, evaluation maps, and quantifier domains before adding algebra or model theory. The diagonal theorem prevents an exhaustive class-code shortcut, but leaves many useful uniform families available.

[merge] Status of source 11’s questions after the merge: all remain open. Source 12’s dichotomy (theorem 8.3) computes the binary coding length of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$ in every case, but not the cut spectrum or the point characters that research question 34.7 asks for.

34.2 Source 12’s research questions and prioritized program

The following are proposed continuations. Except where an older question is explicitly attributed, they are not claimed to be recognized open problems in the literature.

Research question 34.11 (All fixed enumeration types). Determine $\ell_2(\mathcal{W}_\alpha(\mathbf{No}_{<\kappa}))$ for every ordinal α of cardinality $2^{<\kappa}$, not only the minimal type. Can the answer be expressed by a precise ordinal-sum invariant of admissible entry-code lengths, together with the finite terminal permutation factor?

The singular theorem identifies the first obstruction that any such formula must capture. A useful initial target is $\alpha = \mu \cdot 2$ and then $\mu \cdot \gamma + n$ for finite n .

Research question 34.12 (Noncardinal birthday bounds). For an arbitrary infinite ordinal θ , classify the minimal permutation order of $\mathbf{No}_{<\theta}$. Which features of the ordinal tail of θ , beyond $|\theta|$ and $|\mathbf{No}_{<\theta}|$, survive in its coding rank and local character?

The cone argument for an initial cardinal relies on closure under $i + 1 + \alpha$ below the cutoff. At a noncardinal cutoff that closure can fail and should be replaced by a stratified family of cones.

Research question 34.13 (Isomorphism, not merely equimorphism). Classify the order topology and isomorphism invariants of $\mathcal{W}_\mu(\mathbf{No}_{<\kappa})$. At regular cutoffs, and at singular strong limits, what are its density, cellularity, cut-character spectra, and degrees of homogeneity? Which distinguish it from other orders of the same binary rank?

The binary cube itself cannot be the literal answer: it has endpoints and adjacent pairs, while the surreal permutation order has different local behavior. A dense suborder equimorphic with a cube can retain substantial information not measured by rank.

Research question 34.14 (The hierarchy of proper-class blocks). Study the subpredicates of $\mathfrak{W}_{\text{all}}(\mathbf{No})$ with types $\text{Ord} \cdot \gamma + \alpha$ for fixed set ordinals γ, α . Give explicit, size-safe embedding and nonembedding theorems between these subpredicates and the set-like order.

The inaccessible-model calculation provides concrete bounded targets and shows that the full order is strictly more complex than the set-like order there. Extending binary fusion to uniform transformations on class predicates may require stronger comprehension; that dependence should be proved, not assumed away.

Remark 34.15 (Source 12’s question “The exact weak-choice strength”, answered). *Source 12’s text:* Remove set-level AC from the base theory in [theorem 13.2](#). What principle is equivalent to a class well-order of the sign-sequence surreals then? Compare it with coherent well-ordering of power sets of ordinals and with uniform ordinal presentations of transitive closures.

[merge] *Answered (2 October 2026)*. Source 11’s [theorem 13.1](#) proves the equivalence over GB with no set-level choice: a class well-order of the sign-sequence surreals is equivalent to global choice. Its proof is precisely a uniform well-ordering of the power sets $V_{\beta+1}$, obtained by set-valued recursion from the given well-order on the sign sequences of each length θ ; see [remark 13.4](#). (Corrected on 3 October 2026 after the Hilbert’s-tenth review `review_surreal_synthesis_0be9b9134.md`: “uniform” replaces “coherent”, since these orders need not extend one another; [remark 13.4](#) gives a finite example.)

This is separate from the question whether surreal universality for class linear orders itself implies global choice, raised by Hamkins [21]. The present equivalence concerns a well-order of all surreal labels and does not settle that universality question.

Research question 34.16 (Elementary common-prefix theory). Formalize [theorem 14.7](#) over a precisely chosen weak class theory. How little comprehension and class well-foundedness are needed? Can one characterize the same relation directly by a formula avoiding explicit mention of the common-prefix class?

The definition already expands to elementary quantifiers. A mechanized proof would make the separation from abstract class comparability particularly transparent and yield a reusable theorem for other carriers.

Research question 34.17 (Automorphisms and finite condensation). Determine natural automorphism groups of the full lexicographic order, and of the order after collapsing finite residual blocks. Which operations arise from numerical automorphisms or permutations of the underlying surreal carrier, and which genuinely act on enumeration structure?

Finite-tail blocks are now characterized intrinsically by adjacency. That makes condensation a plausible first invariant, but no complete classification of its quotient is claimed here.

Research question 34.18 (Dependence on the class realization). Fix a transitive set universe M and vary its admissible class collection $\mathcal{X}$. Which external order invariants of the admitted global well-orders detect this change? Can the order reconstruct useful information about available truth predicates or class choice functions?

The diagonal obstruction persists internally, while external cardinality can vary. This makes the distinction between full and Henkin semantics a source of mathematical questions rather than merely a formal warning.

Research question 34.19 (Compatibility between birthday cutoffs). Construct and analyze coherent embeddings from the well-order spaces at smaller birthday cutoffs into those at larger cutoffs. Which choices of reserve labels and tail enumerations are functorial, and which preserve finite condensation, coding ranks, or canonical cylinders?

Naively extending every enumeration by an arbitrary tail can change minimal length, and fixed choices need not be compatible across three cutoffs. A coherence theorem would make universe-lifting arguments much more useful in a proof assistant.

[merge] Status of source 12’s questions after the merge: [remark 34.15](#) is answered (it is printed above as a remark). [Research question 34.11](#) continues the reals report’s question “Other ground orders” for the surreal cutoffs, whose minimal type [theorem 8.3](#) settles. The others remain open.

34.3 Source 09’s research directions

The following are proposed problems emerging from the development, not a claim that all are already catalogued open problems. Each is stated with a foundational level so that its intended meaning is clear.

34.3.1 Formal proof of the surreal core theorem

Framework: universe-indexed Lean with explicit smallness and a baseline parameter. Complete the prefix-extension and interpolation modules, then prove the back-and-forth isomorphism. The difficult part is expected to be the bookkeeping of small histories and variable ordinal prefixes, not the finite first-disagreement arguments. A useful first milestone is the inaccessible or regular-cardinal set-sized version, parameterized by an abstract saturated alphabet.

34.3.2 Exact axiomatic strength of the global conclusions

Framework: GB with ordinary set choice but without global choice. [Proposition 13.7](#) settles the baseline question: a set-like global well-order of **No** is equivalent to global choice. Determine the exact strength of alternative universality and homogeneity assertions that do not already assume such a baseline. In particular, which conclusions can be stated and proved for individually definable class orders without supplying a single uniform global enumeration? Distinguish uniform existence assertions from a schema treating one fixed definition at a time; the latter is not automatically a uniform class construction.

34.3.3 Which cuts of the core represent exhaustive branches?

Framework: class cut parameters over $\mathcal{P}_s(b)$ or an explicitly higher-sorted completion. Every global enumeration determines a cut in the dense core, and density makes this representation injective. Characterize its image intrinsically. The missing-branch construction shows that injectivity of a class word is insufficient: exhaustiveness must be detected. A promising formulation separates coherence of set prefixes, realizability of each next-coordinate cut, and eventual coverage of every surreal value.

34.3.4 Automorphisms beyond set-sized partial maps

Framework: GBC for the core; higher-sort or universe-bounded versions for the full comparison object. Set-sized partial isomorphisms extend in the core, but proper-class partial maps need not. Classify which preserve enough prefix or support structure to extend. Also distinguish automorphisms of the bare order from those preserving the preferred filtration by support bounds. These are different classification problems even though the bare core is always **No**.

34.3.5 Non-set-like well-order relations

Framework: GBC or GBC + ETR, with a fixed relation-code coordinate order. Study the alternative binary comparison of all global well-order relations. Which order properties depend on the coordinate enumeration? Does it contain a natural set-coded dense core, and when are its cuts represented by actual class well-orders? No conclusion about that different comparison follows merely from the enumeration results proved here.

34.3.6 Sharp birthday and support bounds

Framework: ZFC on bounded fragments. Given bounds on the birthdays and first-disagreement coordinates of a set family, determine optimal bounds for an interpolating prefix and its supported completion. The present proof bounds the forced prefix length by $\sup\{\Delta(E_i, F_j) + 1 : i \in I, j \in J\} + 1$ and uses an oversized cardinal for support. That bound is safe but not generally minimal. This direction could yield concrete complexity estimates for symbolic surreal order calculations.

34.3.7 What survives away from inaccessible truncations?

Framework: ZFC with specified cardinal arithmetic. Replace the inaccessible κ by a regular cardinal and a set alphabet with appropriate cut saturation. Separate the roles of regularity, $\kappa^{<\kappa} = \kappa$, and closure under the required cut constructor. At singular cardinals, the first-disagreement bound can reach the ambient length even for relatively small families; identify the resulting interpolation spectrum and construct sharp counterexamples.

34.3.8 Other termination conventions for set words

Framework: GBC or bounded ZFC. Our dictionary order makes proper prefixes smaller and therefore creates the limit-prefix suprema of [note 11.4](#). Study a distinguished internal termination value, restricted alphabets, or prefix-free codes of set well-orders. Determine exactly which variants have the set-cut property and hence are isomorphic to **No**. The convention must be part of the statement, not an implicit analogy with surreal signs.

34.3.9 Uniform infinitary constructions indexed by redundant codes

Framework: bounded universes first. Kanovei–Shelah-style redundant indexing suggests avoiding a choice of one representative for each well-order or auxiliary object. Investigate constructions using the support core or bounded strata as indices, while checking that the resulting products, functions, and quotient relations remain at the declared level. The diagonal theorem forbids a uniform same-level list of all global enumerations, but does not preclude purpose-built redundant families adequate for a particular theorem.

[merge] Status of source 09’s directions after the merge. Direction 2: the baseline question it already settles over $\text{GB} + \text{AC}$ needs no set choice ([theorem 13.1](#)); the strength of the other universality and homogeneity assertions remains open. Direction 3 is source 11’s [research question 34.1](#). Direction 5: source 08’s [proposition 17.1](#) gives the equimorphism type of a relation-code order; the dense core and the representation of its cuts remain open. Direction 7: source 12’s dichotomy treats coding lengths at singular cutoffs, not the interpolation spectrum the direction asks for. The others remain open.

34.4 Source 08’s research questions and topics

The following are directions not resolved by the present manuscript. They are not blanket assertions that no relevant answer exists in the literature. Each should begin with a targeted literature search and a precise choice of size level.

34.4.1 Set-sized enumeration orders

1. Isomorphism classification beyond the common spectrum. The ordinal spectrum and cardinality of $\mathcal{W}(X)$ depend only on $|X|$, while its local cylinder structure remembers the numerical order of residual alphabets. Which additional invariants of X can be recovered from $\mathcal{W}(X)$? Determine sufficient and necessary conditions for $\mathcal{W}(X) \cong \mathcal{W}(Y)$, rather than merely embeddings in both directions. Dense alphabets, well-orders, and mixed sums provide natural test cases.

2. Exact invariants at regular cutoffs ([merge] partly answered; see the note after this list). For regular κ with $2^{<\kappa} = \kappa$, compute the density, cellularity, point characters, and cut cofinalities of $\mathcal{W}_\kappa(\mathbf{No}_{<\kappa})$. The binary equimorphism in ([S08.6](#)) does not by itself determine its order-topological isomorphism type. Investigate whether the support skeleton construction has a natural κ -sized dense realization inside this set order under weaker assumptions than strong inaccessibility.

3. Singular birthdays and code lengths ([merge] answered in part; see the note after this list). At singular κ , exactly which cuts of size below κ fail to be realized in $\mathbf{No}_{<\kappa}$ or in its fixed-length enumeration order? Determine the least binary coding lengths for these orders and how they depend on cofinality and cardinal arithmetic. The three specific uses of regularity in [note 7.6](#) identify where a new argument is needed.

4. Prescribed noninitial lengths. For α of cardinality κ , classify $\mathcal{W}_\alpha(\mathbf{No}_{<\lambda})$. Finite tails are understood through [note 6.25](#), but noninitial limit lengths may preserve finer lexicographic complexity. Compare this problem with the fixed-type analysis proposed in the continuum manuscript [2], checking its hypotheses and proofs independently for nonseparable alphabets.

34.4.2 The skeleton and the virtual ambient order

5. Definable choices of the surreal isomorphism. The back-and-forth isomorphism $\mathcal{P}_s(b) \cong \mathbf{No}$ depends on witness choices. With a specified global well-order, can it be chosen to respect useful support or birthday bounds? What additional normalization gives uniqueness, if any? Distinguish definability from parameters, parameter-free definability, and invariance under a change of baseline.

6. Automorphisms and extension from a skeleton. An order automorphism φ of $\mathbf{No}$ acts on $\mathfrak{W}_{sl}$ by $e \mapsto \varphi \circ e$ and sends O_b onto $O_{\varphi \circ b}$. Which automorphisms of one skeleton extend to uniform automorphisms of the full virtual order? Can order alone recognize eventual-agreement orbits, or are these additional structure supplied by the coordinate presentation?

7. Full virtual lexicographic products. For an ordinary class carrier, every set-indexed lexicographic power of $\mathbf{No}$ embeds into a skeleton. The *full virtual* product $\mathfrak{W}_{sl} \times_{lex} \mathfrak{W}_{sl}$ is a different object. Does it admit a uniform order embedding into $\mathfrak{W}_{sl}$? More generally, determine an absorption spectrum for uniform virtual powers. Simply concatenating two Ord-long words has domain $\text{Ord} + \text{Ord}$, not a set-like Ord domain, so the set-sized concatenation argument is unavailable.

8. Higher-size cuts and completion. The set-cut theorem deliberately leaves out cuts specified by proper class families of enumeration codes or by higher collections of classes. Classify which such cuts are realized once a precise higher-type language is chosen. A proposed Dedekind completion must identify the type of its cut objects and cannot be treated as another ordinary class without a size proof.

9. Classification of relation-code orders. For two global coordinate enumerations q, q' of $\mathbf{No}^2$, the relation-code orders have the same binary equimorphism type. Are there useful hypotheses yielding an isomorphism? Determine which invariants are artifacts of the coordinate order and which reflect genuine properties of class well-orders.

34.4.3 Foundations and formalization

10. Minimal axioms, stated theorem by theorem. Separate the strength of a baseline $\text{Ord} \cong \mathbf{No}$, uniform prefix completion, orbit cut filling, and the global back-and-forth theorem. Over variants of GB with set Choice, which global selection principles are necessary for their uniform formulations? The present proofs establish sufficiency of GBC, not an exact reverse-mathematical classification.

11. Comparison of non-set-like class well-orders ([merge] answered; see the note after this list). Formulate an explicit alignment or comparison principle for arbitrary class well-orders and locate its recursion requirements. Which restricted domains, such as finite sums of Ord and set ordinals, admit direct elementary algorithms? A theorem about all class domains should not be extrapolated from these examples without proof.

12. Dependence on the class collection. For fixed M and $\mathcal{C} \subseteq \mathcal{D}$, compare the virtual orders of global enumerations in $(M, \mathcal{C})$ and $(M, \mathcal{D})$. The same set-coded skeleton relative to a shared baseline remains available, but new class rows may appear. Study which higher-size cut properties, operator embeddings, and definability properties change.

13. Quantitative and certified prefix extension. Formalize the support bound in [note 19.8](#), the forced-prefix length in [note 19.13](#), and the birthday of its stopping entry. This would make the central theorem a sequence of explicit bounded lemmas rather than one monolithic existence proof. It is a strong first target for the repository’s existing small-cut interface.

14. Increasing-subset and alternate-termination orders. Study lexicographic orders of numerically increasing set sequences, as opposed to arbitrary enumerations, and compare prefix-small, prefix-large, and centered-termination conventions. The limit-prefix counterexample shows that these choices affect local cofinalities even when the alphabet is **No**.

15. An order-theoretic interface for class nonstandard models. Determine which part of a Kanovei–Shelah-style indexing construction can use a set-coded skeleton and which genuinely needs a larger collection. Keep the order embedding problem separate from the existence of a class ultrafilter, satisfaction relation, or transfer principle. A successful construction should include an explicit uniform-family and smallness audit at every step.

[merge] Status of source 08’s questions after the merge, dated 2 October 2026. Question 11 is answered: the labelled comparisons of [theorem 14.2](#) and [theorem 14.7](#) compare arbitrary class well-orders of a labelled class with elementary comprehension, without ETR (see the note after [proposition 17.1](#)). Question 3 is answered in part: the least binary coding length of the fixed-length enumeration order at a singular cutoff is $\kappa \cdot \kappa$ at a strong limit and $2^{<\kappa}$ otherwise ([theorem 8.3](#)); which cuts of size below κ fail to be realized remains open. Question 2 is answered in part by [proposition 12.1](#) (the order is η_κ , and all its global and local cofinalities and coinitalities equal κ). Question 4 is source 12’s [research question 34.11](#). The others remain open.

35 The sources’ conclusions

[merge] Source 11 has no conclusion section; its summary is [section 36](#). The conclusions of sources 12, 09 and 08 are printed whole.

35.1 Source 12’s conclusion

There is a rigorous full lexicographic order of all global surreal well-orders, provided it is understood as a relation on class objects, not as an ordinary class containing those objects. Its comparison needs no universal enumeration of abstract class order types: the largest common *labelled* prefix is elementary and sufficient.

The set-like suborder has a particularly uniform geometry. It fills set-indexed cuts, contains every class linear order even locally, and is equimorphic with binary classes on **Ord**. The unrestricted order adds new proper-class prefix phenomena, including finite adjacent blocks. The diagonal theorem prevents construction from being exhausted by a same-level class of set codes.

At set-sized birthday bounds the theory becomes quantitative. The full well-order space of a carrier of cardinality μ has exact ordinal spectrum below μ^+ and binary rank μ^+ . The minimal surreal slice has rank μ , except at singular strong-limit cutoffs, where it has rank $\kappa \cdot \kappa$. This is an order-theoretic effect of cofinal blocks of code lengths, invisible to cardinal counting alone.

The inaccessible-rank interpretation shows exactly how these facts can coexist: internal classes and the external collection of all such classes occupy different levels. Respecting those levels removes the apparent paradox without weakening the embedding and universality questions into trivialities.

35.2 Source 09's conclusions

The lexicographic well-ordering problem has three different answers, depending on its carrier. On a fixed set of surreals of size κ , the exact ordinal spectrum stops at κ^+ . Allowing arbitrary set-length words gives a class-universal dictionary order, but prefix limits give it local order properties different from those of **No**. For set-like global well-orders, first disagreement again defines a linear comparison. Existence of a global enumeration is equivalent to global choice over **GB** with set choice, and all global enumerations cannot be collected into an ordinary class of set codes.

The eventual-agreement core resolves the constructive side of the global problem: it is a genuine class, it is isomorphic to the surreal line, it realizes every uniformly set-sized cut of global enumerations, and it is dense in every global interval. Bounded reflection explains why set-indexed configurations can always be studied in a suitable bounded permutation stage. The explicit missing branch and binary diagonal identify two independent boundaries: arbitrary class-sized cuts need not have solutions, and no single uniform class-indexed coding covers all global enumerations.

These boundaries are not paradoxes. They are the information needed to state theorems at the correct logical level.

35.3 Source 08's conclusion

The global enumeration problem has a particularly clean answer once the types of objects are separated. Set-like well-orders of **No** are represented by class bijections from **Ord**, and numerical first-disagreement gives a rigorous virtual linear order. That order is large enough to encode every class binary word, yet its set-cut geometry is already witnessed by a single class of set-supported perturbations. This code class is another copy of the surreal line.

At a fixed set carrier, the structure is different: its exact ordinal spectrum stops at the successor of the carrier cardinal, and finite permutation tails account for every jump. At varying set carriers, termination introduces new local cofinalities. For non-set-like well-orders, a further index convention or relation code is necessary. None of these distinctions is optional bookkeeping: each changes a mathematical conclusion.

The principal proofs are elementary in their stated foundations. Their formalization should begin with bounded objects and the repository's existing sign-sequence cut theorem, and only then pass to the uniform class formulation. The dense-skeleton theorem provides a concrete target: it combines first-disagreement combinatorics, controlled set support, small-cut filling, and universe-aware back-and-forth without requiring a paradoxical collection of all classes.

36 Results, assumptions, and verification status

Result	Foundation used here
Fixed-alphabet cardinality and ordinal spectrum	ZFC; existing repository theorem
Explicit set-length word compression	ZF, as a definable class construction
Local word characters and set-cut arguments	Surreal set cuts; ZFC for cardinal-valued characters
Any class well-order of No implies global choice	GB with ZF set part
Raw comparison, canonical initial part, finite-jump criterion	GB, with the given class orders
Bounded-support completion and interpolation	GBC, with a fixed global enumeration
$\mathcal{P}_{\text{bd}} \cong \mathbf{No}$ and class-order universality	GBC, by set-valued back-and-forth
Binary class coding and no exhaustive uniform evaluator	GBC, interpreted with class parameters
Unrestricted order does not embed into its set-like fragment	Full inaccessible universe; or KM + GC for definable uniform maps
All higher-order collections treated as actual objects	An explicitly chosen higher-order or containing-universe setting

The proof dependencies are short. Set cuts yield fresh coefficients after deleting a set. Fresh coefficients and bounded-support completion yield strong interpolation. Strong interpolation yields density, local cofinality statements, the **No**-isomorphism, and cut coding. Independently, the predecessor comparison yields convex cones, the canonical initial-part decomposition, and the finite-residual jump theorem. Pair swaps supply both positive binary embeddings and the diagonal obstruction.

The bundled Python script checks finite instances of the two comparison rules and of the sign encoding. It checks 465 coefficient-code comparisons, 79,800 word-code comparisons, and 266,272 pairs of finite permutations, with the predecessor rule checked in both directions and the adjacency criterion compared with the finite lexicographic order. These checks help detect a reversed delimiter or an incorrect finite comparison. They do not verify an ordinal induction, a class comprehension argument, or a cardinal nonembedding theorem.

The article contains full mathematical proofs for the claims it promotes to theorems. It does not claim that the new results have been checked in Lean, that the whole repository has been audited, or that every research question listed here is a previously published open problem.

[merge] The other sources' ledgers follow; each records the hypotheses its source used, not that each is necessary.

36.1 Source 12's proof-dependency and assumption ledger

Result	Stated setting	Essential input
Common labelled prefix and full lexical comparison	GBC; comparison is elementary	Existing well-orders; class minima; ambient linear order.
Global-choice equivalence	GB + set AC	Ordinal coding of transitive closures; restriction of a surreal well-order to a set of codes.
No exhaustive class-coded family	GBC	A baseline Ord-bijection; elementary diagonal pair orientation.

Result	Stated setting	Essential input
Set-like cut interpolation	GBC	Set-sized discrepancy bound; surreal small-cut filling; prefix extension.
Local class universality	GBC	Set-valued forth and back-and-forth recursions; global enumeration.
Finite-tail adjacency	GBC	Label transpositions; finite residual cylinders; no infinite order well-ordered in both directions.
Set-sized full spectra	ZFC	Pair coding; regularity of μ^+ ; cylinder-fusion nonembedding.
Minimal-slice singular transition	ZFC	Cone partition; scheduler; ordinal block sums; regularity or singular cofinality as appropriate.
Full-class rank model	Ambient ZFC + an inaccessible κ	Closure of V_κ ; all external classes; regularity for descending sequences.

36.2 Source 09's theorem-status ledger

Result	Logical setting and proof input	Status here
Surreal set cuts	Set options; standard sign-sequence simplicity	Classical; repository primitives inspected.
Class universality and back-and-forth	GBC; set-valued recursion	Classical mechanism, proved in full.
Global baseline $\Leftrightarrow$ GC	GB plus set choice; well-founded ordinal codes	Standard coding argument, proved here.
Fixed-set cardinality and ordinal spectrum	ZFC; pair switches and regular-cardinal stabilization	Generalization/reproof of the set-sized route.
Limit-prefix character of $\mathcal{I}$	ZFC schema or GBC class statement	Written proof.
Bounded-support extension	GBC with fixed b ; set cardinal arithmetic	Written proof; finite mechanism tested.
Core interpolation and $\mathcal{P}_s(b) \cong \mathbf{No}$	GBC; uniform families, small cuts, extension	Main construction; written proof only.
Bounded reflection	GBC; Replacement for a uniform set family	Written proof.
Missing (Ord, Ord) branch	GBC; ordinal prefixes and uniform completions	Explicit counterexample with proof.
Uniform no-coding	GBC; pair switches and diagonal class	Written proof; finite diagonal tested.
Inaccessible model	ZFC plus a strongly inaccessible cardinal	Conditional set-sized theorem, proved here.
Proposed Lean modules	Explicit universes, smallness, and baseline parameter	Plan only; not compiled or verified.

36.3 Source 08’s assumption and proof-dependency ledger

Result	Assumptions	Main dependencies
Fixed-carrier cardinality and universality	ZFC, $ X = \kappa$ infinite	Pair blocks and lower-cut bits
Exact ordinal spectrum	Same	Regularity of κ^+ , not of κ
Adjacency classification	Same	Extreme residual enumerations, two-sided well-foundedness
Regular cutoff theorem	κ regular, $2^{<\kappa} = \kappa$	Birthday bounds, terminal coding, active-row cuts
Pairwise global comparison	Class parameters, elementary comprehension	A least disagreement below one witness
Orbit prefix extension	Baseline b , class replacement, set Choice	Enlargement to a bigger set cardinal
Orbit interpolation	Baseline and uniform set families	Cross-disagreement bound, fresh entry, prefix extension
Skeleton isomorphism and class universality	GBC	Global enumerations, uniform set-valued recursion
Virtual binary equimorphism	Baseline, elementary class operations	Pair blocks and ordinal concatenation
No exhaustive class-uniform list	GBC for arbitrary class indices	Pairwise diagonalization
Full-class model example	External strongly inaccessible κ	Relative sets versus external sets

The ledger records the hypotheses actually used in the supplied proofs. It does not claim that every hypothesis is logically necessary. In particular, the foundational optimization questions in [section 34.4](#) remain distinct from the sufficiency results.

A Source crosswalk and reproducibility

The immutable sources used in the targeted audit are recorded in the accompanying source audit. For convenient verification, the most relevant mappings are:

Source	Relevant content
[1]	Labels <code>lwo:lem:prefix-free</code> , <code>lwo:lem:cylinder</code> , <code>lwo:prop:cardinality</code> , <code>lwo:thm:no-long</code> , <code>lwo:thm:ordinal-spectrum</code> ; bounded sign orders in <code>lwo:sp:thm:surreals</code> .
[3]	Set-coded surreals and bounded birthdays; unrestricted cut obstruction; class-quantification and universe distinctions; set-valued recursion.
[7]	Actual sign carrier, ordinal embedding, smallness obstruction, and bounded fragments.
[9]	Small cut operation, separator properties, and birthday bounds.
[5]	A nearby use of a set-like global well-order and ordinary class-parameter recursion.

To compile the delivered standalone source, run `latexmk -pdf surreal_well_orders.tex` from its directory. All article sections and the bibliography are included in that single file; no separate bibliography processor is needed. Run `python3 finite_checks.py` for the finite checks. The supplied PDF was compiled from the included sources and inspected after rendering.

[merge] *Delivery names.* The paragraph above describes source 11’s delivered package. Its `surreal_well_orders.tex` is shipped as `article.tex`, which this merged text replaces; its PDF is not shipped (`article.pdf` is a build of this text); its `finite_checks.py` and `finite_checks_results.txt` are `code/11-raw-orders-finite_checks.py` and `data/11-raw-orders-finite_checks_results.txt`; the “accompanying source audit” is `11-raw-orders-repository_audit.md`. Build this report with `latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex`, and rerun the checks as in [section B](#). The same applies to the reproducibility records of the other sources that follow.

A.1 Source 12: reproducibility and inspection record

The package contains `article.tex`, the compiled `article.pdf`, a build script, an exact finite-check program, its recorded JSON output, and a research-status and source-audit file. The Python program uses only the standard library. The recorded run passed all 180,905 finite comparison cases. Its comparisons are exact; there are no floating-point tolerances or inferred transfinite results. The PDF was compiled and rendered for visual inspection.

The repository audit directly inspected the merged real-well-order README and the sign-sequence simplicity Lean source at the stated commit. It did not independently compile the repository or verify the predecessor report’s proofs. The needed predecessor patterns have therefore been reproved in the present manuscript. Literature checking was focused on surreal foundations, Kanovei–Shelah indexing, and class recursion and well-orders; it was not sufficient to certify historical originality.

A.2 Source 09: repository and source audit

The repository was inspected through its GitHub connector. All cited repository files are pinned to

63bb8b1d99359e7188abf0a33fb152e121858150.

A later search surfaced a newer commit; the cited foundational files were fetched at the earlier pinned snapshot rather than mixing versions. No repository file was changed.

The targeted inspection covered the merged real well-ordering report’s README and the relevant returned source of the sign-sequence and cut-operation modules. The real report identifies itself as unrefereed and not formalized. Its README describes a merged 54-page report and separates its results from nearby Lean developments. That report is used as research context, not as a kernel-certified dependency.

The sign-sequence file provides actual Lean declarations for the universe-indexed carrier, order, ordinal embedding, and smallness properties; the cut file provides the small-cut operation and its birthday bound. Their source statements were read, but this project did not rebuild the repository or audit all transitive dependencies. No claim of independent kernel verification is made.

The primary literature checked includes Kanovei–Shelah’s original index definition, Rangel–Mariano’s explicit set/class framework, and the class-recursion distinctions in Gitman–Hamkins–Holy–Schlicht–Williams. The Kanovei–Shelah index page was also visually inspected. The literature search was targeted rather than exhaustive. General universality and back-and-forth arguments are classical; the support-core organization and the accompanying obstruction statements are proved here without a priority claim.

A.3 Source 09: verification and reproducibility

The accompanying `code/finite_checks.py` uses only Python’s standard library. It checks finite versions of the pair-switch lexicographic embedding, partial-prefix permutation completion, the forced-prefix separator algorithm, support extension preserving first disagreements, and the diagonal switch construction. Exact rational arithmetic is used for the separator tests. A deterministic random seed makes the sampled checks reproducible; other suites exhaust their specified finite ranges.

These checks do not verify any transfinite theorem. Their role is to detect indexing, orientation, and extension mistakes in the finite combinatorial mechanisms. Proofs of the infinite statements are the written arguments in the article. The recorded run passed 164,560 assertions, including 1,101 separator-family cases. The test report is `data/finite_checks.json`. The archive also contains a build script and a theorem-status ledger.

To compile the article, run `bash build.sh` in the archive’s root directory. It makes three passes with `pdflatex`, resolves the internal bibliography and cross-references, and writes build products into `build/` before copying the finished PDF to the root.

A.4 Source 08: finite regression checks

The package includes `verify_finite.py`, requiring Python 3.10 or later and only its standard library. Running it writes `verification_results.json`. The supplied run passed:

Check	Instances
Consecutive permutation pairs, sizes 1 through 7	5,906
All increasing pairs of distinct permutations, sizes through 5	7,432
Pair-code order comparisons, bit lengths 0 through 7	21,845
Sign words of lengths 0 through 6	127
All sign-code order and prefix-free comparisons	16,129

The permutation test checks both directions of the adjacency rule on all manageable pairs, not just a few positive examples. The sign test compares the numerical convention $- < 0 < +$ with the binary blocks $00 < 01 < 11$, including pairs in which one sign word is a proper prefix. The pair-code test checks that every output is a permutation of the correct finite alphabet and that comparisons agree with the input binary order.

These tests can detect a reversed pair orientation, a wrong terminal block, or a mistaken finite-tail rule. They cannot prove regularity arguments, transfinite stabilization, set/class separation, or the existence of a global isomorphism. Those depend on the mathematical proofs in the text.

A.5 Source 08: source audit and reproducibility

Two mathematical source files were inspected. One is `SignSequence.lean`; the other is `SignSequenceCut.lean`. Both were read at the pinned ProveIt commit. The project’s README was used only for its stated organization and audit policy. No build was run, and no claim was imported merely because a research report’s title suggested it. The source files have blob hashes

```
SignSequence      e2c093860f29b0b9ea312d50099fc4aa89a76a99
SignSequenceCut   c5c35afd98a5ad293687367ee9fb0244f94cc91c
```

The Kanovei–Shelah preprint was inspected directly, including the page defining its index. The class-recursion references were consulted through the author’s research summaries. Classical book and journal references provide background attribution; no claim of an exhaustive literature or

priority search is made. The prior continuum manuscript was identified in the user's file library and used as a contextual predecessor, not as a formally certified dependency.

The package contains a self-contained L^AT_EX source with an embedded bibliography, the compiled PDF, a build script, the finite verification script and its output, and a README. No external figure or private font file is needed. Compile with a standard TeX Live installation using `latexmk -pdf surreal_well_orders.tex`, or run `pdflatex` sufficiently many times to resolve cross-references.

B The finite checks as shipped

[merge] All four programs use only the Python standard library and check finite instances; they verify no transfinite, class-theoretic or cardinal statement. Rerun on 2 October 2026 on copies, with Python 3.14.4 on Windows, each reproduced its recorded output exactly apart from line endings: on Windows all four write CRLF, while the records have LF, so compare with `diff -strip-trailing-cr`.

Source	Program and record	Checks	Output
11	<code>code/11-raw-orders-finite_checks.py</code> ; <code>data/11-raw-orders-finite_checks_results.txt</code>	465 sign-code pairs; 79,800 word-code pairs; predecessor comparison and adjacency on all 266,272 pairs of permutations of at most 6 points (867 adjacent)	prints to standard output
12	<code>code/12-singular-finite_checks.py</code> ; <code>data/12-singular-finite_checks.json</code>	180,905 cases: sign code, pair orientation, common prefix and adjacency, larger adjacent pairs, row scheduler	writes the file named by <code>-output</code>
09	<code>code/09-core-finite_checks.py</code> ; <code>data/09-core-finite_checks.json</code>	164,560 assertions, 1,101 separator families, seed 20261002: pair switch, prefix completion, forced separator, bounded reflection, diagonal switch	default <code>-output</code> is <code>data/finite_checks.json</code> beside <code>code/</code> , an unprefixed new file; always pass <code>-output</code>
08	<code>code/08-skeleton-verify_finite.py</code> ; <code>data/08-skeleton-verification_results.json</code>	5,906 consecutive and 7,432 ordered pairs of permutations; 21,845 pair-code comparisons; 16,129 sign-code comparisons on 127 words	writes <code>verification_results.json</code> beside itself

Run them on copies, never in the report directory. From the report directory, with an empty scratch directory `$S`:

```
cp code/08-skeleton-verify_finite.py "$S/"
python "$S/08-skeleton-verify_finite.py"          # writes $S/verification_results.json
python code/09-core-finite_checks.py --output "$S/09.json"
python code/11-raw-orders-finite_checks.py > "$S/11.txt"
python code/12-singular-finite_checks.py --output "$S/12.json"
```

and compare with the four records in `data/`. The delivered build scripts build nothing in this layout: `code/08-skeleton-build.sh` runs `verify_finite.py` and compiles `surreal_well_orders.tex` beside itself, `code/09-core-build.sh` compiles an `article.tex` in `code/`, and `code/12-singular-build.sh` changes into `code/` and then calls `code/finite_checks.py`. They are kept as delivered.

C The other sources' title pages and abstracts

[merge] The results these summaries announce are printed in the body; the summaries are kept for the record.

Source 12

Title-page summary. Main distinction. An arbitrary class well-order of **No** need not have order type Ord . Nevertheless, two such well-orders have an *elementarily definable largest common labelled initial segment*, possibly a proper class. This supplies a lexicographic comparison in GBC without invoking general class well-order comparability or ETR.

Principal quantitative result. Put $\mathbf{No}_{<\kappa} = \{s : \alpha \rightarrow \{-, +\} : \alpha < \kappa\}$ and $\mu = |\mathbf{No}_{<\kappa}| = 2^{<\kappa}$, for an infinite initial cardinal κ . The minimal-length permutation order satisfies

$$\mathcal{W}_\mu(\mathbf{No}_{<\kappa}) \simeq_e \begin{cases} \mathbb{B}_{\kappa \cdot \kappa}, & \mu = \kappa \text{ and } \text{cf } \kappa < \kappa, \\ \mathbb{B}_\mu, & \text{otherwise,} \end{cases}$$

where $\mathbb{B}_\gamma = (2^\gamma, <_{\text{lex}})$ and $\simeq_e$ means mutual order-embeddability, not isomorphism.

Status. This is an AI-assisted, unrefereed research manuscript. Proofs are supplied for the stated results; historical priority is not claimed. The accompanying exact finite tests do not verify the transfinite arguments. No new Lean formalization is claimed.

Abstract. We study the lexicographic order of well-orderings of the surreal numbers, carefully distinguishing bounded sets of surreals, set-like global well-orders, and arbitrary global class well-orders. For the full class problem, ordinal-indexed enumerations are insufficient. We construct the largest common labelled initial segment of two well-orders directly from their relations, prove that comparison of their next elements defines a linear order, and show that this construction needs only elementary class comprehension once the well-orders exist. Over GB with set choice, the existence of any class well-order of the surreals is equivalent to global choice. The collection of all these well-orders is not itself an ordinary class of GBC or Kelley–Morse theory; a diagonal theorem rules out any uniform exhaustive coding by a class of sets.

For set-like global well-orders we prove density, interpolation for set-indexed cuts, local universality for class linear orders, and mutual embeddability with the lexicographic comparison of binary classes on Ord . For unrestricted global well-orders we classify adjacent pairs: they arise precisely from finite residual permutations after a common, necessarily proper-class, prefix. At a birthday cutoff κ , we obtain an exact regular/singular dichotomy for the binary coding rank of minimal-length enumerations. We also establish the exact ordinal and binary-cube embedding spectra for the full set-sized order. Finally, a full-class model based on an inaccessible rank explains exactly how the bounded and global theories fit together, and provides a rigorous example in which the full order and the set-like order have the same cardinality and ordinal spectrum but different coding ranks. Explicit proof-assistant interfaces and further research questions are included.

Source 09

Abstract. The phrase “the lexicographic order of all well-orders of the surreal numbers” has several mathematically different interpretations. We develop three: exhaustive enumerations of a fixed set of surreals; all set-length injective surreal words; and class bijections from the ordinals onto the surreal numbers, representing set-like global well-orders. For an infinite set X of cardinality κ ,

we give a self-contained proof that its lexicographic well-ordering space has cardinality 2^κ , contains every linear order of size at most κ , and contains an ordinal precisely when that ordinal is below κ^+ .

For the global problem, we work in Gödel–Bernays set theory with global choice and do not pretend that a collection of proper classes is itself an ordinary class. Fix a global enumeration b . Its eventually-equal perturbations admit canonical set codes. We prove that their lexicographic order is isomorphic to the surreal line, meets every set-prefix cylinder, and interpolates every uniformly set-indexed separated pair of families of arbitrary global enumerations. Thus every class linear order embeds into this genuine class core. We also prove bounded reflection for set-indexed families, construct an explicit unfilled (Ord, Ord) -configuration, and diagonalize against every uniform class-indexed list of all global enumerations.

A strongly inaccessible truncation gives a fully set-sized model of the size distinction: the core has cardinality κ , whereas all length κ enumerations have cardinality 2^κ . The article includes foundational and Lean formalization plans, a repository audit, finite verification code, and a focused research agenda.

Title-page status. Status: AI-assisted research manuscript; complete written arguments are provided for the stated results. Not peer-reviewed or Lean-verified. Classical ingredients and repository results are distinguished from the constructions developed here. No claim of historical priority is made.

Source 08

Abstract. There is no single unqualified object called “the lexicographic order of all well-orders of the surreal numbers.” We distinguish well-orderings of a fixed set of surreals, well-orderings of varying set-sized subsets, set-like well-orderings of the whole surreal class, and arbitrary class well-orderings. For an infinite ordered set X of cardinality κ , the exhaustive-enumeration order $\mathcal{W}(X)$ has size 2^κ , contains every linear order of size at most κ , and contains an ordinal β exactly when $\beta < \kappa^+$. Its adjacent pairs arise precisely from finite permutation tails.

For the whole surreal class, work in Gödel–Bernays set theory with Global Choice. The numerical first-disagreement comparison of class bijections $\text{Ord} \rightarrow \mathbf{No}$ is meaningful without treating classes as elements of classes. Fix one such bijection b . We prove that every set-sized separated family of global enumerations has a strict separator obtained from b by a permutation of set support. The class of canonical codes for these perturbations is therefore order-isomorphic to $\mathbf{No}$. It is a dense, set-cut-realizing skeleton inside the much larger virtual enumeration order, and every class linear order embeds into it. The full virtual order is equimorphic with the lexicographic order of class binary sequences, but admits no class-uniform exhaustive parametrization. We supply proofs, cutoff versions, counterexamples to tempting identifications, a universe-sensitive Lean implementation plan tied to inspected ProveIt sources, and a research agenda.

Status box.

Status and scope. The results below are mathematical derivations with explicit proofs, not claims of historical priority. The manuscript has not been independently refereed, and its new results have not been checked by a proof assistant. Repository declarations are identified separately from proposed extensions. The accompanying executable checks cover finite coding and adjacency examples only. All uses of “class” are relative to the stated foundational model.

D Crosswalk of the sources' numbered statements

[merge] Each row gives a numbered statement of a source (numbered as in that source's own PDF), and where it is printed or recorded here. A note records a statement proved here by the same argument; a section labelled "route" prints a different argument. Labels are the source's own with its prefix, or `swo:sub:number` for unlabelled statements.

D.1 Source 11 (base)

No.	Statement in source 11	Printed here as
3.1	Definition	Definition 5.1, p. 21
3.2	Lemma (Deleting a set)	Lemma 5.2, p. 21
3.3	Proposition	Proposition 5.13, p. 27
4.1	Theorem (Cardinality and exact ordinal spectrum)	Theorem 6.1, p. 30
4.2	Proposition (Exactly where jumps occur)	Proposition 6.2, p. 31
5.1	Corollary	Corollary 7.1, p. 37
5.2	Lemma	Lemma 7.2, p. 37
6.1	Theorem (Explicit sign coding)	Theorem 9.1, p. 42
6.2	Proposition (Birthday bounds)	Proposition 9.2, p. 43
7.1	Theorem (Exact local character)	Theorem 10.1, p. 44
7.2	Corollary	Corollary 10.2, p. 44
8.1	Theorem (Word-cut classification)	Theorem 11.1, p. 45
8.2	Corollary (Successor words)	Corollary 11.2, p. 45
9.1	Proposition	Proposition 12.1, p. 47
10.1	Theorem (Existence and global choice)	Theorem 13.1, p. 48
11.1	Lemma (Unique first disagreement)	Lemma 14.1, p. 52
11.2	Theorem (Raw lexicographic order)	Theorem 14.2, p. 53
11.3	Proposition (Convex prefix cones)	Proposition 14.3, p. 53
12.1	Theorem (Canonical initial part)	Theorem 15.1, p. 55
12.2	Example (Two proper blocks)	Example 15.2, p. 56
12.3	Proposition (Convex fibers)	Proposition 15.3, p. 56
12.4	Remark (Singleton fibers and iteration)	Remark 15.4, p. 56
13.1	Lemma (Extreme enumerations)	Lemma 16.1, p. 57
13.2	Theorem (Exact adjacency criterion)	Theorem 16.2, p. 57
13.3	Example (An adjacent pair of type $\text{Ord} + 2$)	Example 16.3, p. 58
14.1	Definition	Definition 19.1, p. 62
14.2	Definition	Definition 19.3, p. 63
14.3	Lemma (Completion of an arbitrary set prefix)	Lemma 19.4, p. 63
14.4	Theorem (Prefix interpolation)	Theorem 19.10, p. 66
14.5	Corollary	Corollary 19.15, p. 67
14.6	Corollary (The class of realizations of a set cut)	Corollary 19.18, p. 68
14.7	Proposition	Proposition 19.20, p. 68
14.8	Proposition (Prefix cylinders)	Proposition 19.25, p. 70
14.9	Proposition (Approximants at every point)	Proposition 19.27, p. 70
14.10	Corollary (Set saturation is not Dedekind completeness)	Corollary 19.29, p. 71
14.11	Theorem	Theorem 19.32, p. 73
14.12	Corollary	Corollary 19.36, p. 74
14.13	Theorem (No exhaustive uniform class of codes)	Theorem 19.37, p. 75
14.14	Theorem (An omitted-range cut)	Theorem 19.42, p. 77
14.15	Example (Two omitted points produce adjacent raw orders)	Example 19.45, p. 78

No.	Statement in source 11	Printed here as
14.16	Corollary (Exactly which raw set cuts admit bounded completions)	Corollary 19.46, p. 78
15.1	Theorem (Nonembedding in a full universe)	Theorem 20.1, p. 81
15.2	Theorem	Theorem 20.2, p. 81
16.1	Research question	Research question 34.1, p. 128
16.2	Research question	Research question 34.2, p. 128
16.3	Research question	Research question 34.3, p. 128
16.4	Research question	Research question 34.4, p. 129
16.5	Research question	Research question 34.5, p. 129
16.6	Research question	Research question 34.6, p. 129
16.7	Research question	Research question 34.7, p. 129
16.8	Research question	Research question 34.8, p. 130
16.9	Research question	Research question 34.9, p. 130
16.10	Research question	Research question 34.10, p. 130

D.2 Source 12

No.	Statement in source 12	Printed here as
3.1	Definition (Common labelled prefix)	Definition 14.4, p. 54
3.2	Lemma (Maximal-prefix lemma)	Lemma 14.5, p. 54
3.3	Definition (Full lexicographic comparison)	Definition 14.6, p. 54
3.4	Theorem (Full class lexicographic order)	Theorem 14.7, p. 54
3.5	Remark (Why this does not prove class order-type comparability)	Remark 14.8, p. 55
3.6	Example (A disagreement after every ordinary ordinal)	Example 14.9, p. 55
3.7	Corollary (Agreement with enumeration lexicography)	Corollary 14.10, p. 55
4.1	Theorem (Choice equivalence)	Theorem 13.2, p. 49
4.2	Remark (The explicit set-choice hypothesis)	Remark 13.3, p. 50
5.1	Theorem (No exhaustive class-coded family)	Theorem 19.38, p. 75
5.2	Corollary (No uniform set-code solution)	Corollary 19.39, p. 76
6.1	Lemma (Extending a set prefix)	Lemma 19.5, p. 64
6.2	Lemma (Avoiding a forbidden set)	Lemma 5.3, p. 21
6.3	Theorem (Set-indexed interpolation)	Note 19.11, p. 66
6.4	Proposition (Cofinality and local character)	Note 19.28, p. 71
6.5	Lemma (Surreal universality, with the size of the recursion exposed)	Lemma 5.14, p. 28
6.6	Lemma (Deleting a set does not change the numerical class order)	Lemma 5.15, p. 28
6.7	Theorem (Local class universality and cylinders)	Note 19.21, p. 69
7.1	Lemma (A self-delimiting numerical code)	Lemma 19.33, p. 73
7.2	Theorem (Binary-class equimorphism)	Theorem 19.34, p. 73
8.1	Definition (A prefix cylinder in the full order)	Definition 16.4, p. 58
8.2	Proposition (Residual-order decomposition)	Proposition 16.5, p. 58
8.3	Theorem (Exact adjacency criterion)	Theorem 16.6, p. 59
8.4	Corollary (Proper-class prefixes are responsible)	Corollary 16.7, p. 59
9.1	Lemma (Prefix-freeness of exhaustive enumerations)	Lemma 6.3, p. 32
9.2	Lemma (Strict binary hierarchy)	Lemma 6.4, p. 33
9.3	Lemma (Characteristic-cut coding)	Lemma 6.5, p. 33

No.	Statement in source 12	Printed here as
9.4	Lemma (Pair orientation coding)	Lemma 6.6, p. 33
9.5	Lemma (No μ^+ -long monotone chain)	Lemma 6.7, p. 33
9.6	Theorem (Cardinality and exact spectra)	Theorem 6.8, p. 34
9.7	Remark (What the rank does not say)	Remark 6.9, p. 34
10.1	Lemma (Row scheduler)	Lemma 8.1, p. 40
10.2	Lemma (Surreal cone decomposition)	Lemma 8.2, p. 40
10.3	Theorem (Exact minimal-slice dichotomy)	Theorem 8.3, p. 40
10.4	Corollary (A complete cube test for the minimal slice)	Corollary 8.4, p. 41
10.5	Example (An unconditional singular example)	Example 8.5, p. 41
10.6	Example (Regular cutoffs)	Example 8.6, p. 41
11.1	Theorem (Exact external identification)	Theorem 21.1, p. 84
14.1	Research question (All fixed enumeration types)	Research question 34.11, p. 130
14.2	Research question (Noncardinal birthday bounds)	Research question 34.12, p. 131
14.3	Research question (Isomorphism, not merely equimorphism)	Research question 34.13, p. 131
14.4	Research question (The hierarchy of proper-class blocks)	Research question 34.14, p. 131
14.5	Research question (The exact weak-choice strength)	Remark 34.15, p. 131
14.6	Research question (Elementary common-prefix theory)	Research question 34.16, p. 131
14.7	Research question (Automorphisms and finite condensation)	Research question 34.17, p. 131
14.8	Research question (Dependence on the class realization)	Research question 34.18, p. 132
14.9	Research question (Compatibility between birthday cutoffs)	Research question 34.19, p. 132

D.3 Source 09

No.	Statement in source 09	Printed here as
2.1	Definition	Definition 5.9, p. 25
2.2	Lemma (Avoiding a prescribed set)	Lemma 5.10, p. 26
3.1	Theorem (Class universality)	Theorem 5.16, p. 28
3.2	Theorem (Uniqueness and set-homogeneity)	Theorem 5.17, p. 29
3.3	Corollary	Corollary 5.18, p. 29
4.1	Definition	Note 6.10, p. 34
4.2	Lemma (Exhaustiveness removes the prefix ambiguity)	Note 6.11, p. 34
4.3	Theorem (Cardinality and universality)	Note 6.12, p. 34
4.4	Lemma (Eventual stabilization)	Lemma 6.13, p. 35
4.5	Theorem (Exact ordinal spectrum)	Theorem 6.14, p. 35
4.6	Remark	Remark 6.15, p. 35
4.7	Proposition	Note 6.16, p. 35
4.8	Proposition (Finite residuals are exactly the adjacency mechanism)	Note 6.17, p. 35
5.1	Proposition	Proposition 7.3, p. 38
6.1	Proposition	Proposition 11.3, p. 46
6.2	Proposition (Limit prefixes create small one-sided character)	Note 11.4, p. 46

No.	Statement in source 09	Printed here as
7.1	Definition	Definition 13.5, p. 51
7.2	Proposition	Proposition 13.6, p. 51
7.3	Proposition (The baseline already entails global choice)	Proposition 13.7, p. 51
8.1	Lemma (Bounded-support prefix completion)	Lemma 19.6, p. 65
8.2	Remark (Why extra unused positions matter)	Remark 19.7, p. 65
9.1	Theorem (Uniform set-indexed interpolation)	Note 19.12, p. 67
9.2	Corollary (The surreal core theorem)	Note 19.16, p. 67
9.3	Corollary (Dense copies in every interval)	Note 19.19, p. 68
10.1	Theorem (Bounded reflection)	Theorem 19.30, p. 71
10.2	Theorem	Note 19.22, p. 69
10.3	Proposition (Representable suborders do not exceed surreal universality)	Proposition 19.24, p. 69
10.4	Proposition (Cylinder self-similarity)	Note 19.26, p. 70
11.1	Theorem (An explicit unfilled (Ord, Ord) configuration)	Theorem 19.43, p. 78
11.2	Remark (A general missing-branch mechanism)	Remark 19.44, p. 78
12.1	Theorem (Uniform no-coding theorem)	Note 19.40, p. 76
13.1	Theorem (Inaccessible set-sized model)	Theorem 22.1, p. 85

D.4 Source 08

No.	Statement in source 08	Printed here as
2.1	Proposition	Proposition 5.4, p. 22
2.2	Lemma	Lemma 5.5, p. 22
2.3	Theorem (Birthday-controlled cut filling)	Theorem 5.6, p. 22
2.4	Corollary (Fresh separators)	Corollary 5.7, p. 23
2.5	Definition	Definition 5.8, p. 23
3.1	Lemma (No prefix ambiguity)	Note 6.18, p. 36
3.2	Lemma (Pair coding)	Note 6.19, p. 36
3.3	Lemma	Note 6.20, p. 36
3.4	Theorem	Note 6.21, p. 36
4.1	Theorem (Prefix stabilization)	Note 6.22, p. 36
4.2	Corollary (Exact spectra)	Note 6.23, p. 36
4.3	Proposition (A uniform binary upper coding)	Note 6.24, p. 36
5.1	Theorem (Adjacency classification)	Note 6.25, p. 36
5.2	Corollary (Finite condensation blocks)	Note 6.26, p. 37
6.1	Proposition	Note 7.4, p. 39
6.2	Lemma	Note 7.5, p. 39
6.3	Theorem (The initial-length cutoff order)	Note 7.6, p. 39
6.4	Remark	Note 7.7, p. 39
7.1	Proposition	Note 11.5, p. 47
8.1	Definition	Definition 5.11, p. 27
8.2	Theorem	Note 5.12, p. 27
9.1	Proposition	Note 19.2, p. 62
10.1	Lemma (Prefix extension inside one orbit)	Note 19.8, p. 65
10.2	Corollary	Note 19.9, p. 65
11.1	Theorem (Orbit interpolation)	Note 19.13, p. 67
11.2	Corollary	Note 19.14, p. 67
12.1	Theorem (Uniqueness of a class saturated line)	Note 5.19, p. 29

No.	Statement in source 08	Printed here as
12.2	Theorem (Surreal-sized skeleton)	Note 19.17, p. 67
12.3	Theorem (Universality for class linear orders)	Note 5.20, p. 29
12.4	Corollary	Note 19.23, p. 69
12.5	Proposition	Proposition 19.31, p. 72
13.1	Theorem (Virtual binary equimorphism)	Note 19.35, p. 74
13.2	Theorem (No class-uniform exhaustive list)	Note 19.41, p. 76
14.1	Proposition	Proposition 17.1, p. 60

References

- [1] VladimirReshetnikov/ProveIt repository, *Lexicographic orders of the well-orderings of the reals*, merged research report, 2 October 2026, `SetTheory/Cardinals/docs/reports/ordinals-and-order-types/lexicographic-well-orderings-of-reals/`. Source 11 read its article at revision `ae7c1e6aa`; sources 09 and 12 read its README at revision `63bb8b1d9`; source 08 did not read it (see [2]). The article has not changed since `603f6ed31`.
- [2] *Lexicographic Orders of Well-Orderings of the Continuum: Ordinal spectra, lexicographic complexity, finite-tail geometry, and self-absorption*, research manuscript prepared for Vladimir Reshetnikov, 2 October 2026, 24 pages; batch-76 manuscript 02, the base source 02 of [1]. Source 08 cites it as a “prior working manuscript in the user’s library, supplied there as `article(20261002-170841).pdf` and corresponding L^AT_EX source”; unrefereed.
- [3] VladimirReshetnikov/ProveIt repository, *Foundations for surreal and surcomplex mathematics*, research report, `Algebra/SurrealNumbers/docs/foundations-and-computation/foundations/`; read by source 11 at revision `ae7c1e6aa`.
- [4] VladimirReshetnikov/ProveIt repository, *Birthday cutoffs and hereditary sets*, merged research report, `Algebra/SurrealNumbers/docs/foundations-and-computation/birthday-cutoffs-and-hereditary-sets/`.
- [5] VladimirReshetnikov/ProveIt repository, *The surreal numbers as a real vector space*, research report, `Algebra/SurrealNumbers/docs/surreal/real-vector-space-structure/`; read by source 11 at revision `ae7c1e6aa`.
- [6] VladimirReshetnikov/ProveIt repository, surreal-number project README, `Algebra/SurrealNumbers/README.md`; read by source 08 at revision `ae7c1e6aa`.
- [7] ProveIt contributors, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequence.lean`; read by sources 11 and 08 at revision `ae7c1e6aa` (blob `e2c093860f29b0b9ea312d50099fc4aa89a76a99`) and by source 09 at `63bb8b1d9`.
- [8] ProveIt contributors, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceCut.lean`; read by source 08 at revision `ae7c1e6aa` (blob `c5c35afd98a5ad293687367ee9fb0244f94cc91c`).
- [9] ProveIt contributors, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceCutOperation.lean`; read by source 11 at revision `ae7c1e6aa` and by source 09 at `63bb8b1d9`.
- [10] ProveIt contributors, `Algebra/SurrealNumbers/Surreal/Foundations/SignSequenceSimplicity.lean`; read by source 12 at revision `63bb8b1d9` and by source 11 at `ae7c1e6aa`. It proves minimum-birthday and simplest-separator uniqueness; source 12 did not rebuild it.
- [11] J. H. Conway, *On Numbers and Games*, Academic Press, 1976; second edition, A K Peters, 2001.
- [12] Harry Gonshor, *An Introduction to the Theory of Surreal Numbers*, London Mathematical Society Lecture Note Series 110, Cambridge University Press, 1986. doi:10.1017/CBO9780511629143.

- [13] N. L. Alling, Conway’s field of surreal numbers, *Transactions of the American Mathematical Society* **287** (1985), 365–386. doi:10.1090/S0002-9947-1985-0766225-7.
- [14] Vincent Bagayoko and Joris van der Hoeven, *Surreal substructures*, *Fundamenta Mathematicae* 266 (2024), 25–96. doi:10.4064/fm231020-23-2. arXiv:2305.02001.
- [15] Philip Ehrlich, *The absolute arithmetic continuum and the unification of all numbers great and small*, *Bulletin of Symbolic Logic* 18 (2012), no. 1, 1–45. doi:10.2178/bsl/1327328438.
- [16] Vladimir Kanovei and Saharon Shelah, *A definable nonstandard model of the reals*, *Journal of Symbolic Logic* 69 (2004), no. 1, 159–164. doi:10.2178/jsl/1080938834. arXiv:math/0311165; the index is defined in Section 1, page 2 of the preprint.
- [17] D. Rocha Rangel and H. L. Mariano, An algebraic (set) theory of surreal numbers, I, 2019. arXiv:1911.12726. Research preprint.
- [18] Joel David Hamkins, *Surreal and ordinal numbers*, author answer, *Mathematics Stack Exchange*, 5 November 2010. <https://math.stackexchange.com/a/9038>.
- [19] Joel David Hamkins, *The global choice principle in Gödel–Bernays set theory*, 3 December 2014. <https://jdh.hamkins.org/the-global-choice-principle-in-godel-bernays-set-theory/>.
- [20] J. D. Hamkins, Answer to “Universal order type”, *MathOverflow*, 6 March 2011, <https://mathoverflow.net/questions/57597/universal-order-type>. An author-given account of the forth construction for surreal universality.
- [21] J. D. Hamkins, “Is the universality of the surreal number line a weak global choice principle?”, *MathOverflow*, 7 January 2016, <https://mathoverflow.net/questions/227849/>. Cited as a historical question, not as a certified 2026 status claim.
- [22] Victoria Gitman and Joel David Hamkins, *Open determinacy for class games*, research paper. arXiv:1509.01099.
- [23] J. D. Hamkins, “Open determinacy for class games”, author’s research overview of joint work with V. Gitman, 4 September 2015. [Author’s source page](#).
- [24] V. Gitman, J. D. Hamkins, P. Holy, P. Schlicht, and K. J. Williams, “The exact strength of the class forcing theorem,” *Journal of Symbolic Logic* **85** (2020), 869–905. arXiv:1707.03700; version 2, 10 March 2021.
- [25] J. D. Hamkins, “The exact strength of the class forcing theorem,” author’s research overview of joint work with V. Gitman, P. Holy, P. Schlicht, and K. J. Williams, 12 July 2017. [Author’s source page](#).
- [26] J. D. Hamkins, “Second-order transfinite recursion is equivalent to Kelley–Morse set theory over GBC,” author’s research exposition, 23 July 2017. [Author’s webpage](#).
- [27] Joel David Hamkins and W. Hugh Woodin, *Open class determinacy is preserved by forcing*, research paper, 2018. arXiv:1806.11180. Especially Sections 2–3 and Theorem 6.
- [28] Carolin Antos and Sy-David Friedman, *Hyperclass forcing in Morse–Kelley class theory*, research paper. arXiv:1510.04082.
- [29] Lean language reference, *Universes*, official documentation, accessed October 2026. <https://lean-lang.org/doc/reference/latest/The-Type-System/Universes/>.
- [30] Mathlib contributors, *Mathlib.SetTheory.Ordinal.Basic*, official API documentation, accessed October 2026. https://leanprover-community.github.io/mathlib4_docs/Mathlib/SetTheory/Ordinal/Basic.html.